

Physlib Monthly Report

1–30 June 2026

Contributors: Aleksei Bykov, Andrea Pari, Bjørn Kjos-Hanssen, Christian Krause, doxtor6, Florian Wiesner, giuseppe.sorge, Gregory Loges, Joseph Tooby-Smith, juanjfndz, Justin Findlay, Lazar Milikic, LibertasSpZ, nateabr, Nicola Bernini, Nikolai Kashcheev, Paul Lezeau, Ranjan Sharma, Robby Sneiderman, Shibashish, Shlok Vaibhav Singh, Tracy McSheery, Vasily Ilin, Wouter Deconinck, Yanting Teng, Yidi Qi, Zhi Kai Pong

Reviewers: Alex Meiburg, Daniel Morrison, doxtor6, Gregory Loges, Joseph Tooby-Smith, Justin Findlay, Matteo Cipollina, Nikolai Kashcheev, Wouter Deconinck, Zhi Kai Pong

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Abstract

Physlib is a community library of formalised physics for the [Lean 4](#) proof assistant: definitions and results from across physics, stated in a formal language and checked by machine. This report is an automatically generated record of one month’s additions to it.

It covers 1–30 June 2026 on branch `master` of [leanprover-community/physlib](#), between commits `d44dd71` and `05930fe`. Over that period 27 contributors landed 154 commits, changing 46,837 lines across 492 files and adding 1,418 new Lean declarations, each catalogued with its statement in Section 2. The complete diff is available at [GitHub compare](#).

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1 Introduction

During Jun 2026, 27 contributors submitted 154 commits that changed 46,837 lines (+28,676 / -18,161) across 492 files spanning 47 top-level areas of the repository. Of those, 86 files were newly added and 5 were removed outright. This report catalogues 1,418 newly added Lean declarations: 923 lemmas, 303 defs, 105 instances, 43 theorems, 28 abbrevs, 13 structures, 1 inductive, 1 informal definition, and 1 informal lemma.

See the [full compare diff](#) and the [list of commits](#) on GitHub for the raw source.

1.1 Contributors

Ordered by lines changed.

Contributor	Lines Changed	Commits
Joseph Tooby-Smith	15,350	21
Vasily Ilin	8,962	1
Zhi Kai Pong	5,540	9
juanjfndz	3,103	6
Gregory Loges	2,913	12
Andrea Pari	2,515	4
Bjørn Kjos-Hanssen	1,993	3
Robby Sneiderman	1,736	15
nateabr	1,433	7
Nikolai Kashcheev	1,308	7
Yidi Qi	1,194	13
LibertasSpZ	806	2
Lazar Milikic	742	5
Nicola Bernini	722	20
Justin Findlay	595	3
doxtor6	450	4
Aleksei Bykov	315	2
giuseppe.sorge	309	7
Shibashish	140	3
Yanting Teng	135	1
Christian Krause	131	2
Shlok Vaibhav Singh	61	1
Florian Wiesner	54	2
Tracy McSheery	33	1
Wouter Deconinck	32	1
Ranjan Sharma	16	1
Paul Lezeau	13	1

1.2 Reviewers

Ordered by pull requests reviewed.

Reviewer	PRs Reviewed
Joseph Tooby-Smith	132
Zhi Kai Pong	11
Daniel Morrison	10

doxtor6	2
Gregory Loges	2
Justin Findlay	1
Nikolai Kashcheev	1
Matteo Cipollina	1
Alex Meiburg	1
Wouter Deconinck	1

1.3 Modified subfolders

Subfolder	Files	New	Deleted	Additions	Deletions
Physlib/Relativity	85	+12	-3	+5,706	-7,342
Physlib/SpaceAndTime	40	+9	—	+3,364	-1,393
Physlib/Mathematics	35	+5	—	+2,642	-1,564
Physlib/Particles	44	+2	—	+2,087	-1,693
PhyslibAlpha/ClassicalFieldTheory	18	+18	—	+3,083	-0
Physlib/QuantumMechanics	33	+4	—	+1,916	-791
Physlib/ClassicalMechanics	15	+1	—	+1,393	-1,014
Physlib/Electromagnetism	31	+1	—	+1,034	-1,107
scripts	9	+4	—	+1,683	-10
PhyslibAlpha/QuantumMechanics	2	+2	—	+1,308	-0
QuantumInfo/ForMathlib	38	—	—	+472	-581
Physlib/StringTheory	4	—	—	+306	-655
PhyslibAlpha/SpaceAndTime	7	+7	—	+903	-0
PhyslibAlpha/Mathematics	1	+1	—	+565	-0
Physlib/QFT	43	—	—	+153	-389
Physlib/StatisticalMechanics	6	—	—	+116	-365
Physlib/Thermodynamics	3	—	—	+81	-330
QuantumInfo/Entropy	4	—	—	+150	-173
Physlib/Units	9	—	—	+173	-128
Physlib/Cosmology	7	+6	—	+264	-1
scripts/PhyslibAlpha	5	+5	—	+242	-0
QuantumInfo/States	6	—	—	+73	-102
.github/workflows	3	+2	-1	+118	-54
scripts/MetaPrograms	12	—	—	+133	-38
QuantumInfo/ResourceTheory	3	—	—	+73	-81
lakefile.lean	1	—	-1	+0	-125
lakefile.toml	1	+1	—	+125	-0
PhyslibAlpha/ClassicalMechanics	1	+1	—	+122	-0
QuantumInfo/Channels	6	—	—	+51	-57
Physlib/CondensedMatter	1	—	—	+21	-71
AGENTS.md	1	+1	—	+85	-0
Physlib/FluidDynamics	1	—	—	+14	-41
lake-manifest.json	1	—	—	+27	-26
Physlib.lean	1	—	—	+39	-4
QuantumInfo/ClassicalInfo	3	—	—	+18	-15
PhyslibAlpha.lean	1	+1	—	+32	-0
AI-POLICY.md	1	+1	—	+31	-0
PhyslibAlpha	1	+1	—	+30	-0
docs	1	—	—	+23	-0

README.md	1	—	—	+7	-1
CITATION.cff	1	+1	—	+7	-0
QuantumInfo/Measurements	1	—	—	+1	-4
CONTRIBUTING.md	1	—	—	+2	-2
QuantumInfo/Operators	1	—	—	+1	-2
lean-toolchain	1	—	—	+1	-1
.codespellignore	1	—	—	+1	-0
Physlib/Meta	1	—	—	+0	-1

1.4 New declarations by kind

Kind	Count
lemma	923
def	303
instance	105
theorem	43
abbrev	28
structure	13
inductive	1
informal_definition	1
informal_lemma	1

2 Details by file

2.1 Physlib.ClassicalMechanics.FreeParticle.Basic

[Physlib/ClassicalMechanics/FreeParticle/Basic.lean](#) on GitHub.

def linearMomentum

[\[source\]](#)

The linear momentum of the free particle along a trajectory.

This is given by the classical one-dimensional expression $p = m v$, where m is the particle mass and v is the velocity.

noncomputable

```
def linearMomentum (s : FreeParticle) (q : Trajectory) (t : Time) : ℝ
```

lemma linearMomentum_conserved_of_velocity_const

[\[source\]](#)

If a free-particle trajectory has constant velocity, then its linear momentum is constant.

```
lemma linearMomentum_conserved_of_velocity_const (s : FreeParticle) (q : Trajectory)
  (h : ∃ v₀, ∀ t, s.velocity q t = v₀) :
  ∃ p, ∀ t, s.linearMomentum q t = p
```

theorem linearMomentum_conserved

[\[source\]](#)

A free particle satisfying the equation of motion conserves linear momentum.

Newton's second law implies that the acceleration vanishes, so the velocity is constant. Since the particle mass is fixed, the linear momentum is constant in time.

```
theorem linearMomentum_conserved (s : FreeParticle) (q : Trajectory)
  (h : ∀ t, s.NewtonsSecondLaw q t) (hcont : ContDiff ℝ 2 q) :
  ∃ p, ∀ t, s.linearMomentum q t = p
```

2.2 Physlib.ClassicalMechanics.HarmonicOscillator.ConfigurationSpace

[Physlib/ClassicalMechanics/HarmonicOscillator/ConfigurationSpace.lean](#) on GitHub.

lemma coe_apply [\[source\]](#)

```
lemma coe_apply (x : ConfigurationSpace) (i : Fin 1) : x i = x.val i
```

instance TopologicalSpace ConfigurationSpace [\[source\]](#)

The topology on configuration space, induced by the chosen coordinate ConfigurationSpace.val into EuclideanSpace ℝ (Fin 1).

```
instance : TopologicalSpace ConfigurationSpace
```

def valEquiv [\[source\]](#)

The coordinate equivalence between configuration space and its EuclideanSpace ℝ (Fin 1) model, given by ConfigurationSpace.val with its wrapper inverse.

```
def valEquiv : ConfigurationSpace ≈ EuclideanSpace ℝ (Fin 1) where
```

instance T2Space ConfigurationSpace [\[source\]](#)

Configuration space is Hausdorff, transported from EuclideanSpace ℝ (Fin 1) across the coordinate homeomorphism.

```
instance : T2Space ConfigurationSpace
```

instance SecondCountableTopology ConfigurationSpace [\[source\]](#)

Configuration space is second countable, transported from EuclideanSpace ℝ (Fin 1) across the coordinate homeomorphism.

```
instance : SecondCountableTopology ConfigurationSpace
```

instance ChartedSpace (EuclideanSpace ℝ (Fin 1)) ConfigurationSpace [\[source\]](#)

The structure of a charted space on ConfigurationSpace, modeled on its EuclideanSpace ℝ (Fin 1) coordinate via the single global chart valHomeomorphism.

```
instance : ChartedSpace (EuclideanSpace ℝ (Fin 1)) ConfigurationSpace where
```

instance IsManifold $\mathcal{I}(\mathbb{R}, \text{EuclideanSpace } \mathbb{R} (\text{Fin } 1)) \omega$ ConfigurationSpace [\[source\]](#)

The structure of a smooth (indeed analytic) manifold on ConfigurationSpace. With a single global chart, the only coordinate change is the chart's self-transition, which is analytic.


```
instance : IsManifold  $\mathcal{I}(\mathbb{R}, \text{EuclideanSpace } \mathbb{R} \text{ (Fin 1)}) \omega \text{ ConfigurationSpace where}$ 
```

2.3 Physlib.ClassicalMechanics.HarmonicOscillator.Solution

[Physlib/ClassicalMechanics/HarmonicOscillator/Solution.lean](#) on GitHub.

def InitialConditionsFromTwoPositions.toInitialConditions

[\[source\]](#)

Convert two-position initial conditions to standard initial conditions at $t = 0$.

Obtained by solving the 2×2 linear system from the trajectory formula at t_1 and t_2 .

*See `toInitialConditions_trajectory_at_t1` and `toInitialConditions_trajectory_at_t2` in section D.2 for the correctness proofs (valid under $\sin(S.\omega * (t_2 - t_1)) \neq 0$).*

```
noncomputable def toInitialConditions (S : HarmonicOscillator)
  (IC : InitialConditionsFromTwoPositions) : InitialConditions where
```

def InitialConditionsFromTwoVelocities.toInitialConditions

[\[source\]](#)

Convert two-velocity initial conditions to standard initial conditions at $t = 0$.

Obtained by solving the 2×2 linear system from the velocity formula at t_1 and t_2 .

*See `toInitialConditions_velocity_at_t1` and `toInitialConditions_velocity_at_t2` in section D.3 for the correctness proofs (valid under $\sin(S.\omega * (t_2 - t_1)) \neq 0$).*

```
noncomputable def toInitialConditions (S : HarmonicOscillator)
  (IC : InitialConditionsFromTwoVelocities) : InitialConditions where
```

lemma toInitialConditions_trajectory_at_t1

[\[source\]](#)

*The trajectory from `toInitialConditions` passes through x_{t_1} at time t_1 , provided $\sin(S.\omega * (t_2 - t_1)) \neq 0$.*

```
lemma toInitialConditions_trajectory_at_t1 (S : HarmonicOscillator)
  (IC : InitialConditionsFromTwoPositions)
  (hΔ : sin (S.ω * (IC.t2 - IC.t1)) ≠ 0) :
  (IC.toInitialConditions S).trajectory S IC.t1 = IC.x_t1
```

lemma toInitialConditions_trajectory_at_t2

[\[source\]](#)

*The trajectory from `toInitialConditions` passes through x_{t_2} at time t_2 , provided $\sin(S.\omega * (t_2 - t_1)) \neq 0$.*

```
lemma toInitialConditions_trajectory_at_t2 (S : HarmonicOscillator)
  (IC : InitialConditionsFromTwoPositions)
  (hΔ : sin (S.ω * (IC.t2 - IC.t1)) ≠ 0) :
  (IC.toInitialConditions S).trajectory S IC.t2 = IC.x_t2
```

lemma toInitialConditions_velocity_at_t1

[\[source\]](#)

*The trajectory from `toInitialConditions` has velocity v_{t_1} at time t_1 , provided $\sin(S.\omega * (t_2 - t_1)) \neq 0$.*

```

lemma toInitialConditions_velocity_at_t1 (S : HarmonicOscillator)
  (IC : InitialConditionsFromTwoVelocities)
  (hΔ : sin (S.ω * (IC.t2 - IC.t1)) ≠ 0) :
  ∂t ((IC.toInitialConditions S).trajectory S) IC.t1 = IC.v_t1

```

lemma toInitialConditions_velocity_at_t2

[\[source\]](#)

*The trajectory from toInitialConditions has velocity v_{t_2} at time t_2 , provided $\sin (S.\omega * (t_2 - t_1)) \neq 0$.*

```

lemma toInitialConditions_velocity_at_t2 (S : HarmonicOscillator)
  (IC : InitialConditionsFromTwoVelocities)
  (hΔ : sin (S.ω * (IC.t2 - IC.t1)) ≠ 0) :
  ∂t ((IC.toInitialConditions S).trajectory S) IC.t2 = IC.v_t2

```

def AmplitudePhase.toInitialConditions

[\[source\]](#)

Convert amplitude–phase initial conditions to standard initial conditions at $t = 0$, via $x_0 = A \cos \varphi$ and $v_0 = A \omega \sin \varphi$.

See toInitialConditions_trajectory_eq_cos and toInitialConditions_velocity_eq_sin in section E.3 for the correctness proofs.

```

noncomputable def toInitialConditions (S : HarmonicOscillator) (IC : AmplitudePhase) :
  InitialConditions where

```

lemma toInitialConditions_trajectory_eq_cos

[\[source\]](#)

The trajectory of amplitude–phase initial conditions is the cosine normal form $x(t) = A \cos (\omega t - \varphi)$.

```

lemma toInitialConditions_trajectory_eq_cos (S : HarmonicOscillator) (IC : AmplitudePhase)
  (t : Time) :
  (IC.toInitialConditions S).trajectory S t
  = EuclideanSpace.single 0 (IC.A * cos (S.ω * t - IC.φ))

```

lemma toInitialConditions_velocity_eq_sin

[\[source\]](#)

The velocity of the amplitude–phase trajectory is $v(t) = -A \omega \sin (\omega t - \varphi)$.

```

lemma toInitialConditions_velocity_eq_sin (S : HarmonicOscillator) (IC : AmplitudePhase)
  (t : Time) :
  ∂t ((IC.toInitialConditions S).trajectory S) t
  = EuclideanSpace.single 0 (-(IC.A * S.ω * sin (S.ω * t.val - IC.φ)))

```

def fromInitialConditions

[\[source\]](#)

Recover amplitude–phase data from standard initial conditions, as the polar coordinates of the phase vector $(x_0, v_0 / \omega)$ embedded as $z = x_0 + (v_0 / \omega) i$: the amplitude is $\|z\|$ and the phase is $\text{Complex.arg } z$.

See toInitialConditions_fromInitialConditions for the right-inverse identity.

```

noncomputable def fromInitialConditions (S : HarmonicOscillator) (IC : InitialConditions) :
  AmplitudePhase where

```

lemma toInitialConditions_fromInitialConditions[\[source\]](#)

fromInitialConditions is a right inverse of toInitialConditions: converting initial conditions to amplitude-phase form and back recovers them exactly.

```
lemma toInitialConditions_fromInitialConditions (S : HarmonicOscillator)
  (IC : InitialConditions) :
  (fromInitialConditions S IC).toInitialConditions S = IC
```

lemma trajectory_eq_cos[\[source\]](#)

Every trajectory of the harmonic oscillator is a single shifted cosine after converting its initial conditions to amplitude-phase form.

```
lemma trajectory_eq_cos (IC : InitialConditions) (t : Time) :
  IC.trajectory S t =
    EuclideanSpace.single 0 ((AmplitudePhase.fromInitialConditions S IC).A *
      cos (S.ω * t - (AmplitudePhase.fromInitialConditions S IC).φ))
```

lemma trajectory_velocity_eq_sin[\[source\]](#)

The velocity of every trajectory is the corresponding shifted sine in amplitude-phase form.

```
lemma trajectory_velocity_eq_sin (IC : InitialConditions) (t : Time) :
  ∂t (IC.trajectory S) t =
    EuclideanSpace.single 0 (-((AmplitudePhase.fromInitialConditions S IC).A * S.ω *
      sin (S.ω * t.val - (AmplitudePhase.fromInitialConditions S IC).φ)))
```

lemma trajectory_velocity_eq_zero_iff_sin_eq_zero[\[source\]](#)

For nonzero amplitude, the velocity vanishes exactly when the sine factor in amplitude-phase form vanishes.

```
lemma trajectory_velocity_eq_zero_iff_sin_eq_zero (IC : InitialConditions)
  (hA : (AmplitudePhase.fromInitialConditions S IC).A ≠ 0) (t : Time) :
  ∂t (IC.trajectory S) t = 0 ↔
    sin (S.ω * t.val - (AmplitudePhase.fromInitialConditions S IC).φ) = 0
```

lemma trajectory_velocity_eq_zero_iff_exists_int[\[source\]](#)

For nonzero amplitude, the velocity is zero exactly at phase times $\phi + n\pi$.

```
lemma trajectory_velocity_eq_zero_iff_exists_int (IC : InitialConditions)
  (hA : (AmplitudePhase.fromInitialConditions S IC).A ≠ 0) (t : Time) :
  ∂t (IC.trajectory S) t = 0 ↔
    ∃ n : ℤ,
      (t : ℝ) =
        ((AmplitudePhase.fromInitialConditions S IC).φ + n * π) / S.ω
```

lemma trajectory_velocity_eq_zero_iff_norm_eq_amplitude[\[source\]](#)

The velocity vanishes exactly when the trajectory has norm equal to the amplitude.

```
lemma trajectory_velocity_eq_zero_iff_norm_eq_amplitude (IC : InitialConditions)
  (t : Time) :
```

```

∂ₜ (IC.trajectory S) t = 0 ↔
||IC.trajectory S t|| = (AmplitudePhase.fromInitialConditions S IC).A

```

lemma trajectory_eq_zero_iff_cos_eq_zero

[\[source\]](#)

For nonzero amplitude, the trajectory passes through zero exactly when the cosine factor in amplitude-phase form vanishes.

```

lemma trajectory_eq_zero_iff_cos_eq_zero (IC : InitialConditions)
(hA : (AmplitudePhase.fromInitialConditions S IC).A ≠ 0) (t : Time) :
IC.trajectory S t = 0 ↔
cos (S.ω * t.val - (AmplitudePhase.fromInitialConditions S IC).φ) = 0

```

lemma trajectory_eq_zero_iff_exists_int

[\[source\]](#)

For nonzero amplitude, the trajectory passes through zero exactly at phase times $\varphi + (2n + 1)\pi / 2$.

```

lemma trajectory_eq_zero_iff_exists_int (IC : InitialConditions)
(hA : (AmplitudePhase.fromInitialConditions S IC).A ≠ 0) (t : Time) :
IC.trajectory S t = 0 ↔
∃ n : ℤ,
(t : ℝ) =
((AmplitudePhase.fromInitialConditions S IC).φ + (2 * n + 1) * π / 2) / S.ω

```

lemma return_time

[\[source\]](#)

Assuming that the initial coordinate and velocity are not simultaneously zero, the time stamps when the harmonic oscillator returns to its initial coordinate and velocity is a multiple of its period

```

lemma return_time (IC : InitialConditions) (non_trivial : IC.x₀ ≠ 0 ∨ IC.v₀ ≠ 0)
(t : Time) (ht : IC.trajectory S t = IC.x₀ ∧ ∂ₜ (IC.trajectory S) t = IC.v₀) :
∃ n : ℤ, (n : ℝ) * (T S) = t

```

2.4 Physlib.ClassicalMechanics.Lagrangian.TotalDerivativeEquivalence

[Physlib/ClassicalMechanics/Lagrangian/TotalDerivativeEquivalence.lean](#) on GitHub.

lemma isTotalTimeDerivative_explicit

[\[source\]](#)

Explicit reformulation (by the chain rule): $\delta L(t, q, d_t q) = \partial F / \partial t(t, q) + \langle \nabla_r F(t, q), d_t q \rangle$

or

$\delta L(t, q, d_t q) = fderiv \mathbb{R} F(t, q)(1, d_t q)$

```

lemma isTotalTimeDerivative_explicit {δL : Time → X → X → ℝ} :
IsTotalTimeDerivative δL ↔ (∃ (F : Time → X → ℝ) (_ : ContDiff ℝ ∞ 1 F),
∀ t q v, δL t q v = fderiv ℝ 1 F(t, q)((1 : Time), v))

```

2.5 Physlib.Electromagnetism.Distributional.Basic

[Physlib/Electromagnetism/Distributional/Basic.lean](#) on GitHub.

def ofStaticScalarPotentialFunction[\[source\]](#)

The creation of an electromagnetic potential from a static scalar-potential function.

If the function is distribution-bounded, it is first promoted to a distribution using `Space.distOfFunction`. Otherwise the constructor returns zero.

```
noncomputable def ofStaticScalarPotentialFunction {d} (c : SpeedOfLight)
  (φ : Space d → ℝ) : DistElectromagneticPotential d
```

lemma ofStaticScalarPotentialFunction_of_isDistBounded[\[source\]](#)

```
@[simp]
lemma ofStaticScalarPotentialFunction_of_isDistBounded {d} (c : SpeedOfLight)
  (φ : Space d → ℝ) (hφ : Space.IsDistBounded φ) :
  ofStaticScalarPotentialFunction c φ =
    ofStaticScalarPotential c (Space.distOfFunction φ hφ)
```

lemma ofStaticScalarPotentialFunction_eq_zero_of_not_isDistBounded[\[source\]](#)

```
@[simp]
lemma ofStaticScalarPotentialFunction_eq_zero_of_not_isDistBounded {d}
  (c : SpeedOfLight) (φ : Space d → ℝ) (hφ : ¬ Space.IsDistBounded φ) :
  ofStaticScalarPotentialFunction c φ = 0
```

def ofVectorPotential[\[source\]](#)

The creation of an electromagnetic potential from a vector potential.

```
noncomputable def ofVectorPotential {d} (c : SpeedOfLight) :
  ((Time × Space d) → d[ℝ] EuclideanSpace ℝ (Fin d)) →l [ℝ]
  DistElectromagneticPotential d where
```

def ofStaticVectorPotential[\[source\]](#)

The creation of an electromagnetic potential from a static vector potential.

```
noncomputable def ofStaticVectorPotential {d} (c : SpeedOfLight) :
  ((Space d) → d[ℝ] EuclideanSpace ℝ (Fin d)) →l [ℝ] DistElectromagneticPotential d
```

lemma distTensorDeriv_eq_sum_sum[\[source\]](#)

```
lemma distTensorDeriv_eq_sum_sum {d} (A : DistElectromagneticPotential d)
  (ε : S(SpaceTime d, ℝ)) :
  distTensorDeriv A ε = ∑ μ, ∑ ν, (SpaceTime.distDeriv μ A ε ν) •
    Lorentz.CoVector.basis μ ⊗t [ℝ] Lorentz.Vector.basis ν
```

lemma distTensorDeriv_basis_repr_apply[\[source\]](#)

```
@[simp]
lemma distTensorDeriv_basis_repr_apply {d} {μν : (Fin 1 ⊕ Fin d) × (Fin 1 ⊕ Fin d)}
  (A : DistElectromagneticPotential d)
  (ε : S(SpaceTime d, ℝ)) :
  (Lorentz.CoVector.basis.tensorProduct Lorentz.Vector.basis).repr (distTensorDeriv A ε) μν =
    distDeriv μν.1 A ε μν.2
```

lemma toTensor_distTensorDeriv_basis_repr_apply[\[source\]](#)

```

lemma toTensor_distTensorDeriv_basis_repr_apply {d} (A : DistElectromagneticPotential d)
  (ε :  $\mathcal{S}(\text{SpaceTime } d, \mathbb{R})$ ) (b : ComponentIdx (S := realLorentzTensor d)
    (Fin.append ![Color.down] ![Color.up])) :
  (Tensor.basis _).repr (Tensorial.toTensor (distTensorDeriv A ε)) b =
    distDeriv (b 0) A ε (b 1)

```

2.6 Physlib.Electromagnetism.Kinematics.EMPotential

[Physlib/Electromagnetism/Kinematics/EMPotential.lean](#) on GitHub.

instance Zero (ElectromagneticPotential d)[\[source\]](#)

```

instance {d} : Zero (ElectromagneticPotential d) where

```

lemma zero_val[\[source\]](#)

```

@[simp]
lemma zero_val {d} : (0 : ElectromagneticPotential d).val = 0

```

lemma zero_apply[\[source\]](#)

```

@[simp]
lemma zero_apply {d} (x : SpaceTime d) : (0 : ElectromagneticPotential d) x = 0

```

instance Neg (ElectromagneticPotential d)[\[source\]](#)

```

instance {d} : Neg (ElectromagneticPotential d) where

```

lemma neg_val[\[source\]](#)

```

@[simp]
lemma neg_val {d} (A : ElectromagneticPotential d) :
  (- A).val = - A.val

```

lemma neg_apply[\[source\]](#)

```

@[simp]
lemma neg_apply {d} (A : ElectromagneticPotential d) (x : SpaceTime d) :
  (- A) x = - A x

```

instance Sub (ElectromagneticPotential d)[\[source\]](#)

```

instance {d} : Sub (ElectromagneticPotential d) where

```

lemma sub_val[\[source\]](#)

```

@[simp]
lemma sub_val {d} (A B : ElectromagneticPotential d) :
  (A - B).val = A.val - B.val

```

lemma sub_apply[\[source\]](#)

```
@[simp]
lemma sub_apply {d} (A B : ElectromagneticPotential d) (x : SpaceTime d) :
  (A - B) x = A x - B x
```

instance AddCommGroup (ElectromagneticPotential d)[\[source\]](#)

```
instance {d} : AddCommGroup (ElectromagneticPotential d) where
```

lemma action_apply[\[source\]](#)

```
@[simp]
lemma action_apply {d} (Λ : LorentzGroup d) (A : ElectromagneticPotential d)
  (x : SpaceTime d) :
  (Λ • A) x = Λ • A (Λ-1 • x)
```

instance DistribMulAction (LorentzGroup d) (ElectromagneticPotential d)[\[source\]](#)

```
noncomputable instance {d} :
  DistribMulAction (LorentzGroup d) (ElectromagneticPotential d) where
```

2.7 Physlib.Electromagnetism.Kinematics.FieldStrength

[Physlib/Electromagnetism/Kinematics/FieldStrength.lean](#) on GitHub.

lemma toFieldStrength_eval_eq_basis_repr[\[source\]](#)

Evaluating both tensor indices of the field strength gives the coefficient in the standard tensor-product basis.

```
lemma toFieldStrength_eval_eq_basis_repr {d} (A : ElectromagneticPotential d)
  (x : SpaceTime d) (μ ν : Fin 1 ⊗ Fin d) :
  toField {A.toFieldStrength x | [μ] [ν]}T =
  (Lorentz.CoVector.basis.tensorProduct Lorentz.Vector.basis).repr
  (A.toFieldStrength x) (μ, ν)
```

lemma toFieldStrength_eval_apply[\[source\]](#)

The evaluated components of the field strength tensor in terms of derivatives of the electromagnetic potential.

```
lemma toFieldStrength_eval_apply {d} (A : ElectromagneticPotential d)
  (x : SpaceTime d) (μ ν : Fin 1 ⊗ Fin d) :
  toField {A.toFieldStrength x | [μ] [ν]}T =
  ∑ κ, (η μ κ * ∂_κ A x ν - η ν κ * ∂_κ A x μ)
```

lemma toFieldStrength_eval_apply_eq_single[\[source\]](#)

The evaluated components of the field strength tensor after using diagonal form of the Minkowski metric.

```
lemma toFieldStrength_eval_apply_eq_single {d} (A : ElectromagneticPotential d)
  (x : SpaceTime d) (μ ν : Fin 1 ⊗ Fin d) :
```

$$\text{toField } \{A.\text{toFieldStrength } x \mid [\mu] \text{ } [\nu]\}^T =$$

$$\eta_{\mu\mu} * \partial_{\mu} A \times \nu - \eta_{\nu\nu} * \partial_{\nu} A \times \mu$$

lemma toFieldStrength_eval_eq_fieldStrengthMatrix

[source]

Index evaluation of the field strength tensor agrees with the corresponding component of the field strength matrix.

```
lemma toFieldStrength_eval_eq_fieldStrengthMatrix {d} (A : ElectromagneticPotential d)
  (x : SpaceTime d) (μ ν : Fin 1 ⊕ Fin d) :
  toField {A.toFieldStrength x | [μ] [ν]}T = A.fieldStrengthMatrix x (μ, ν)
```

2.8 Physlib.Electromagnetism.Kinematics.GaugeTransformation

[Physlib/Electromagnetism/Kinematics/GaugeTransformation.lean](#) on GitHub.

def ofGradient

[source]

*The pure-gauge electromagnetic potential $A^\mu = \partial^\mu \chi = \eta^{\{\mu\nu\}} \partial_\nu \chi$ built from a gauge function χ . The metric-contracted form $\sum \kappa, \eta_{\mu\kappa} * \partial_\kappa \chi$ is the definition; the diagonal form (which equals $\eta_{\mu\mu} * \partial_\mu \chi$ since η is diagonal) is derived in `ofGradient_apply`.*

```
noncomputable def ofGradient {d} (χ : SpaceTime d → ℝ) : ElectromagneticPotential d where
```

lemma ofGradient_apply_sum

[source]

Unfolding of the summed definition of `ofGradient`.

```
lemma ofGradient_apply_sum {d} (χ : SpaceTime d → ℝ) (x : SpaceTime d) (μ : Fin 1 ⊕ Fin d) :
  ofGradient χ x μ = ∑ κ, η_{μκ} * ∂_κ χ x
```

lemma ofGradient_apply

[source]

Evaluation of `ofGradient` in the diagonal form; the off-diagonal entries of η vanish so only the $\kappa = \mu$ term survives.

```
lemma ofGradient_apply {d} (χ : SpaceTime d → ℝ) (x : SpaceTime d) (μ : Fin 1 ⊕ Fin d) :
  ofGradient χ x μ = η_{μμ} * ∂_μ χ x
```

lemma ofGradient_zero

[source]

The pure-gauge potential built from the zero gauge function has all components zero.

```
lemma ofGradient_zero {d} (x : SpaceTime d) (μ : Fin 1 ⊕ Fin d) :
  ofGradient (0 : SpaceTime d → ℝ) x μ = 0
```

lemma ofGradient_add

[source]

`ofGradient` is additive in the gauge function (when both summands are differentiable).

```
lemma ofGradient_add {d} {χ₁ χ₂ : SpaceTime d → ℝ}
  (hχ₁ : Differentiable ℝ χ₁) (hχ₂ : Differentiable ℝ χ₂) :
  ofGradient (χ₁ + χ₂) = ofGradient χ₁ + ofGradient χ₂
```


lemma differentiable_ofGradient[\[source\]](#)

The pure-gauge potential is differentiable when χ is C^2 .

```
lemma differentiable_ofGradient {d} {χ : SpaceTime d → ℝ} (hχ : ContDiff ℝ 2 χ) :
  Differentiable ℝ (ofGradient χ)
```

lemma contDiff_ofGradient[\[source\]](#)

The pure-gauge potential is C^n when χ is C^{n+1} .

```
lemma contDiff_ofGradient {n} {d} {χ : SpaceTime d → ℝ} (hχ : ContDiff ℝ (n + 1) χ) :
  ContDiff ℝ n (ofGradient χ)
```

lemma toFieldStrength_ofGradient[\[source\]](#)

A pure-gauge potential has vanishing field strength.

```
lemma toFieldStrength_ofGradient {d} {χ : SpaceTime d → ℝ} (hχ : ContDiff ℝ 2 χ)
  (x : SpaceTime d) : (ofGradient χ).toFieldStrength x = 0
```

lemma ofGradient_equivariant[\[source\]](#)

ofGradient intertwines the Lorentz action on potentials with composition by Λ^{-1} on the gauge function: $\Lambda \cdot \text{ofGradient } \chi = \text{ofGradient } (\chi \circ (\Lambda^{-1} \cdot \cdot))$.

```
lemma ofGradient_equivariant {d} (χ : SpaceTime d → ℝ) (hχ : Differentiable ℝ χ)
  (Λ : LorentzGroup d) :
  Λ • ofGradient χ = ofGradient (χ ∘ (Λ⁻¹ • ·))
```

def gaugeTransform[\[source\]](#)

The gauge transformation $A^\mu \mapsto A^\mu + \partial^\mu \chi$ of an electromagnetic potential.

```
noncomputable def gaugeTransform {d} (χ : SpaceTime d → ℝ) (A : ElectromagneticPotential d) :
  ElectromagneticPotential d
```

lemma gaugeTransform_apply[\[source\]](#)

Evaluation of gaugeTransform.

```
lemma gaugeTransform_apply {d} (χ : SpaceTime d → ℝ) (A : ElectromagneticPotential d)
  (x : SpaceTime d) : gaugeTransform χ A x = A x + ofGradient χ x
```

lemma toFieldStrength_gaugeTransform[\[source\]](#)

The field strength tensor is invariant under gauge transformations.

```
lemma toFieldStrength_gaugeTransform {d} (A : ElectromagneticPotential d)
  (χ : SpaceTime d → ℝ) (hA : Differentiable ℝ A) (hχ : ContDiff ℝ 2 χ) (x : SpaceTime d) :
  (gaugeTransform χ A).toFieldStrength x = A.toFieldStrength x
```

lemma fieldStrengthMatrix_gaugeTransform[\[source\]](#)

The field strength matrix is invariant under gauge transformations.

```
lemma fieldStrengthMatrix_gaugeTransform {d} (A : ElectromagneticPotential d)
  (χ : SpaceTime d → ℝ) (hA : Differentiable ℝ A) (hχ : ContDiff ℝ 2 χ) (x : SpaceTime d) :
  (gaugeTransform χ A).fieldStrengthMatrix x = A.fieldStrengthMatrix x
```

lemma gaugeTransform_zero[\[source\]](#)

Shifting by the zero gauge function is the identity.

```
lemma gaugeTransform_zero {d} (A : ElectromagneticPotential d) :
  gaugeTransform (0 : SpaceTime d → ℝ) A = A
```

lemma gaugeTransform_gaugeTransform[\[source\]](#)

Two successive gauge shifts compose: shifting by χ_2 then χ_1 equals shifting by $\chi_1 + \chi_2$. This upgrades one-step F -invariance to invariance along any finite chain of gauge shifts.

```
lemma gaugeTransform_gaugeTransform {d} (A : ElectromagneticPotential d)
  (χ₁ χ₂ : SpaceTime d → ℝ) (hχ₁ : Differentiable ℝ χ₁) (hχ₂ : Differentiable ℝ χ₂) :
  gaugeTransform χ₁ (gaugeTransform χ₂ A) = gaugeTransform (χ₁ + χ₂) A
```

lemma gaugeTransform_equivariant[\[source\]](#)

Gauge transformations commute with Lorentz transformations: applying Λ and then performing a gauge transformation by χ equals performing a gauge transformation by $\chi \circ (\Lambda^{-1} \bullet \cdot)$ and then applying Λ .

```
lemma gaugeTransform_equivariant {d} (A : ElectromagneticPotential d)
  (χ : SpaceTime d → ℝ) (hχ : Differentiable ℝ χ) (Λ : LorentzGroup d) :
  Λ • gaugeTransform χ A = gaugeTransform (χ ∘ (Λ⁻¹ • ·)) (Λ • A)
```

lemma fieldStrengthMatrix_bareGradient_inl_inr[\[source\]](#)

The $(inl\ 0, inr\ i)$ component of the field strength matrix of the bare-gradient potential $B^\mu := \partial_\mu \chi$ for $\chi(x) = x^0 \cdot x^i$ equals 2. This witnesses that the bare covariant gradient does not produce a gauge-invariant field strength, so the raised-index contraction $\eta^{\mu\nu} \partial_\nu \chi$ in `ofGradient` is necessary (see the module overview).

```
lemma fieldStrengthMatrix_bareGradient_inl_inr {d : ℕ} (i : Fin d)
  (x : SpaceTime d) :
  let χ : SpaceTime d → ℝ
```

lemma toFieldStrength_bareGradient_ne_zero[\[source\]](#)

The field strength of the bare-gradient potential $B^\mu := \partial_\mu \chi$ for $\chi(x) = x^0 \cdot x^i$ is nonzero (follows from `fieldStrengthMatrix_bareGradient_inl_inr`).

```
lemma toFieldStrength_bareGradient_ne_zero {d : ℕ} (i : Fin d)
  (x : SpaceTime d) :
  let χ : SpaceTime d → ℝ
```

2.9 Physlib.Electromagnetism.Kinematics.MagneticField

[Physlib/Electromagnetism/Kinematics/MagneticField.lean](#) on GitHub.

lemma magneticField_coord_eq_fieldStrengthMatrix

[\[source\]](#)

```
lemma magneticField_coord_eq_fieldStrengthMatrix {i : Fin 3} {c : SpeedOfLight}
  (A : ElectromagneticPotential) (t : Time)
  (x : Space) (hA : Differentiable ℝ A) :
  A.magneticField c t x i =
  - A.fieldStrengthMatrix ((toTimeAndSpace c).symm (t, x)) (Sum.inr (i+1), Sum.inr (i+2))
```

2.10 Physlib.Mathematics.Calculus.Wirtinger.Basic

[Physlib/Mathematics/Calculus/Wirtinger/Basic.lean](#) on GitHub.

def weightedDirDeriv

[\[source\]](#)

The base-point field $p \mapsto (1/2)(d_{\{b_1\}} f + c \cdot d_{\{b_2\}} f)$, a weighted combination of the real Fréchet derivative of f along two directions b_1, b_2 . The directional Wirtinger operators are its two specializations: $dWirtingerDir f v$ at $c = -i$, $(b_1, b_2) = (v, i \cdot v)$, and $dWirtingerAntiDir f v$ at $c = i$. Keeping b_1, b_2 free lets $\S G$ differentiate it a second time once and reuse the bridge for both operators.

```
def weightedDirDeriv (f : V → ℂ) (c : ℂ) (b1 b2 : V) : V → ℂ
```

def dWirtingerDir

[\[source\]](#)

The holomorphic directional Wirtinger derivative $\partial_v f = (1/2)(d_v f - i \cdot d_{\{i \cdot v\}} f)$ of $f : V \rightarrow \mathbb{C}$ along the direction vector $v : V$, the `weightedDirDeriv` at $c = -i$.

```
def dWirtingerDir (f : V → ℂ) (v u : V) : ℂ
```

def dWirtingerAntiDir

[\[source\]](#)

The anti-holomorphic directional Wirtinger derivative $\partial^-_v f = (1/2)(d_v f + i \cdot d_{\{i \cdot v\}} f)$ of $f : V \rightarrow \mathbb{C}$ along the direction vector $v : V$, the `weightedDirDeriv` at $c = i$.

```
def dWirtingerAntiDir (f : V → ℂ) (v u : V) : ℂ
```

lemma dWirtingerDir_apply

[\[source\]](#)

Definitional unfolding of `dWirtingerDir` to the explicit Wirtinger combination, used to expand the outer operator of a composition without touching the inner one.

```
lemma dWirtingerDir_apply (g : V → ℂ) (v u : V) :
  dWirtingerDir g v u
  = (1 / 2 : ℂ) * (fderiv ℝ g u v - Complex.I * fderiv ℝ g u (Complex.I • v))
```

lemma dWirtingerAntiDir_apply

[\[source\]](#)

Definitional unfolding of `dWirtingerAntiDir` to the explicit Wirtinger combination.

```

lemma dWirtingerAntiDir_apply (g : V → ℂ) (v u : V) :
  dWirtingerAntiDir g v u
    = (1 / 2 : ℂ) * (fderiv ℝ g u v + Complex.I * fderiv ℝ g u (Complex.I • v))

```

lemma dWirtingerDir_add

[\[source\]](#)

Additivity of dWirtingerDir, $\partial_v(g + h) = \partial_v g + \partial_v h$.

```

lemma dWirtingerDir_add {g h : V → ℂ} (hg : DifferentiableAt ℝ g u)
  (hh : DifferentiableAt ℝ h u) (v : V) :
  dWirtingerDir (g + h) v u = dWirtingerDir g v u + dWirtingerDir h v u

```

lemma dWirtingerAntiDir_add

[\[source\]](#)

Additivity of dWirtingerAntiDir, $\partial^-_v(g + h) = \partial^-_v g + \partial^-_v h$.

```

lemma dWirtingerAntiDir_add {g h : V → ℂ} (hg : DifferentiableAt ℝ g u)
  (hh : DifferentiableAt ℝ h u) (v : V) :
  dWirtingerAntiDir (g + h) v u = dWirtingerAntiDir g v u + dWirtingerAntiDir h v u

```

lemma dWirtingerDir_smul

[\[source\]](#)

Compatibility of dWirtingerDir with complex scalar multiplication, $\partial_v(c \cdot g) = c \cdot \partial_v g$.

```

lemma dWirtingerDir_smul (c : ℂ) {g : V → ℂ} (hg : DifferentiableAt ℝ g u) (v : V) :
  dWirtingerDir (c • g) v u = c • dWirtingerDir g v u

```

lemma dWirtingerAntiDir_smul

[\[source\]](#)

Compatibility of dWirtingerAntiDir with complex scalar multiplication, $\partial^-_v(c \cdot g) = c \cdot \partial^-_v g$.

```

lemma dWirtingerAntiDir_smul (c : ℂ) {g : V → ℂ} (hg : DifferentiableAt ℝ g u) (v : V) :
  dWirtingerAntiDir (c • g) v u = c • dWirtingerAntiDir g v u

```

lemma fderiv_mul_apply

[\[source\]](#)

The real Fréchet derivative of a product, evaluated at a tangent v.

```

private lemma fderiv_mul_apply {g h : V → ℂ} (hg : DifferentiableAt ℝ g u)
  (hh : DifferentiableAt ℝ h u) (v : V) :
  fderiv ℝ (g * h) u v = g u * fderiv ℝ h u v + h u * fderiv ℝ g u v

```

lemma dWirtingerDir_mul

[\[source\]](#)

The Wirtinger Leibniz rule for dWirtingerDir, $\partial_v(g \cdot h) = \partial_v g \cdot h + g \cdot \partial_v h$.

```

lemma dWirtingerDir_mul {g h : V → ℂ} (hg : DifferentiableAt ℝ g u)
  (hh : DifferentiableAt ℝ h u) (v : V) :
  dWirtingerDir (g * h) v u = dWirtingerDir g v u * h u + g u * dWirtingerDir h v u

```

lemma dWirtingerAntiDir_mul[\[source\]](#)

The Wirtinger Leibniz rule for $dWirtingerAntiDir$, $\partial^-_v(g \cdot h) = \partial^-_v g \cdot h + g \cdot \partial^-_v h$.

```
lemma dWirtingerAntiDir_mul {g h : V → ℂ} (hg : DifferentiableAt ℝ g u)
  (hh : DifferentiableAt ℝ h u) (v : V) :
  dWirtingerAntiDir (g * h) v u =
    dWirtingerAntiDir g v u * h u + g u * dWirtingerAntiDir h v u
```

lemma dWirtingerDir_fun_sum[\[source\]](#)

Finite-sum rule for $dWirtingerDir$, $\partial^-_v(\sum_a F_a) = \sum_a \partial^-_v F_a$.

```
lemma dWirtingerDir_fun_sum {α : Type*} {s : Finset α} {F : α → V → ℂ}
  (hF : ∀ a ∈ s, DifferentiableAt ℝ (F a) u) (v : V) :
  dWirtingerDir (fun p => ∑ a ∈ s, F a p) v u = ∑ a ∈ s, dWirtingerDir (F a) v u
```

lemma dWirtingerAntiDir_fun_sum[\[source\]](#)

Finite-sum rule for $dWirtingerAntiDir$, $\partial^-_v(\sum_a F_a) = \sum_a \partial^-_v F_a$.

```
lemma dWirtingerAntiDir_fun_sum {α : Type*} {s : Finset α} {F : α → V → ℂ}
  (hF : ∀ a ∈ s, DifferentiableAt ℝ (F a) u) (v : V) :
  dWirtingerAntiDir (fun p => ∑ a ∈ s, F a p) v u = ∑ a ∈ s, dWirtingerAntiDir (F a) v u
```

lemma fderiv_star_eq[\[source\]](#)

Differentiation commutes with conjugation: the real Fréchet derivative of the pointwise conjugate $p \mapsto \text{star } (f p)$ is conjCLE (conjugation on $\mathbb{C}$) composed with $fderiv \mathbb{R} f u$; in physicists' notation, $d f^- = \text{conj}(d f)$. Conjugation is $\mathbb{R}$ -linear, so it slides through the real derivative unchanged, whereas it does *not* commute with the holomorphic Wirtinger derivative ∂^-_v . The star conjugates the *output* $f p$, so this is not a derivative in a conjugate variable.

This is the analytic core of the conjugation lemmas below ($dWirtingerDir_star_comp$ and its dual): distributed over the Wirtinger split of $fderiv \mathbb{R} f u$ ($\text{realLinear_apply_eq_wirtinger}$, §D), the outer conjCLE conjugates the two coefficients and swaps the holomorphic and anti-holomorphic parts.

```
lemma fderiv_star_eq {E : Type*} [NormedAddCommGroup E] [NormedSpace ℝ E]
  {f : E → ℂ} {u : E} (hf : DifferentiableAt ℝ f u) :
  fderiv ℝ (fun p : E => star (f p)) u =
    Complex.conjCLE.toContinuousLinearMap.comp (fderiv ℝ f u)
```

lemma dWirtingerDir_star_comp[\[source\]](#)

Conjugating the function swaps the operators up to an outer conjugation: $\partial^-_v f^- = \text{conj } (\partial^-_v f)$.

```
lemma dWirtingerDir_star_comp (hf : DifferentiableAt ℝ f u) (v : V) :
  dWirtingerDir (fun p => star (f p)) v u = star (dWirtingerAntiDir f v u)
```

lemma dWirtingerAntiDir_star_comp[\[source\]](#)

Conjugating the function swaps the operators up to an outer conjugation: $\partial^-_v f^- = \text{conj } (\partial^-_v f)$. Dual of `dWirtingerDir_star_comp`.

```
lemma dWirtingerAntiDir_star_comp (hf : DifferentiableAt ℝ f u) (v : V) :
  dWirtingerAntiDir (fun p => star (f p)) v u = star (dWirtingerDir f v u)
```

lemma realLinear_apply_eq_wirtinger[\[source\]](#)

Split a real-linear map $\mathbb{C} \rightarrow \mathbb{C}$ into its Wirtinger components. Any real-linear $L : \mathbb{C} \rightarrow \mathbb{C}$ splits into a holomorphic and an anti-holomorphic part with the Wirtinger coefficients $a = \frac{1}{2}(L \ 1 - i * L \ i)$, $b = \frac{1}{2}(L \ 1 + i * L \ i)$ as weights:

$$L \ w = a * w + b * \text{star } w.$$

This is purely algebraic: L is an arbitrary real-linear map, no derivative involved. Its use is the Wirtinger chain rule (`dWirtingerDir_comp` below), where the weights of the outer differential $L = \text{fderiv } \mathbb{R} \ g \ (f \ u)$ are the coefficients $\partial g / \partial f$, $\partial g / \partial f^-$.

```
lemma realLinear_apply_eq_wirtinger (L : ℂ →L[ℝ] ℂ) (w : ℂ) :
  L w =
    ((1 / 2 : ℂ) * (L 1 - Complex.I * L Complex.I)) * w
    + ((1 / 2 : ℂ) * (L 1 + Complex.I * L Complex.I)) * star w
```

lemma dWirtingerDir_comp[\[source\]](#)

The two-term Wirtinger chain rule for `dWirtingerDir`, outer $g : \mathbb{C} \rightarrow \mathbb{C}$ and inner $f : V \rightarrow \mathbb{C}$:

$$\partial^-_v (g \circ f) = (\partial g / \partial f) \cdot \partial^-_v f + (\partial g / \partial f^-) \cdot \partial^-_v f^-.$$

`realLinear_apply_eq_wirtinger` splits the chain rule's outer $\mathbb{R}$ -linear factor into the $\partial g / \partial f$, $\partial g / \partial f^-$ coefficients, each multiplying its inner directional derivative $\partial^-_v f$, $\partial^-_v f^-$.

```
lemma dWirtingerDir_comp {g : ℂ → ℂ} (hg : DifferentiableAt ℝ g (f u))
  (hf : DifferentiableAt ℝ f u) (v : V) :
  dWirtingerDir (fun p => g (f p)) v u =
    dWirtingerDir g 1 (f u) * dWirtingerDir f v u
    + dWirtingerAntiDir g 1 (f u) * dWirtingerDir (fun p => star (f p)) v u
```

lemma dWirtingerAntiDir_comp[\[source\]](#)

The two-term Wirtinger chain rule for `dWirtingerAntiDir`, the anti-holomorphic dual of `dWirtingerDir_comp`:

$$\partial^-_v (g \circ f) = (\partial g / \partial f) \cdot \partial^-_v f + (\partial g / \partial f^-) \cdot \partial^-_v f^-.$$

Same outer $\partial g / \partial f$, $\partial g / \partial f^-$ coefficients, now each multiplying its anti-holomorphic inner derivative $\partial^-_v f$, $\partial^-_v f^-$; same proof as `dWirtingerDir_comp`.

```
lemma dWirtingerAntiDir_comp {g : ℂ → ℂ} (hg : DifferentiableAt ℝ g (f u))
  (hf : DifferentiableAt ℝ f u) (v : V) :
  dWirtingerAntiDir (fun p => g (f p)) v u =
    dWirtingerDir g 1 (f u) * dWirtingerAntiDir f v u
    + dWirtingerAntiDir g 1 (f u) * dWirtingerAntiDir (fun p => star (f p)) v u
```

lemma fderiv_comp_clm_apply[\[source\]](#)

Chain rule for an inner continuous linear map L . Because the derivative of a linear map is the map itself, the real Fréchet derivative of $g \circ L$ at u , applied to x , equals the derivative of g at $L u$ applied to $L x$.

```
private lemma fderiv_comp_clm_apply {g : V' → ℂ} {L : V →L[ℝ] V'} {u : V}
(hg : DifferentiableAt ℝ g (L u)) (x : V) :
fderiv ℝ (fun p => g (L p)) u x = fderiv ℝ g (L u) (L x)
```

lemma dWirtingerDir_comp_conjLinear[\[source\]](#)

Domain conjugation swaps the operators: precomposing with a conjugate- $\mathbb{C}$ -linear L turns the holomorphic derivative of $g \circ L$ at u into the anti-holomorphic derivative of g at the mapped point $L u$, in the mapped direction $L v$: $\partial_v(g \circ L)$ at u equals $\partial_{\bar{L} v} g$ at $L u$.

```
lemma dWirtingerDir_comp_conjLinear {g : V' → ℂ} {L : V →L[ℝ] V'} {u : V}
(hL : ∀ x : V, L (Complex.I • x) = -(Complex.I • L x))
(hg : DifferentiableAt ℝ g (L u)) (v : V) :
dWirtingerDir (fun p => g (L p)) v u = dWirtingerAntiDir g (L v) (L u)
```

lemma dWirtingerAntiDir_comp_conjLinear[\[source\]](#)

Dual of `dWirtingerDir_comp_conjLinear`: the anti-holomorphic derivative of $g \circ L$ at u is the holomorphic derivative of g at the mapped point $L u$, in the mapped direction $L v$: $\partial_v(g \circ L)$ at u equals $\partial_{L v} g$ at $L u$.

```
lemma dWirtingerAntiDir_comp_conjLinear {g : V' → ℂ} {L : V →L[ℝ] V'} {u : V}
(hL : ∀ x : V, L (Complex.I • x) = -(Complex.I • L x))
(hg : DifferentiableAt ℝ g (L u)) (v : V) :
dWirtingerAntiDir (fun p => g (L p)) v u = dWirtingerDir g (L v) (L u)
```

lemma dWirtingerDir_eq_of_clinear[\[source\]](#)

Holomorphic collapse: along a direction where the real derivative is $\mathbb{C}$ -linear, the holomorphic derivative is the full real derivative, $\partial_v f = d_v f$.

```
lemma dWirtingerDir_eq_of_clinear {v : V}
(h : fderiv ℝ f u (Complex.I • v) = Complex.I • fderiv ℝ f u v) :
dWirtingerDir f v u = fderiv ℝ f u v
```

lemma dWirtingerAntiDir_eq_zero_of_clinear[\[source\]](#)

Holomorphic collapse: the anti-holomorphic derivative vanishes along a direction of $\mathbb{C}$ -linearity, $\partial_v f = 0$.

```
lemma dWirtingerAntiDir_eq_zero_of_clinear {v : V}
(h : fderiv ℝ f u (Complex.I • v) = Complex.I • fderiv ℝ f u v) :
dWirtingerAntiDir f v u = 0
```

lemma dWirtingerDir_eq_zero_of_antilinear[\[source\]](#)

Anti-holomorphic collapse: a direction of conjugate- $\mathbb{C}$ -linearity kills the holomorphic derivative, $\partial_v f = 0$.

```

lemma dWirtingerDir_eq_zero_of_antilinear {v : V}
  (h : fderiv ℝ f u (Complex.I • v) = -(Complex.I • fderiv ℝ f u v)) :
  dWirtingerDir f v u = 0

```

lemma dWirtingerAntiDir_eq_of_antilinear

[\[source\]](#)

Anti-holomorphic collapse: the anti-holomorphic derivative is the full real derivative along a direction of conjugate- $\mathbb{C}$ -linearity, $\partial^-_v f = d_v f$.

```

lemma dWirtingerAntiDir_eq_of_antilinear {v : V}
  (h : fderiv ℝ f u (Complex.I • v) = -(Complex.I • fderiv ℝ f u v)) :
  dWirtingerAntiDir f v u = fderiv ℝ f u v

```

lemma hasFDerivAt_fderiv_apply

[\[source\]](#)

The field $p \mapsto d_b f$ is the evaluation map $\cdot b$ composed with $fderiv \mathbb{R} f$, so when $fderiv \mathbb{R} f$ is differentiable its derivative is $fderiv \mathbb{R} (fderiv \mathbb{R} f) u$ post-composed with that evaluation.

```

private lemma hasFDerivAt_fderiv_apply (hf' : DifferentiableAt ℝ (fderiv ℝ f) u)
  (b : V) :
  HasFDerivAt (fun p => fderiv ℝ f p b)
    ((ContinuousLinearMap.apply ℝ ℂ b).comp (fderiv ℝ (fderiv ℝ f) u)) u

```

lemma hasFDerivAt_weightedDirDeriv

[\[source\]](#)

The `weightedDirDeriv` is differentiable wherever $fderiv \mathbb{R} f$ is.

```

private lemma hasFDerivAt_weightedDirDeriv (hf' : DifferentiableAt ℝ (fderiv ℝ f) u)
  (c : ℂ) (b₁ b₂ : V) :
  HasFDerivAt (weightedDirDeriv f c b₁ b₂)
    ((1 / 2 : ℂ) • ((ContinuousLinearMap.apply ℝ ℂ b₁).comp (fderiv ℝ (fderiv ℝ f) u)
      + c • (ContinuousLinearMap.apply ℝ ℂ b₂).comp (fderiv ℝ (fderiv ℝ f) u))) u

```

lemma fderiv_weightedDirDeriv

[\[source\]](#)

The bridge: differentiating a `weightedDirDeriv` along a third direction a lands on the second real Fréchet derivative $fderiv \mathbb{R} (fderiv \mathbb{R} f) u a b$ in the two slots.

```

private lemma fderiv_weightedDirDeriv (hf' : DifferentiableAt ℝ (fderiv ℝ f) u)
  (c : ℂ) (b₁ b₂ a : V) :
  fderiv ℝ (weightedDirDeriv f c b₁ b₂) u a
    = (1 / 2 : ℂ) * (fderiv ℝ (fderiv ℝ f) u a b₁
      + c * fderiv ℝ (fderiv ℝ f) u a b₂)

```

lemma dWirtingerAntiDir_eq_weightedDirDeriv

[\[source\]](#)

A directional derivative is a `weightedDirDeriv`: anti-holomorphic with $c = i$.

```

private lemma dWirtingerAntiDir_eq_weightedDirDeriv (w : V) :
  (fun p => dWirtingerAntiDir f w p) = weightedDirDeriv f Complex.I w (Complex.I • w)

```


lemma dWirtingerDir_eq_weightedDirDeriv[\[source\]](#)

A directional derivative is a weightedDirDeriv: holomorphic with $c = -i$.

```
private lemma dWirtingerDir_eq_weightedDirDeriv (v : V) :
  (fun p => dWirtingerDir f v p) = weightedDirDeriv f (-Complex.I) v (Complex.I • v)
```

lemma fderiv_dWirtingerAntiDir[\[source\]](#)

Differentiating the anti-holomorphic directional derivative lands on the second real Fréchet derivative in the two slots.

```
private lemma fderiv_dWirtingerAntiDir (hf' : DifferentiableAt ℝ (fderiv ℝ f) u)
  (w a : V) :
  fderiv ℝ (fun p => dWirtingerAntiDir f w p) u a
    = (1 / 2 : ℂ) * (fderiv ℝ (fderiv ℝ f) u a w
      + Complex.I * fderiv ℝ (fderiv ℝ f) u a (Complex.I • w))
```

lemma fderiv_dWirtingerDir[\[source\]](#)

Differentiating the holomorphic directional derivative lands on the second real Fréchet derivative in the two slots.

```
private lemma fderiv_dWirtingerDir (hf' : DifferentiableAt ℝ (fderiv ℝ f) u)
  (v a : V) :
  fderiv ℝ (fun p => dWirtingerDir f v p) u a
    = (1 / 2 : ℂ) * (fderiv ℝ (fderiv ℝ f) u a v
      - Complex.I * fderiv ℝ (fderiv ℝ f) u a (Complex.I • v))
```

lemma differentiableAt_dWirtingerDir[\[source\]](#)

On a C^2 field the holomorphic directional derivative is itself real-differentiable.

```
lemma differentiableAt_dWirtingerDir (hf2 : ContDiffAt ℝ 2 f u) (v : V) :
  DifferentiableAt ℝ (fun p => dWirtingerDir f v p) u
```

lemma differentiableAt_dWirtingerAntiDir[\[source\]](#)

On a C^2 field the anti-holomorphic directional derivative is itself real-differentiable.

```
lemma differentiableAt_dWirtingerAntiDir (hf2 : ContDiffAt ℝ 2 f u) (w : V) :
  DifferentiableAt ℝ (fun p => dWirtingerAntiDir f w p) u
```

lemma dWirtingerDir_congr_of_eventuallyEq[\[source\]](#)

The holomorphic directional derivative depends only on the field near the point: fields agreeing on a neighbourhood have equal derivative.

```
lemma dWirtingerDir_congr_of_eventuallyEq {f₁ f₂ : V → ℂ} {u : V}
  (h : f₁ =f[nhds u] f₂) (v : V) :
  dWirtingerDir f₁ v u = dWirtingerDir f₂ v u
```

lemma dWirtingerAntiDir_congr_of_eventuallyEq[\[source\]](#)

The anti-holomorphic directional derivative depends only on the field near the point; dual of `dWirtingerDir_congr_of_eventuallyEq`.

```
lemma dWirtingerAntiDir_congr_of_eventuallyEq {f₁ f₂ : V → ℂ} {u : V}
(h : f₁ =f[nhds u] f₂) (v : V) :
dWirtingerAntiDir f₁ v u = dWirtingerAntiDir f₂ v u
```

theorem dWirtingerDir_dWirtingerAntiDir_comm

[\[source\]](#)

***Schwarz's theorem** for the directional Wirtinger operators: on a $\mathbb{C}^2$ field f the holomorphic and anti-holomorphic directional derivatives commute in any two directions, $\partial_v \partial_w f = \partial_w \partial_v f$.*

The commutation adds no analytic input. By the $\S G$ bridge each order expands into a real-linear combination of the second real Fréchet derivative `fderiv ℝ (fderiv ℝ f) u` on the four directions $v, i \cdot v, w, i \cdot w$. The two orders give the same combination up to transposing the two slots of that second derivative, and `ContDiffAt.isSymmSndFDerivAt`, the symmetry of ordinary mixed second partials, equates them.

```
theorem dWirtingerDir_dWirtingerAntiDir_comm (hf2 : ContDiffAt ℝ 2 f u) (v w : V) :
dWirtingerDir (fun p => dWirtingerAntiDir f w p) v u
= dWirtingerAntiDir (fun p => dWirtingerDir f v p) w u
```

2.11 Physlib.Mathematics.Calculus.Wirtinger.Coordinate

[Physlib/Mathematics/Calculus/Wirtinger/Coordinate.lean](#) on GitHub.

def dWirtingerCoord

[\[source\]](#)

Holomorphic Wirtinger derivative along the I -th coordinate of $\iota \rightarrow \mathbb{C}$.

```
def dWirtingerCoord (f : (ι → ℂ) → ℂ) (I : ι) : (ι → ℂ) → ℂ
```

def dWirtingerAntiCoord

[\[source\]](#)

Anti-holomorphic Wirtinger derivative along the I -th coordinate of $\iota \rightarrow \mathbb{C}$.

```
def dWirtingerAntiCoord (f : (ι → ℂ) → ℂ) (I : ι) : (ι → ℂ) → ℂ
```

lemma dWirtingerCoord_apply

[\[source\]](#)

Real-Fréchet form of `dWirtingerCoord`: `dWirtingerCoord f I u = (1/2)($\partial_x - i \cdot \partial_y$) f, the derivatives along the slot- I real and imaginary coordinate directions. Unconditional — the directional definition makes it definitional (Complex.I • Pi.single I 1 = Pi.single I Complex.I); no differentiability hypothesis is needed.`

```
lemma dWirtingerCoord_apply {f : (ι → ℂ) → ℂ}
{u : (ι → ℂ)} (I : ι) :
dWirtingerCoord f I u = (1 / 2 : ℂ) * (fderiv ℝ f u (Pi.single I 1)
- Complex.I * fderiv ℝ f u (Pi.single I Complex.I))
```

lemma dWirtingerAntiCoord_apply

[\[source\]](#)

Real-Fréchet form of `dWirtingerAntiCoord`: $\partial_{-I} f = (1/2)(\partial_x + i \cdot \partial_y) f$, mirror of `dWirtingerCoord_apply` with the sign flip on the imaginary-direction term. Unconditional, as for `dWirtingerCoord_apply`.

```
lemma dWirtingerAntiCoord_apply {f : (ι → ℂ) → ℂ}
  {u : (ι → ℂ)} (I : ι) :
  dWirtingerAntiCoord f I u = (1 / 2 : ℂ) * (fderiv ℝ f u (Pi.single I 1)
    + Complex.I * fderiv ℝ f u (Pi.single I Complex.I))
```

lemma clinear_of_holomorphic

[\[source\]](#)

The real derivative of a holomorphic $f : E \rightarrow \mathbb{C}$ is $\mathbb{C}$ -linear along every direction — the hypothesis the foundation collapse `dWirtingerDir_eq_of_clinear` consumes. A holomorphic f has $fderiv \mathbb{R} f u = (fderiv \mathbb{C} f u).restrictScalars \mathbb{R} (DifferentiableAt.fderiv_restrictScalars f u)$ so its real derivative agrees with the $\mathbb{C}$ -linear complex derivative on every direction. Used at $E = \iota \rightarrow \mathbb{C}$ for the coordinate collapse and at $E = \mathbb{C}$ for the outer-g chain rule (§D).

```
private lemma clinear_of_holomorphic {E : Type*} [NormedAddCommGroup E] [NormedSpace ℝ E]
  [NormedSpace ℂ E] [IsScalarTower ℝ ℂ E] {f : E → ℂ} {u : E}
  (hf : DifferentiableAt ℂ f u) (d : E) :
  fderiv ℝ f u (Complex.I • d) = Complex.I • fderiv ℝ f u d
```

lemma dWirtingerCoord_congr_of_eventuallyEq_apply

[\[source\]](#)

`dWirtingerCoord` is local: functions agreeing on a neighbourhood of u have equal holomorphic Wirtinger derivative at u ($f_1 \equiv^f [nhds u] f_2 \implies \partial_{-I} f_1 u = \partial_{-I} f_2 u$).

```
lemma dWirtingerCoord_congr_of_eventuallyEq_apply {f₁ f₂ : (ι → ℂ) → ℂ}
  {u : (ι → ℂ)} (h : f₁ ≡f [nhds u] f₂) (I : ι) :
  dWirtingerCoord f₁ I u = dWirtingerCoord f₂ I u
```

lemma dWirtingerCoord_neg_apply

[\[source\]](#)

Pointwise negation rule for the holomorphic coordinate derivative at u : $\partial_{-I} (-f) = -\partial_{-I} f$. Used with `dWirtingerCoord_add_apply` to assemble the coordinate-difference rule `dWirtingerCoord_coordDiff` (§C).

```
lemma dWirtingerCoord_neg_apply {u : (ι → ℂ)} (I : ι) :
  dWirtingerCoord (fun v => -(f v)) I u = -(dWirtingerCoord f I u)
```

lemma fderiv_coordProj

[\[source\]](#)

The real Fréchet derivative of the J -th coordinate projection $v \mapsto v J$. Consumed by the Kronecker coordinate-value lemmas `dWirtingerCoord_coordProj` / `dWirtingerAntiCoord_coordProj`, which feed it the slot- I real and imaginary directions.

```
private lemma fderiv_coordProj (J : ι) (u d : (ι → ℂ)) :
  fderiv ℝ (fun v : (ι → ℂ) => v J) u d = d J
```

lemma dWirtingerCoord_add_apply

[\[source\]](#)

Pointwise additivity of the holomorphic coordinate derivative at u : $\partial_I (f + g) = \partial_I f + \partial_I g$. Used to assemble the coordinate-difference rule `dWirtingerCoord_coordDiff` (§C).

```
lemma dWirtingerCoord_add_apply {u : (ι → ℂ)}
(hf : DifferentiableAt ℝ f u) (hg : DifferentiableAt ℝ g u) (I : ι) :
dWirtingerCoord (f + g) I u = dWirtingerCoord f I u + dWirtingerCoord g I u
```

lemma dWirtingerCoord_smul_apply

[\[source\]](#)

Pointwise compatibility with complex scalar multiplication at u : $\partial_I (c \cdot f) = c \cdot \partial_I f$.

```
lemma dWirtingerCoord_smul_apply {u : (ι → ℂ)}
(c : ℂ) (hf : DifferentiableAt ℝ f u) (I : ι) :
dWirtingerCoord (c • f) I u = c • dWirtingerCoord f I u
```

lemma dWirtingerCoord_mul_apply

[\[source\]](#)

Pointwise Leibniz rule for the holomorphic coordinate derivative at u : $\partial_I (f \cdot g) = \partial_I f \cdot g + f \cdot \partial_I g$.

```
lemma dWirtingerCoord_mul_apply {u : (ι → ℂ)}
(hf : DifferentiableAt ℝ f u) (hg : DifferentiableAt ℝ g u) (I : ι) :
dWirtingerCoord (f * g) I u =
dWirtingerCoord f I u * g u + f u * dWirtingerCoord g I u
```

lemma dWirtingerCoord_fun_sum_apply

[\[source\]](#)

Pointwise finite-sum rule for holomorphic coordinate derivatives at u : $\partial_I (\sum a \in s, F a) = \sum a \in s, \partial_I (F a)$.

```
lemma dWirtingerCoord_fun_sum_apply {α : Type*} {s : Finset α}
{F : α → (ι → ℂ) → ℂ} {u : (ι → ℂ)}
(hF : ∀ a ∈ s, DifferentiableAt ℝ (F a) u) (I : ι) :
dWirtingerCoord (fun v => ∑ a ∈ s, F a v) I u =
∑ a ∈ s, dWirtingerCoord (F a) I u
```

lemma dWirtingerCoord_eq_complex_fderiv_apply

[\[source\]](#)

For a holomorphic function, `dWirtingerCoord` is the complex Fréchet derivative in the corresponding coordinate direction: $\partial_I f = \text{fderiv } \mathbb{C} f u (\text{Pi.single } I \ 1)$.

```
lemma dWirtingerCoord_eq_complex_fderiv_apply {u : (ι → ℂ)}
(hf : DifferentiableAt ℂ f u) (I : ι) :
dWirtingerCoord f I u = fderiv ℂ f u (Pi.single I 1)
```

def conjCLM

[\[source\]](#)

Pointwise conjugation bundled as an $\mathbb{R}$ -linear CLM (conjugate- $\mathbb{C}$ -linear): the I -th component is conjugation of the I -th coordinate `Complex.conjCLE ∘ proj I`. Its underlying function is the star of $\iota \rightarrow \mathbb{C}$ (`conjCLM_apply`). Bundling star as a $\rightarrow L[\mathbb{R}]$ is what lets the anti-holomorphic lemmas below feed it to the foundation

$dWirtingerDir_comp_conjLinear$, which requires its domain map as a continuous linear map.

```
private def conjCLM : (ι → ℂ) → L[ℝ] (ι → ℂ)
```

lemma conjCLM_smul_I

[\[source\]](#)

$conjCLM$ is conjugate- $\mathbb{C}$ -linear: $conj (i \cdot d) = -(i \cdot conj d)$. The hL hypothesis the foundation $dWirtingerDir_comp_conjLinear$ / $dWirtingerAntiDir_comp_conjLinear$ consume.

```
private lemma conjCLM_smul_I (d : ι → ℂ) :
  conjCLM (Complex.I • d) = -(Complex.I • conjCLM (ι := ι) d)
```

lemma dWirtingerCoord_eq_zero_of_antiHolomorphic_apply

[\[source\]](#)

An *anti-holomorphic* function $v \mapsto g (star v)$ (a holomorphic g precomposed with conjugation) depends on the coordinates only through their conjugates z^- . Its holomorphic derivative therefore vanishes, $\partial_I (g \circ star) = 0$, since ∂_I differentiates w.r.t. z and treats z^- as constant. Dual to $\partial_I^- f = 0$ for holomorphic f .

```
lemma dWirtingerCoord_eq_zero_of_antiHolomorphic_apply {u : (ι → ℂ)}
  (hg : DifferentiableAt ℂ g (star u)) (I : ι) :
  dWirtingerCoord (fun v : (ι → ℂ) => g (star v)) I u = 0
```

lemma dWirtingerAntiCoord_congr_of_eventuallyEq_apply

[\[source\]](#)

$dWirtingerAntiCoord$ is local ($f_1 =^f [nhds u] f_2 \implies \partial_I^- f_1 u = \partial_I^- f_2 u$).

```
lemma dWirtingerAntiCoord_congr_of_eventuallyEq_apply {f1 f2 : (ι → ℂ) → ℂ}
  {u : (ι → ℂ)} (h : f1 =^f [nhds u] f2) (I : ι) :
  dWirtingerAntiCoord f1 I u = dWirtingerAntiCoord f2 I u
```

lemma dWirtingerAntiCoord_neg_apply

[\[source\]](#)

Pointwise negation rule for the anti-holomorphic coordinate derivative at u : $\partial_I^- (-f) = -\partial_I^- f$. Used with $dWirtingerAntiCoord_add_apply$ to assemble the coordinate-difference rule $dWirtingerAntiCoord_coordDiff$ (§C).

```
lemma dWirtingerAntiCoord_neg_apply {u : (ι → ℂ)} (I : ι) :
  dWirtingerAntiCoord (fun v => -(f v)) I u = -(dWirtingerAntiCoord f I u)
```

lemma dWirtingerCoord_conjCoord

[\[source\]](#)

$\partial_I z^{-J} = 0$. The conjugate of $dWirtingerAntiCoord_coordProj$ ($z^{-J} = star z^J$), read off through $dWirtingerDir_star_comp$ rather than recomputed. Used to assemble the coordinate-difference rule $dWirtingerCoord_coordDiff$ (§C).

```
lemma dWirtingerCoord_conjCoord (I J : ι) :
  dWirtingerCoord (fun u : (ι → ℂ) => star (u J)) I = 0
```

lemma dWirtingerAntiCoord_conjCoord

[\[source\]](#)

$\partial^-_I z^J = \delta_{IJ}$. The conjugate of `dWirtingerCoord_coordProj`, read off through `dWirtingerAntiDir_star_comp` rather than recomputed. Used to assemble the coordinate-difference rule `dWirtingerAntiCoord_coordDiff` (§C).

```
lemma dWirtingerAntiCoord_conjCoord (I J : ι) :
  dWirtingerAntiCoord (fun u : (ι → ℂ) => star (u J)) I =
    fun _ => if I = J then 1 else 0
```

lemma dWirtingerAntiCoord_add_apply

[\[source\]](#)

Pointwise additivity of the anti-holomorphic coordinate derivative at u : $\partial^-_I (f + g) = \partial^-_I f + \partial^-_I g$. Used to assemble the coordinate-difference rule `dWirtingerAntiCoord_coordDiff` (§C).

```
lemma dWirtingerAntiCoord_add_apply {u : (ι → ℂ)}
  (hf : DifferentiableAt ℝ f u) (hg : DifferentiableAt ℝ g u) (I : ι) :
  dWirtingerAntiCoord (f + g) I u = dWirtingerAntiCoord f I u + dWirtingerAntiCoord g I u
```

lemma dWirtingerAntiCoord_smul_apply

[\[source\]](#)

Pointwise compatibility with complex scalar multiplication at u : $\partial^-_I (c \cdot f) = c \cdot \partial^-_I f$.

```
lemma dWirtingerAntiCoord_smul_apply {u : (ι → ℂ)}
  (c : ℂ) (hf : DifferentiableAt ℝ f u) (I : ι) :
  dWirtingerAntiCoord (c • f) I u = c • dWirtingerAntiCoord f I u
```

lemma dWirtingerAntiCoord_mul_apply

[\[source\]](#)

Pointwise Leibniz rule for the anti-holomorphic coordinate derivative at u : $\partial^-_I (f \cdot g) = \partial^-_I f \cdot g + f \cdot \partial^-_I g$.

```
lemma dWirtingerAntiCoord_mul_apply {u : (ι → ℂ)}
  (hf : DifferentiableAt ℝ f u) (hg : DifferentiableAt ℝ g u) (I : ι) :
  dWirtingerAntiCoord (f * g) I u =
    dWirtingerAntiCoord f I u * g u + f u * dWirtingerAntiCoord g I u
```

lemma dWirtingerAntiCoord_fun_sum_apply

[\[source\]](#)

Pointwise finite-sum rule for anti-holomorphic coordinate derivatives at u : $\partial^-_I (\sum a \in s, F a) = \sum a \in s, \partial^-_I (F a)$.

```
lemma dWirtingerAntiCoord_fun_sum_apply {α : Type*} {s : Finset α}
  {F : α → (ι → ℂ) → ℂ} {u : (ι → ℂ)}
  (hF : ∀ a ∈ s, DifferentiableAt ℝ (F a) u) (I : ι) :
  dWirtingerAntiCoord (fun v => ∑ a ∈ s, F a v) I u =
    ∑ a ∈ s, dWirtingerAntiCoord (F a) I u
```

lemma dWirtingerAntiCoord_eq_complex_fderiv_apply

[\[source\]](#)

For an anti-holomorphic function the anti-holomorphic Wirtinger derivative equals the complex Fréchet derivative of g at $\text{star } u$ along the slot- I real coordinate direction, $\partial^-_I (g \circ \text{star}) = \text{fderiv } \mathbb{C} g (\text{star } u) (\text{Pi.single } I \ 1)$. Dual of `dWirtingerCoord_eq_complex_fderiv`.

```

lemma dWirtingerAntiCoord_eq_complex_fderiv_apply {u : (ι → ℂ)}
  (hg : DifferentiableAt ℂ g (star u)) (I : ι) :
  dWirtingerAntiCoord (fun v : (ι → ℂ) => g (star v)) I u =
    fderiv ℂ g (star u) (Pi.single I 1)

```

lemma dWirtingerAntiCoord_eq_zero_of_holomorphic_apply

[\[source\]](#)

Holomorphic functions have zero anti-holomorphic coordinate derivative: $\partial^-_I f = 0$.

```

lemma dWirtingerAntiCoord_eq_zero_of_holomorphic_apply {u : (ι → ℂ)}
  (hf : DifferentiableAt ℂ f u) (I : ι) :
  dWirtingerAntiCoord f I u = 0

```

lemma coordDiff_eq_add_neg

[\[source\]](#)

The coordinate difference $z^J - z^{\bar{J}}$ as a sum of the holomorphic coordinate and the negated conjugate coordinate — the form on which the `dWirtingerCoord` / `dWirtingerAntiCoord` additivity rules apply.

```

private lemma coordDiff_eq_add_neg (J : ι) :
  (fun v : (ι → ℂ) => v J - star (v J))
    = (fun v => v J) + (fun v => -(star (v J)))

```

lemma dWirtingerCoord_coordDiff

[\[source\]](#)

$\partial_I (z^J - z^{\bar{J}}) = \delta_{IJ}$, from $\partial_I z^J = \delta_{IJ}$ and $\partial_I z^{\bar{J}} = 0$.

```

lemma dWirtingerCoord_coordDiff (I J : ι) :
  dWirtingerCoord (fun v : (ι → ℂ) => v J - star (v J)) I
    = fun _ => if I = J then 1 else 0

```

lemma dWirtingerAntiCoord_coordDiff

[\[source\]](#)

$\partial^-_I (z^J - z^{\bar{J}}) = -\delta_{IJ}$, from $\partial^-_I z^J = 0$ and $\partial^-_I z^{\bar{J}} = \delta_{IJ}$.

```

lemma dWirtingerAntiCoord_coordDiff (I J : ι) :
  dWirtingerAntiCoord (fun v : (ι → ℂ) => v J - star (v J)) I
    = fun _ => if I = J then -1 else 0

```

lemma dWirtingerCoord_comp_apply

[\[source\]](#)

The two-term coordinate chain rule for a real-differentiable outer g , pointwise at u : $\partial_I (g \circ f) = (\partial g / \partial f) \cdot \partial_I f + (\partial g / \partial f^-) \cdot \partial_I f^-$, with coefficients $\partial g / \partial f = dWirtingerDir g 1 (f u)$ and $\partial g / \partial f^- = dWirtingerAntiDir g 1 (f u)$. The $d = \text{Pi.single } I 1$ case of the foundation `dWirtingerDir_comp`. The single-term holomorphic specialization `dWirtingerCoord_comp_holomorphic_apply` is proved from this.

```

lemma dWirtingerCoord_comp_apply {u : (ι → ℂ)}
  (hg : DifferentiableAt ℝ g (f u)) (hf : DifferentiableAt ℝ f u) (I : ι) :
  dWirtingerCoord (fun v => g (f v)) I u =
    dWirtingerDir g 1 (f u) * dWirtingerCoord f I u
    + dWirtingerAntiDir g 1 (f u) *
      dWirtingerCoord (fun v : (ι → ℂ) => star (f v)) I u

```

lemma dWirtingerCoord_comp_holomorphic_apply[\[source\]](#)

The single-term coordinate chain rule for a holomorphic outer g , pointwise at u : $\partial_I (g \circ f) = \text{deriv } g (f u) \cdot \partial_I f$. From the two-term `dWirtingerCoord_comp_apply`: for holomorphic g the anti-holomorphic coefficient `dWirtingerAntiDir g 1 (f u)` vanishes and `dWirtingerDir g 1 (f u)` collapses to `deriv g (f u)`, both off the $\mathbb{C}$ -linearity `clinear_of_holomorphic` (§B).

```
lemma dWirtingerCoord_comp_holomorphic_apply {u : (ι → ℂ)}
(hg : DifferentiableAt ℂ g (f u)) (hf : DifferentiableAt ℝ f u) (I : ι) :
dWirtingerCoord (fun v => g (f v)) I u =
  deriv g (f u) * dWirtingerCoord f I u
```

lemma dWirtingerCoord_star_comp_apply[\[source\]](#)

Conjugating the inner field swaps the two operators: $\partial_I f^- = \text{conj } (\partial_I f)$. The $d = \text{Pi.single } I \ 1$ case of the foundation `dWirtingerDir_star_comp`.

```
lemma dWirtingerCoord_star_comp_apply {u : (ι → ℂ)}
(hf : DifferentiableAt ℝ f u) (I : ι) :
dWirtingerCoord (fun v : (ι → ℂ) => star (f v)) I u =
  star (dWirtingerAntiCoord f I u)
```

lemma dWirtingerAntiCoord_comp_apply[\[source\]](#)

The two-term coordinate chain rule for a real-differentiable outer g , anti-holomorphic version, pointwise at u : $\partial_I^- (g \circ f) = (\partial g / \partial f) \cdot \partial_I^- f + (\partial g / \partial f^-) \cdot \partial_I^- f^-$. Same outer coefficients as `dWirtingerCoord_comp_apply`, now multiplying the anti-holomorphic inner derivatives. The $d = \text{Pi.single } I \ 1$ case of the foundation `dWirtingerAntiDir_comp`. The single-term holomorphic specialization `dWirtingerAntiCoord_comp_holomorphic_apply` is proved from this.

```
lemma dWirtingerAntiCoord_comp_apply {u : (ι → ℂ)}
(hg : DifferentiableAt ℝ g (f u)) (hf : DifferentiableAt ℝ f u) (I : ι) :
dWirtingerAntiCoord (fun v => g (f v)) I u =
  dWirtingerDir g 1 (f u) * dWirtingerAntiCoord f I u
  + dWirtingerAntiDir g 1 (f u) *
    dWirtingerAntiCoord (fun v : (ι → ℂ) => star (f v)) I u
```

lemma dWirtingerAntiCoord_comp_holomorphic_apply[\[source\]](#)

The single-term coordinate chain rule for a holomorphic outer g , anti-holomorphic version, pointwise at u : $\partial_I^- (g \circ f) = \text{deriv } g (f u) \cdot \partial_I^- f$. As in `dWirtingerCoord_comp_holomorphic_apply` the $\partial g / \partial f^-$ channel vanishes and $\partial g / \partial f$ collapses to `deriv g (f u)`.

```
lemma dWirtingerAntiCoord_comp_holomorphic_apply {u : (ι → ℂ)}
(hg : DifferentiableAt ℂ g (f u)) (hf : DifferentiableAt ℝ f u) (I : ι) :
dWirtingerAntiCoord (fun v => g (f v)) I u =
  deriv g (f u) * dWirtingerAntiCoord f I u
```

lemma dWirtingerAntiCoord_star_comp_apply[\[source\]](#)

Conjugating the inner field swaps the two operators: $\partial_I^- f^- = \text{conj } (\partial_I f)$. Dual of `dWirtingerCoord_star_comp_apply`, the $d = \text{Pi.single } I \ 1$ case of the foundation `dWirtingerAntiDir_star_comp`.


```

lemma dWirtingerAntiCoord_star_comp_apply {u : (ι → ℂ)}
(hf : DifferentiableAt ℝ f u) (I : ι) :
dWirtingerAntiCoord (fun v : (ι → ℂ) => star (f v)) I u =
star (dWirtingerCoord f I u)

```

lemma differentiableAt_dWirtingerCoord

[source]

On a C^2 function the holomorphic coordinate Wirtinger derivative is itself real-differentiable ($\text{DifferentiableAt } \mathbb{R} (\partial_I f) u$) — the $d = \text{Pi.single } I \ 1$ case of $\text{differentiableAt_dWirtingerCoord}$.

```

lemma differentiableAt_dWirtingerCoord (hf2 : ContDiffAt ℝ 2 f u) (I : ι) :
DifferentiableAt ℝ (fun v => dWirtingerCoord f I v) u

```

lemma differentiableAt_dWirtingerAntiCoord

[source]

On a C^2 function the anti-holomorphic coordinate Wirtinger derivative is itself real-differentiable ($\text{DifferentiableAt } \mathbb{R} (\partial_{\bar{J}} f) u$) — the $d = \text{Pi.single } J \ 1$ case of $\text{differentiableAt_dWirtingerAntiDir}$.

```

lemma differentiableAt_dWirtingerAntiCoord (hf2 : ContDiffAt ℝ 2 f u) (J : ι) :
DifferentiableAt ℝ (fun v => dWirtingerAntiCoord f J v) u

```

theorem dWirtingerCoord_dWirtingerAntiCoord_comm

[source]

***Schwarz's theorem** for the coordinate Wirtinger operators: on a C^2 function the holomorphic and anti-holomorphic derivatives commute, $\partial_I \partial_{\bar{J}} f = \partial_{\bar{J}} \partial_I f$. The $d = \text{Pi.single } I \ 1, e = \text{Pi.single } J \ 1$ case of the general $\text{dWirtingerDir_dWirtingerAntiDir_comm}$.*

```

theorem dWirtingerCoord_dWirtingerAntiCoord_comm (hf2 : ContDiffAt ℝ 2 f u) (I J : ι) :
dWirtingerCoord (fun v => dWirtingerAntiCoord f J v) I u
= dWirtingerAntiCoord (fun v => dWirtingerCoord f I v) J u

```

2.12 Physlib.Mathematics.ConjModule

[Physlib/Mathematics/ConjModule.lean](#) on GitHub.

def ConjModule

[source]

The conjugate module of M : the same additive group with the scalar action twisted by conjugation, $r \bullet v = \text{star } r \bullet v$. A type synonym so the twisted action stays off M .

```

def ConjModule (M : Type*)

```

instance AddCommGroup (ConjModule M)

[source]

```

instance : AddCommGroup (ConjModule M)

```

instance instModule

[source]

The twisted action $r \bullet v = \text{star } r \bullet v$, obtained by restricting scalars along the conjugation ring endomorphism $\text{starRingEnd } k$.

```

instance instModule : Module k (ConjModule M)

```

def conjEquiv[\[source\]](#)

The canonical conjugate-linear equivalence $M \approx_{sl}[\text{starRingEnd } k] \text{ ConjModule } M$, the identity on the underlying additive group.

```
def conjEquiv : M ≈sl[starRingEnd k] ConjModule M where
```

def involution[\[source\]](#)

Conjugating twice returns the original module: the k -linear isomorphism $\text{ConjModule } (\text{ConjModule } M) \approx_l[k] M$. It is k -linear, not merely semilinear, because $\text{starRingEnd } k$ composed with itself is the identity.

```
def involution : ConjModule (ConjModule M) ≈l[k] M
```

def starFinsupp[\[source\]](#)

Coordinate-wise conjugation on $\iota \rightarrow_0 k$, a conjugate-linear self-equivalence.

```
noncomputable def starFinsupp : (ι →0 k) ≈sl[starRingEnd k] (ι →0 k) where
```

def conj[\[source\]](#)

A basis of M transported to a basis of $\text{ConjModule } M$: the same basis vectors, with coordinates conjugated $((\text{Basis.conj } b).repr \ v = \text{star} \circ b.repr \ v)$.

```
noncomputable def _root_.Basis.conj (b : Basis ι k M) : Basis ι k (ConjModule M)
```

2.13 Physlib.Mathematics.CrossProductMatrix

[Physlib/Mathematics/CrossProductMatrix.lean](#) on GitHub.

def crossProductMatrix[\[source\]](#)

The hat map $[\omega]_\times$: the skew-symmetric 3×3 matrix acting as the cross product with ω , i.e. $[\omega]_\times *_{\mathbb{V}} v = \omega \times_{\mathbb{V}} v$.

```
def crossProductMatrix (ω : Fin 3 → ℝ) : Matrix (Fin 3) (Fin 3) ℝ
```

lemma crossProductMatrix_mulVec[\[source\]](#)

The hat matrix acts on a vector as the cross product.

```
@[simp]
```

```
lemma crossProductMatrix_mulVec (ω v : Fin 3 → ℝ) :  
  crossProductMatrix ω *v v = ω ×3 v
```

lemma crossProductMatrix_transpose[\[source\]](#)

The hat map produces skew-symmetric matrices.

```
@[simp]
```

```
lemma crossProductMatrix_transpose (ω : Fin 3 → ℝ) :  
  (crossProductMatrix ω)T = - crossProductMatrix ω
```

def crossProductVee[\[source\]](#)

The vee map: reads the vector off a 3×3 matrix. It is a left inverse of the hat map.

```
def crossProductVee (A : Matrix (Fin 3) (Fin 3) ℝ) : Fin 3 → ℝ
```

lemma crossProductVee_crossProductMatrix[\[source\]](#)

The vee map is a left inverse of the hat map.

```
@[simp]
```

```
lemma crossProductVee_crossProductMatrix (ω : Fin 3 → ℝ) :  
  crossProductVee (crossProductMatrix ω) = ω
```

2.14 Physlib.Mathematics.KroneckerDelta

[Physlib/Mathematics/KroneckerDelta.lean](#) on GitHub.

def generalizedKroneckerDelta[\[source\]](#)

Generalized Kronecker delta: $\delta^{\{\mu_1 \dots \mu_n\}}_{\{v_1 \dots v_n\}} = \det (\delta[\mu_i, v_j])$.

This is defined for any finite type α with decidable equality.

```
def generalizedKroneckerDelta {α ι : Type} [DecidableEq α]  
  [DecidableEq ι] [Fintype ι]  
  (μ : ι → α) (ν : ι → α) : ℤ
```

lemma generalizedKroneckerDelta_swap[\[source\]](#)

Swapping two of the upper indices of the generalized Kronecker delta negates it. This is one row transposition of the underlying determinant.

```
lemma generalizedKroneckerDelta_swap {α ι : Type} [DecidableEq α] [DecidableEq ι] [Fintype ι]  
  (μ ν : ι → α) {i j : ι} (hij : i ≠ j) :  
  generalizedKroneckerDelta (μ ∘ Equiv.swap i j) ν = - generalizedKroneckerDelta μ ν
```

2.15 Physlib.Mathematics.LinearPMap

[Physlib/Mathematics/LinearPMap.lean](#) on GitHub.

lemma sub_le_zero[\[source\]](#)

```
lemma sub_le_zero : f - f ≤ 0
```

lemma neg_add_le_zero[\[source\]](#)

```
lemma neg_add_le_zero : -f + f ≤ 0
```

lemma le_iff_neg_le_neg[\[source\]](#)

```
lemma le_iff_neg_le_neg : g₁ ≤ g₂ ↔ -g₁ ≤ -g₂
```

lemma le_neg_iff_neg_le[\[source\]](#)

lemma le_neg_iff_neg_le : $g_1 \leq -g_2 \leftrightarrow -g_1 \leq g_2$

lemma add_sub_le_cancel[\[source\]](#)

lemma add_sub_le_cancel : $f_1 + (f_2 - f_1) \leq f_2$

lemma add_sub_le_cancel_left[\[source\]](#)

lemma add_sub_le_cancel_left : $f_1 + f_2 - f_1 \leq f_2$

lemma add_sub_le_cancel_right[\[source\]](#)

lemma add_sub_le_cancel_right : $f_1 + f_2 - f_2 \leq f_1$

lemma add_add_sub_le_cancel[\[source\]](#)

lemma add_add_sub_le_cancel : $f_1 + f_2 + (f_3 - f_2) \leq f_1 + f_3$

lemma add_sub_sub_le_cancel[\[source\]](#)

lemma add_sub_sub_le_cancel : $f_1 + f_2 - (f_1 - f_3) \leq f_2 + f_3$

lemma sub_sub_sub_le_cancel_right[\[source\]](#)

lemma sub_sub_sub_le_cancel_right : $f_1 - f_2 - (f_3 - f_2) \leq f_1 - f_3$

lemma sub_sub_sub_le_cancel_left[\[source\]](#)

lemma sub_sub_sub_le_cancel_left : $f_1 - f_2 - (f_1 - f_3) \leq f_3 - f_2$

lemma sub_le_of_le_add[\[source\]](#)

lemma sub_le_of_le_add (h : $g \leq g_1 + g_2$) : $g - g_2 \leq g_1$

lemma sub_add_le_cancel[\[source\]](#)

lemma sub_add_le_cancel : $f_1 - f_2 + f_2 \leq f_1$

lemma add_le_of_le_sub[\[source\]](#)

lemma add_le_of_le_sub (h : $g \leq g_1 - g_2$) : $g + g_2 \leq g_1$

lemma add_left_le_of_le[\[source\]](#)

lemma add_left_le_of_le (h : $g_1 \leq g_2$) : $f + g_1 \leq f + g_2$

lemma add_right_le_of_le[\[source\]](#)

```
lemma add_right_le_of_le (h : g1 ≤ g2) : g1 + f ≤ g2 + f
```

lemma sub_right_le_of_le[\[source\]](#)

```
lemma sub_right_le_of_le (h : g1 ≤ g2) : g1 - f ≤ g2 - f
```

lemma sub_left_le_of_le[\[source\]](#)

```
lemma sub_left_le_of_le (h : g1 ≤ g2) : f - g1 ≤ f - g2
```

lemma zero_smul_le[\[source\]](#)

```
lemma zero_smul_le (f : E →ℓ.[ℓ] F) : (0 : ℓ) • f ≤ 0
```

lemma zero_smul_eq[\[source\]](#)

```
@[simp]
lemma zero_smul_eq {f : E →ℓ.[ℓ] F} (h : f.domain = τ) : (0 : ℓ) • f = 0
```

lemma compRestricted_domain[\[source\]](#)

```
lemma compRestricted_domain : (g ∘r f).domain = (g.domain.comap f.toFun).map f.domain.subtype
```

lemma mem_compRestricted_domain_iff'[\[source\]](#)

```
lemma mem_compRestricted_domain_iff' {x : E} :
  x ∈ (v ∘r u).domain ↔ ∃ y : u.domain, x = y ∧ ∃ y' : v.domain, u y = y'
```

lemma neg_compRestricted[\[source\]](#)

```
@[simp]
lemma neg_compRestricted : (-g) ∘r f = -g ∘r f
```

lemma compRestricted_neg[\[source\]](#)

```
@[simp]
lemma compRestricted_neg : g ∘r (-f) = -g ∘r f
```

lemma add_compRestricted[\[source\]](#)

```
lemma add_compRestricted : (g1 + g2) ∘r f = g1 ∘r f + g2 ∘r f
```

lemma sub_compRestricted[\[source\]](#)

```
lemma sub_compRestricted : (g1 - g2) ∘r f = g1 ∘r f - g2 ∘r f
```

lemma compRestricted_add_ge[\[source\]](#)

```
lemma compRestricted_add_ge : g ∘r f1 + g ∘r f2 ≤ g ∘r (f1 + f2)
```

lemma compRestricted_sub_ge[\[source\]](#)

```
lemma compRestricted_sub_ge : g ∘r f1 - g ∘r f2 ≤ g ∘r (f1 - f2)
```

lemma compRestricted_smul[\[source\]](#)

```
lemma compRestricted_smul {S : Type*} [DivisionRing S]
  [Module S E] [Module S F] [Module S G] [SMulCommClass S S F] [SMulCommClass S S G]
  {c : S} (hc : c ≠ 0) (g : F →l.[S] G) (f : E →l.[S] F) :
  g ∘r (c • f) = c • (g ∘r f)
```

lemma smul_compRestricted[\[source\]](#)

```
@[simp]
lemma smul_compRestricted {M : Type*} [Monoid M] [DistribMulAction M G] [SMulCommClass R M G]
  (c : M) (g : F →l.[R] G) (f : E →l.[R] F) :
  (c • g) ∘r f = c • (g ∘r f)
```

lemma one_toFun[\[source\]](#)

```
@[simp]
lemma one_toFun : (1 : E →l.[R] E).toFun = topEquiv.toLinearMap
```

lemma one_coe[\[source\]](#)

```
@[simp]
lemma one_coe : (1 : E →l.[R] E).toFun' = ↑topEquiv.toLinearMap
```

lemma inverse_ker[\[source\]](#)

```
lemma inverse_ker : f.inverse.toFun.ker = ⊥
```

lemma inverse_inverse[\[source\]](#)

```
lemma inverse_inverse : f.inverse.inverse = f
```

lemma inverse_compRestricted_eq[\[source\]](#)

```
lemma inverse_compRestricted_eq : f.inverse ∘r f = domRestrict 1 f.domain
```

lemma compRestricted_inverse_eq[\[source\]](#)

```
lemma compRestricted_inverse_eq : f ∘r f.inverse = domRestrict 1 f.inverse.domain
```

2.16 Physlib.Particles.BeyondTheStandardModel.RHN.AnomalyCancellation.Basic

[Physlib/Particles/BeyondTheStandardModel/RHN/AnomalyCancellation/Basic.lean](#) on GitHub.

lemma sum_one[\[source\]](#)

```
lemma sum_one [AddCommMonoid M] (f : Fin (SMvSpecies 1).numberCharges → M) :
  ∑ i, f i = f ⟨0, by simp⟩
```

2.17 Physlib.Particles.StandardModel.AnomalyCancellation.Basic

[Physlib/Particles/StandardModel/AnomalyCancellation/Basic.lean](#) on GitHub.

lemma sum_SMSpecies_numberCharges_one[\[source\]](#)

```
lemma sum_SMSpecies_numberCharges_one {M} [AddCommMonoid M]
  (f : Fin (SMSpecies 1).numberCharges → M) :
  ∑ i, f i = f ⟨0, by simp⟩
```

lemma toSpecies_apply_eq[\[source\]](#)

```
lemma toSpecies_apply_eq (i : Fin 5) (S : (SMCharges n).Charges) :
  toSpecies i S = fun j => toSpeciesEquiv S i j
```

2.18 Physlib.Particles.StandardModel.AnomalyCancellation.NoGrav.One.LinearParameterization

[Physlib/Particles/StandardModel/AnomalyCancellation/NoGrav/One/LinearParameterization.lean](#) on GitHub.

lemma toSpecies_apply_asCharges[\[source\]](#)

```
lemma toSpecies_apply_asCharges (S : linearParameters) (i : Fin 5) :
  toSpecies i S.asCharges = fun _ => S.asCharges i
```

2.19 Physlib.Particles.StandardModel.Basic

[Physlib/Particles/StandardModel/Basic.lean](#) on GitHub.

def gaugeGroupZ6UnitaryOfRoot[\[source\]](#)

The unitary complex number associated to a sixth root of unity.

```
noncomputable def gaugeGroupZ6UnitaryOfRoot (α : rootsOfUnity 6 ℂ) : unitary ℂ
```

lemma gaugeGroupZ6UnitaryOfRoot_coe[\[source\]](#)

```
@[simp]
lemma gaugeGroupZ6UnitaryOfRoot_coe (α : rootsOfUnity 6 ℂ) :
  (gaugeGroupZ6UnitaryOfRoot α : ℂ) = ((α : ℂ×) : ℂ)
```

def gaugeGroupZ6SU3ofRoot[\[source\]](#)

The SU(3) scalar matrix associated to a sixth root of unity.

```
noncomputable def gaugeGroupZ6SU3ofRoot (α : rootsOfUnity 6 ℂ) :
  specialUnitaryGroup (Fin 3) ℂ
```

def gaugeGroup $\mathbb{Z}_6$ SU20fRoot[\[source\]](#)*The $SU(2)$ scalar matrix associated to a sixth root of unity.*

```
noncomputable def gaugeGroup $\mathbb{Z}_6$ SU20fRoot (α : rootsOfUnity 6  $\mathbb{C}$ ) :
  specialUnitaryGroup (Fin 2)  $\mathbb{C}$ 
```

def gaugeGroup $\mathbb{Z}_6$ 0fRoot[\[source\]](#)*The element of GaugeGroupI associated to a sixth root of unity.*

```
noncomputable def gaugeGroup $\mathbb{Z}_6$ 0fRoot (α : rootsOfUnity 6  $\mathbb{C}$ ) : GaugeGroupI
```

lemma gaugeGroup $\mathbb{Z}_6$ 0fRoot_toSU3[\[source\]](#)

```
@[simp]
lemma gaugeGroup $\mathbb{Z}_6$ 0fRoot_toSU3 (α : rootsOfUnity 6  $\mathbb{C}$ ) :
  GaugeGroupI.toSU3 (gaugeGroup $\mathbb{Z}_6$ 0fRoot α) = gaugeGroup $\mathbb{Z}_6$ SU30fRoot α
```

lemma gaugeGroup $\mathbb{Z}_6$ 0fRoot_toSU2[\[source\]](#)

```
@[simp]
lemma gaugeGroup $\mathbb{Z}_6$ 0fRoot_toSU2 (α : rootsOfUnity 6  $\mathbb{C}$ ) :
  GaugeGroupI.toSU2 (gaugeGroup $\mathbb{Z}_6$ 0fRoot α) = gaugeGroup $\mathbb{Z}_6$ SU20fRoot α
```

lemma gaugeGroup $\mathbb{Z}_6$ 0fRoot_toU1[\[source\]](#)

```
@[simp]
lemma gaugeGroup $\mathbb{Z}_6$ 0fRoot_toU1 (α : rootsOfUnity 6  $\mathbb{C}$ ) :
  GaugeGroupI.toU1 (gaugeGroup $\mathbb{Z}_6$ 0fRoot α) = gaugeGroup $\mathbb{Z}_6$ Unitary0fRoot α
```

lemma gaugeGroup $\mathbb{Z}_6$ 0fRoot_mem_center[\[source\]](#)

```
lemma gaugeGroup $\mathbb{Z}_6$ 0fRoot_mem_center (α : rootsOfUnity 6  $\mathbb{C}$ ) :
  gaugeGroup $\mathbb{Z}_6$ 0fRoot α ∈ Subgroup.center GaugeGroupI
```

def gaugeGroup $\mathbb{Z}_6$ Hom[\[source\]](#)*The homomorphism from sixth roots of unity to GaugeGroupI.*

```
noncomputable def gaugeGroup $\mathbb{Z}_6$ Hom : rootsOfUnity 6  $\mathbb{C}$  →* GaugeGroupI where
```

lemma gaugeGroup $\mathbb{Z}_6$ Hom_apply[\[source\]](#)

```
@[simp]
lemma gaugeGroup $\mathbb{Z}_6$ Hom_apply (α : rootsOfUnity 6  $\mathbb{C}$ ) :
  gaugeGroup $\mathbb{Z}_6$ Hom α = gaugeGroup $\mathbb{Z}_6$ 0fRoot α
```

lemma gaugeGroup $\mathbb{Z}_6$ Hom_toSU3[\[source\]](#)

```
@[simp]
lemma gaugeGroup $\mathbb{Z}_6$ Hom_toSU3 (α : rootsOfUnity 6  $\mathbb{C}$ ) :
  GaugeGroupI.toSU3 (gaugeGroup $\mathbb{Z}_6$ Hom α) = gaugeGroup $\mathbb{Z}_6$ SU30fRoot α
```


lemma gaugeGroup $\mathbb{Z}_6$ Hom_toSU2[\[source\]](#)

@[simp]

```
lemma gaugeGroup $\mathbb{Z}_6$ Hom_toSU2 ( $\alpha$  : rootsOfUnity 6  $\mathbb{C}$ ) :
  GaugeGroupI.toSU2 (gaugeGroup $\mathbb{Z}_6$ Hom  $\alpha$ ) = gaugeGroup $\mathbb{Z}_6$ SU2OfRoot  $\alpha$ 
```

lemma gaugeGroup $\mathbb{Z}_6$ Hom_toU1[\[source\]](#)

@[simp]

```
lemma gaugeGroup $\mathbb{Z}_6$ Hom_toU1 ( $\alpha$  : rootsOfUnity 6  $\mathbb{C}$ ) :
  GaugeGroupI.toU1 (gaugeGroup $\mathbb{Z}_6$ Hom  $\alpha$ ) = gaugeGroup $\mathbb{Z}_6$ UnitaryOfRoot  $\alpha$ 
```

lemma gaugeGroup $\mathbb{Z}_6$ OfRoot_mem[\[source\]](#)

```
lemma gaugeGroup $\mathbb{Z}_6$ OfRoot_mem ( $\alpha$  : rootsOfUnity 6  $\mathbb{C}$ ) :
  gaugeGroup $\mathbb{Z}_6$ OfRoot  $\alpha$   $\in$  gaugeGroup $\mathbb{Z}_6$ SubGroup
```

lemma mem_gaugeGroup $\mathbb{Z}_6$ SubGroup_iff[\[source\]](#)

```
lemma mem_gaugeGroup $\mathbb{Z}_6$ SubGroup_iff (g : GaugeGroupI) :
  g  $\in$  gaugeGroup $\mathbb{Z}_6$ SubGroup  $\leftrightarrow \exists \alpha$  : rootsOfUnity 6  $\mathbb{C}$ , gaugeGroup $\mathbb{Z}_6$ OfRoot  $\alpha$  = g
```

lemma gaugeGroup $\mathbb{Z}_6$ SubGroup_le_center[\[source\]](#)

```
lemma gaugeGroup $\mathbb{Z}_6$ SubGroup_le_center :
  gaugeGroup $\mathbb{Z}_6$ SubGroup  $\leq$  Subgroup.center GaugeGroupI
```

instance gaugeGroup $\mathbb{Z}_6$ SubGroup_normal[\[source\]](#)

```
instance gaugeGroup $\mathbb{Z}_6$ SubGroup_normal : gaugeGroup $\mathbb{Z}_6$ SubGroup.Normal where
```

instance Group GaugeGroup $\mathbb{Z}_6$ [\[source\]](#)

```
noncomputable instance : Group GaugeGroup $\mathbb{Z}_6$ 
```

def GaugeGroup $\mathbb{Z}_6$.mk[\[source\]](#)

The quotient map from GaugeGroupI to GaugeGroup $\mathbb{Z}_6$.

```
noncomputable def mk : GaugeGroupI  $\rightarrow^*$  GaugeGroup $\mathbb{Z}_6$ 
```

lemma mk_gaugeGroup $\mathbb{Z}_6$ OfRoot[\[source\]](#)

@[simp]

```
lemma mk_gaugeGroup $\mathbb{Z}_6$ OfRoot ( $\alpha$  : rootsOfUnity 6  $\mathbb{C}$ ) :
  mk (gaugeGroup $\mathbb{Z}_6$ OfRoot  $\alpha$ ) = 1
```

def gaugeGroup $\mathbb{Z}_2$ RootTo $\mathbb{Z}_6$ Root[\[source\]](#)

The inclusion of second roots of unity into sixth roots of unity.

```
noncomputable def gaugeGroup $\mathbb{Z}_2$ RootTo $\mathbb{Z}_6$ Root : rootsOfUnity 2  $\mathbb{C}$   $\rightarrow^*$  rootsOfUnity 6  $\mathbb{C}$ 
```

def gaugeGroupZzOfRoot[\[source\]](#)

The element of GaugeGroupI associated to a second root of unity.

```
noncomputable def gaugeGroupZzOfRoot (α : rootsOfUnity 2 ℂ) : GaugeGroupI
```

lemma gaugeGroupZzOfRoot_toSU3[\[source\]](#)

```
@[simp]
```

```
lemma gaugeGroupZzOfRoot_toSU3 (α : rootsOfUnity 2 ℂ) :  
  GaugeGroupI.toSU3 (gaugeGroupZzOfRoot α) =  
    gaugeGroupZ6SU3OfRoot (gaugeGroupZzRootToZ6Root α)
```

lemma gaugeGroupZzOfRoot_toSU2[\[source\]](#)

```
@[simp]
```

```
lemma gaugeGroupZzOfRoot_toSU2 (α : rootsOfUnity 2 ℂ) :  
  GaugeGroupI.toSU2 (gaugeGroupZzOfRoot α) =  
    gaugeGroupZ6SU2OfRoot (gaugeGroupZzRootToZ6Root α)
```

lemma gaugeGroupZzOfRoot_toU1[\[source\]](#)

```
@[simp]
```

```
lemma gaugeGroupZzOfRoot_toU1 (α : rootsOfUnity 2 ℂ) :  
  GaugeGroupI.toU1 (gaugeGroupZzOfRoot α) =  
    gaugeGroupZ6UnitaryOfRoot (gaugeGroupZzRootToZ6Root α)
```

lemma gaugeGroupZzOfRoot_mem_center[\[source\]](#)

```
lemma gaugeGroupZzOfRoot_mem_center (α : rootsOfUnity 2 ℂ) :  
  gaugeGroupZzOfRoot α ∈ Subgroup.center GaugeGroupI
```

def gaugeGroupZzHom[\[source\]](#)

The homomorphism from second roots of unity to GaugeGroupI.

```
noncomputable def gaugeGroupZzHom : rootsOfUnity 2 ℂ →* GaugeGroupI
```

lemma gaugeGroupZzHom_apply[\[source\]](#)

```
@[simp]
```

```
lemma gaugeGroupZzHom_apply (α : rootsOfUnity 2 ℂ) :  
  gaugeGroupZzHom α = gaugeGroupZzOfRoot α
```

lemma gaugeGroupZzHom_toSU3[\[source\]](#)

```
@[simp]
```

```
lemma gaugeGroupZzHom_toSU3 (α : rootsOfUnity 2 ℂ) :  
  GaugeGroupI.toSU3 (gaugeGroupZzHom α) =  
    gaugeGroupZ6SU3OfRoot (gaugeGroupZzRootToZ6Root α)
```

lemma gaugeGroup $\mathbb{Z}_2$ Hom_toSU2[\[source\]](#)

```
@[simp]
lemma gaugeGroup $\mathbb{Z}_2$ Hom_toSU2 ( $\alpha$  : rootsOfUnity 2  $\mathbb{C}$ ) :
  GaugeGroupI.toSU2 (gaugeGroup $\mathbb{Z}_2$ Hom  $\alpha$ ) =
    gaugeGroup $\mathbb{Z}_6$ SU2OfRoot (gaugeGroup $\mathbb{Z}_2$ RootTo $\mathbb{Z}_6$ Root  $\alpha$ )
```

lemma gaugeGroup $\mathbb{Z}_2$ Hom_toU1[\[source\]](#)

```
@[simp]
lemma gaugeGroup $\mathbb{Z}_2$ Hom_toU1 ( $\alpha$  : rootsOfUnity 2  $\mathbb{C}$ ) :
  GaugeGroupI.toU1 (gaugeGroup $\mathbb{Z}_2$ Hom  $\alpha$ ) =
    gaugeGroup $\mathbb{Z}_6$ UnitaryOfRoot (gaugeGroup $\mathbb{Z}_2$ RootTo $\mathbb{Z}_6$ Root  $\alpha$ )
```

lemma gaugeGroup $\mathbb{Z}_2$ OfRoot_mem[\[source\]](#)

```
lemma gaugeGroup $\mathbb{Z}_2$ OfRoot_mem ( $\alpha$  : rootsOfUnity 2  $\mathbb{C}$ ) :
  gaugeGroup $\mathbb{Z}_2$ OfRoot  $\alpha$   $\in$  gaugeGroup $\mathbb{Z}_2$ SubGroup
```

lemma mem_gaugeGroup $\mathbb{Z}_2$ SubGroup_iff[\[source\]](#)

```
lemma mem_gaugeGroup $\mathbb{Z}_2$ SubGroup_iff ( $g$  : GaugeGroupI) :
   $g \in$  gaugeGroup $\mathbb{Z}_2$ SubGroup  $\leftrightarrow \exists \alpha$  : rootsOfUnity 2  $\mathbb{C}$ , gaugeGroup $\mathbb{Z}_2$ OfRoot  $\alpha$  =  $g$ 
```

lemma gaugeGroup $\mathbb{Z}_2$ SubGroup_le_gaugeGroup $\mathbb{Z}_6$ SubGroup[\[source\]](#)

```
lemma gaugeGroup $\mathbb{Z}_2$ SubGroup_le_gaugeGroup $\mathbb{Z}_6$ SubGroup :
  gaugeGroup $\mathbb{Z}_2$ SubGroup  $\leq$  gaugeGroup $\mathbb{Z}_6$ SubGroup
```

lemma gaugeGroup $\mathbb{Z}_2$ SubGroup_le_center[\[source\]](#)

```
lemma gaugeGroup $\mathbb{Z}_2$ SubGroup_le_center :
  gaugeGroup $\mathbb{Z}_2$ SubGroup  $\leq$  Subgroup.center GaugeGroupI
```

instance gaugeGroup $\mathbb{Z}_2$ SubGroup_normal[\[source\]](#)

```
instance gaugeGroup $\mathbb{Z}_2$ SubGroup_normal : gaugeGroup $\mathbb{Z}_2$ SubGroup.Normal where
```

instance Group GaugeGroup $\mathbb{Z}_2$ [\[source\]](#)

```
noncomputable instance : Group GaugeGroup $\mathbb{Z}_2$ 
```

def GaugeGroup $\mathbb{Z}_2$.mk[\[source\]](#)

The quotient map from GaugeGroupI to GaugeGroup $\mathbb{Z}_2$.

```
noncomputable def mk : GaugeGroupI  $\rightarrow^*$  GaugeGroup $\mathbb{Z}_2$ 
```

lemma mk_gaugeGroup $\mathbb{Z}_2$ OfRoot[\[source\]](#)

```
@[simp]
lemma mk_gaugeGroupZ2OfRoot (α : rootsOfUnity 2 ℂ) :
  mk (gaugeGroupZ2OfRoot α) = 1
```

def gaugeGroupZ₃RootToZ₆Root

[\[source\]](#)

The inclusion of third roots of unity into sixth roots of unity.

```
noncomputable def gaugeGroupZ3RootToZ6Root : rootsOfUnity 3 ℂ →* rootsOfUnity 6 ℂ
```

def gaugeGroupZ₃OfRoot

[\[source\]](#)

The element of GaugeGroupI associated to a third root of unity.

```
noncomputable def gaugeGroupZ3OfRoot (α : rootsOfUnity 3 ℂ) : GaugeGroupI
```

lemma gaugeGroupZ₃OfRoot_toSU3

[\[source\]](#)

```
@[simp]
lemma gaugeGroupZ3OfRoot_toSU3 (α : rootsOfUnity 3 ℂ) :
  GaugeGroupI.toSU3 (gaugeGroupZ3OfRoot α) =
    gaugeGroupZ6SU3OfRoot (gaugeGroupZ3RootToZ6Root α)
```

lemma gaugeGroupZ₃OfRoot_toSU2

[\[source\]](#)

```
@[simp]
lemma gaugeGroupZ3OfRoot_toSU2 (α : rootsOfUnity 3 ℂ) :
  GaugeGroupI.toSU2 (gaugeGroupZ3OfRoot α) =
    gaugeGroupZ6SU2OfRoot (gaugeGroupZ3RootToZ6Root α)
```

lemma gaugeGroupZ₃OfRoot_toU1

[\[source\]](#)

```
@[simp]
lemma gaugeGroupZ3OfRoot_toU1 (α : rootsOfUnity 3 ℂ) :
  GaugeGroupI.toU1 (gaugeGroupZ3OfRoot α) =
    gaugeGroupZ6UnitaryOfRoot (gaugeGroupZ3RootToZ6Root α)
```

lemma gaugeGroupZ₃OfRoot_mem_center

[\[source\]](#)

```
lemma gaugeGroupZ3OfRoot_mem_center (α : rootsOfUnity 3 ℂ) :
  gaugeGroupZ3OfRoot α ∈ Subgroup.center GaugeGroupI
```

def gaugeGroupZ₃Hom

[\[source\]](#)

The homomorphism from third roots of unity to GaugeGroupI.

```
noncomputable def gaugeGroupZ3Hom : rootsOfUnity 3 ℂ →* GaugeGroupI
```

lemma gaugeGroupZ₃Hom_apply

[\[source\]](#)

```
@[simp]
lemma gaugeGroupZ3Hom_apply (α : rootsOfUnity 3 ℂ) :
```

```
gaugeGroupZ₃Hom α = gaugeGroupZ₃OfRoot α
```

lemma gaugeGroupZ₃Hom_toSU3

[\[source\]](#)

```
@[simp]
lemma gaugeGroupZ₃Hom_toSU3 (α : rootsOfUnity 3 ℂ) :
  GaugeGroupI.toSU3 (gaugeGroupZ₃Hom α) =
    gaugeGroupZ₆SU3OfRoot (gaugeGroupZ₃RootToZ₆Root α)
```

lemma gaugeGroupZ₃Hom_toSU2

[\[source\]](#)

```
@[simp]
lemma gaugeGroupZ₃Hom_toSU2 (α : rootsOfUnity 3 ℂ) :
  GaugeGroupI.toSU2 (gaugeGroupZ₃Hom α) =
    gaugeGroupZ₆SU2OfRoot (gaugeGroupZ₃RootToZ₆Root α)
```

lemma gaugeGroupZ₃Hom_toU1

[\[source\]](#)

```
@[simp]
lemma gaugeGroupZ₃Hom_toU1 (α : rootsOfUnity 3 ℂ) :
  GaugeGroupI.toU1 (gaugeGroupZ₃Hom α) =
    gaugeGroupZ₆UnitaryOfRoot (gaugeGroupZ₃RootToZ₆Root α)
```

lemma gaugeGroupZ₃OfRoot_mem

[\[source\]](#)

```
lemma gaugeGroupZ₃OfRoot_mem (α : rootsOfUnity 3 ℂ) :
  gaugeGroupZ₃OfRoot α ∈ gaugeGroupZ₃SubGroup
```

lemma mem_gaugeGroupZ₃SubGroup_iff

[\[source\]](#)

```
lemma mem_gaugeGroupZ₃SubGroup_iff (g : GaugeGroupI) :
  g ∈ gaugeGroupZ₃SubGroup ↔ ∃ α : rootsOfUnity 3 ℂ, gaugeGroupZ₃OfRoot α = g
```

lemma gaugeGroupZ₃SubGroup_le_gaugeGroupZ₆SubGroup

[\[source\]](#)

```
lemma gaugeGroupZ₃SubGroup_le_gaugeGroupZ₆SubGroup :
  gaugeGroupZ₃SubGroup ≤ gaugeGroupZ₆SubGroup
```

lemma gaugeGroupZ₃SubGroup_le_center

[\[source\]](#)

```
lemma gaugeGroupZ₃SubGroup_le_center :
  gaugeGroupZ₃SubGroup ≤ Subgroup.center GaugeGroupI
```

instance gaugeGroupZ₃SubGroup_normal

[\[source\]](#)

```
instance gaugeGroupZ₃SubGroup_normal : gaugeGroupZ₃SubGroup.Normal where
```

instance Group GaugeGroupZ₃

[\[source\]](#)

```
noncomputable instance : Group GaugeGroupZ₃
```

def GaugeGroup $\mathbb{Z}_3$.mk[\[source\]](#)*The quotient map from GaugeGroupI to GaugeGroup $\mathbb{Z}_3$.*

```
noncomputable def mk : GaugeGroupI →* GaugeGroup $\mathbb{Z}_3$ 
```

lemma mk_gaugeGroup $\mathbb{Z}_3$ OfRoot[\[source\]](#)

```
@[simp]
```

```
lemma mk_gaugeGroup $\mathbb{Z}_3$ OfRoot (α : rootsOfUnity 3  $\mathbb{C}$ ) :  
  mk (gaugeGroup $\mathbb{Z}_3$ OfRoot α) = 1
```

instance Group (GaugeGroup q)[\[source\]](#)

```
noncomputable instance (q : GaugeGroupQuot) : Group (GaugeGroup q)
```

def subgroup[\[source\]](#)*The central subgroup of GaugeGroupI quotiented by a gauge-group quotient choice.*

```
noncomputable def subgroup : GaugeGroupQuot → Subgroup GaugeGroupI  
| . $\mathbb{Z}_6$  => gaugeGroup $\mathbb{Z}_6$ SubGroup  
| . $\mathbb{Z}_2$  => gaugeGroup $\mathbb{Z}_2$ SubGroup  
| . $\mathbb{Z}_3$  => gaugeGroup $\mathbb{Z}_3$ SubGroup  
| .I => 1
```

lemma subgroup_le_center[\[source\]](#)*The subgroup attached to a gauge-group quotient choice lies in the center of GaugeGroupI.*

```
lemma subgroup_le_center (q : GaugeGroupQuot) :  
  subgroup q ≤ Subgroup.center GaugeGroupI
```

instance subgroup_normal[\[source\]](#)*The subgroup attached to a gauge-group quotient choice is normal in GaugeGroupI.*

```
instance subgroup_normal (q : GaugeGroupQuot) : (subgroup q).Normal
```

def quotientMap[\[source\]](#)*The quotient map from GaugeGroupI to the gauge group selected by a quotient choice.*

```
noncomputable def quotientMap (q : GaugeGroupQuot) : GaugeGroupI →* GaugeGroup q
```

lemma quotientMap_I_apply[\[source\]](#)

```
@[simp]
```

```
lemma quotientMap_I_apply (g : GaugeGroupI) :  
  quotientMap .I g = g
```

lemma quotientMap_ℤ₆_gaugeGroupℤ₆OfRoot[\[source\]](#)

@[simp]

```
lemma quotientMap_ℤ₆_gaugeGroupℤ₆OfRoot (α : rootsOfUnity 6 ℂ) :
  quotientMap .ℤ₆ (gaugeGroupℤ₆OfRoot α) = 1
```

lemma quotientMap_ℤ₂_gaugeGroupℤ₂OfRoot[\[source\]](#)

@[simp]

```
lemma quotientMap_ℤ₂_gaugeGroupℤ₂OfRoot (α : rootsOfUnity 2 ℂ) :
  quotientMap .ℤ₂ (gaugeGroupℤ₂OfRoot α) = 1
```

lemma quotientMap_ℤ₃_gaugeGroupℤ₃OfRoot[\[source\]](#)

@[simp]

```
lemma quotientMap_ℤ₃_gaugeGroupℤ₃OfRoot (α : rootsOfUnity 3 ℂ) :
  quotientMap .ℤ₃ (gaugeGroupℤ₃OfRoot α) = 1
```

lemma mem_subgroup_iff_quotientMap_eq_one[\[source\]](#)

The kernel of the quotient map is the subgroup selected by the quotient choice.

```
lemma mem_subgroup_iff_quotientMap_eq_one (q : GaugeGroupQuot) (g : GaugeGroupI) :
  g ∈ subgroup q ↔ quotientMap q g = 1
```

lemma quotientMap_eq_iff[\[source\]](#)

Two representatives have the same image under the selected quotient map exactly when their quotient lies in the subgroup selected by the quotient choice.

```
lemma quotientMap_eq_iff (q : GaugeGroupQuot) (g h : GaugeGroupI) :
  quotientMap q g = quotientMap q h ↔ g / h ∈ subgroup q
```

2.20 Physlib.Particles.StandardModel.HiggsBoson.Basic

[Physlib/Particles/StandardModel/HiggsBoson/Basic.lean](#) on GitHub.

instance SMulCommClass ℂ GaugeGroupI HiggsVec[\[source\]](#)

```
instance : SMulCommClass ℂ GaugeGroupI HiggsVec where
```

instance SMulCommClass ℝ GaugeGroupI HiggsVec[\[source\]](#)

```
instance : SMulCommClass ℝ GaugeGroupI HiggsVec where
```

def toRealScalars[\[source\]](#)

The underlying real values of the Higgs vector.

```
def toRealScalars : HiggsVec →ₗ[ℝ] (Fin 4 → ℝ) where
```

lemma toRealScalars_smul_real[\[source\]](#)

```
lemma toRealScalars_smul_real (a : ℝ) (φ : HiggsVec) :
  toRealScalars (a • φ) = a • toRealScalars φ
```

lemma ofReal_toRealScalars[\[source\]](#)

```
lemma ofReal_toRealScalars (a : ℝ) :
  toRealScalars (ofReal a) = ![Real.sqrt a, 0, 0, 0]
```

lemma ofReal_toRealScalars_norm[\[source\]](#)

```
lemma ofReal_toRealScalars_norm (φ : HiggsVec) :
  toRealScalars (ofReal (||φ|| ^ 2)) = ![||φ||, 0, 0, 0]
```

2.21 Physlib.Particles.StandardModel.HiggsBoson.EffectivePotential

[Physlib/Particles/StandardModel/HiggsBoson/EffectivePotential.lean](#) on GitHub.

abbrev EffectivePotential[\[source\]](#)

A general potential of the Higgs field.

```
abbrev EffectivePotential : Type
```

def IsInvariant[\[source\]](#)

The proposition that the general potential is invariant under the global action of the gauge group.

```
def IsInvariant (V : EffectivePotential) : Prop
```

lemma eq_on_orbits[\[source\]](#)

An invariant potential is equal on gauge orbits.

```
lemma eq_on_orbits {φ1 φ2 : HiggsVec} {V : EffectivePotential} (h : IsInvariant V)
  (hφ : φ1 ∈ MulAction.orbit GaugeGroupI φ2) :
  V φ1 = V φ2
```

lemma eq_of_norm_eq[\[source\]](#)

An invariant potential is equal on Higgs vectors with identical norms.

```
lemma eq_of_norm_eq {φ1 φ2 : HiggsVec} {V : EffectivePotential} (h : IsInvariant V)
  (hφ : ||φ1|| = ||φ2||) :
  V φ1 = V φ2
```

lemma factors_through_norm[\[source\]](#)

```
lemma factors_through_norm {V : EffectivePotential} (h : IsInvariant V) :
  ∃ (f : ℝ → ℝ), V = f ∘ norm
```


def HasMaxMassDimLE[\[source\]](#)

The proposition that the potential V has a maximum mass dimension less then or equal to n - also implying it is a polynomial.

```
def HasMaxMassDimLE (V : EffectivePotential) (n : ℕ) : Prop
```

def polynomial[\[source\]](#)

The polynomial associated to a potential V with a maximum mass dimension less than or equal to n .

```
def polynomial (V : EffectivePotential) {n : ℕ} (h : HasMaxMassDimLE V n) :  
  MvPolynomial (Fin 4) ℝ
```

lemma polynomial_totalDegree[\[source\]](#)

```
lemma polynomial_totalDegree {V : EffectivePotential} {n : ℕ} (h : HasMaxMassDimLE V n) :  
  (polynomial V h).totalDegree ≤ n
```

lemma apply_eq_polynomial[\[source\]](#)

```
lemma apply_eq_polynomial {V : EffectivePotential} {n : ℕ} (h : HasMaxMassDimLE V n)  
  (φ : HiggsVec) : V φ = (polynomial V h).eval φ.toRealScalars
```

def termOfMassDim[\[source\]](#)

The part of a potential at a given mass-dimension.

```
def termOfMassDim (V : EffectivePotential) {n : ℕ} (h : HasMaxMassDimLE V n) (m : ℕ) :  
  HiggsVec → ℝ
```

lemma termOfMassDim_eq_zero_of_max_lt[\[source\]](#)

```
lemma termOfMassDim_eq_zero_of_max_lt {V : EffectivePotential} {n : ℕ} (h : HasMaxMassDimLE V n)  
  )  
  {m : ℕ} (hm : n < m) (φ : HiggsVec) :  
  termOfMassDim V h m φ = 0
```

lemma termOfMassDim_homogeneity[\[source\]](#)

```
lemma termOfMassDim_homogeneity {V : EffectivePotential} {n : ℕ} (h : HasMaxMassDimLE V n) (m :  
  ℕ)  
  (φ : HiggsVec) (t : ℝ) : termOfMassDim V h m (t • φ) = t ^ m * termOfMassDim V h m φ
```

lemma apply_eq_sum_termOfMassDim[\[source\]](#)

```
lemma apply_eq_sum_termOfMassDim {V : EffectivePotential} {n : ℕ} (h : HasMaxMassDimLE V n)  
  (φ : HiggsVec) :  
  V φ = ∑ m ∈ Finset.range (n + 1), termOfMassDim V h m φ
```

lemma apply_smul_eq_sum_termOfMassDim[\[source\]](#)

```
lemma apply_smul_eq_sum_termOfMassDim {V : EffectivePotential} {n : ℕ} (h : HasMaxMassDimLE V n)
  (φ : HiggsVec) (t : ℝ) :
  V (t • φ) = ∑ m ∈ Finset.range (n + 1), t ^ m * termOfMassDim V h m φ
```

lemma termOfMassDim_isInvariant[\[source\]](#)

```
lemma termOfMassDim_isInvariant {V : EffectivePotential} {n : ℕ} (h : HasMaxMassDimLE V n)
  (m : ℕ) (hV : IsInvariant V) : IsInvariant (termOfMassDim V h m)
```

lemma termOfMassDim_eq_mul_norm[\[source\]](#)

```
lemma termOfMassDim_eq_mul_norm {V : EffectivePotential} {n : ℕ}
  (h : HasMaxMassDimLE V n) (m : ℕ) (hV : IsInvariant V) (φ : HiggsVec) :
  ∃ c, termOfMassDim V h m φ = c * ||φ|| ^ m
```

lemma termOfMassDim_zero_of_odd[\[source\]](#)

```
lemma termOfMassDim_zero_of_odd {V : EffectivePotential} {n : ℕ} (h : HasMaxMassDimLE V n) (m :
  ℕ)
  (hV : IsInvariant V) (φ : HiggsVec) (hodd : Odd m) :
  termOfMassDim V h m φ = 0
```

lemma apply_eq_sum_even_termOfMassDim[\[source\]](#)

```
lemma apply_eq_sum_even_termOfMassDim {V : EffectivePotential} {n : ℕ} (h : HasMaxMassDimLE V n)
  (hV : IsInvariant V) (φ : HiggsVec) :
  V φ = ∑ m ∈ Finset.range (n / 2 + 1), termOfMassDim V h (2 * m) φ
```

lemma apply_eq_sum_even_termOfMassDim_fin[\[source\]](#)

```
lemma apply_eq_sum_even_termOfMassDim_fin {V : EffectivePotential} {n : ℕ} (h : HasMaxMassDimLE
  V n)
  (hV : IsInvariant V) (φ : HiggsVec) :
  V φ = ∑ m : Fin (n/2 + 1), termOfMassDim V h (2 * m) φ
```

lemma apply_eq_sum_norm_pow[\[source\]](#)

The potential is equal to the sum of norms to even powers.

```
lemma apply_eq_sum_norm_pow {V : EffectivePotential} {n : ℕ} (h : HasMaxMassDimLE V n)
  (hV : IsInvariant V) (φ : HiggsVec) :
  ∃ c : Fin (n/2 + 1) → ℝ, V φ = ∑ m, c m • ||φ|| ^ (2 * m.1)
```

2.22 Physlib.Particles.SuperSymmetry.MSSMNu.AnomalyCancellation.Basic

[Physlib/Particles/SuperSymmetry/MSSMNu/AnomalyCancellation/Basic.lean](#) on GitHub.

lemma sum_MSSMSpecies_numberCharges_eq_expand[\[source\]](#)

```

lemma sum_MSSMSpecies_numberCharges_eq_expand [AddCommMonoid M]
  (f : Fin MSSMSpecies.numberCharges → M) :
  ∑ i, f i = f ⟨0, by simp⟩ + f ⟨1, by simp⟩ + f ⟨2, by simp⟩

```

2.23 Physlib.Particles.SuperSymmetry.N1.Basic

[Physlib/Particles/SuperSymmetry/N1/Basic.lean](#) on GitHub.

abbrev ChiralScalarConfiguration[\[source\]](#)

*The chiral scalar configuration: a complex value for each chiral label. This is the sector's only field data. Declared as an **abbrev** so that unification sees through it to `ChiralIndexingType → ℂ` and applies Mathlib's function-space calculus lemmas directly.*

```

abbrev ChiralScalarConfiguration (ChiralIndexingType : Type*)

```

inductive ChiralColor[\[source\]](#)

*The four colours carried by a chiral-sector index: holomorphy (**chiral** versus **anti**, a scalar versus its complex conjugate) crossed with variance (**up** versus **down**, contravariant versus covariant). Carrying both axes here lets the single index type `ι` label the scalars.*

```

inductive ChiralColor | chiralUp | chiralDown | antiUp | antiDown

```

def tau[\[source\]](#)

The dual colour: flips variance and preserves holomorphy. Two indices may contract exactly when their colours are τ -related, so V^I pairs only with V_I (same holomorphy, opposite variance) and never with a conjugate index.

```

def tau : ChiralColor → ChiralColor
| chiralUp => chiralDown
| chiralDown => chiralUp
| antiUp => antiDown
| antiDown => antiUp

```

def bar[\[source\]](#)

*The conjugate colour: flips holomorphy (**chiral** ↔ **anti**) and preserves variance. Complex conjugation sends an index to its conjugate carrier, so **bar** swaps **chiral*** with **anti***. Distinct from the variance dual **tau**; the two commute (**bar_tau**).*

```

def bar : ChiralColor → ChiralColor
| chiralUp => antiUp
| antiUp => chiralUp
| chiralDown => antiDown
| antiDown => chiralDown

```

abbrev chiralModule[\[source\]](#)

The carrier module of each colour, distinct for all four: the holomorphic vectors $\iota \rightarrow \mathbb{C}$ and their dual $\text{Module.Dual } \mathbb{C} (\iota \rightarrow \mathbb{C})$ on the chiral side, and the conjugate module $\text{ConjModule } \dots$ of each (where i acts as $-i$) on the anti side. Variance is the vector/dual axis, holomorphy the conjugate-module axis; both are genuine carrier data, not labels tracked separately.

```
abbrev chiralModule : ChiralColor → Type
| .chiralUp    =>  $\iota \rightarrow \mathbb{C}$ 
| .chiralDown =>  $\text{Module.Dual } \mathbb{C} (\iota \rightarrow \mathbb{C})$ 
| .antiUp      =>  $\text{ConjModule } (\iota \rightarrow \mathbb{C})$ 
| .antiDown    =>  $\text{ConjModule } (\text{Module.Dual } \mathbb{C} (\iota \rightarrow \mathbb{C}))$ 
```

instance instAddCommGroupChiralModule[\[source\]](#)

```
instance instAddCommGroupChiralModule :  $\forall c, \text{AddCommGroup } (\text{chiralModule } (\iota := \iota) c)$ 
| .chiralUp | .chiralDown | .antiUp | .antiDown => inferInstance
```

instance instModuleChiralModule[\[source\]](#)

```
noncomputable instance instModuleChiralModule :  $\forall c, \text{Module } \mathbb{C} (\text{chiralModule } (\iota := \iota) c)$ 
| .chiralUp | .chiralDown | .antiUp | .antiDown => inferInstance
```

def chiralRep[\[source\]](#)

The representation on each colour, taken trivial over the trivial group Unit : the chiral scalars carry no charge in this sector.

```
def chiralRep : (c : ChiralColor) →  $\text{Representation } \mathbb{C} \text{ Unit } (\text{chiralModule } (\iota := \iota) c)$ 
```

def piBasis[\[source\]](#)

The standard basis of the vector carrier $\iota \rightarrow \mathbb{C}$ (the indicator functions); the basis of chiralUp and the reference basis for the δ pairing.

```
def piBasis :  $\text{Basis } \iota \mathbb{C} (\iota \rightarrow \mathbb{C})$ 
```

def chiralBasis[\[source\]](#)

The basis of each colour's carrier, all indexed by ι : piBasis on the holomorphic vectors, its dual piBasis.dualBasis on the holomorphic covectors, and the Basis.conj of each on the anti-holomorphic side (coordinates *star-ed*).

```
noncomputable def chiralBasis : (c : ChiralColor) →  $\text{Basis } \iota \mathbb{C} (\text{chiralModule } (\iota := \iota) c)$ 
| .chiralUp    => piBasis
| .chiralDown => piBasis.dualBasis
| .antiUp      =>  $\text{Basis.conj } \text{piBasis}$ 
| .antiDown    =>  $\text{Basis.conj } \text{piBasis.dualBasis}$ 
```

def deltaCap[\[source\]](#)

The δ cap $\sum_I b_I \otimes b_I$: the rank-2 tensor in $M \otimes M$ with two upper indices, whose components in the basis b are δ^{ij} . It is an element of $M \otimes M$ (the inverse-metric

"cap"), not a linear map, and serves as the metric field of the species (whose two slots share a colour).

```
def deltaCap (b : Basis ι ℂ M) : M ⊗[ℂ] M
```

def deltaBil₂

[\[source\]](#)

The δ pairing between two based modules sharing the index ι : the dot product of coordinate vectors $(x, y) \mapsto \sum_I (b\ x)_I (b'\ y)_I$. Built from Mathlib's `Basis.toDual` b (the canonical δ map $M \rightarrow \text{Module.Dual } M$, sending b to its dual basis) precomposed on the second slot with the basis transport $b' = b$.

```
def deltaBil2 (b : Basis ι ℂ M) (b' : Basis ι ℂ N) : M →l[ℂ] N →l[ℂ] ℂ
```

lemma deltaBil₂_apply

[\[source\]](#)

$\text{deltaBil}_2\ b\ b'\ x\ y = \sum_I (b\ x)_I (b'\ y)_I$.

```
lemma deltaBil2_apply (b : Basis ι ℂ M) (b' : Basis ι ℂ N) (x : M) (y : N) :
  deltaBil2 b b' x y = ∑ I, b.equivFun x I * b'.equivFun y I
```

def deltaContr₂

[\[source\]](#)

The two-module δ contraction $M \otimes N \rightarrow \mathbb{C}$.

```
def deltaContr2 (b : Basis ι ℂ M) (b' : Basis ι ℂ N) : M ⊗[ℂ] N →l[ℂ] ℂ
```

lemma deltaContr₂_tmul

[\[source\]](#)

$\text{deltaContr}_2\ b\ b'\ (x \otimes_t y) = \sum_I (b\ x)_I (b'\ y)_I$.

```
lemma deltaContr2_tmul (b : Basis ι ℂ M) (b' : Basis ι ℂ N) (x : M) (y : N) :
  deltaContr2 b b' (x ⊗t[ℂ] y) = ∑ I, b.equivFun x I * b'.equivFun y I
```

lemma deltaContr₂_tmul_basis

[\[source\]](#)

$\text{deltaContr}_2\ b\ b'\ (x \otimes_t b'\ J) = x_J$: pairing with the second basis reads off a coordinate.

```
lemma deltaContr2_tmul_basis (b : Basis ι ℂ M) (b' : Basis ι ℂ N) (x : M) (J : ι) :
  deltaContr2 b b' (x ⊗t[ℂ] b' J) = b.equivFun x J
```

lemma deltaContr₂_basis_basis

[\[source\]](#)

$\text{deltaContr}_2\ b\ b'\ (b\ I \otimes_t b'\ J) = \delta_{IJ}$: the two bases are δ -dual.

```
lemma deltaContr2_basis_basis (b : Basis ι ℂ M) (b' : Basis ι ℂ N) (I J : ι) :
  deltaContr2 b b' (b I ⊗t[ℂ] b' J) = if I = J then 1 else 0
```

lemma deltaContr₂_comm

[\[source\]](#)

$\text{deltaContr}_2\ b\ b'\ (x \otimes_t y) = \text{deltaContr}_2\ b'\ b\ (y \otimes_t x)$: swapping slots swaps the two bases.

```
lemma deltaContr2_comm (b : Basis ι ℂ M) (b' : Basis ι ℂ N) (x : M) (y : N) :
  deltaContr2 b b' (x ⊗ℂ y) = deltaContr2 b' b (y ⊗ℂ x)
```

def deltaCap2

[\[source\]](#)

The two-module $\delta \text{ cap } \sum_I b_I \otimes b'_I \in M \otimes N$.

```
def deltaCap2 (b : Basis ι ℂ M) (b' : Basis ι ℂ N) : M ⊗ℂ N
```

lemma deltaCap2_comm

[\[source\]](#)

comm (deltaCap2 b b') = deltaCap2 b' b: swapping the two factors swaps the two bases.

```
lemma deltaCap2_comm (b : Basis ι ℂ M) (b' : Basis ι ℂ N) :
  TensorProduct.comm ℂ M N (deltaCap2 b b') = deltaCap2 b' b
```

lemma deltaUnit2_symm

[\[source\]](#)

The unit_symm law (two-module, toSpanSingleton form): deltaCap2 b' b is the swap of deltaCap2 b b'.

```
lemma deltaUnit2_symm (b : Basis ι ℂ M) (b' : Basis ι ℂ N) :
  LinearMap.toSpanSingleton ℂ _ (deltaCap2 b' b) 1 =
  LinearMap.lTensor N (LinearEquiv.refl ℂ M).toLinearMap
    (TensorProduct.comm ℂ M N (LinearMap.toSpanSingleton ℂ _ (deltaCap2 b b') 1))
```

lemma deltaContr2_unit

[\[source\]](#)

The snake identity (two-module, contr_unit law): contracting $x \in M$ into the M-leg of deltaCap2 b' b $\in N \otimes M$ returns x .

```
lemma deltaContr2_unit (b : Basis ι ℂ M) (b' : Basis ι ℂ N) (x : M) :
  (TensorProduct.lid ℂ M) ((deltaContr2 b b').rTensor M
    ((TensorProduct.assoc ℂ M N M).symm
      (x ⊗ℂ LinearMap.toSpanSingleton ℂ _ (deltaCap2 b' b) 1))) = x
```

lemma deltaContr2_metric

[\[source\]](#)

The contr_metric law (two-module): contracting the inner M/N legs of deltaCap b $\otimes$ deltaCap b' yields deltaCap2 b' b.

```
lemma deltaContr2_metric (b : Basis ι ℂ M) (b' : Basis ι ℂ N) :
  (TensorProduct.comm ℂ M N ((TensorProduct.lid ℂ N).lTensor M
    (((deltaContr2 b b').rTensor N).lTensor M
      (((TensorProduct.assoc ℂ M N N).symm.toLinearMap.lTensor M)
        ((TensorProduct.assoc ℂ M M (N ⊗ℂ N))
          (LinearMap.toSpanSingleton ℂ _ (deltaCap b) 1 ⊗ℂ
            LinearMap.toSpanSingleton ℂ _ (deltaCap b') 1)))))) =
  LinearMap.toSpanSingleton ℂ _ (deltaCap2 b' b) 1
```

def chiralTensor[\[source\]](#)

The chiral-index tensor species, bundled with its conjugation. Its four colours *chiral/anti* $\times$ *up/down* carry the four distinct carriers of $\mathcal{SC}$. *contr c* is the two-module δ pairing of a colour against its variance dual $\tau\ c\ (V\ c \otimes V\ (\tau\ c) \rightarrow \mathbb{C})$; *unit c* is the δ cap across those two carriers; *metric c* is the single-colour δ cap $\sum_I b_I \otimes b_I$. Each *TensorSpecies* coherence law reduces, by case analysis on the colour, to the corresponding abstract two-module δ lemma of $\mathcal{SD}$. The conjugation flips holomorphy (*ChiralColor.bar*) while preserving variance; every basis is indexed by $\mathfrak{I}$, so *barIdx_eq* is *rfl*, and *conj_contrComm* is *star* $\delta = \delta$. Instantiating *ConjTensorSpecies* this way gives the chiral sector both the framework's generic tensor API and its conjugation API (*conjT* and its laws) on one object.

```
def chiralTensor : ConjTensorSpecies  $\mathbb{C}$  ChiralColor Unit (chiralModule ( $\mathfrak{I} := \mathfrak{I}$ )) (fun _ =>  $\mathfrak{I}$ )
  (chiralRep ( $\mathfrak{I} := \mathfrak{I}$ )) (chiralBasis ( $\mathfrak{I} := \mathfrak{I}$ )) where
```

def conjScalar[\[source\]](#)

Conjugation of a scalar tensor, normalized back to the scalar colour list $![]$.

```
def conjScalar (t : (chiralTensor ( $\mathfrak{I} := \mathfrak{I}$ )).Tensor  $![]$ ) :
  (chiralTensor ( $\mathfrak{I} := \mathfrak{I}$ )).Tensor  $![]$ 
```

def conjChiralCovector[\[source\]](#)

Conjugation of a holomorphic covector, normalized to the anti-holomorphic covector colour list $![\text{antiDown}]$.

```
def conjChiralCovector
  (t : (chiralTensor ( $\mathfrak{I} := \mathfrak{I}$ )).Tensor  $![\text{chiralDown}]$ ) :
  (chiralTensor ( $\mathfrak{I} := \mathfrak{I}$ )).Tensor  $![\text{antiDown}]$ 
```

lemma toField_conjScalar[\[source\]](#)

For scalar tensors, *toField* of the normalized tensor conjugate is the complex conjugate of *toField*.

```
lemma toField_conjScalar (t : (chiralTensor ( $\mathfrak{I} := \mathfrak{I}$ )).Tensor  $![]$ ) :
  (conjScalar t).toField = star t.toField
```

lemma repr_conjChiralCovector[\[source\]](#)

Component formula for the holomorphic covector conjugate: the $![I]$ basis component of *conjChiralCovector t* is the complex conjugate of the $![I]$ component of *t*.

```
lemma repr_conjChiralCovector
  (t : (chiralTensor ( $\mathfrak{I} := \mathfrak{I}$ )).Tensor  $![\text{chiralDown}]$ ) (I :  $\mathfrak{I}$ ) :
  (basis (S := (chiralTensor ( $\mathfrak{I} := \mathfrak{I}$ )).toTensorSpecies)  $![\text{antiDown}]$ ).repr
    (conjChiralCovector t)  $![I]$  =
    star ((basis (S := (chiralTensor ( $\mathfrak{I} := \mathfrak{I}$ )).toTensorSpecies)  $![\text{chiralDown}]$ ).repr t  $![I]$ )
```

lemma conjChiralCovector_add[\[source\]](#)

Conjugation of a holomorphic covector is additive.

```
@[simp]
lemma conjChiralCovector_add
  (t₁ t₂ : (chiralTensor (ι := ι)).Tensor ![chiralDown]) :
  conjChiralCovector (t₁ + t₂) = conjChiralCovector t₁ + conjChiralCovector t₂
```

lemma conjChiralCovector_smul

[\[source\]](#)

Conjugation of a holomorphic covector is conjugate-linear: a scalar r pulls out as $\text{star } r$.

```
@[simp]
lemma conjChiralCovector_smul (r : ℂ)
  (t : (chiralTensor (ι := ι)).Tensor ![chiralDown]) :
  conjChiralCovector (r • t) = star r • conjChiralCovector t
```

2.24 Physlib.Particles.SuperSymmetry.SU5.ChargeSpectrum.AllowsTerm

[Physlib/Particles/SuperSymmetry/SU5/ChargeSpectrum/AllowsTerm.lean](#) on GitHub.

instance Decidable (AllowsTermQ5 x q5 T)

[\[source\]](#)

```
instance (x : ChargeSpectrum ℤ) (q5 : ℤ) (T : PotentialTerm) :
  Decidable (AllowsTermQ5 x q5 T)
```

instance Decidable (AllowsTermQ10 x q10 T)

[\[source\]](#)

```
instance (x : ChargeSpectrum ℤ) (q10 : ℤ) (T : PotentialTerm) :
  Decidable (AllowsTermQ10 x q10 T)
```

2.25 Physlib.Particles.SuperSymmetry.SU5.ChargeSpectrum.PhenoClosed

[Physlib/Particles/SuperSymmetry/SU5/ChargeSpectrum/PhenoClosed.lean](#) on GitHub.

instance Decidable (ContainsPhenoCompletionsOfMinimallyAllows S5 S10 charges)

[\[source\]](#)

```
instance {S5 S10 : Finset ℤ} {charges : Multiset (ChargeSpectrum ℤ)} :
  Decidable (ContainsPhenoCompletionsOfMinimallyAllows S5 S10 charges)
```

2.26 Physlib.QuantumMechanics.DDimensions.Operators.SpectralTheory.Basic

[Physlib/QuantumMechanics/DDimensions/Operators/SpectralTheory/Basic.lean](#) on GitHub.

abbrev resolvent

[\[source\]](#)

The resolvent, $(T - z \cdot 1)^{-1}$.

```
abbrev resolvent (T : H →ℓ.[ℂ] H) (z : ℂ) : H →ℓ.[ℂ] H
```


def IsLowerBound[\[source\]](#)

*IsLowerBound T z c is the property that $c * \|x\| \leq \|T x - z \cdot x\|$ for all $x : T.domain$.*

```
def IsLowerBound (T : H →l.[C] H) (z : C) (c : ℝ) : Prop
```

lemma isLowerBound_neg[\[source\]](#)

```
lemma isLowerBound_neg {T : H →l.[C] H} {z : C} {c : ℝ} (h : IsLowerBound T z c) :
  IsLowerBound (-T) (-z) c
```

lemma isLowerBound_of_right_le[\[source\]](#)

```
lemma isLowerBound_of_right_le
  {T : H →l.[C] H} {z : C} {c1 c2 : ℝ} (hle : c1 ≤ c2) (h : IsLowerBound T z c2) :
  IsLowerBound T z c1
```

lemma isLowerBound_of_left_le[\[source\]](#)

```
lemma isLowerBound_of_left_le
  {T1 T2 : H →l.[C] H} (hle : T1 ≤ T2) {z : C} {c : ℝ} (h : IsLowerBound T2 z c) :
  IsLowerBound T1 z c
```

lemma isLowerBound_closure[\[source\]](#)

```
lemma isLowerBound_closure
  {T : H →l.[C] H} {z : C} {c : ℝ} (h : IsLowerBound T z c) : IsLowerBound T.closure z c
```

def regularityDomain[\[source\]](#)

The regular points for T.

*z : C is a regular point for T iff there exists a constant $c > 0$ such that $c * \|x\| \leq \|(T - z \cdot 1) x\|$ for all $x \in T.domain$.*

```
def regularityDomain (T : H →l.[C] H) : Set C
```

lemma regularityDomain_neg[\[source\]](#)

```
@[simp]
lemma regularityDomain_neg (T : H →l.[C] H) : (-T).regularityDomain = -T.regularityDomain
```

lemma regularityDomain_smul[\[source\]](#)

```
@[simp]
lemma regularityDomain_smul (T : H →l.[C] H) {w : C} (hw : w ≠ 0) :
  (w • T).regularityDomain = w • T.regularityDomain
```

lemma regularityDomain_antitone[\[source\]](#)

T ≤ T' implies T'.regularityDomain ⊆ T.regularityDomain.

```
lemma regularityDomain_antitone : Antitone (regularityDomain (H := H))
```

lemma mem_regularityDomain_iff[\[source\]](#)

z is a regular point for T iff $T - z \cdot 1$ has a continuous (equivalently, bounded) inverse.

```
lemma mem_regularityDomain_iff {T : H →l.[C] H} {z : C} :
  z ∈ T.regularityDomain ↔ (T - z • 1).toFun.ker = ⊥ ∧ Continuous (R T z)
```

lemma ball_subset_regularityDomain[\[source\]](#)

The regularity domain of T contains open balls with radii controlled by the lower bounds.

```
lemma ball_subset_regularityDomain
  {T : H →l.[C] H} {z : C} {c : ℝ} (h : IsLowerBound T z c) : ball z c ⊆ T.regularityDomain
```

lemma regularityDomain_isOpen[\[source\]](#)

The regularity domain is an open set.

```
lemma regularityDomain_isOpen (T : H →l.[C] H) : IsOpen T.regularityDomain
```

lemma regularityDomain_closure[\[source\]](#)

T and $T.\text{closure}$ have the same regularity domain.

```
lemma regularityDomain_closure (T : H →l.[C] H) :
  T.closure.regularityDomain = T.regularityDomain
```

lemma closure_range_sub_eq_range_closure_sub[\[source\]](#)

```
lemma IsClosable.closure_range_sub_eq_range_closure_sub [CompleteSpace H]
  {T : H →l.[C] H} (hT : T.IsClosable) {z : C} (hz : z ∈ T.regularityDomain) :
  (T - z • 1).toFun.range.closure = (T.closure - z • 1).toFun.range
```

lemma sub_range_isClosed[\[source\]](#)

```
lemma IsClosed.sub_range_isClosed [CompleteSpace H]
  {T : H →l.[C] H} (hT : T.IsClosed) {z : C} (hz : z ∈ T.regularityDomain) :
  _root_.IsClosed ((T - z • 1).toFun.range : Set H)
```

lemma orthogonal_closure_sub_range[\[source\]](#)

$$(T.\text{closure} - z \cdot 1).\text{range}^\perp = (T^\dagger - \text{conj } z \cdot 1).\text{ker}$$

```
lemma IsUnbounded.orthogonal_closure_sub_range [CompleteSpace H]
  {T : H →l.[C] H} (hT : T.IsUnbounded) (z : C) :
  (T.closure - z • 1).toFun.range⊥
  = (T† - conj z • 1).toFun.ker.map (T† - conj z • 1).domain.subtype
```

lemma orthogonal_adjoint_sub_ker[\[source\]](#)

$$(T^\dagger - \text{conj } z \cdot 1).\text{ker}^\perp = (T.\text{closure} - z \cdot 1).\text{range}$$

```

lemma IsUnbounded.orthogonal_adjoint_sub_ker [CompleteSpace H]
  {T : H →l.[C] H} (hT : T.IsUnbounded) {z : C} (hz : z ∈ T.regularityDomain) :
  ((T† - conj z • 1).toFun.ker.map (T† - conj z • 1).domain.subtype)⊥
    = (T.closure - z • 1).toFun.range

```

def deficiencySubspace

[\[source\]](#)

For a partial linear map T and any complex number z , the closed submodule which is orthogonal to the range of $T - z \cdot 1$.

$T.defectNumber\ z$ is defined as the rank of this subspace.

```

def deficiencySubspace (T : H →l.[C] H) (z : C) : ClosedSubmodule C H

```

lemma deficiencySubspace_coe

[\[source\]](#)

```

@[simp]
lemma deficiencySubspace_coe (T : H →l.[C] H) (z : C) :
  T.deficiencySubspace z = (T - z • 1).toFun.range⊥

```

def defectNumber

[\[source\]](#)

The rank of $T.deficiencySubspace\ z = (T - z \cdot 1).range^{\perp}$.

```

def defectNumber (T : H →l.[C] H) (z : C) : Cardinal

```

lemma defectNumber_eq

[\[source\]](#)

```

lemma defectNumber_eq (T : H →l.[C] H) (z : C) :
  T.defectNumber z = Module.rank C (T.deficiencySubspace z)

```

lemma defectNumber_eq_zero_iff

[\[source\]](#)

```

lemma IsClosed.defectNumber_eq_zero_iff [CompleteSpace H]
  {T : H →l.[C] H} (hT : T.IsClosed) {z : C} (hz : z ∈ T.regularityDomain) :
  T.defectNumber z = 0 ↔ (T - z • 1).toFun.range = T

```

lemma defectNumber_closure

[\[source\]](#)

T and $T.closure$ have the same defect number at points in their regularity domain.

```

lemma defectNumber_closure [CompleteSpace H]
  {T : H →l.[C] H} {z : C} (hz : z ∈ T.regularityDomain) :
  T.closure.defectNumber z = T.defectNumber z

```

lemma inf_ne_bot_of_rank_lt

[\[source\]](#)

```

lemma _root_.Submodule.inf_ne_bot_of_rank_lt
  {E F : Submodule C H} [E.HasOrthogonalProjection] (h_rank : Module.rank C E < Module.rank
    C F) :
  E⊥ ⊓ F ≠ ⊥

```

lemma exists_inner_eq_zero_of_defectNumber_lt[\[source\]](#)

```

lemma IsClosed.exists_inner_eq_zero_of_defectNumber_lt [CompleteSpace H]
  {T : H →l.[C] H} (hT : T.IsClosed)
  {z₁ z₂ : C} (hz₁ : z₁ ∈ T.regularityDomain) (h : T.defectNumber z₁ < T.defectNumber z₂) :
  ∃ x : T.domain, x ≠ 0 ∧ ⟨⟨T x - z₁ • x, T x - z₂ • x⟩⟩_C = 0

```

lemma defectNumber_eq_of_mem_ball[\[source\]](#)

```

lemma IsClosable.defectNumber_eq_of_mem_ball [CompleteSpace H] {T : H →l.[C] H} (hT : T.
  IsClosable)
  {z₁ z₂ : C} {c : ℝ} (h : IsLowerBound T z₁ c) (h_ball : z₂ ∈ ball z₁ c) :
  T.defectNumber z₁ = T.defectNumber z₂

```

lemma defectNumber_const[\[source\]](#)

The defect number is constant on each connected component of the regularity domain.

```

lemma IsClosable.defectNumber_const [CompleteSpace H]
  {T : H →l.[C] H} (hT : T.IsClosable)
  {z₁ z₂ : C} (hz : z₂ ∈ connectedComponentIn T.regularityDomain z₁) :
  T.defectNumber z₁ = T.defectNumber z₂

```

def numericalRange[\[source\]](#)

The set $\{\langle x, T x \rangle\rangle_C \mid x \in T.\text{domain} \wedge \|x\| = 1\} \subseteq C$.

```

def numericalRange (T : H →l.[C] H) : Set C

```

lemma numericalRange_eq[\[source\]](#)

```

lemma numericalRange_eq (T : H →l.[C] H) : 0 T = (fun x ↦ ⟨⟨T x, T x⟩⟩_C) '' {x | \|x\| = 1}

```

lemma mem_numericalRange[\[source\]](#)

```

lemma mem_numericalRange {T : H →l.[C] H} {x : T.domain} (hx : x ≠ 0) :
  ( \|x\| ^ 2 )-1 * ⟨⟨T x, T x⟩⟩_C ∈ 0 T

```

lemma numericalRange_nonempty[\[source\]](#)

```

lemma numericalRange_nonempty {T : H →l.[C] H} (hT : T.domain ≠ ⊥) : (0 T).Nonempty

```

lemma numericalRange_neg[\[source\]](#)

```

@[simp]
lemma numericalRange_neg (T : H →l.[C] H) : 0 (-T) = -0 T

```

lemma numericalRange_smul[\[source\]](#)

```

@[simp]
lemma numericalRange_smul (T : H →l.[C] H) (c : C) : 0 (c • T) = c • 0 T

```

lemma numericalRange_sub_const[\[source\]](#)

```
lemma numericalRange_sub_const (T : H →l.[ℂ] H) (c : ℂ) : 0 (T - c • 1) = 0 T - {c}
```

lemma compl_closure_numericalRange_subset_regularityDomain[\[source\]](#)

The regularity domain contains the exterior of the numerical range.

```
lemma compl_closure_numericalRange_subset_regularityDomain (T : H →l.[ℂ] H) :  
  (_root_.closure (0 T))c ⊆ T.regularityDomain
```

lemma exists_phase_add_im_eq_zero[\[source\]](#)

```
private lemma exists_phase_add_im_eq_zero (z1 z2 : ℂ) :  
  ∃ θ : ℝ, (exp (I * θ) * z1 + exp (-I * θ) * z2).im = 0
```

theorem numericalRange_convex[\[source\]](#)

The Toeplitz-Hausdorff theorem.

```
theorem numericalRange_convex (T : H →l.[ℂ] H) : Convex ℝ (0 T)
```

def resolventSet[\[source\]](#)

The resolvent set, ρ , of a partial linear map.

A complex number z is in ρT iff the linear map $T - z • 1$ from $T.domain$ to H is a bijection with continuous (equivalently, bounded) inverse.

```
def resolventSet (T : H →l.[ℂ] H) : Set ℂ
```

lemma LinearPMap.resolventSet_eq[\[source\]](#)

```
lemma resolventSet_eq (T : H →l.[ℂ] H) :  
  ρ T = {z | (T - z • 1).toFun.ker = ⊥ ∧ (T - z • 1).toFun.range = T ∧ Continuous (R T z)}
```

lemma mem_resolventSet_iff[\[source\]](#)

```
lemma mem_resolventSet_iff {T : H →l.[ℂ] H} {z : ℂ} :  
  z ∈ ρ T ↔ (T - z • 1).toFun.ker = ⊥ ∧ (T - z • 1).toFun.range = T ∧ Continuous (R T z)
```

lemma resolventSet_eq_empty[\[source\]](#)

If an operator is not closed then its resolvent set is empty.

```
lemma resolventSet_eq_empty [CompleteSpace H] {T : H →l.[ℂ] H} (h : ¬T.IsClosed) : ρ T = ∅
```

lemma resolventSet_subset_regularityDomain[\[source\]](#)

```
lemma resolventSet_subset_regularityDomain (T : H →l.[ℂ] H) : ρ T ⊆ T.regularityDomain
```

lemma IsClosed.resolventSet_eq[\[source\]](#)

For a closed operator the continuity of the resolvent is redundant in the definition of the resolvent set.

```
lemma IsClosed.resolventSet_eq [CompleteSpace H] {T : H →l.[C] H} (hT : T.IsClosed) :
  ρ T = {z : C | (T - z • 1).toFun.ker = ⊥ ∧ (T - z • 1).toFun.range = T}
```

lemma resolventSet_eq'[\[source\]](#)

For a closed operator the resolvent set consists of those regular points for which the defect number is zero.

```
lemma IsClosed.resolventSet_eq' [CompleteSpace H] {T : H →l.[C] H} (hT : T.IsClosed) :
  ρ T = T.regularityDomain ∩ T.defectNumber-1 {0}
```

lemma resolventSet_isOpen[\[source\]](#)

The resolvent set is an open subset of $\mathbb{C}$.

```
lemma resolventSet_isOpen [CompleteSpace H] (T : H →l.[C] H) : IsOpen (ρ T)
```

def spectrum[\[source\]](#)

The spectrum, σ , of a partial linear map.

σT is the complement of ρT . A complex number z is in σT iff the linear map $T - z \cdot 1$ from $T.\text{domain}$ to H fails to be bijective or $(T - z \cdot 1)^{-1}$ is not continuous (equivalently, is not bounded).

```
def spectrum (T : H →l.[C] H) : Set C
```

lemma LinearPMap.spectrum_eq[\[source\]](#)

```
lemma spectrum_eq (T : H →l.[C] H) : σ T = (ρ T)c
```

lemma mem_spectrum_iff[\[source\]](#)

```
lemma mem_spectrum_iff {T : H →l.[C] H} {z : C} :
  z ∈ σ T ↔ (T - z • 1).toFun.ker ≠ ⊥ ∨ (T - z • 1).toFun.range ≠ T ∨ ¬Continuous (R T z)
```

lemma spectrum_eq_univ[\[source\]](#)

If an operator is not closed then its spectrum is all of $\mathbb{C}$.

```
lemma spectrum_eq_univ [CompleteSpace H] {T : H →l.[C] H} (h : ¬T.IsClosed) : σ T = univ
```

lemma spectrum_isClosed[\[source\]](#)

The spectrum is a closed subset of $\mathbb{C}$.

```
lemma spectrum_isClosed [CompleteSpace H] (T : H →l.[C] H) : _root_.IsClosed (σ T)
```

def pointSpectrum[\[source\]](#)

The point spectrum, σ^p , of a partial linear map.

A complex number z is in $\sigma^p T$ iff $T - z \cdot 1$ is not injective.

```
def pointSpectrum (T : H →l.[C] H) : Set C
```

lemma pointSpectrum_eq[\[source\]](#)

```
lemma pointSpectrum_eq (T : H →l.[C] H) :  $\sigma^p T = \{z \mid (T - z \cdot 1).toFun.ker \neq \perp\}$ 
```

lemma mem_pointSpectrum_iff[\[source\]](#)

```
lemma mem_pointSpectrum_iff {T : H →l.[C] H} {z : C} :  $z \in \sigma^p T \leftrightarrow (T - z \cdot 1).toFun.ker \neq \perp$ 
```

lemma pointSpectrum_subset_spectrum[\[source\]](#)

```
lemma pointSpectrum_subset_spectrum (T : H →l.[C] H) :  $\sigma^p T \subseteq \sigma T$ 
```

def residualSpectrum[\[source\]](#)

The residual spectrum, σ^r , of a partial linear map.

A complex number z is in $\sigma^r T$ iff $T - z \cdot 1$ is injective but not surjective and $(T - z \cdot 1)^{-1}$ is continuous (equivalently, bounded).

```
def residualSpectrum (T : H →l.[C] H) : Set C
```

lemma residualSpectrum_eq[\[source\]](#)

```
lemma residualSpectrum_eq (T : H →l.[C] H) :  
   $\sigma^r T = \{z \mid (T - z \cdot 1).toFun.ker = \perp \wedge (T - z \cdot 1).toFun.range \neq T \wedge \text{Continuous } (R T z)\}$ 
```

lemma mem_residualSpectrum_iff[\[source\]](#)

```
lemma mem_residualSpectrum_iff {T : H →l.[C] H} {z : C} :  
   $z \in \sigma^r T \leftrightarrow (T - z \cdot 1).toFun.ker = \perp \wedge (T - z \cdot 1).toFun.range \neq T \wedge \text{Continuous } (R T z)$ 
```

lemma residualSpectrum_subset_spectrum[\[source\]](#)

```
lemma residualSpectrum_subset_spectrum (T : H →l.[C] H) :  $\sigma^r T \subseteq \sigma T$ 
```

lemma residualSpectrum_subset_regularityDomain[\[source\]](#)

```
lemma residualSpectrum_subset_regularityDomain (T : H →l.[C] H) :  $\sigma^r T \subseteq T.\text{regularityDomain}$ 
```

def continuousSpectrum[\[source\]](#)

The continuous spectrum, σ^c , of a partial linear map.

A complex number z is in $\sigma^c T$ iff the range of $T - z \cdot 1$ is not closed.

```
def continuousSpectrum (T : H →l.[C] H) : Set C
```

lemma continuousSpectrum_eq

[\[source\]](#)

```
lemma continuousSpectrum_eq (T : H →l.[C] H) :
  σc T = {z | ¬_root_.IsClosed ((T - z • 1).toFun.range : Set H)}
```

lemma mem_continuousSpectrum_iff

[\[source\]](#)

```
lemma mem_continuousSpectrum_iff {T : H →l.[C] H} {z : C} :
  z ∈ σc T ↔ ¬_root_.IsClosed ((T - z • 1).toFun.range : Set H)
```

lemma continuousSpectrum_subset_spectrum

[\[source\]](#)

```
lemma continuousSpectrum_subset_spectrum (T : H →l.[C] H) : σc T ⊆ σ T
```

lemma IsClosed.spectrum_eq

[\[source\]](#)

```
lemma IsClosed.spectrum_eq [CompleteSpace H] {T : H →l.[C] H} (hT : T.IsClosed) :
  σ T = σp T ∪ σr T ∪ σc T
```

lemma pointSpectrum_inter_residualSpectrum

[\[source\]](#)

```
lemma pointSpectrum_inter_residualSpectrum (T : H →l.[C] H) : σp T ∩ σr T = ∅
```

lemma resolvent_sub

[\[source\]](#)

```
lemma resolvent_sub
  {T1 T2 : H →l.[C] H} (hT : T2.domain ≤ T1.domain) {z : C} (hz1 : z ∈ ρ T1) (hz2 : z ∈ ρ T2)
  :
  R T1 z - R T2 z = R T1 z * (T2 - T1) * R T2 z
```

lemma resolvent_sub'

[\[source\]](#)

```
lemma resolvent_sub' {T : H →l.[C] H} (z1 z2 : C) (hz1 : z1 ∈ ρ T) (hz2 : z2 ∈ ρ T) :
  R T z1 - R T z2 = (z1 - z2) • (R T z1 * R T z2)
```

2.27 Physlib.QuantumMechanics.DDimensions.Operators.SpectralTheory.SelfAdjoint

[Physlib/QuantumMechanics/DDimensions/Operators/SpectralTheory/SelfAdjoint.lean](#) on GitHub.

lemma resolventSet_eq_regularityDomain

[\[source\]](#)

```
lemma resolventSet_eq_regularityDomain : ρ T = T.regularityDomain
```

lemma mem_resolventSet_of_range_eq_top

[\[source\]](#)

(T - z • 1).range = τ is a sufficient condition for z ∈ ρ T (and it is a necessary condition by definition of ρ).


```
lemma mem_resolventSet_of_range_eq_top {z : ℂ} (h : (T - z • 1).toFun.range = T) : z ∈ ρ T
```

lemma spectrum_real

[\[source\]](#)

The spectrum of a self-adjoint operator is real.

```
lemma spectrum_real : σ T ⊆ range ofReal
```

lemma residualSpectrum_eq_empty

[\[source\]](#)

The residual spectrum of a self-adjoint operator is empty.

```
lemma residualSpectrum_eq_empty : σr T = ∅
```

2.28 Physlib.QuantumMechanics.DDimensions.Operators.SpectralTheory.SpectralMeasure

[Physlib/QuantumMechanics/DDimensions/Operators/SpectralTheory/SpectralMeasure.lean](#) on GitHub.

instance IsAddTorsionFree (H →L[ℂ] H)

[\[source\]](#)

```
instance (H : Type*) [SeminormedAddCommGroup H] [InnerProductSpace ℂ H] :
  IsAddTorsionFree (H →L[ℂ] H) where
```

structure SpectralMeasure

[\[source\]](#)

A spectral measure on a measurable space α is a σ -additive function $\text{Set } \alpha \rightarrow H \rightarrow L[\mathbb{C}] H$ such that each set is mapped to a star projection on H , the empty set and non-measurable sets are mapped to zero, and univ is mapped to the identity.

```
structure SpectralMeasure
  (α : Type*) [MeasurableSpace α]
  (H : Type*) [NormedAddCommGroup H] [InnerProductSpace ℂ H] [CompleteSpace H]
  extends VectorMeasure α (H →L[ℂ] H) where
  isStarProjection' : ∀ A, IsStarProjection (measureOf' A)
  univ' : measureOf' univ = 1
```

instance instCoeVectorMeasure

[\[source\]](#)

```
instance instCoeVectorMeasure : Coe (SpectralMeasure α H) (VectorMeasure α (H →L[ℂ] H))
```

instance instCoeFun

[\[source\]](#)

```
instance instCoeFun : CoeFun (SpectralMeasure α H) fun _ ↦ Set α → H →L[ℂ] H
```

lemma isStarProjection

[\[source\]](#)

```
lemma isStarProjection (A : Set α) : IsStarProjection (μS A)
```

lemma univ[\[source\]](#)

```
@[simp]
lemma univ :  $\mu S$  univ = 1
```

lemma comp_self[\[source\]](#)

```
@[simp]
lemma comp_self (A : Set  $\alpha$ ) :  $\mu S$  A  $\circ_L$   $\mu S$  A =  $\mu S$  A
```

lemma comp_of_disjoint[\[source\]](#)

```
lemma comp_of_disjoint
  {A B : Set  $\alpha$ } (h : Disjoint A B) (hA : MeasurableSet A) (hB : MeasurableSet B) :
   $\mu S$  A  $\circ_L$   $\mu S$  B = 0
```

lemma comp_eq_of_inter[\[source\]](#)

```
lemma comp_eq_of_inter {A B : Set  $\alpha$ } (hA : MeasurableSet A) (hB : MeasurableSet B) :
   $\mu S$  A  $\circ_L$   $\mu S$  B =  $\mu S$  (A  $\cap$  B)
```

lemma commute[\[source\]](#)

```
lemma commute (A B : Set  $\alpha$ ) : Commute ( $\mu S$  A) ( $\mu S$  B)
```

2.29 Physlib.QuantumMechanics.DDimensions.Operators.SpectralTheory.Symmetric

[Physlib/QuantumMechanics/DDimensions/Operators/SpectralTheory/Symmetric.lean](#) on GitHub.

def realNumericalRange[\[source\]](#)

The projection of the numerical range onto the real axis.

```
def realNumericalRange (T : H  $\rightarrow_L$  [ $\mathbb{C}$ ] H) : Set  $\mathbb{R}$ 
```

lemma realNumericalRange_eq[\[source\]](#)

```
lemma realNumericalRange_eq (T : H  $\rightarrow_L$  [ $\mathbb{C}$ ] H) :  $\theta_{re}$  T = re ''  $\theta$  T
```

lemma realNumericalRange_neg[\[source\]](#)

```
@[simp]
lemma realNumericalRange_neg (T : H  $\rightarrow_L$  [ $\mathbb{C}$ ] H) :  $\theta_{re}$  (-T) = - $\theta_{re}$  T
```

lemma im_eq_zero_of_mem_numericalRange[\[source\]](#)

The numerical range of a symmetric operator is contained in the real axis.

```
lemma im_eq_zero_of_mem_numericalRange {z :  $\mathbb{C}$ } (hz : z  $\in$   $\theta$  T) : z.im = 0
```

lemma numericalRange_subset[\[source\]](#)

The numerical range of a symmetric operator is contained in the real axis.

```
lemma numericalRange_subset : 0 T ≤ range ofReal
```

lemma numericalRange_eq[\[source\]](#)

The numerical range of a symmetric operator is equal to its projection onto the real axis.

```
lemma numericalRange_eq : 0 T = ofReal '' 0_{r,e} T
```

lemma closure_numericalRange_subset[\[source\]](#)

```
lemma closure_numericalRange_subset : _root_.closure (0 T) ≤ range ofReal
```

lemma mem_regularityDomain_of_im_ne_zero[\[source\]](#)

The regularity domain of a symmetric operator contains all complex numbers with non-zero imaginary part.

```
lemma mem_regularityDomain_of_im_ne_zero {z : ℂ} (hz : z.im ≠ 0) : z ∈ T.regularityDomain
```

lemma compl_ofReal_subset_regularityDomain[\[source\]](#)

The regularity domain of a symmetric operator contains all complex numbers with non-zero imaginary part.

```
lemma compl_ofReal_subset_regularityDomain : (range ofReal)c ≤ T.regularityDomain
```

lemma Iio_subset_regularityDomain[\[source\]](#)

If m is a lower bound on the numerical range then the regularity domain contains $(-\infty, m)$.

```
lemma Iio_subset_regularityDomain {m : ℝ} (h : m ∈ lowerBounds (0_{r,e} T)) :  
  ofReal '' Iio m ≤ T.regularityDomain
```

lemma Ioi_subset_regularityDomain[\[source\]](#)

If m is an upper bound on the numerical range then the regularity domain contains (m, ∞) .

```
lemma Ioi_subset_regularityDomain {m : ℝ} (h : m ∈ upperBounds (0_{r,e} T)) :  
  ofReal '' Ioi m ≤ T.regularityDomain
```

lemma regularityDomain_isConnected_iff[\[source\]](#)

The regularity domain of a symmetric operator is connected iff it contains a real number.

```
lemma regularityDomain_isConnected_iff :  
  IsConnected T.regularityDomain ↔ (ofReal -1 T.regularityDomain).Nonempty
```

lemma regularityDomain_isConnected_of_bddBelow[\[source\]](#)

The regularity domain of a lower semibounded symmetric operator is connected.

```
lemma regularityDomain_isConnected_of_bddBelow (h : BddBelow (0, e T)) :
  IsConnected T.regularityDomain
```

lemma regularityDomain_isConnected_of_bddAbove[\[source\]](#)

The regularity domain of an upper semibounded symmetric operator is connected.

```
lemma regularityDomain_isConnected_of_bddAbove (h : BddAbove (0, e T)) :
  IsConnected T.regularityDomain
```

lemma pointSpectrum_real[\[source\]](#)

Eigenvalues of a symmetric unbounded operator are real.

```
lemma pointSpectrum_real : σp T ⊆ range ofReal
```

2.30 Physlib.QuantumMechanics.DDimensions.Operators.Unbounded

[Physlib/QuantumMechanics/DDimensions/Operators/Unbounded.lean](#) on GitHub.

lemma orthogonal_range[\[source\]](#)

$$U.\text{range}^\perp = U\dagger.\text{ker}$$

c.f. LinearMap.orthogonal_range and ContinuousLinearMap.orthogonal_range

```
lemma HasDenseDomain.orthogonal_range [CompleteSpace H] (h : U.HasDenseDomain) :
  U.toFun.range⊥ = U\dagger.toFun.ker.map U\dagger.domain.subtype
```

lemma orthogonal_adjoint_ker[\[source\]](#)

$$U\dagger.\text{ker}^\perp = U.\text{range.closure}$$

```
lemma HasDenseDomain.orthogonal_adjoint_ker [CompleteSpace H] [CompleteSpace H']
  (h : U.HasDenseDomain) :
  (U\dagger.toFun.ker.map U\dagger.domain.subtype)⊥ = U.toFun.range.closure
```

lemma closure_eq[\[source\]](#)

```
lemma IsClosed.closure_eq (h : U.IsClosed) : U.closure = U
```

lemma adjoint_zero[\[source\]](#)

@[simp]

```
lemma adjoint_zero [CompleteSpace H] : (0 : H →l [C] H')† = 0
```

lemma adjoint_sub_le_sub_adjoint[\[source\]](#)

```
lemma adjoint_sub_le_sub_adjoint [CompleteSpace H]
  (U1 U2 : H →l [C] H') (h12 : (U1 - U2).HasDenseDomain) : U1† - U2† ≤ (U1 - U2)†
```

lemma continuous_iff_bounded[\[source\]](#)

$f : E \rightarrow_l[\mathbb{K}] F$ is continuous iff there exists $M > 0$ s.t. $\|f x\| \leq M * \|x\|$ for all $x : E$.

This is a (convenient) immediate consequence of `IsBoundedLinearMap.isLinearMap_and_continuous_iff_is`

```
lemma _root_.LinearMap.continuous_iff_bounded {ℓ E F : Type*} [NontriviallyNormedField ℓ]
  [SeminormedAddCommGroup E] [NormedSpace ℓ E] [SeminormedAddCommGroup F] [NormedSpace ℓ F]
  {f : E →ℓ[ℓ] F} : Continuous f ↔ ∃ M, 0 < M ∧ ∀ x : E, ‖f x‖ ≤ M * ‖x‖
```

lemma isClosable_of_continuous[\[source\]](#)

Continuous operators are closable.

```
lemma isClosable_of_continuous (h : Continuous U) : U.IsClosable
```

lemma closure_domain_eq_domain_closure_of_continuous[\[source\]](#)

A strengthening of `closure_domain_le_domain_closure` for continuous operators.

```
lemma closure_domain_eq_domain_closure_of_continuous [CompleteSpace H'] (h : Continuous U) :
  U.closure.domain = U.domain.closure
```

lemma isClosed_iff_isClosed_domain_of_continuous[\[source\]](#)

A continuous operator is closed iff its domain is closed.

```
lemma isClosed_iff_isClosed_domain_of_continuous [CompleteSpace H'] (h : Continuous U) :
  U.IsClosed ↔ _root_.IsClosed (U.domain : Set H)
```

lemma isClosed_toFun_graph[\[source\]](#)

```
lemma IsClosed.isClosed_toFun_graph (hU : U.IsClosed) :
  _root_.IsClosed (U.toFun.graph : Set (U.domain × H'))
```

lemma continuous_of_isClosed_domain[\[source\]](#)

The ***closed graph theorem*** for partial linear maps: a closed operator with closed domain is continuous.

This follows immediately from `LinearMap.continuous_of_isClosed_graph` and the completeness of H and H' .

```
lemma IsClosed.continuous_of_isClosed_domain [CompleteSpace H] [CompleteSpace H']
  (hU : U.IsClosed) (h : _root_.IsClosed (U.domain : Set H)) :
  Continuous U
```

lemma IsClosable.add_continuous[\[source\]](#)

Closability is preserved upon adding a continuous operator.

```
lemma IsClosable.add_continuous
  (h1 : U1.IsClosable) (h2 : Continuous U2) (h : U1.domain ≤ U2.domain) :
  (U1 + U2).IsClosable
```

lemma IsClosable.sub_continuous[\[source\]](#)*Closability is preserved upon subtracting a continuous operator.*

```
lemma IsClosable.sub_continuous
  (h₁ : U₁.IsClosable) (h₂ : Continuous U₂) (h : U₁.domain ≤ U₂.domain) : (U₁ - U₂).
  IsClosable
```

lemma IsClosed.add_continuous[\[source\]](#)*Closedness is preserved upon adding a continuous operator.*

```
lemma IsClosed.add_continuous [CompleteSpace H']
  (h₁ : U₁.IsClosed) (h₂ : Continuous U₂) (h : U₁.domain ≤ U₂.domain) : (U₁ + U₂).IsClosed
```

lemma IsClosed.sub_continuous[\[source\]](#)*Closedness is preserved upon subtracting a continuous operator.*

```
lemma IsClosed.sub_continuous [CompleteSpace H']
  (h₁ : U₁.IsClosed) (h₂ : Continuous U₂) (h : U₁.domain ≤ U₂.domain) : (U₁ - U₂).IsClosed
```

lemma orthogonal_adjoint_range[\[source\]](#) $U\dagger.range^\perp = U.closure.ker$

```
lemma IsUnbounded.orthogonal_adjoint_range [CompleteSpace H] [CompleteSpace H']
  (h : U.IsUnbounded) : U\dagger.toFun.range^\perp = U.closure.toFun.ker.map U.closure.domain.subtype
```

lemma orthogonal_closure_ker[\[source\]](#) $U.closure.ker^\perp = U\dagger.range$

```
lemma IsUnbounded.orthogonal_closure_ker [CompleteSpace H] [CompleteSpace H'] (h : U.
  IsUnbounded) :
  (U.closure.toFun.ker.map U.closure.domain.subtype)^\perp = U\dagger.toFun.range.closure
```

2.31 Physlib.QuantumMechanics.FiniteTarget.HilbertSpace

Physlib/QuantumMechanics/FiniteTarget/HilbertSpace.lean on GitHub.

def equivEuclidean[\[source\]](#)*The equivalence between FiniteHilbertSpace d and EuclideanSpace ℂ d given by val.*

```
def equivEuclidean : FiniteHilbertSpace d ≈ EuclideanSpace ℂ d where
```

instance AddCommGroup (FiniteHilbertSpace d)[\[source\]](#)

```
noncomputable instance : AddCommGroup (FiniteHilbertSpace d)
```

instance Module ℂ (FiniteHilbertSpace d)[\[source\]](#)

```
noncomputable instance : Module ℂ (FiniteHilbertSpace d)
```

lemma val_add[\[source\]](#)

@[simp]

lemma val_add ($\psi \ \phi : \text{FiniteHilbertSpace } d$) : $(\psi + \phi).\text{val} = \psi.\text{val} + \phi.\text{val}$ **lemma val_smul**[\[source\]](#)

@[simp]

lemma val_smul ($c : \mathbb{C}$) ($\psi : \text{FiniteHilbertSpace } d$) : $(c \cdot \psi).\text{val} = c \cdot \psi.\text{val}$ **lemma val_zero**[\[source\]](#)

@[simp]

lemma val_zero : $(0 : \text{FiniteHilbertSpace } d).\text{val} = 0$ **def linearEquivEuclidean**[\[source\]](#)

The equivalence between `FiniteHilbertSpace d` and `EuclideanSpace ℂ d` as a $\mathbb{C}$ -linear equivalence, upgrading `equivEuclidean`.

noncomputable def linearEquivEuclidean : $\text{FiniteHilbertSpace } d \approx_l[\mathbb{C}] \text{EuclideanSpace } \mathbb{C} \ d$ **instance NormedAddCommGroup (FiniteHilbertSpace d)**[\[source\]](#)**noncomputable instance** : NormedAddCommGroup (FiniteHilbertSpace d)**lemma norm_eq_val**[\[source\]](#)

@[simp]

lemma norm_eq_val ($\psi : \text{FiniteHilbertSpace } d$) : $\|\psi\| = \|\psi.\text{val}\|$ **instance InnerProductSpace ℂ (FiniteHilbertSpace d)**[\[source\]](#)**noncomputable instance** : InnerProductSpace $\mathbb{C}$ (FiniteHilbertSpace d)**lemma inner_eq_val**[\[source\]](#)

@[simp]

lemma inner_eq_val ($\psi \ \phi : \text{FiniteHilbertSpace } d$) : $\text{inner } \mathbb{C} \ \psi \ \phi = \text{inner } \mathbb{C} \ \psi.\text{val} \ \phi.\text{val}$ **instance FiniteDimensional ℂ (FiniteHilbertSpace d)**[\[source\]](#)**instance** : FiniteDimensional $\mathbb{C}$ (FiniteHilbertSpace d)**instance CompleteSpace (FiniteHilbertSpace d)**[\[source\]](#)**instance** : CompleteSpace (FiniteHilbertSpace d)

def isometryEquivEuclidean[\[source\]](#)

The equivalence between `FiniteHilbertSpace d` and `EuclideanSpace ℂ d` as a linear isometry equivalence, upgrading `linearEquivEuclidean`.

```
noncomputable def isometryEquivEuclidean : FiniteHilbertSpace d ≈li[ℂ] EuclideanSpace ℂ d where
```

def basisFun[\[source\]](#)

The standard orthonormal basis of `FiniteHilbertSpace d`, indexed by `d`.

```
noncomputable def basisFun (d : Type*) [Fintype d] [DecidableEq d] :  
  OrthonormalBasis d ℂ (FiniteHilbertSpace d)
```

lemma basisFun_apply[\[source\]](#)

```
lemma basisFun_apply (i : d) : basisFun d i = (EuclideanSpace.single i 1)
```

2.32 Physlib.Relativity.LorentzGroup.Basic

[Physlib/Relativity/LorentzGroup/Basic.lean](#) on GitHub.

lemma continuous_self_mul_dual[\[source\]](#)

```
lemma continuous_self_mul_dual {d : ℕ} :  
  Continuous (fun Λ : Matrix (Fin 1 ⊕ Fin d) (Fin 1 ⊕ Fin d) ℝ => Λ * dual Λ)
```

lemma isClosed[\[source\]](#)

The Lorentz group is closed in the ambient matrix space.

```
lemma isClosed (d : ℕ) :  
  IsClosed (LorentzGroup d : Set (Matrix (Fin 1 ⊕ Fin d) (Fin 1 ⊕ Fin d) ℝ))
```

lemma isClosedEmbedding_val[\[source\]](#)

The coercion from the Lorentz group to its ambient matrix space is a closed embedding.

```
lemma isClosedEmbedding_val {d : ℕ} :  
  Topology.IsClosedEmbedding  
    (Subtype.val : LorentzGroup d → Matrix (Fin 1 ⊕ Fin d) (Fin 1 ⊕ Fin d) ℝ)
```

2.33 Physlib.Relativity.Tensors.Basic

[Physlib/Relativity/Tensors/Basic.lean](#) on GitHub.

lemma update_eq_function_update[\[source\]](#)

```
lemma update_eq_function_update {n : ℕ} {c : Fin n → C} [inst : DecidableEq (Fin n)]  
  (p : Pure S c) (i : Fin n)  
  (x : V (c i)) : update p i x = Function.update p i x
```


lemma update_diff[\[source\]](#)

```
lemma update_diff {n : ℕ} {c : Fin n → C} [inst : DecidableEq (Fin n)] (p : Pure S c) (i j :
  Fin n)
  (x : V (c i)) (hij : i ≠ j) : (update p i x) j = p j
```

lemma componentMap_eq_repr[\[source\]](#)

componentMap is the basis representation (basis c).repr definitionally; this bridges the two notations wherever a component computation meets a repr rewrite.

```
lemma componentMap_eq_repr {n : ℕ} (c : Fin n → C) (t : S.Tensor c)
  (ψ : ComponentIdx (S := S) c) : componentMap c t ψ = (basis c).repr t ψ
```

lemma componentMap_ext[\[source\]](#)

Two tensors with the same colour sequence are equal when all their components agree.

```
lemma componentMap_ext {n : ℕ} {c : Fin n → C} {t₁ t₂ : S.Tensor c}
  (h : ∀ b, componentMap c t₁ b = componentMap c t₂ b) : t₁ = t₂
```

instance FiniteDimensional k (S.Tensor c)[\[source\]](#)

```
instance {k : Type} [Field k] {C : Type} {G : Type} [Group G]
  {V : C → Type} [V c, AddCommGroup (V c)] [V c, Module k (V c)]
  {basisIdx : C → Type} [V c, Fintype (basisIdx c)] [V c, DecidableEq (basisIdx c)]
  {rep : (c : C) → Representation k G (V c)} {b : (c : C) → Basis (basisIdx c) k (V c)}
  (S : TensorSpecies k C G V basisIdx rep b)
  {c : Fin n → C} : FiniteDimensional k (S.Tensor c)
```

instance TopologicalSpace (S.Tensor c)[\[source\]](#)

```
noncomputable instance {k : Type} [RCLike k] {C : Type} {G : Type} [Group G]
  {V : C → Type} [V c, AddCommGroup (V c)] [V c, Module k (V c)]
  {basisIdx : C → Type} [V c, Fintype (basisIdx c)] [V c, DecidableEq (basisIdx c)]
  {rep : (c : C) → Representation k G (V c)} {b : (c : C) → Basis (basisIdx c) k (V c)}
  (S : TensorSpecies k C G V basisIdx rep b)
  {c : Fin n → C} : TopologicalSpace (S.Tensor c)
```

instance IsTopologicalAddGroup (S.Tensor c)[\[source\]](#)

```
instance {k : Type} [RCLike k] {C : Type} {G : Type} [Group G]
  {V : C → Type} [V c, AddCommGroup (V c)] [V c, Module k (V c)]
  {basisIdx : C → Type} [V c, Fintype (basisIdx c)] [V c, DecidableEq (basisIdx c)]
  {rep : (c : C) → Representation k G (V c)} {b : (c : C) → Basis (basisIdx c) k (V c)}
  (S : TensorSpecies k C G V basisIdx rep b)
  {c : Fin n → C} : IsTopologicalAddGroup (S.Tensor c)
```

lemma rep_cast[\[source\]](#)

```
lemma rep_cast {g : G} {c c₁ : C} (p : V c) (h : c = c₁) :
  LinearEquiv.cast (R := k) h (rep c g p) = rep c₁ g (LinearEquiv.cast (R := k) h p)
```

lemma actionP_cast[\[source\]](#)

```
lemma actionP_cast {g : G} {p : Pure S c} (h : c = c1) :
  LinearEquiv.cast (R := k) h (g • p) = g • LinearEquiv.cast (R := k) h p
```

instance SMul G (S.Tensor c)[\[source\]](#)

```
noncomputable instance : SMul G (S.Tensor c) where
```

instance DistribMulAction G (S.Tensor c)[\[source\]](#)

```
noncomputable instance : DistribMulAction G (S.Tensor c) where
```

lemma permP_equivariant[\[source\]](#)

```
lemma Pure.permP_equivariant {n m : ℕ} {c : Fin n → C} {c1 : Fin m → C}
  {σ : Fin m → Fin n} (h : IsReindexing c c1 σ) (g : G) (p : Pure S c) :
  Pure.permP σ h (g • p) = g • Pure.permP σ h p
```

lemma toField_injective[\[source\]](#)

```
lemma toField_injective {c : Fin 0 → C} :
  Function.Injective (toField : S.Tensor c → k)
```

lemma toField_basis[\[source\]](#)

```
lemma toField_basis {c : Fin 0 → C} (b : ComponentIdx (S := S) c) :
  toField (basis c b) = 1
```

lemma eq_smul_toField[\[source\]](#)

```
lemma eq_smul_toField {c : Fin 0 → C} (t : Tensor S c) :
  t = toField t • (basis c (@default (ComponentIdx (S := S) c) Unique.instInhabited))
```

2.34 Physlib.Relativity.Tensors.ComplexTensor.Basic

[Physlib/Relativity/Tensors/ComplexTensor/Basic.lean](#) on GitHub.

abbrev modules[\[source\]](#)

The modules associated with each of the different types of complex Lorentz vector space.

```
abbrev modules : Color → Type
| Color.upL => Fermion.LeftHandedWeyl
| Color.downL => Fermion.DualLeftHandedWeyl
| Color.upR => Fermion.RightHandedWeyl
| Color.downR => Fermion.DualRightHandedWeyl
| Color.up => Lorentz.ContrCModule
| Color.down => Lorentz.CoCModule
```

instance modulesAddCommGroup[\[source\]](#)

```
instance modulesAddCommGroup : ∀ c, AddCommGroup (modules c)
| Color.upL => inferInstance
| Color.downL => inferInstance
| Color.upR => inferInstance
| Color.downR => inferInstance
| Color.up => inferInstance
| Color.down => inferInstance
```

instance modulesModule[\[source\]](#)

```
noncomputable instance modulesModule : ∀ c, Module ℂ (modules c)
| Color.upL => inferInstance
| Color.downL => inferInstance
| Color.upR => inferInstance
| Color.downR => inferInstance
| Color.up => inferInstance
| Color.down => inferInstance
```

abbrev basis[\[source\]](#)

The basis associated with each of the different types of complex Lorentz vector space.

```
abbrev basis (c : Color) : Module.Basis (Fin (repDim c)) ℂ (modules c)
```

abbrev rep[\[source\]](#)

The reps associated with each of the different types of complex Lorentz vector space.

```
abbrev rep (c : Color) : Representation ℂ SL(2, ℂ) (modules c)
```

lemma contrPCoeff_basis[\[source\]](#)

```
lemma contrPCoeff_basis {n : ℕ} {c : Fin n → complexLorentzTensor.Color} (i j : Fin n)
(hij : i ≠ j ∧ (complexLorentzTensor.τ (c i) = c j))
(b : ComponentIdx (S := complexLorentzTensor) c) :
Pure.contrPCoeff i j hij (Pure.basisVector c b) = if b i =
  Fin.cast (by simp [← hij.2, repDim_tau]) (b j)
then 1 else 0
```

2.35 Physlib.Relativity.Tensors.ComplexTensor.Metrics.Basic

[Physlib/Relativity/Tensors/ComplexTensor/Metrics/Basic.lean](#) on GitHub.

abbrev dualLeftMetric[\[source\]](#)

The metric ε_{aa} as a complex Lorentz tensor.

```
abbrev dualLeftMetric : CT[.downL, .downL]
```

abbrev dualRightMetric[\[source\]](#)

The metric $\varepsilon_{\dot{a}\dot{a}}$ as a complex Lorentz tensor.

```
abbrev dualRightMetric :  $\mathbb{C}T[.downR, .downR]$ 
```

```
lemma dualLeftMetric_eq_fromConstPair
```

[\[source\]](#)

```
lemma dualLeftMetric_eq_fromConstPair :  $\varepsilon L' = \text{fromConstPair Fermion.dualLeftMetric}$ 
```

```
lemma dualRightMetric_eq_fromConstPair
```

[\[source\]](#)

```
lemma dualRightMetric_eq_fromConstPair :  $\varepsilon R' = \text{fromConstPair Fermion.dualRightMetric}$ 
```

```
lemma dualLeftMetric_eq_fromPairT
```

[\[source\]](#)

```
lemma dualLeftMetric_eq_fromPairT :  $\varepsilon L' = \text{fromPairT (Fermion.dualLeftMetricVal)}$ 
```

```
lemma dualRightMetric_eq_fromPairT
```

[\[source\]](#)

```
lemma dualRightMetric_eq_fromPairT :  $\varepsilon R' = \text{fromPairT (Fermion.dualRightMetricVal)}$ 
```

```
lemma dualLeftMetric_eq_dualLeftBasis
```

[\[source\]](#)

```
lemma dualLeftMetric_eq_dualLeftBasis :  $\varepsilon L' =$   

  fromPairT (dualLeftBasis 0  $\otimes_{\mathbb{C}}$  dualLeftBasis 1)  

  - fromPairT (dualLeftBasis 1  $\otimes_{\mathbb{C}}$  dualLeftBasis 0)
```

```
lemma dualRightMetric_eq_dualRightBasis
```

[\[source\]](#)

```
lemma dualRightMetric_eq_dualRightBasis :  $\varepsilon R' =$   

  fromPairT (dualRightBasis 0  $\otimes_{\mathbb{C}}$  dualRightBasis 1)  

  - fromPairT (dualRightBasis 1  $\otimes_{\mathbb{C}}$  dualRightBasis 0)
```

```
lemma dualLeftMetric_eq_basis
```

[\[source\]](#)

```
lemma dualLeftMetric_eq_basis :  $\varepsilon L' =$   

  (Tensor.basis (S := complexLorentzTensor) ![Color.downL, Color.downL]  

    (fun | 0 => (0 : Fin 2) | 1 => (1 : Fin 2)))  

  - (Tensor.basis (S := complexLorentzTensor)  

    ![Color.downL, Color.downL] (fun | 0 => (1 : Fin 2) | 1 => (0 : Fin 2)))
```

```
lemma dualRightMetric_eq_basis
```

[\[source\]](#)

```
lemma dualRightMetric_eq_basis :  $\varepsilon R' =$   

  (Tensor.basis (S := complexLorentzTensor)  

    ![Color.downR, Color.downR] (fun | 0 => (0 : Fin 2) | 1 => (1 : Fin 2)))  

  - (Tensor.basis (S := complexLorentzTensor)  

    ![Color.downR, Color.downR] (fun | 0 => (1 : Fin 2) | 1 => (0 : Fin 2)))
```

```
lemma dualLeftMetric_eq_ofRat
```

[\[source\]](#)

```
lemma dualLeftMetric_eq_ofRat :  $\varepsilon L' = \text{ofRat fun f =>}$   

  if f 0 = Fin.cast (by rfl) (0 : Fin 2)  $\wedge$  f 1 = Fin.cast (by rfl) (1 : Fin 2) then 1 else
```

```

if f 1 = Fin.cast (by rfl) (0 : Fin 2) ∧ f 0 = Fin.cast (by rfl) (1 : Fin 2) then
  - 1 else 0

```

lemma dualRightMetric_eq_ofRat[\[source\]](#)

```

lemma dualRightMetric_eq_ofRat : εR' = ofRat fun f =>
  if f 0 = Fin.cast (by rfl) (0 : Fin 2) ∧ f 1 = Fin.cast (by rfl) (1 : Fin 2) then 1 else
  if f 1 = Fin.cast (by rfl) (0 : Fin 2) ∧ f 0 = Fin.cast (by rfl) (1 : Fin 2) then
    - 1 else 0

```

lemma actionT_dualLeftMetric[\[source\]](#)

The tensor dualLeftMetric is invariant under the action of $SL(2, \mathbb{C})$.

```

lemma actionT_dualLeftMetric (g : SL(2, ℂ)) : g • εL' = εL'

```

lemma actionT_dualRightMetric[\[source\]](#)

The tensor dualRightMetric is invariant under the action of $SL(2, \mathbb{C})$.

```

lemma actionT_dualRightMetric (g : SL(2, ℂ)) : g • εR' = εR'

```

2.36 Physlib.Relativity.Tensors.ComplexTensor.Metrics.Lemmas

[Physlib/Relativity/Tensors/ComplexTensor/Metrics/Lemmas.lean](#) on GitHub.

lemma dualLeftMetric_antisymm[\[source\]](#)

The dual-left metric is antisymmetric $\{\varepsilon L' \mid \alpha \alpha' = - \varepsilon L' \mid \alpha' \alpha\}^T$.

```

lemma dualLeftMetric_antisymm : {εL' | α α' = - (εL' | α' α)}T

```

lemma dualRightMetric_antisymm[\[source\]](#)

The dual-right metric is antisymmetric $\{\varepsilon R' \mid \beta \beta' = - \varepsilon R' \mid \beta' \beta\}^T$.

```

lemma dualRightMetric_antisymm : {εR' | α α' = - (εR' | α' α)}T

```

lemma leftMetric_contr_dualLeftMetric[\[source\]](#)

The contraction of the left metric with the dual-left metric is the unit $\{\varepsilon L \mid \alpha \beta \otimes \varepsilon L' \mid \beta \gamma = \delta L \mid \alpha \gamma\}^T$.

```

lemma leftMetric_contr_dualLeftMetric : {εL | α β ⊗ εL' | β γ = δL | α γ}T

```

lemma rightMetric_contr_dualRightMetric[\[source\]](#)

The contraction of the right metric with the dual-right metric is the unit $\{\varepsilon R \mid \alpha \beta \otimes \varepsilon R' \mid \beta \gamma = \delta R \mid \alpha \gamma\}^T$.

```

lemma rightMetric_contr_dualRightMetric : {εR | α β ⊗ εR' | β γ = δR | α γ}T

```

lemma dualLeftMetric_contr_leftMetric[\[source\]](#)

The contraction of the dual-left metric with the left metric is the unit $\{\varepsilon L' \mid \alpha \beta \otimes \varepsilon L \mid \beta \gamma = \delta L' \mid \alpha \gamma\}^T$.

lemma dualLeftMetric_contr_leftMetric : $\{\varepsilon L' \mid \alpha \beta \otimes \varepsilon L \mid \beta \gamma = \delta L' \mid \alpha \gamma\}^T$

lemma dualRightMetric_contr_rightMetric[\[source\]](#)

The contraction of the dual-right metric with the right metric is the unit $\{\varepsilon R' \mid \alpha \beta \otimes \varepsilon R \mid \beta \gamma = \delta R' \mid \alpha \gamma\}^T$.

lemma dualRightMetric_contr_rightMetric : $\{\varepsilon R' \mid \alpha \beta \otimes \varepsilon R \mid \beta \gamma = \delta R' \mid \alpha \gamma\}^T$

2.37 Physlib.Relativity.Tensors.ComplexTensor.Units.Basic

[Physlib/Relativity/Tensors/ComplexTensor/Units/Basic.lean](#) on GitHub.

abbrev dualLeftLeftUnit[\[source\]](#)

The unit δ_a^a as a complex Lorentz tensor.

abbrev dualLeftLeftUnit : $\mathbb{C}T[.downL, .upL]$

abbrev leftDualLeftUnit[\[source\]](#)

The unit δ^a_a as a complex Lorentz tensor.

abbrev leftDualLeftUnit : $\mathbb{C}T[.upL, .downL]$

abbrev dualRightRightUnit[\[source\]](#)

The unit $\delta_{\dot{a}}^{\dot{a}}$ as a complex Lorentz tensor.

abbrev dualRightRightUnit : $\mathbb{C}T[.downR, .upR]$

abbrev rightDualRightUnit[\[source\]](#)

The unit $\delta^{\dot{a}}_{\dot{a}}$ as a complex Lorentz tensor.

abbrev rightDualRightUnit : $\mathbb{C}T[.upR, .downR]$

lemma dualLeftLeftUnit_eq_fromConstPair[\[source\]](#)

lemma dualLeftLeftUnit_eq_fromConstPair : $\delta L' = \text{fromConstPair Fermion.dualLeftLeftUnit}$

lemma leftDualLeftUnit_eq_fromConstPair[\[source\]](#)

lemma leftDualLeftUnit_eq_fromConstPair : $\delta L = \text{fromConstPair Fermion.leftDualLeftUnit}$

lemma dualRightRightUnit_eq_fromConstPair[\[source\]](#)

lemma dualRightRightUnit_eq_fromConstPair : $\delta R' = \text{fromConstPair Fermion.dualRightRightUnit}$

lemma rightDualRightUnit_eq_fromConstPair[\[source\]](#)

```
lemma rightDualRightUnit_eq_fromConstPair : δR = fromConstPair Fermion.rightDualRightUnit
```

lemma dualLeftLeftUnit_eq_fromPairT[\[source\]](#)

```
lemma dualLeftLeftUnit_eq_fromPairT : δL' = fromPairT (Fermion.dualLeftLeftUnitVal)
```

lemma leftDualLeftUnit_eq_fromPairT[\[source\]](#)

```
lemma leftDualLeftUnit_eq_fromPairT : δL = fromPairT (Fermion.leftDualLeftUnitVal)
```

lemma dualRightRightUnit_eq_fromPairT[\[source\]](#)

```
lemma dualRightRightUnit_eq_fromPairT : δR' = fromPairT (Fermion.dualRightRightUnitVal)
```

lemma rightDualRightUnit_eq_fromPairT[\[source\]](#)

```
lemma rightDualRightUnit_eq_fromPairT : δR = fromPairT (Fermion.rightDualRightUnitVal)
```

lemma dualLeftLeftUnit_eq_dualLeftBasis_leftBasis[\[source\]](#)

```
lemma dualLeftLeftUnit_eq_dualLeftBasis_leftBasis : δL' =
  ∑ i, fromPairT (dualLeftBasis i ⊗ₜ[C] leftBasis i)
```

lemma leftDualLeftUnit_eq_leftBasis_dualLeftBasis[\[source\]](#)

```
lemma leftDualLeftUnit_eq_leftBasis_dualLeftBasis : δL =
  ∑ i, fromPairT (leftBasis i ⊗ₜ[C] dualLeftBasis i)
```

lemma dualRightRightUnit_eq_dualRightBasis_rightBasis[\[source\]](#)

```
lemma dualRightRightUnit_eq_dualRightBasis_rightBasis : δR' =
  ∑ i, fromPairT (dualRightBasis i ⊗ₜ[C] rightBasis i)
```

lemma rightDualRightUnit_eq_rightBasis_dualRightBasis[\[source\]](#)

```
lemma rightDualRightUnit_eq_rightBasis_dualRightBasis : δR =
  ∑ i, fromPairT (rightBasis i ⊗ₜ[C] dualRightBasis i)
```

lemma dualLeftLeftUnit_eq_basis[\[source\]](#)

```
lemma dualLeftLeftUnit_eq_basis : δL' =
  ∑ i, Tensor.basis (S := complexLorentzTensor)
    ![Color.downL, Color.upL] (fun | 0 => i | 1 => i)
```

lemma leftDualLeftUnit_eq_basis[\[source\]](#)

```
lemma leftDualLeftUnit_eq_basis : δL =
  ∑ i, Tensor.basis (S := complexLorentzTensor)
    ![Color.upL, Color.downL] (fun | 0 => i | 1 => i)
```

lemma dualRightRightUnit_eq_basis[\[source\]](#)

```
lemma dualRightRightUnit_eq_basis : δR' =
  ∑ i, Tensor.basis (S := complexLorentzTensor)
    ![Color.downR, Color.upR] (fun | 0 => i | 1 => i)
```

lemma rightDualRightUnit_eq_basis[\[source\]](#)

```
lemma rightDualRightUnit_eq_basis : δR =
  ∑ i, Tensor.basis (S := complexLorentzTensor)
    ![Color.upR, Color.downR] (fun | 0 => i | 1 => i)
```

lemma dualLeftLeftUnit_eq_ofRat[\[source\]](#)

```
lemma dualLeftLeftUnit_eq_ofRat : δL' = ofRat fun f =>
  if f 0 = f 1 then 1 else 0
```

lemma leftDualLeftUnit_eq_ofRat[\[source\]](#)

```
lemma leftDualLeftUnit_eq_ofRat : δL = ofRat fun f =>
  if f 0 = f 1 then 1 else 0
```

lemma dualRightRightUnit_eq_ofRat[\[source\]](#)

```
lemma dualRightRightUnit_eq_ofRat : δR' = ofRat fun f =>
  if f 0 = f 1 then 1 else 0
```

lemma rightDualRightUnit_eq_ofRat[\[source\]](#)

```
lemma rightDualRightUnit_eq_ofRat : δR = ofRat fun f =>
  if f 0 = f 1 then 1 else 0
```

lemma actionT_dualLeftLeftUnit[\[source\]](#)

The tensor dualLeftLeftUnit is invariant under the action of $SL(2, \mathbb{C})$.

```
lemma actionT_dualLeftLeftUnit (g : SL(2, ℂ)) : g • δL' = δL'
```

lemma actionT_leftDualLeftUnit[\[source\]](#)

The tensor leftDualLeftUnit is invariant under the action of $SL(2, \mathbb{C})$.

```
lemma actionT_leftDualLeftUnit (g : SL(2, ℂ)) : g • δL = δL
```

lemma actionT_dualRightRightUnit[\[source\]](#)

The tensor dualRightRightUnit is invariant under the action of $SL(2, \mathbb{C})$.

```
lemma actionT_dualRightRightUnit (g : SL(2, ℂ)) : g • δR' = δR'
```


lemma actionT_rightDualRightUnit[\[source\]](#)

The tensor rightDualRightUnit is invariant under the action of $SL(2, \mathbb{C})$.

```
lemma actionT_rightDualRightUnit (g : SL(2, ℂ)) : g • δR = δR
```

2.38 Physlib.Relativity.Tensors.ComplexTensor.Units.Symm

[Physlib/Relativity/Tensors/ComplexTensor/Units/Symm.lean](#) on GitHub.

lemma dualLeftLeftUnit_symm[\[source\]](#)

Swapping indices of dualLeftLeftUnit returns leftDualLeftUnit: $\{\delta L' \mid \alpha \alpha' = \delta L \mid \alpha' \alpha\}^T$.

```
lemma dualLeftLeftUnit_symm : {δL' | α α' = δL | α' α}^T
```

lemma leftDualLeftUnit_symm[\[source\]](#)

Swapping indices of leftDualLeftUnit returns dualLeftLeftUnit: $\{\delta L \mid \alpha \alpha' = \delta L' \mid \alpha' \alpha\}^T$.

```
lemma leftDualLeftUnit_symm : {δL | α α' = δL' | α' α}^T
```

lemma dualRightRightUnit_symm[\[source\]](#)

Swapping indices of dualRightRightUnit returns rightDualRightUnit: $\{\delta R' \mid \beta \beta' = \delta R \mid \beta' \beta\}^T$.

```
lemma dualRightRightUnit_symm : {δR' | β β' = δR | β' β}^T
```

lemma rightDualRightUnit_symm[\[source\]](#)

Swapping indices of rightDualRightUnit returns dualRightRightUnit: $\{\delta R \mid \beta \beta' = \delta R' \mid \beta' \beta\}^T$.

```
lemma rightDualRightUnit_symm : {δR | β β' = δR' | β' β}^T
```

2.39 Physlib.Relativity.Tensors.ComplexTensor.Vector.Pre.Basic

[Physlib/Relativity/Tensors/ComplexTensor/Vector/Pre/Basic.lean](#) on GitHub.

lemma SL2CRep_val[\[source\]](#)

```
lemma CoCModule.SL2CRep_val (M : SL(2, ℂ)) (v : CoCModule) :
  ((CoCModule.SL2CRep M) v).val =
  (LorentzGroup.toComplex (SL2C.toLorentzGroup M))⁻¹ᵀ *ᵥ v.val
```

def inclCoRealLorentz[\[source\]](#)

The semilinear map including real Lorentz co-vectors into complex covariant Lorentz vectors.

```
def inclCoRealLorentz : CoMod 3 →sl [Complex.ofRealHom] CoCModule where
```

lemma inclCoRealLorentz_val[\[source\]](#)

```
lemma inclCoRealLorentz_val (v : CoMod 3) :
  (inclCoRealLorentz v).val = ofRealHom ∘ v.toFinIdR
```

lemma complexCoBasis_of_real[\[source\]](#)

```
lemma complexCoBasis_of_real (i : Fin 1 ⊗ Fin 3) :
  (complexCoBasis i) = inclCoRealLorentz (CoMod.stdBasis i)
```

lemma inclCoRealLorentz_p[\[source\]](#)

```
lemma inclCoRealLorentz_p (M : SL(2, ℂ)) (v : CoMod 3) :
  (CoCModule.SL2CRep M) (inclCoRealLorentz v) =
  inclCoRealLorentz ((Co 3).p (SL2C.toLorentzGroup M) v)
```

2.40 Physlib.Relativity.Tensors.ComplexTensor.Vector.Pre.Modules

[Physlib/Relativity/Tensors/ComplexTensor/Vector/Pre/Modules.lean](#) on GitHub.

lemma ext[\[source\]](#)

```
@[ext]
lemma ext (ψ ψ' : CoCModule) (h : ψ.val = ψ'.val) : ψ = ψ'
```

lemma val_add[\[source\]](#)

```
@[simp]
lemma val_add (ψ ψ' : CoCModule) : (ψ + ψ').val = ψ.val + ψ'.val
```

lemma val_smul[\[source\]](#)

```
@[simp]
lemma val_smul (r : ℂ) (ψ : CoCModule) : (r • ψ).val = r • ψ.val
```

2.41 Physlib.Relativity.Tensors.ComplexTensor.Weyl.Basic

[Physlib/Relativity/Tensors/ComplexTensor/Weyl/Basic.lean](#) on GitHub.

def leftHandedRep[\[source\]](#)

The vector space $\mathbb{C}^2$ carrying the fundamental representation of $SL(2, \mathbb{C})$. In index notation corresponds to a Weyl fermion with indices ψ^a .

```
def leftHandedRep : Representation ℂ SL(2, ℂ) LeftHandedWeyl where
```

def dualLeftHandedRep[\[source\]](#)

The vector space $\mathbb{C}^2$ carrying the representation of $SL(2, \mathbb{C})$ given by $M \rightarrow (M^{-1})^T$. In index notation corresponds to a Weyl fermion with indices ψ_a .

```
def dualLeftHandedRep : Representation ℂ SL(2, ℂ) DualLeftHandedWeyl where
```

def dualLeftBasis[\[source\]](#)

The standard basis on dual-left-handed Weyl fermions.

```
def dualLeftBasis : Basis (Fin 2) ℂ DualLeftHandedWeyl
```

lemma dualLeftBasis_toFin2ℂ[\[source\]](#)

```
@[simp]
```

```
lemma dualLeftBasis_toFin2ℂ (i : Fin 2) : (dualLeftBasis i).toFin2ℂ = Pi.single i 1
```

lemma dualLeftBasis_p_apply[\[source\]](#)

```
@[simp]
```

```
lemma dualLeftBasis_p_apply (M : SL(2,ℂ)) (i j : Fin 2) :  
  (LinearMap.toMatrix dualLeftBasis dualLeftBasis) (dualLeftHandedRep M) i j = (M.1-1)T i j
```

def rightHandedRep[\[source\]](#)

The vector space $\mathbb{C}^2$ carrying the conjugate representation of $SL(2, \mathbb{C})$. In index notation corresponds to a Weyl fermion with indices $\psi^{\dot{a}}$.

```
def rightHandedRep : Representation ℂ SL(2,ℂ) RightHandedWeyl where
```

def dualRightHandedRep[\[source\]](#)

The vector space $\mathbb{C}^2$ carrying the representation of $SL(2, \mathbb{C})$ given by $M \rightarrow (M^{-1})^t$. In index notation this corresponds to a Weyl fermion with index $\psi_{\dot{a}}$.

```
def dualRightHandedRep : Representation ℂ SL(2,ℂ) DualRightHandedWeyl where
```

def dualRightBasis[\[source\]](#)

The standard basis on dual-right-handed Weyl fermions.

```
def dualRightBasis : Basis (Fin 2) ℂ DualRightHandedWeyl
```

lemma dualRightBasis_toFin2ℂ[\[source\]](#)

```
@[simp]
```

```
lemma dualRightBasis_toFin2ℂ (i : Fin 2) : (dualRightBasis i).toFin2ℂ = Pi.single i 1
```

lemma dualRightBasis_p_apply[\[source\]](#)

```
@[simp]
```

```
lemma dualRightBasis_p_apply (M : SL(2,ℂ)) (i j : Fin 2) :  
  (LinearMap.toMatrix dualRightBasis dualRightBasis) (dualRightHandedRep M) i j =  
  ((M.1-1).conjTranspose) i j
```

def leftHandedToDual[\[source\]](#)

The morphism between the representation `leftHanded` and the representation `dualLeftHanded` defined by multiplying an element of `leftHanded` by the matrix $\varepsilon^{a\theta a_1} = !![0, 1; -1, 0]$.

```
def leftHandedToDual : leftHandedRep.IntertwiningMap dualLeftHandedRep where
```

lemma leftHandedToDual_hom_apply

[\[source\]](#)

```
lemma leftHandedToDual_hom_apply (ψ : LeftHandedWeyl) :
  leftHandedToDual ψ =
  DualLeftHandedWeyl.toFin2CEquiv.symm (!! [0, 1; -1, 0] *v ψ.toFin2C)
```

def leftHandedDualTo

[\[source\]](#)

The morphism from `dualLeftHanded` to `leftHanded` defined by multiplying an element of `DualLeftHandedWeyl` by the matrix $\varepsilon_{a_1 a_2} = !![0, -1; 1, 0]$.

```
def leftHandedDualTo : dualLeftHandedRep.IntertwiningMap leftHandedRep where
```

lemma leftHandedDualTo_hom_apply

[\[source\]](#)

```
lemma leftHandedDualTo_hom_apply (ψ : DualLeftHandedWeyl) :
  leftHandedDualTo ψ =
  LeftHandedWeyl.toFin2CEquiv.symm (!! [0, -1; 1, 0] *v ψ.toFin2C)
```

def leftHandedDualEquiv

[\[source\]](#)

The equivalence between the representation `leftHanded` and the representation `dualLeftHanded` defined by multiplying an element of `leftHanded` by the matrix $\varepsilon^{a\theta a_1} = !![0, 1; -1, 0]$.

```
def leftHandedDualEquiv : leftHandedRep.Equiv dualLeftHandedRep
```

lemma leftHandedDualEquiv_hom_hom_apply

[\[source\]](#)

leftHandedDualEquiv acting on an element $\psi : \text{leftHanded}$ corresponds to multiplying ψ by the matrix $!![0, 1; -1, 0]$.

```
lemma leftHandedDualEquiv_hom_hom_apply (ψ : LeftHandedWeyl) :
  leftHandedDualEquiv ψ =
  DualLeftHandedWeyl.toFin2CEquiv.symm (!! [0, 1; -1, 0] *v ψ.toFin2C)
```

lemma leftHandedDualEquiv_inv_hom_apply

[\[source\]](#)

The inverse of `leftHandedDualEquiv` acting on an element $\psi : \text{dualLeftHanded}$ corresponds to multiplying ψ by the matrix $!![0, -1; 1, 0]$.

```
lemma leftHandedDualEquiv_inv_hom_apply (ψ : DualLeftHandedWeyl) :
  leftHandedDualEquiv.symm ψ =
  LeftHandedWeyl.toFin2CEquiv.symm (!! [0, -1; 1, 0] *v ψ.toFin2C)
```

informal_definition rightHandedWeylDualEquiv[\[source\]](#)

The linear equivalence between `rightHandedWeyl` and `DualRightHandedWeyl` given by multiplying an element of `rightHandedWeyl` by the matrix $\varepsilon^{\alpha\theta\alpha_1} = !![0, 1; -1, 0]$.

```
informal_definition rightHandedWeylDualEquiv where
```

```
  deps := [``rightHandedRep, ``dualRightHandedRep]
  tag := "6VZR4"
```

informal_lemma rightHandedWeylDualEquiv_equivariant[\[source\]](#)

The linear equivalence `rightHandedWeylDualEquiv` is equivariant with respect to the action of $SL(2, \mathbb{C})$ on `rightHandedWeyl` and `DualRightHandedWeyl`.

```
informal_lemma rightHandedWeylDualEquiv_equivariant where
```

```
  deps := [``rightHandedWeylDualEquiv]
  tag := "6VZSG"
```

2.42 Physlib.Relativity.Tensors.ComplexTensor.Weyl.Contraction

[Physlib/Relativity/Tensors/ComplexTensor/Weyl/Contraction.lean](#) on GitHub.

def leftDualBi[\[source\]](#)

The bi-linear map corresponding to contraction of a left-handed Weyl fermion with a dual-left-handed Weyl fermion.

```
def leftDualBi : LeftHandedWeyl →l[ℂ] DualLeftHandedWeyl →l[ℂ] ℂ where
```

def dualLeftBi[\[source\]](#)

The bi-linear map corresponding to contraction of a dual-left-handed Weyl fermion with a left-handed Weyl fermion.

```
def dualLeftBi : DualLeftHandedWeyl →l[ℂ] LeftHandedWeyl →l[ℂ] ℂ where
```

def rightDualBi[\[source\]](#)

The bi-linear map corresponding to contraction of a right-handed Weyl fermion with a dual-right-handed Weyl fermion.

```
def rightDualBi : RightHandedWeyl →l[ℂ] DualRightHandedWeyl →l[ℂ] ℂ where
```

def dualRightBi[\[source\]](#)

The bi-linear map corresponding to contraction of a dual-right-handed Weyl fermion with a right-handed Weyl fermion.

```
def dualRightBi : DualRightHandedWeyl →l[ℂ] RightHandedWeyl →l[ℂ] ℂ where
```

def leftDualContraction[\[source\]](#)

The linear map from `leftHandedWeyl` $\otimes$ `DualLeftHandedWeyl` to $\mathbb{C}$ given by summing over components of `leftHandedWeyl` and `DualLeftHandedWeyl` in the standard basis

(i.e. the dot product). Physically, the contraction of a left-handed Weyl fermion with a dual-left-handed Weyl fermion. In index notation this is $\psi^a \varphi_a$.

```
def leftDualContraction : (leftHandedRep.tprod dualLeftHandedRep).IntertwiningMap
  (Representation.trivial ℂ SL(2,ℂ) ℂ) where
```

lemma leftDualContraction_hom_tmul

[\[source\]](#)

```
lemma leftDualContraction_hom_tmul (ψ : LeftHandedWeyl)
  (φ : DualLeftHandedWeyl) :
  leftDualContraction (ψ ⊗ₜ φ) = ψ.toFin2ℂ ·ᵥ φ.toFin2ℂ
```

lemma leftDualContraction_basis

[\[source\]](#)

```
lemma leftDualContraction_basis (i j : Fin 2) :
  leftDualContraction (leftBasis i ⊗ₜ dualLeftBasis j) = if i.1 = j.1 then (1 : ℂ) else 0
```

def dualLeftContraction

[\[source\]](#)

The linear map from $\text{DualLeftHandedWeyl} \otimes \text{leftHandedWeyl}$ to $\mathbb{C}$ given by summing over components of $\text{DualLeftHandedWeyl}$ and leftHandedWeyl in the standard basis (i.e. the dot product). Physically, the contraction of a dual-left-handed Weyl fermion with a left-handed Weyl fermion. In index notation this is $\varphi_a \psi^a$.

```
def dualLeftContraction : (dualLeftHandedRep.tprod leftHandedRep).IntertwiningMap
  (Representation.trivial ℂ SL(2,ℂ) ℂ) where
```

lemma dualLeftContraction_hom_tmul

[\[source\]](#)

```
lemma dualLeftContraction_hom_tmul (φ : DualLeftHandedWeyl) (ψ : LeftHandedWeyl) :
  dualLeftContraction (φ ⊗ₜ ψ) = φ.toFin2ℂ ·ᵥ ψ.toFin2ℂ
```

lemma dualLeftContraction_basis

[\[source\]](#)

```
lemma dualLeftContraction_basis (i j : Fin 2) :
  dualLeftContraction (dualLeftBasis i ⊗ₜ leftBasis j) = if i.1 = j.1 then (1 : ℂ) else 0
```

def rightDualContraction

[\[source\]](#)

The linear map from $\text{rightHandedWeyl} \otimes \text{DualRightHandedWeyl}$ to $\mathbb{C}$ given by summing over components of rightHandedWeyl and $\text{DualRightHandedWeyl}$ in the standard basis (i.e. the dot product). The contraction of a right-handed Weyl fermion with a left-handed Weyl fermion. In index notation this is $\psi^{\dot{a}} \varphi_{\dot{a}}$.

```
def rightDualContraction : (rightHandedRep.tprod dualRightHandedRep).IntertwiningMap
  (Representation.trivial ℂ SL(2,ℂ) ℂ) where
```

lemma rightDualContraction_hom_tmul

[\[source\]](#)

```
lemma rightDualContraction_hom_tmul (ψ : RightHandedWeyl)
  (φ : DualRightHandedWeyl) :
  rightDualContraction (ψ ⊗ₜ φ) = ψ.toFin2ℂ ·ᵥ φ.toFin2ℂ
```

lemma rightDualContraction_basis[\[source\]](#)

```

lemma rightDualContraction_basis (i j : Fin 2) :
  rightDualContraction (rightBasis i ⊗ₜ dualRightBasis j) =
  if i.1 = j.1 then (1 : ℂ) else 0

```

def dualRightContraction[\[source\]](#)

The linear map from $DualRightHandedWeyl \otimes rightHandedWeyl$ to $\mathbb{C}$ given by summing over components of $DualRightHandedWeyl$ and $rightHandedWeyl$ in the standard basis (i.e. the dot product). The contraction of a right-handed Weyl fermion with a left-handed Weyl fermion. In index notation this is $\varphi_{\dot{a}} \psi^{\dot{a}}$.

```

def dualRightContraction : (dualRightHandedRep.tprod rightHandedRep).IntertwiningMap
  (Representation.trivial ℂ SL(2,ℂ) ℂ) where

```

lemma dualRightContraction_hom_tmul[\[source\]](#)

```

lemma dualRightContraction_hom_tmul (φ : DualRightHandedWeyl)
  (ψ : RightHandedWeyl) :
  dualRightContraction (φ ⊗ₜ ψ) = φ.toFin2ℂ ·v ψ.toFin2ℂ

```

lemma dualRightContraction_basis[\[source\]](#)

```

lemma dualRightContraction_basis (i j : Fin 2) :
  dualRightContraction (dualRightBasis i ⊗ₜ rightBasis j) =
  if i.1 = j.1 then (1 : ℂ) else 0

```

lemma leftDualContraction_tmul_symm[\[source\]](#)

```

lemma leftDualContraction_tmul_symm (ψ : LeftHandedWeyl) (φ : DualLeftHandedWeyl) :
  leftDualContraction (ψ ⊗ₜ [ℂ] φ) = dualLeftContraction (φ ⊗ₜ [ℂ] ψ)

```

lemma dualLeftContraction_tmul_symm[\[source\]](#)

```

lemma dualLeftContraction_tmul_symm (φ : DualLeftHandedWeyl) (ψ : LeftHandedWeyl) :
  dualLeftContraction (φ ⊗ₜ [ℂ] ψ) = leftDualContraction (ψ ⊗ₜ [ℂ] φ)

```

lemma rightDualContraction_tmul_symm[\[source\]](#)

```

lemma rightDualContraction_tmul_symm (ψ : RightHandedWeyl) (φ : DualRightHandedWeyl) :
  rightDualContraction (ψ ⊗ₜ [ℂ] φ) = dualRightContraction (φ ⊗ₜ [ℂ] ψ)

```

lemma dualRightContraction_tmul_symm[\[source\]](#)

```

lemma dualRightContraction_tmul_symm (φ : DualRightHandedWeyl) (ψ : RightHandedWeyl) :
  dualRightContraction (φ ⊗ₜ [ℂ] ψ) = rightDualContraction (ψ ⊗ₜ [ℂ] φ)

```

2.43 Physlib.Relativity.Tensors.ComplexTensor.Weyl.Metric

[Physlib/Relativity/Tensors/ComplexTensor/Weyl/Metric.lean](#) on GitHub.

lemma leftMetricVal_expand_tmul'

[\[source\]](#)

```
lemma leftMetricVal_expand_tmul' : leftMetricVal = leftBasis 1 ⊗t[C] leftBasis 0
  - leftBasis 0 ⊗t[C] leftBasis 1
```

def dualLeftMetricVal

[\[source\]](#)

The metric ε_{aa} as an element of $(\text{dualLeftHanded} \otimes \text{dualLeftHanded}).V$.

```
def dualLeftMetricVal : (DualLeftHandedWeyl ⊗[C] DualLeftHandedWeyl)
```

lemma dualLeftMetricVal_expand_tmul

[\[source\]](#)

Expansion of dualLeftMetricVal into the left basis.

```
lemma dualLeftMetricVal_expand_tmul : dualLeftMetricVal =
  dualLeftBasis 0 ⊗t[C] dualLeftBasis 1 - dualLeftBasis 1 ⊗t[C] dualLeftBasis 0
```

def dualLeftMetric

[\[source\]](#)

The metric ε_{aa} as a morphism $\mathcal{K}_- (\text{Rep } \mathbb{C} \text{ } SL(2, \mathbb{C})) \rightarrow \text{dualLeftHanded} \otimes \text{dualLeftHanded}$, making manifest its invariance under the action of $SL(2, \mathbb{C})$.

```
def dualLeftMetric : (Representation.trivial C SL(2, C) C).IntertwiningMap
  (dualLeftHandedRep.tprod dualLeftHandedRep) where
```

lemma dualLeftMetric_apply_one

[\[source\]](#)

```
lemma dualLeftMetric_apply_one : dualLeftMetric (1 : C) = dualLeftMetricVal
```

lemma rightMetricVal_expand_tmul'

[\[source\]](#)

```
lemma rightMetricVal_expand_tmul' : rightMetricVal = rightBasis 1 ⊗t[C] rightBasis 0
  - rightBasis 0 ⊗t[C] rightBasis 1
```

def dualRightMetricVal

[\[source\]](#)

The metric $\varepsilon_{\{dot a\}\{dot a\}}$ as an element of $(\text{dualRightHanded} \otimes \text{dualRightHanded}).V$.

```
def dualRightMetricVal : DualRightHandedWeyl ⊗[C] DualRightHandedWeyl
```

lemma dualRightMetricVal_expand_tmul

[\[source\]](#)

Expansion of rightMetricVal into the left basis.

```
lemma dualRightMetricVal_expand_tmul : dualRightMetricVal =
  dualRightBasis 0 ⊗t[C] dualRightBasis 1 - dualRightBasis 1 ⊗t[C] dualRightBasis 0
```

def dualRightMetric

[\[source\]](#)

The metric $\varepsilon_{\{dot a\}\{dot a\}}$ as a morphism $\mathcal{K}_- (\text{Rep } \mathbb{C} \text{ } SL(2, \mathbb{C})) \rightarrow \text{dualRightHanded} \otimes \text{dualRightHanded}$, making manifest its invariance under the action of $SL(2, \mathbb{C})$.


```
def dualRightMetric : (Representation.trivial  $\mathbb{C}$  SL(2, $\mathbb{C}$ )  $\mathbb{C}$ ).IntertwiningMap
  (dualRightHandedRep.tprod dualRightHandedRep) where
```

lemma dualRightMetric_apply_one

[\[source\]](#)

```
lemma dualRightMetric_apply_one : dualRightMetric (1 :  $\mathbb{C}$ ) = dualRightMetricVal
```

lemma leftDualContraction_apply_metric

[\[source\]](#)

```
lemma leftDualContraction_apply_metric :
  (TensorProduct.comm  $\mathbb{C}$  _ _ <|
   (TensorProduct.lid  $\mathbb{C}$  _).lTensor _ <|
   (leftDualContraction.toLinearMap.rTensor _).lTensor _ <|
   (TensorProduct.assoc  $\mathbb{C}$  _ _ _).symm.toLinearMap.lTensor _ <|
   TensorProduct.assoc  $\mathbb{C}$  _ _ (_  $\otimes$  [ $\mathbb{C}$ ] _) <|
   (leftMetric 1)  $\otimes$  [ $\mathbb{C}$ ] (dualLeftMetric 1)) = dualLeftLeftUnit (1 :  $\mathbb{C}$ )
```

lemma dualLeftContraction_apply_metric

[\[source\]](#)

```
lemma dualLeftContraction_apply_metric :
  (TensorProduct.comm  $\mathbb{C}$  _ _ <|
   (TensorProduct.lid  $\mathbb{C}$  _).lTensor _ <|
   (dualLeftContraction.toLinearMap.rTensor _).lTensor _ <|
   (TensorProduct.assoc  $\mathbb{C}$  _ _ _).symm.toLinearMap.lTensor _ <|
   TensorProduct.assoc  $\mathbb{C}$  _ _ (_  $\otimes$  [ $\mathbb{C}$ ] _) <|
   (dualLeftMetric 1)  $\otimes$  [ $\mathbb{C}$ ] (leftMetric 1)) = leftDualLeftUnit (1 :  $\mathbb{C}$ )
```

lemma rightDualContraction_apply_metric

[\[source\]](#)

```
lemma rightDualContraction_apply_metric :
  (TensorProduct.comm  $\mathbb{C}$  _ _ <|
   (TensorProduct.lid  $\mathbb{C}$  _).lTensor _ <|
   (rightDualContraction.toLinearMap.rTensor _).lTensor _ <|
   (TensorProduct.assoc  $\mathbb{C}$  _ _ _).symm.toLinearMap.lTensor _ <|
   TensorProduct.assoc  $\mathbb{C}$  _ _ (_  $\otimes$  [ $\mathbb{C}$ ] _) <|
   (rightMetric 1)  $\otimes$  [ $\mathbb{C}$ ] (dualRightMetric 1)) = dualRightRightUnit (1 :  $\mathbb{C}$ )
```

lemma dualRightContraction_apply_metric

[\[source\]](#)

```
lemma dualRightContraction_apply_metric :
  (TensorProduct.comm  $\mathbb{C}$  _ _ <|
   (TensorProduct.lid  $\mathbb{C}$  _).lTensor _ <|
   (dualRightContraction.toLinearMap.rTensor _).lTensor _ <|
   (TensorProduct.assoc  $\mathbb{C}$  _ _ _).symm.toLinearMap.lTensor _ <|
   TensorProduct.assoc  $\mathbb{C}$  _ _ (_  $\otimes$  [ $\mathbb{C}$ ] _) <|
   (dualRightMetric 1)  $\otimes$  [ $\mathbb{C}$ ] (rightMetric 1)) = rightDualRightUnit (1 :  $\mathbb{C}$ )
```

2.44 Physlib.Relativity.Tensors.ComplexTensor.Weyl.Modules

[Physlib/Relativity/Tensors/ComplexTensor/Weyl/Modules.lean](#) on GitHub.

structure LeftHandedWeyl[\[source\]](#)

The module in which left handed fermions live. This is equivalent to $\text{Fin } 2 \rightarrow \mathbb{C}$.

```
structure LeftHandedWeyl where
  val : Fin 2 → ℂ
```

def LeftHandedWeyl.toFin2CFun[\[source\]](#)

The equivalence between LeftHandedWeyl and $\text{Fin } 2 \rightarrow \mathbb{C}$.

```
def toFin2CFun : LeftHandedWeyl ≈ (Fin 2 → ℂ) where
```

instance AddCommMonoid LeftHandedWeyl[\[source\]](#)

The instance of AddCommMonoid on LeftHandedWeyl defined via its equivalence with $\text{Fin } 2 \rightarrow \mathbb{C}$.

```
instance : AddCommMonoid LeftHandedWeyl
```

instance AddCommGroup LeftHandedWeyl[\[source\]](#)

The instance of AddCommGroup on LeftHandedWeyl defined via its equivalence with $\text{Fin } 2 \rightarrow \mathbb{C}$.

```
instance : AddCommGroup LeftHandedWeyl
```

instance Module ℂ LeftHandedWeyl[\[source\]](#)

The instance of Module on LeftHandedWeyl defined via its equivalence with $\text{Fin } 2 \rightarrow \mathbb{C}$.

```
instance : Module ℂ LeftHandedWeyl
```

def LeftHandedWeyl.toFin2CEquiv[\[source\]](#)

The linear equivalence between LeftHandedWeyl and $(\text{Fin } 2 \rightarrow \mathbb{C})$.

```
@[simps!]
def toFin2CEquiv : LeftHandedWeyl ≈ℓ[ℂ] (Fin 2 → ℂ) where
```

abbrev LeftHandedWeyl.toFin2C[\[source\]](#)

The underlying element of $\text{Fin } 2 \rightarrow \mathbb{C}$ of a element in LeftHandedWeyl defined through the linear equivalence toFin2CEquiv.

```
abbrev toFin2C (ψ : LeftHandedWeyl)
```

structure DualLeftHandedWeyl[\[source\]](#)

The module in which dual-left handed fermions live. This is equivalent to $\text{Fin } 2 \rightarrow \mathbb{C}$.

```
structure DualLeftHandedWeyl where
  val : Fin 2 → ℂ
```

def DualLeftHandedWeyl.toFin2CFun[\[source\]](#)

The equivalence between DualLeftHandedWeyl and $\text{Fin } 2 \rightarrow \mathbb{C}$.

```
def toFin2CFun : DualLeftHandedWeyl ≈ (Fin 2 → ℂ) where
```

instance AddCommMonoid DualLeftHandedWeyl[\[source\]](#)

The instance of AddCommMonoid on DualLeftHandedWeyl defined via its equivalence with $\text{Fin } 2 \rightarrow \mathbb{C}$.

```
instance : AddCommMonoid DualLeftHandedWeyl
```

instance AddCommGroup DualLeftHandedWeyl[\[source\]](#)

The instance of AddCommGroup on DualLeftHandedWeyl defined via its equivalence with $\text{Fin } 2 \rightarrow \mathbb{C}$.

```
instance : AddCommGroup DualLeftHandedWeyl
```

instance Module ℂ DualLeftHandedWeyl[\[source\]](#)

The instance of Module on DualLeftHandedWeyl defined via its equivalence with $\text{Fin } 2 \rightarrow \mathbb{C}$.

```
instance : Module ℂ DualLeftHandedWeyl
```

def DualLeftHandedWeyl.toFin2CEquiv[\[source\]](#)

The linear equivalence between DualLeftHandedWeyl and $(\text{Fin } 2 \rightarrow \mathbb{C})$.

```
@[simps!]
```

```
def toFin2CEquiv : DualLeftHandedWeyl ≈l[ℂ] (Fin 2 → ℂ) where
```

abbrev DualLeftHandedWeyl.toFin2C[\[source\]](#)

The underlying element of $\text{Fin } 2 \rightarrow \mathbb{C}$ of a element in DualLeftHandedWeyl defined through the linear equivalence toFin2CEquiv.

```
abbrev toFin2C (ψ : DualLeftHandedWeyl)
```

structure RightHandedWeyl[\[source\]](#)

The module in which right handed fermions live. This is equivalent to $\text{Fin } 2 \rightarrow \mathbb{C}$.

```
structure RightHandedWeyl where
```

```
  val : Fin 2 → ℂ
```

def RightHandedWeyl.toFin2CFun[\[source\]](#)

The equivalence between RightHandedWeyl and $\text{Fin } 2 \rightarrow \mathbb{C}$.

```
def toFin2CFun : RightHandedWeyl ≈ (Fin 2 → ℂ) where
```

instance AddCommMonoid RightHandedWeyl[\[source\]](#)

The instance of AddCommMonoid on RightHandedWeyl defined via its equivalence with $\text{Fin } 2 \rightarrow \mathbb{C}$.

```
instance : AddCommMonoid RightHandedWeyl
```

instance AddCommGroup RightHandedWeyl[\[source\]](#)

The instance of AddCommGroup on RightHandedWeyl defined via its equivalence with $\text{Fin } 2 \rightarrow \mathbb{C}$.

```
instance : AddCommGroup RightHandedWeyl
```

instance Module ℂ RightHandedWeyl[\[source\]](#)

The instance of Module on RightHandedWeyl defined via its equivalence with $\text{Fin } 2 \rightarrow \mathbb{C}$.

```
instance : Module ℂ RightHandedWeyl
```

def RightHandedWeyl.toFin2CEquiv[\[source\]](#)

The linear equivalence between RightHandedWeyl and $(\text{Fin } 2 \rightarrow \mathbb{C})$.

```
@[simps!]
```

```
def toFin2CEquiv : RightHandedWeyl ≈l[ℂ] (Fin 2 → ℂ) where
```

abbrev RightHandedWeyl.toFin2C[\[source\]](#)

The underlying element of $\text{Fin } 2 \rightarrow \mathbb{C}$ of a element in RightHandedWeyl defined through the linear equivalence toFin2CEquiv.

```
abbrev toFin2C (ψ : RightHandedWeyl)
```

structure DualRightHandedWeyl[\[source\]](#)

The module in which dual-right handed fermions live. This is equivalent to $\text{Fin } 2 \rightarrow \mathbb{C}$.

```
structure DualRightHandedWeyl where
```

```
  val : Fin 2 → ℂ
```

def DualRightHandedWeyl.toFin2CFun[\[source\]](#)

The equivalence between DualRightHandedWeyl and $\text{Fin } 2 \rightarrow \mathbb{C}$.

```
def toFin2CFun : DualRightHandedWeyl ≈ (Fin 2 → ℂ) where
```

instance AddCommMonoid DualRightHandedWeyl[\[source\]](#)

The instance of AddCommMonoid on DualRightHandedWeyl defined via its equivalence with $\text{Fin } 2 \rightarrow \mathbb{C}$.

```
instance : AddCommMonoid DualRightHandedWeyl
```

instance AddCommGroup DualRightHandedWeyl

[\[source\]](#)

The instance of AddCommGroup on DualRightHandedWeyl defined via its equivalence with $\text{Fin } 2 \rightarrow \mathbb{C}$.

```
instance : AddCommGroup DualRightHandedWeyl
```

instance Module $\mathbb{C}$ DualRightHandedWeyl

[\[source\]](#)

The instance of Module on DualRightHandedWeyl defined via its equivalence with $\text{Fin } 2 \rightarrow \mathbb{C}$.

```
instance : Module  $\mathbb{C}$  DualRightHandedWeyl
```

def DualRightHandedWeyl.toFin2CEquiv

[\[source\]](#)

The linear equivalence between DualRightHandedWeyl and $(\text{Fin } 2 \rightarrow \mathbb{C})$.

```
@[simps!]
```

```
def toFin2CEquiv : DualRightHandedWeyl  $\approx_l$  [ $\mathbb{C}$ ] ( $\text{Fin } 2 \rightarrow \mathbb{C}$ ) where
```

abbrev DualRightHandedWeyl.toFin2C

[\[source\]](#)

The underlying element of $\text{Fin } 2 \rightarrow \mathbb{C}$ of a element in DualRightHandedWeyl defined through the linear equivalence toFin2CEquiv.

```
abbrev toFin2C ( $\psi$  : DualRightHandedWeyl)
```

2.45 Physlib.Relativity.Tensors.ComplexTensor.Weyl.Two

[Physlib/Relativity/Tensors/ComplexTensor/Weyl/Two.lean](#) on GitHub.

def dualLeftdualLeftToMatrix

[\[source\]](#)

Equivalence of dualLeftHanded $\otimes$ dualLeftHanded to 2×2 complex matrices.

```
def dualLeftdualLeftToMatrix : (DualLeftHandedWeyl  $\otimes$  [ $\mathbb{C}$ ] DualLeftHandedWeyl)  $\approx_l$  [ $\mathbb{C}$ ]
  Matrix (Fin 2) (Fin 2)  $\mathbb{C}$ 
```

lemma dualLeftdualLeftToMatrix_symm_expand_tmul

[\[source\]](#)

Expanding dualLeftdualLeftToMatrix in terms of the standard basis.

```
lemma dualLeftdualLeftToMatrix_symm_expand_tmul (M : Matrix (Fin 2) (Fin 2)  $\mathbb{C}$ ) :
  dualLeftdualLeftToMatrix.symm M =  $\sum i, \sum j, M i j \cdot$ 
    (dualLeftBasis i  $\otimes_t$  [ $\mathbb{C}$ ] dualLeftBasis j)
```

def leftDualLeftToMatrix

[\[source\]](#)

Equivalence of leftHanded $\otimes$ dualLeftHanded to 2×2 complex matrices.

```
def leftDualLeftToMatrix : (LeftHandedWeyl ⊗[C] DualLeftHandedWeyl) ≈l[C]
  Matrix (Fin 2) (Fin 2) C
```

lemma leftDualLeftToMatrix_symm_expand_tmul

[\[source\]](#)

Expanding leftDualLeftToMatrix in terms of the standard basis.

```
lemma leftDualLeftToMatrix_symm_expand_tmul (M : Matrix (Fin 2) (Fin 2) C) :
  leftDualLeftToMatrix.symm M = ∑ i, ∑ j, M i j • (leftBasis i ⊗t[C] dualLeftBasis j)
```

def dualLeftLeftToMatrix

[\[source\]](#)

Equivalence of dualLeftHanded ⊗ leftHanded to 2 × 2 complex matrices.

```
def dualLeftLeftToMatrix : (DualLeftHandedWeyl ⊗[C] LeftHandedWeyl) ≈l[C]
  Matrix (Fin 2) (Fin 2) C
```

lemma dualLeftLeftToMatrix_symm_expand_tmul

[\[source\]](#)

Expanding dualLeftLeftToMatrix in terms of the standard basis.

```
lemma dualLeftLeftToMatrix_symm_expand_tmul (M : Matrix (Fin 2) (Fin 2) C) :
  dualLeftLeftToMatrix.symm M = ∑ i, ∑ j, M i j • (dualLeftBasis i ⊗t[C] leftBasis j)
```

def dualRightDualRightToMatrix

[\[source\]](#)

Equivalence of dualRightHanded ⊗ dualRightHanded to 2 × 2 complex matrices.

```
def dualRightDualRightToMatrix : (DualRightHandedWeyl ⊗[C] DualRightHandedWeyl) ≈l[C]
  Matrix (Fin 2) (Fin 2) C
```

lemma dualRightDualRightToMatrix_symm_expand_tmul

[\[source\]](#)

Expanding dualRightDualRightToMatrix in terms of the standard basis.

```
lemma dualRightDualRightToMatrix_symm_expand_tmul (M : Matrix (Fin 2) (Fin 2) C) :
  dualRightDualRightToMatrix.symm M =
  ∑ i, ∑ j, M i j • (dualRightBasis i ⊗t[C] dualRightBasis j)
```

def rightDualRightToMatrix

[\[source\]](#)

Equivalence of rightHanded ⊗ dualRightHanded to 2 × 2 complex matrices.

```
def rightDualRightToMatrix : (RightHandedWeyl ⊗[C] DualRightHandedWeyl) ≈l[C]
  Matrix (Fin 2) (Fin 2) C
```

lemma rightDualRightToMatrix_symm_expand_tmul

[\[source\]](#)

Expanding rightDualRightToMatrix in terms of the standard basis.

```
lemma rightDualRightToMatrix_symm_expand_tmul (M : Matrix (Fin 2) (Fin 2) C) :
  rightDualRightToMatrix.symm M = ∑ i, ∑ j, M i j • (rightBasis i ⊗t[C] dualRightBasis j)
```

def dualRightRightToMatrix[\[source\]](#)

Equivalence of dualRightHanded $\otimes$ rightHanded to 2×2 complex matrices.

```
def dualRightRightToMatrix : (DualRightHandedWeyl  $\otimes$  [C] RightHandedWeyl)  $\approx_l$  [C]
Matrix (Fin 2) (Fin 2) C
```

lemma dualRightRightToMatrix_symm_expand_tmul[\[source\]](#)

Expanding dualRightRightToMatrix in terms of the standard basis.

```
lemma dualRightRightToMatrix_symm_expand_tmul (M : Matrix (Fin 2) (Fin 2) C) :
dualRightRightToMatrix.symm M =  $\sum_i \sum_j M_{ij} \cdot (\text{dualRightBasis } i \otimes [\text{C}] \text{ rightBasis } j)$ 
```

def dualLeftDualRightToMatrix[\[source\]](#)

Equivalence of dualLeftHanded $\otimes$ dualRightHanded to 2×2 complex matrices.

```
def dualLeftDualRightToMatrix : (DualLeftHandedWeyl  $\otimes$  [C] DualRightHandedWeyl)  $\approx_l$  [C]
Matrix (Fin 2) (Fin 2) C
```

lemma dualLeftDualRightToMatrix_symm_expand_tmul[\[source\]](#)

Expanding dualLeftDualRightToMatrix in terms of the standard basis.

```
lemma dualLeftDualRightToMatrix_symm_expand_tmul (M : Matrix (Fin 2) (Fin 2) C) :
dualLeftDualRightToMatrix.symm M =  $\sum_i \sum_j M_{ij} \cdot$ 
(dualLeftBasis  $i \otimes [\text{C}] \text{ dualRightBasis } j$ )
```

lemma coe_linearEquivFunOnFinite[\[source\]](#)

The coercion of Finsupp.linearEquivFunOnFinite to a function is the underlying finitely-supported function, used to bridge it with Matrix.mulVec.

```
private lemma coe_linearEquivFunOnFinite (g : (Fin 2  $\times$  Fin 2)  $\rightarrow_0$  C) :
Finsupp.linearEquivFunOnFinite C C (Fin 2  $\times$  Fin 2) g =  $\uparrow$ g
```

lemma dualLeftdualLeftToMatrix_p[\[source\]](#)

*The group action of $SL(2, \mathbb{C})$ on dualLeftHanded $\otimes$ dualLeftHanded is equivalent to $(M.1^{-1})^T * \text{leftLeftToMatrix } v * (M.1^{-1})$.*

```
lemma dualLeftdualLeftToMatrix_p (v : (DualLeftHandedWeyl  $\otimes$  [C] DualLeftHandedWeyl)) (M : SL(2,
C)) :
dualLeftdualLeftToMatrix (TensorProduct.map (dualLeftHandedRep M) (dualLeftHandedRep M) v)
=
(M.1 $^{-1}$ ) $^T$  * dualLeftdualLeftToMatrix v * (M.1 $^{-1}$ )
```

lemma leftDualLeftToMatrix_p[\[source\]](#)

*The group action of $SL(2, \mathbb{C})$ on leftHanded $\otimes$ dualLeftHanded is equivalent to $M.1 * \text{leftDualLeftToMatrix } v * (M.1^{-1})$.*

```
lemma leftDualLeftToMatrix_p (v : (LeftHandedWeyl  $\otimes$  [C] DualLeftHandedWeyl)) (M : SL(2, C)) :
leftDualLeftToMatrix (TensorProduct.map (leftHandedRep M) (dualLeftHandedRep M) v) =
M.1 * leftDualLeftToMatrix v * (M.1 $^{-1}$ )
```

lemma dualLeftLeftToMatrix_p[\[source\]](#)

*The group action of $SL(2, \mathbb{C})$ on $\text{dualLeftHanded} \otimes \text{leftHanded}$ is equivalent to $(M.1^{-1})^T * \text{leftDualLeftToMatrix } v * (M.1)^T$.*

```
lemma dualLeftLeftToMatrix_p (v : (DualLeftHandedWeyl ⊗[ℂ] LeftHandedWeyl)) (M : SL(2, ℂ)) :
  dualLeftLeftToMatrix (TensorProduct.map (dualLeftHandedRep M) (leftHandedRep M) v) =
  (M.1-1)T * dualLeftLeftToMatrix v * (M.1)T
```

lemma dualRightDualRightToMatrix_p[\[source\]](#)

*The group action of $SL(2, \mathbb{C})$ on $\text{dualRightHanded} \otimes \text{dualRightHanded}$ is equivalent to $((M.1^{-1}).\text{conjTranspose} * \text{rightRightToMatrix } v * ((M.1^{-1}).\text{conjTranspose}))^T$.*

```
lemma dualRightDualRightToMatrix_p (v : (DualRightHandedWeyl ⊗[ℂ] DualRightHandedWeyl))
  (M : SL(2, ℂ)) :
  dualRightDualRightToMatrix (TensorProduct.map (dualRightHandedRep M) (dualRightHandedRep M)
    v) =
  ((M.1-1).conjTranspose) * dualRightDualRightToMatrix v * ((M.1-1).conjTranspose)T
```

lemma rightDualRightToMatrix_p[\[source\]](#)

*The group action of $SL(2, \mathbb{C})$ on $\text{rightHanded} \otimes \text{dualRightHanded}$ is equivalent to $(M.1.\text{map star}) * \text{rightDualRightToMatrix } v * ((M.1^{-1}).\text{conjTranspose})^T$.*

```
lemma rightDualRightToMatrix_p (v : (RightHandedWeyl ⊗[ℂ] DualRightHandedWeyl)) (M : SL(2, ℂ)) :
  rightDualRightToMatrix (TensorProduct.map (rightHandedRep M) (dualRightHandedRep M) v) =
  (M.1.map star) * rightDualRightToMatrix v * ((M.1-1).conjTranspose)T
```

lemma dualRightRightToMatrix_p[\[source\]](#)

*The group action of $SL(2, \mathbb{C})$ on $\text{dualRightHanded} \otimes \text{rightHanded}$ is equivalent to $((M.1^{-1}).\text{conjTranspose} * \text{rightDualRightToMatrix } v * (M.1.\text{map star}))^T$.*

```
lemma dualRightRightToMatrix_p (v : (DualRightHandedWeyl ⊗[ℂ] RightHandedWeyl)) (M : SL(2, ℂ)) :
  dualRightRightToMatrix (TensorProduct.map (dualRightHandedRep M) (rightHandedRep M) v) =
  ((M.1-1).conjTranspose) * dualRightRightToMatrix v * (M.1.map star)T
```

lemma dualLeftDualRightToMatrix_p[\[source\]](#)

```
lemma dualLeftDualRightToMatrix_p (v : (DualLeftHandedWeyl ⊗[ℂ] DualRightHandedWeyl))
  (M : SL(2, ℂ)) :
  dualLeftDualRightToMatrix (TensorProduct.map (dualLeftHandedRep M) (dualRightHandedRep M) v) =
  (M.1-1)T * dualLeftDualRightToMatrix v * ((M.1-1).conjTranspose)T
```

lemma dualLeftdualLeftToMatrix_p_symm[\[source\]](#)

```
lemma dualLeftdualLeftToMatrix_p_symm (v : Matrix (Fin 2) (Fin 2) ℂ) (M : SL(2, ℂ)) :
  TensorProduct.map (dualLeftHandedRep M) (dualLeftHandedRep M)
    (dualLeftdualLeftToMatrix.symm v) =
  dualLeftdualLeftToMatrix.symm ((M.1-1)T * v * (M.1-1))
```


lemma leftDualLeftToMatrix_p_symm[\[source\]](#)

```
lemma leftDualLeftToMatrix_p_symm (v : Matrix (Fin 2) (Fin 2) ℂ) (M : SL(2,ℂ)) :
  TensorProduct.map (leftHandedRep M) (dualLeftHandedRep M) (leftDualLeftToMatrix.symm v) =
  leftDualLeftToMatrix.symm (M.1 * v * (M.1-1))
```

lemma dualLeftLeftToMatrix_p_symm[\[source\]](#)

```
lemma dualLeftLeftToMatrix_p_symm (v : Matrix (Fin 2) (Fin 2) ℂ) (M : SL(2,ℂ)) :
  TensorProduct.map (dualLeftHandedRep M) (leftHandedRep M) (dualLeftLeftToMatrix.symm v) =
  dualLeftLeftToMatrix.symm ((M.1-1)T * v * (M.1)T)
```

lemma dualRightDualRightToMatrix_p_symm[\[source\]](#)

```
lemma dualRightDualRightToMatrix_p_symm (v : Matrix (Fin 2) (Fin 2) ℂ) (M : SL(2,ℂ)) :
  TensorProduct.map (dualRightHandedRep M) (dualRightHandedRep M)
    (dualRightDualRightToMatrix.symm v) =
  dualRightDualRightToMatrix.symm (((M.1-1).conjTranspose) * v * ((M.1-1).conjTranspose)T)
```

lemma rightDualRightToMatrix_p_symm[\[source\]](#)

```
lemma rightDualRightToMatrix_p_symm (v : Matrix (Fin 2) (Fin 2) ℂ) (M : SL(2,ℂ)) :
  TensorProduct.map (rightHandedRep M) (dualRightHandedRep M) (rightDualRightToMatrix.symm v)
    =
  rightDualRightToMatrix.symm ((M.1.map star) * v * (((M.1-1).conjTranspose)T))
```

lemma dualRightRightToMatrix_p_symm[\[source\]](#)

```
lemma dualRightRightToMatrix_p_symm (v : Matrix (Fin 2) (Fin 2) ℂ) (M : SL(2,ℂ)) :
  TensorProduct.map (dualRightHandedRep M) (rightHandedRep M) (dualRightRightToMatrix.symm v)
    =
  dualRightRightToMatrix.symm (((M.1-1).conjTranspose) * v * (M.1.map star)T)
```

lemma dualLeftDualRightToMatrix_p_symm[\[source\]](#)

```
lemma dualLeftDualRightToMatrix_p_symm (v : Matrix (Fin 2) (Fin 2) ℂ) (M : SL(2,ℂ)) :
  TensorProduct.map (dualLeftHandedRep M) (dualRightHandedRep M)
    (dualLeftDualRightToMatrix.symm v) =
  dualLeftDualRightToMatrix.symm ((M.1-1)T * v * ((M.1-1).conjTranspose)T)
```

lemma dualLeftDualRightToMatrix_p_symm_selfAdjoint[\[source\]](#)

```
lemma dualLeftDualRightToMatrix_p_symm_selfAdjoint (v : Matrix (Fin 2) (Fin 2) ℂ)
  (hv : IsSelfAdjoint v) (M : SL(2,ℂ)) :
  TensorProduct.map (dualLeftHandedRep M) (dualRightHandedRep M)
    (dualLeftDualRightToMatrix.symm v) =
  dualLeftDualRightToMatrix.symm (SL2C.toSelfAdjointMap (M.transpose-1) (v, hv))
```

2.46 Physlib.Relativity.Tensors.ComplexTensor.Weyl.Unit

[Physlib/Relativity/Tensors/ComplexTensor/Weyl/Unit.lean](#) on GitHub.

def leftDualLeftUnitVal[\[source\]](#)

The left-dual-left unit δ_a^a as an element of $(\text{leftHanded} \otimes \text{dualLeftHanded}).V$.

```
def leftDualLeftUnitVal : (LeftHandedWeyl ⊗[C] DualLeftHandedWeyl)
```

lemma leftDualLeftUnitVal_expand_tmul[\[source\]](#)

Expansion of `leftDualLeftUnitVal` into the basis.

```
lemma leftDualLeftUnitVal_expand_tmul : leftDualLeftUnitVal =  
  leftBasis 0 ⊗ₜ[C] dualLeftBasis 0 + leftBasis 1 ⊗ₜ[C] dualLeftBasis 1
```

def leftDualLeftUnit[\[source\]](#)

The left-dual-left unit δ_a^a as a morphism $\mathcal{K}_- (\text{Rep } \mathbb{C} \text{ } SL(2, \mathbb{C})) \rightarrow \text{leftHanded} \otimes \text{dualLeftHanded}$, manifesting the invariance under the $SL(2, \mathbb{C})$ action.

```
def leftDualLeftUnit : (Representation.trivial C SL(2,C) C).IntertwiningMap  
  (leftHandedRep.tprod dualLeftHandedRep) where
```

lemma leftDualLeftUnit_apply_one[\[source\]](#)

```
lemma leftDualLeftUnit_apply_one : leftDualLeftUnit (1 : C) = leftDualLeftUnitVal
```

def dualLeftLeftUnitVal[\[source\]](#)

The dual-left-left unit δ_a^a as an element of $(\text{dualLeftHanded} \otimes \text{leftHanded}).V$.

```
def dualLeftLeftUnitVal : (DualLeftHandedWeyl ⊗[C] LeftHandedWeyl)
```

lemma dualLeftLeftUnitVal_expand_tmul[\[source\]](#)

Expansion of `dualLeftLeftUnitVal` into the basis.

```
lemma dualLeftLeftUnitVal_expand_tmul : dualLeftLeftUnitVal =  
  dualLeftBasis 0 ⊗ₜ[C] leftBasis 0 + dualLeftBasis 1 ⊗ₜ[C] leftBasis 1
```

def dualLeftLeftUnit[\[source\]](#)

The dual-left-left unit δ_a^a as a morphism $\mathcal{K}_- (\text{Rep } \mathbb{C} \text{ } SL(2, \mathbb{C})) \rightarrow \text{dualLeftHanded} \otimes \text{leftHanded}$, manifesting the invariance under the $SL(2, \mathbb{C})$ action.

```
def dualLeftLeftUnit :  
  (Representation.trivial C SL(2,C) C).IntertwiningMap  
  (dualLeftHandedRep.tprod leftHandedRep) where
```

lemma dualLeftLeftUnit_apply_one[\[source\]](#)

Applying the morphism `dualLeftLeftUnit` to 1 returns `dualLeftLeftUnitVal`.

```
lemma dualLeftLeftUnit_apply_one : dualLeftLeftUnit (1 : C) = dualLeftLeftUnitVal
```

def rightDualRightUnitVal[\[source\]](#)

The right-dual-right unit $\delta^{\{\text{dot } a\}}_{\{\text{dot } a\}}$ as an element of $(\text{rightHanded} \otimes \text{dualRightHanded}).V$.

```
def rightDualRightUnitVal : RightHandedWeyl ⊗ [C] DualRightHandedWeyl
```

lemma rightDualRightUnitVal_expand_tmul[\[source\]](#)

Expansion of `rightDualRightUnitVal` into the basis.

```
lemma rightDualRightUnitVal_expand_tmul : rightDualRightUnitVal =  
  rightBasis 0 ⊗ [C] dualRightBasis 0 + rightBasis 1 ⊗ [C] dualRightBasis 1
```

def rightDualRightUnit[\[source\]](#)

The right-dual-right unit $\delta^{\{\text{dot } a\}}_{\{\text{dot } a\}}$ as a morphism $\mathcal{K}_- (\text{Rep } \mathbb{C} \text{ } SL(2, \mathbb{C})) \rightarrow \text{rightHanded} \otimes \text{dualRightHanded}$, manifesting the invariance under the $SL(2, \mathbb{C})$ action.

```
def rightDualRightUnit : (Representation.trivial C SL(2,C) C).IntertwiningMap  
  (rightHandedRep.tprod dualRightHandedRep) where
```

lemma rightDualRightUnit_apply_one[\[source\]](#)

```
lemma rightDualRightUnit_apply_one : rightDualRightUnit (1 : C) = rightDualRightUnitVal
```

def dualRightRightUnitVal[\[source\]](#)

The dual-right-right unit $\delta_{\{\text{dot } a\}}^{\{\text{dot } a\}}$ as an element of $(\text{rightHanded} \otimes \text{dualRightHanded}).V$.

```
def dualRightRightUnitVal : (DualRightHandedWeyl ⊗ [C] RightHandedWeyl)
```

lemma dualRightRightUnitVal_expand_tmul[\[source\]](#)

Expansion of `dualRightRightUnitVal` into the basis.

```
lemma dualRightRightUnitVal_expand_tmul : dualRightRightUnitVal =  
  dualRightBasis 0 ⊗ [C] rightBasis 0 + dualRightBasis 1 ⊗ [C] rightBasis 1
```

def dualRightRightUnit[\[source\]](#)

The dual-right-right unit $\delta_{\{\text{dot } a\}}^{\{\text{dot } a\}}$ as a morphism $\mathcal{K}_- (\text{Rep } \mathbb{C} \text{ } SL(2, \mathbb{C})) \rightarrow \text{dualRightHanded} \otimes \text{rightHanded}$, manifesting the invariance under the $SL(2, \mathbb{C})$ action.

```
def dualRightRightUnit : (Representation.trivial C SL(2,C) C).IntertwiningMap  
  (dualRightHandedRep.tprod rightHandedRep) where
```

lemma dualRightRightUnit_apply_one[\[source\]](#)

```
lemma dualRightRightUnit_apply_one : dualRightRightUnit (1 : C) = dualRightRightUnitVal
```

lemma contr_dualLeftLeftUnit[\[source\]](#)*Contraction on the right with dualLeftLeftUnit does nothing.*

```

lemma contr_dualLeftLeftUnit (x : LeftHandedWeyl) :
  (TensorProduct.lid ℂ _ <|
   leftDualContraction.toLinearMap.rTensor _ <|
   (TensorProduct.assoc ℂ _ _).symm <|
   x ⊗ₑ[ℂ] (dualLeftLeftUnit (1 : ℂ))) = x

```

lemma contr_leftDualLeftUnit[\[source\]](#)*Contraction on the right with leftDualLeftUnit does nothing.*

```

lemma contr_leftDualLeftUnit (x : DualLeftHandedWeyl) :
  (TensorProduct.lid ℂ _ <|
   dualLeftContraction.toLinearMap.rTensor _ <|
   (TensorProduct.assoc ℂ _ _).symm <|
   x ⊗ₑ[ℂ] (leftDualLeftUnit (1 : ℂ))) = x

```

lemma contr_dualRightRightUnit[\[source\]](#)*Contraction on the right with dualRightRightUnit does nothing.*

```

lemma contr_dualRightRightUnit (x : RightHandedWeyl) :
  (TensorProduct.lid ℂ _ <|
   rightDualContraction.toLinearMap.rTensor _ <|
   (TensorProduct.assoc ℂ _ _).symm <|
   x ⊗ₑ[ℂ] (dualRightRightUnit (1 : ℂ))) = x

```

lemma contr_rightDualRightUnit[\[source\]](#)*Contraction on the right with rightDualRightUnit does nothing.*

```

lemma contr_rightDualRightUnit (x : DualRightHandedWeyl) :
  (TensorProduct.lid ℂ _ <|
   dualRightContraction.toLinearMap.rTensor _ <|
   (TensorProduct.assoc ℂ _ _).symm <|
   x ⊗ₑ[ℂ] (rightDualRightUnit (1 : ℂ))) = x

```

lemma dualLeftLeftUnit_symm[\[source\]](#)

```

lemma dualLeftLeftUnit_symm :
  dualLeftLeftUnit (1 : ℂ) = LinearMap.lTensor _ (LinearEquiv.refl _ _).toLinearMap
    (TensorProduct.comm ℂ _ _ (leftDualLeftUnit (1 : ℂ)))

```

lemma leftDualLeftUnit_symm[\[source\]](#)

```

lemma leftDualLeftUnit_symm :
  leftDualLeftUnit (1 : ℂ) = LinearMap.lTensor _ (LinearEquiv.refl _ _).toLinearMap
    (TensorProduct.comm ℂ _ _ (dualLeftLeftUnit (1 : ℂ)))

```

lemma dualRightRightUnit_symm[\[source\]](#)

```

lemma dualRightRightUnit_symm :
  dualRightRightUnit (1 :  $\mathbb{C}$ ) = LinearMap.lTensor _ (LinearEquiv.refl _).toLinearMap
    (TensorProduct.comm  $\mathbb{C}$  _ _ (rightDualRightUnit (1 :  $\mathbb{C}$ )))

```

lemma rightDualRightUnit_symm

[\[source\]](#)

```

lemma rightDualRightUnit_symm :
  rightDualRightUnit (1 :  $\mathbb{C}$ ) = LinearMap.lTensor _ (LinearEquiv.refl _).toLinearMap
    (TensorProduct.comm  $\mathbb{C}$  _ _ (dualRightRightUnit (1 :  $\mathbb{C}$ )))

```

2.47 Physlib.Relativity.Tensors.ComponentIdx.Product

[Physlib/Relativity/Tensors/ComponentIdx/Product.lean](#) on GitHub.

lemma prod_symm_natAdd

[\[source\]](#)

```

@[simp]
lemma ComponentIdx.prod_symm_natAdd {n1 n2 :  $\mathbb{N}$ } {c : Fin n1 →  $\mathbb{C}$ } {c1 : Fin n2 →  $\mathbb{C}$ }
  (p : ComponentIdx (S := S) c) (q : ComponentIdx (S := S) c1) (i : Fin n2) :
  ComponentIdx.prod.symm (p, q) (Fin.natAdd n1 i) =
    basisIdxCongr (by simp) (q i)

```

lemma prod_symm_castAdd

[\[source\]](#)

```

@[simp]
lemma ComponentIdx.prod_symm_castAdd {n1 n2 :  $\mathbb{N}$ } {c : Fin n1 →  $\mathbb{C}$ } {c1 : Fin n2 →  $\mathbb{C}$ }
  (p : ComponentIdx (S := S) c) (q : ComponentIdx (S := S) c1) (i : Fin n1) :
  ComponentIdx.prod.symm (p, q) (Fin.castAdd n2 i) =
    basisIdxCongr (by simp) (p i)

```

2.48 Physlib.Relativity.Tensors.ComponentIdx.Single

[Physlib/Relativity/Tensors/ComponentIdx/Single.lean](#) on GitHub.

def single

[\[source\]](#)

The equivalence between component indices for a single color and the basis indices of that color.

```

def ComponentIdx.single {c :  $\mathbb{C}$ } :
  ComponentIdx (S := S) ![c] = basisIdx c where

```

lemma single_apply

[\[source\]](#)

```

@[simp]
lemma ComponentIdx.single_apply {c :  $\mathbb{C}$ } (b : ComponentIdx (S := S) ![c]) :
  ComponentIdx.single (S := S) b = basisIdxCongr (by simp) (b 0)

```

lemma single_symm_apply

[\[source\]](#)

```

@[simp]
lemma ComponentIdx.single_symm_apply {c :  $\mathbb{C}$ } (b : basisIdx c) (i : Fin 1) :
  (ComponentIdx.single (S := S) (c := c)).symm b i = basisIdxCongr (by simp) b

```

2.49 Physlib.Relativity.Tensors.Conjugation.Basic

[Physlib/Relativity/Tensors/Conjugation/Basic.lean](#) on GitHub.

structure ConjTensorSpecies

[\[source\]](#)

A tensor species bundled with a conjugation. Beyond the `TensorSpecies` it extends, it carries the conjugate-colour involution `bar`, the index-set identification `barIdx_eq`, and the contraction-coherence `conj_contrComm`. Also carries `bar_involution` and `bar_tau` (that `bar` is involutive and commutes with the variance dual).

```
structure ConjTensorSpecies (k : Type) [CommRing k] [StarRing k] (C : Type) (G : Type) [Group G]
  (V : C → Type) [∀ c, AddCommGroup (V c)] [∀ c, Module k (V c)]
  (basisIdx : C → Type) [∀ c, Fintype (basisIdx c)] [∀ c, DecidableEq (basisIdx c)]
  (rep : (c : C) → Representation k G (V c)) (b : (c : C) → Basis (basisIdx c) k (V c))
  extends TensorSpecies k C G V basisIdx rep b where
bar : C → C
bar_involution : Function.Involutive bar
bar_tau : ∀ c, bar (toTensorSpecies.τ c) = toTensorSpecies.τ (bar c)
barIdx_eq : ∀ c, basisIdx (bar c) = basisIdx c
conj_contrComm : ∀ (d : C) (x₁ : basisIdx d) (x₂ : basisIdx (toTensorSpecies.τ d)),
  star (toTensorSpecies.contr d (b d x₁ ⊗ₜ[k] b (toTensorSpecies.τ d) x₂))
    = toTensorSpecies.contr (bar d) (b (bar d) ((Equiv.cast (barIdx_eq d)).symm x₁) ⊗ₜ[k]
      b (toTensorSpecies.τ (bar d)) (basisIdxCongr (bar_tau d)
        ((Equiv.cast (barIdx_eq (toTensorSpecies.τ d)).symm x₂))))
```

def componentReindex

[\[source\]](#)

Reindex component labels of `bar ∘ c` back to `c`, slotwise via the cast `barIdx_eq`.

```
def componentReindex {n : ℕ} (c : Fin n → C) :
  ComponentIdx (S := S.toTensorSpecies) (S.bar ∘ c) = ComponentIdx (S := S.toTensorSpecies) c
```

def conjT

[\[source\]](#)

Conjugation of a tensor: conjugate the components and reindex the basis to the conjugate colours. `conj`-semilinear by construction (see `conjT_smul`).

```
def conjT {n : ℕ} {c : Fin n → C} (t : S.Tensor c) : S.Tensor (S.bar ∘ c)
```

lemma conjT_eq_permT_iff

[\[source\]](#)

Componentwise criterion for `conjT t = permT σ h t'`. The conjugate of `t` equals the recolouring `permT σ h t'` exactly when, at every component, the `star`-conjugated reindexed component of `t` matches the corresponding component of `permT σ h t'`. This packages the `componentMap_conjT` expansion and the `repr/componentMap` bridge that the reality and Hermiticity proofs downstream would otherwise repeat by hand; the caller is left only with the species-specific permutation bookkeeping on the right-hand side.

```
lemma conjT_eq_permT_iff {n m : ℕ} {c : Fin n → C} {c' : Fin m → C}
  {σ : Fin n → Fin m} (h : IsReindexing c' (S.bar ∘ c) σ)
  (t : S.Tensor c) (t' : S.Tensor c') :
  S.conjT t = permT σ h t' ↔
  ∀ φ : ComponentIdx (S := S.toTensorSpecies) (S.bar ∘ c),
```

```
star (componentMap c t (S.componentReindex c φ))
  = (Tensor.basis (S.bar ∘ c)).repr (permT σ h t') φ
```

lemma barIdx_involutive_symm[\[source\]](#)

Undoing the two cast reindexings (at bar c then at c) on a label of the doubly-conjugated colour returns the bar_involution cast. Free from barIdx_eq by cast_cast + proof irrelevance, with no coherence field needed.

```
private lemma barIdx_involutive_symm (c : C) (y : basisIdx (S.bar (S.bar c))) :
  Equiv.cast (S.barIdx_eq c) (Equiv.cast (S.barIdx_eq (S.bar c)) y)
  = basisIdxCongr (S.bar_involution c) y
```

lemma isReindexing_bar_bar[\[source\]](#)

The identity permutation satisfies IsReindexing from c to bar ∘ bar ∘ c, as bar is an involution.

```
lemma isReindexing_bar_bar {n : ℕ} (c : Fin n → C) :
  IsReindexing c (S.bar ∘ S.bar ∘ c) (id : Fin n → Fin n)
```

lemma conjT_conjT[\[source\]](#)

Conjugation is an involution: conjugating twice returns the original tensor, up to the bar_involution recolouring (the identity permutation permT).

```
lemma conjT_conjT {n : ℕ} {c : Fin n → C} (t : S.Tensor c) :
  S.conjT (S.conjT t)
  = permT id (S.isReindexing_bar_bar c) t
```

lemma contrCond_bar[\[source\]](#)

Slots with dual colour in c have dual colour in bar ∘ c, as bar commutes with τ.

```
lemma contrCond_bar {n : ℕ} {c : Fin (n + 1 + 1) → C} {i j : Fin (n + 1 + 1)}
  (h : i ≠ j ∧ S.τ (c i) = c j) :
  i ≠ j ∧ S.τ ((S.bar ∘ c) i) = (S.bar ∘ c) j
```

lemma conjT_contrT[\[source\]](#)

Conjugation commutes with contraction: conjugating a contracted tensor equals contracting the conjugate on the bar-images of the same two slots.

```
lemma conjT_contrT {n : ℕ} {c : Fin (n + 1 + 1) → C} (i j : Fin (n + 1 + 1))
  (h : i ≠ j ∧ S.τ (c i) = c j) (t : S.Tensor c) :
  S.conjT (contrT n i j h t)
  = contrT n i j (S.contrCond_bar h) (S.conjT t)
```

def conjEquiv[\[source\]](#)

The conjugate-linear slot isomorphism $V\ c \approx_{sl}[\text{starRingEnd } k]\ V\ (\text{bar } c)$: conjugate the coordinates in the species basis and relabel to the conjugate colour via barIdx_eq.

```
def conjEquiv {c : C} : V c ≈sl[\text{starRingEnd } k] V (S.bar c)
```

def IsHermitian[\[source\]](#)

A pairing $g : V \text{ c } \otimes V (\text{bar } c) \rightarrow k$ is Hermitian when conjugating and swapping its two slots via `conjEquiv` returns its star: $g (x \otimes y) = \text{star } (g (\text{conjEquiv.symm } y \otimes \text{conjEquiv } x))$.

```
def IsHermitian {c : C} (g : V c ⊗ [k] V (S.bar c) →l[k] k) : Prop
```

2.50 Physlib.Relativity.Tensors.Constructors

[Physlib/Relativity/Tensors/Constructors.lean](#) on GitHub.

lemma fromPairT_eq_pure[\[source\]](#)

```
lemma fromPairT_eq_pure {c1 c2 : C} (x : V c1) (y : V c2) :
  fromPairT (S := S) (x ⊗t[k] y) = Pure.toTensor (fun | 0 => x | 1 => y)
```

2.51 Physlib.Relativity.Tensors.Contraction.SuccSuccAbove

[Physlib/Relativity/Tensors/Contraction/SuccSuccAbove.lean](#) on GitHub.

lemma apply_succSuccAbove_symm[\[source\]](#)

```
@[simp, noint simpVarHead]
lemma apply_succSuccAbove_symm {c : Fin (n + 1 + 1) → C} (i j : Fin (n + 1 + 1))
  (k : Fin n) : c (succSuccAbove i j k) = c (succSuccAbove j i k)
```

2.52 Physlib.Relativity.Tensors.Evaluation

[Physlib/Relativity/Tensors/Evaluation.lean](#) on GitHub.

lemma evalT_basis_single[\[source\]](#)

Evaluating the single-index basis tensor `basis ![c] (single.symm b)` at the index x yields the field element 1 if $b = x$ (transported across `![c]` $\theta = c$) and 0 otherwise: evaluation of a one-index basis tensor is the Kronecker delta.

```
lemma evalT_basis_single {c : C} (b : basisIdx c) (x : basisIdx (![c] 0)) :
  (evalT 0 x (basis (S := S) ![c] (ComponentIdx.single.symm b))).toField =
  if basisIdxCongr (by simp) b = x then 1 else 0
```

lemma eq_sum_evalT_of_single_tensor_basis[\[source\]](#)

Basis expansion of a one-index tensor: every $t : \text{Tensor } S \text{ } ![c]$ is the sum over basis indices i of its evaluation coefficient `toField (evalT 0 i t)` times the corresponding basis tensor.

```
lemma eq_sum_evalT_of_single_tensor_basis {c : C} (t : Tensor S ![c]) :
  t = ∑ i, toField (evalT 0 i t) • basis ![c] (ComponentIdx.single.symm
    (basisIdxCongr (by simp) i))
```


lemma eq_sum_evalT[\[source\]](#)

Reconstruction of a tensor from the evaluations of its last index: every $t : \text{Tensor } S \text{ } c$ is the sum over basis indices i of the evaluation $\text{evalT } (\text{Fin.last } n) \text{ } i \text{ } t$ tensored with the basis covector $\text{basis } ![c \text{ } (\text{Fin.last } n)] \text{ } (\text{single.symm } i)$, with the appended index permuted back into the last slot.

```
lemma eq_sum_evalT {n : ℕ} {c : Fin (n + 1) → C} (t : Tensor S c) :
  t = ∑ i, permT id (IsReindexing.append_succ_last c) (prodT (evalT (Fin.last n) i t)
    (basis ![c (Fin.last n)] (ComponentIdx.single.symm i)))
```

lemma eq_sum_evalT_zero[\[source\]](#)

Reconstruction of a tensor from the evaluations of its first index: every $t : \text{Tensor } S \text{ } c$ is the sum over basis indices i of the basis covector $\text{basis } ![c \text{ } 0] \text{ } (\text{single.symm } i)$ tensored with the evaluation $\text{evalT } 0 \text{ } i \text{ } t$, with the prepended index permuted back into the first slot. This is the first-index analogue of `eq_sum_evalT`.

```
lemma eq_sum_evalT_zero {n : ℕ} {c : Fin (n + 1) → C} (t : Tensor S c) :
  t = ∑ i, permT _ (IsReindexing.append_of_first c)
    (prodT (basis ![c 0] (ComponentIdx.single.symm i)) (evalT 0 i t))
```

2.53 Physlib.Relativity.Tensors.LeviCivita.Basic

[Physlib/Relativity/Tensors/LeviCivita/Basic.lean](#) on GitHub.

def leviCivita[\[source\]](#)

The Levi-Civita tensor $\epsilon^{u v p \delta}$ as a real Lorentz tensor in $d = 3$, with $\epsilon^{0 1 2 3} = 1$.

The component on a multi-index f is the generalized Kronecker delta of f against the identity, i.e. the sign of f when f is a permutation and 0 otherwise.

```
noncomputable def leviCivita : ℝ[3, .up, .up, .up, .up]
```

lemma leviCivita_eq_ofInt[\[source\]](#)

The `TensorInt.toTensor` form of the Levi-Civita tensor.

```
lemma leviCivita_eq_ofInt : ε4 =
  TensorInt.toTensor (S := realLorentzTensor 3)
  (c := ![Color.up, Color.up, Color.up, Color.up]) fun f =>
  generalizedKroneckerDelta (fun i => finSumFinEquiv (f i)) (id : Fin 4 → Fin 4)
```

lemma leviCivita_basis_repr_apply[\[source\]](#)

The components of the Levi-Civita tensor in the standard basis are the generalized Kronecker delta of the multi-index against the identity.

```
lemma leviCivita_basis_repr_apply
  (b : ComponentIdx (S := realLorentzTensor 3) ![Color.up, Color.up, Color.up, Color.up]) :
  (Tensor.basis _).repr ε4 b
  = (generalizedKroneckerDelta (fun i => finSumFinEquiv (b i)) (id : Fin 4 → Fin 4) : ℝ)
```

lemma leviCivita_basis_repr_eq_zero_of_eq[\[source\]](#)

The Levi-Civita tensor vanishes on any multi-index with a repeated value: if two distinct index positions $i \neq j$ carry the same basis index, the component is zero.

```
lemma leviCivita_basis_repr_eq_zero_of_eq
  {b : ComponentIdx (S := realLorentzTensor 3) ![Color.up, Color.up, Color.up, Color.up]}
  {i j : Fin 4} (hij : i ≠ j) (h : b i = b j) :
  (Tensor.basis _).repr ε4 b = 0
```

lemma leviCivita_antisymm[\[source\]](#)

The Levi-Civita tensor is antisymmetric in its first two indices $\{\varepsilon_4 \mid \mu \nu \rho \sigma = -\varepsilon_4 \mid \nu \mu \rho \sigma\}^T$.

```
lemma leviCivita_antisymm : {ε4 \ μ ν ρ σ = - (ε4 \ ν μ ρ σ)}T
```

lemma leviCivita_antisymm_mid[\[source\]](#)

The Levi-Civita tensor is antisymmetric in its middle two indices $\{\varepsilon_4 \mid \mu \nu \rho \sigma = -\varepsilon_4 \mid \mu \rho \nu \sigma\}^T$.

```
lemma leviCivita_antisymm_mid : {ε4 \ μ ν ρ σ = - (ε4 \ μ ρ ν σ)}T
```

lemma leviCivita_antisymm_last[\[source\]](#)

The Levi-Civita tensor is antisymmetric in its last two indices $\{\varepsilon_4 \mid \mu \nu \rho \sigma = -\varepsilon_4 \mid \mu \nu \sigma \rho\}^T$.

```
lemma leviCivita_antisymm_last : {ε4 \ μ ν ρ σ = - (ε4 \ μ ν σ ρ)}T
```

2.54 Physlib.Relativity.Tensors.Product

[Physlib/Relativity/Tensors/Product.lean](#) on GitHub.

lemma prodP_update_left[\[source\]](#)

```
@[simp]
lemma Pure.prodP_update_left {n1 n2} {c : Fin n1 → C} {c1 : Fin n2 → C}
  (p1 : Pure S c) (p2 : Pure S c1) (i : Fin n1) (x : V (c i)) :
  Pure.prodP (Pure.update p1 i x) p2
= Pure.update (Pure.prodP p1 p2) (Fin.castAdd n2 i)
  (LinearEquiv.cast (R := k) (by simp) x)
```

lemma prodP_update_right[\[source\]](#)

```
@[simp]
lemma Pure.prodP_update_right {n1 n2} {c : Fin n1 → C} {c1 : Fin n2 → C}
  (p1 : Pure S c) (p2 : Pure S c1) (i : Fin n2) (x : V (c1 i)) :
  Pure.prodP p1 (Pure.update p2 i x)
= Pure.update (Pure.prodP p1 p2) (Fin.natAdd n1 i)
  (LinearEquiv.cast (R := k) (by simp) x)
```

lemma prodP_update_add_toTensor_left[\[source\]](#)

```

lemma Pure.prodP_update_add_toTensor_left {n1 n2} {c : Fin n1 → C} {c1 : Fin n2 → C}
  (p1 : Pure S c) (p2 : Pure S c1) (i : Fin n2) (x y : V (c1 i)) :
  (Pure.prodP p1 (Pure.update p2 i (x + y))).toTensor
= (Pure.prodP p1 (Pure.update p2 i x)).toTensor
+ (Pure.prodP p1 (Pure.update p2 i y)).toTensor

```

lemma prodT_basis'[\[source\]](#)

```

lemma prodT_basis' {n1 n2} {c : Fin n1 → C} {c1 : Fin n2 → C}
  (b : ComponentIdx c) (b1 : ComponentIdx (S := S) c1) :
  (basis c b).prodT (basis c1 b1) =
  basis (Fin.append c c1) (ComponentIdx.prod.symm (b, b1))

```

lemma prodT_zero_right[\[source\]](#)

```

lemma prodT_zero_right {n} {c : Fin n → C}
  {c1 : Fin 0 → C} (t : S.Tensor c) (t1 : S.Tensor c1) :
  prodT t t1 = (toField t1) • permT id (IsReindexing.append_zero_right) t

```

2.55 Physlib.Relativity.Tensors.RealTensor.Basic

[Physlib/Relativity/Tensors/RealTensor/Basic.lean](#) on GitHub.

abbrev modules[\[source\]](#)

The modules associated with each of the different types of real Lorentz vector space.

```

abbrev modules (d : ℕ) : Color → Type
| Color.up => Lorentz.ContrMod d
| Color.down => Lorentz.CoMod d

```

instance modulesAddCommGroup[\[source\]](#)

```

instance modulesAddCommGroup (d : ℕ) : ∀ c, AddCommGroup (modules d c)
| Color.up => inferInstance
| Color.down => inferInstance

```

instance modulesModule[\[source\]](#)

```

instance modulesModule (d : ℕ) : ∀ c, Module ℝ (modules d c)
| Color.up => inferInstance
| Color.down => inferInstance

```

2.56 Physlib.Relativity.Tensors.RealTensor.CoVector.Basic

[Physlib/Relativity/Tensors/RealTensor/CoVector/Basic.lean](#) on GitHub.

lemma eq_of_apply_eq[\[source\]](#)

```

@[ext]
lemma eq_of_apply_eq {d : ℕ} {v w : CoVector d} (h : ∀ i, v i = w i) : v = w

```

lemma apply_sum[\[source\]](#)

```
@[simp]
lemma apply_sum {d : ℕ} {ι : Type} [Fintype ι] (f : ι → CoVector d) (i : Fin 1 ⊗ Fin d) :
  (∑ j, f j) i = ∑ j, f j i
```

2.57 Physlib.Relativity.Tensors.RealTensor.CoVector.Representation

[Physlib/Relativity/Tensors/RealTensor/CoVector/Representation.lean](#) on GitHub.

def rep[\[source\]](#)

The representation of the Lorentz group on Lorentz vectors.

```
def rep {d : ℕ} : Representation ℝ (LorentzGroup d) (CoVector d) where
```

lemma rep_apply_eq_mulVec[\[source\]](#)

```
lemma rep_apply_eq_mulVec (d : ℕ) (Λ : LorentzGroup d) (v : CoVector d) :
  rep Λ v = (LorentzGroup.transpose Λ⁻¹) *v v
```

lemma rep_apply_eq_sum[\[source\]](#)

```
lemma rep_apply_eq_sum (d : ℕ) (Λ : LorentzGroup d) (v : CoVector d) (k : Fin 1 ⊗ Fin d) :
  rep Λ v k = ∑ j, (Λ⁻¹).1 j k • v j
```

lemma rep_apply_eq_sum_coe[\[source\]](#)

```
lemma rep_apply_eq_sum_coe (d : ℕ) (Λ : LorentzGroup d) (v : CoVector d) (k : Fin 1 ⊗ Fin d) :
  rep Λ v k = ∑ j, Λ.1⁻¹ j k • v j
```

lemma rep_apply_basis[\[source\]](#)

```
lemma rep_apply_basis {d} (μ : Fin 1 ⊗ Fin d) (Λ : LorentzGroup d) :
  rep Λ (basis μ) = ∑ j, Λ.1⁻¹ μ j • basis j
```

lemma rep_toMatrix[\[source\]](#)

```
lemma rep_toMatrix (d : ℕ) (Λ : LorentzGroup d) :
  LinearMap.toMatrix basis basis (rep Λ) = Λ.1⁻¹ᵀ
```

lemma rep_injective[\[source\]](#)

```
lemma rep_injective (d : ℕ) (Λ : LorentzGroup d) : Function.Injective (rep Λ)
```

lemma rep_surjective[\[source\]](#)

```
lemma rep_surjective (d : ℕ) (Λ : LorentzGroup d) : Function.Surjective (rep Λ)
```

lemma rep_bijective[\[source\]](#)

```
lemma rep_bijective (d : ℕ) (Λ : LorentzGroup d) : Function.Bijective (rep Λ)
```

2.58 Physlib.Relativity.Tensors.RealTensor.ToComplex

[Physlib/Relativity/Tensors/RealTensor/ToComplex.lean](#) on GitHub.

def toComplexVector[\[source\]](#)

For a given color, the map turning a real Lorentz vector into a complex one.

```
noncomputable def toComplexVector (c : realLorentzTensor.Color) :  
  realLorentzTensor.modules 3 c →sl [Complex.ofRealHom] complexLorentzTensor.modules  
  (colorToComplex c) where
```

lemma toComplexVector_up_eq_inclCongrRealLorentz[\[source\]](#)

```
lemma toComplexVector_up_eq_inclCongrRealLorentz (v : Lorentz.ContrMod 3) :  
  toComplexVector Color.up v = Lorentz.inclCongrRealLorentz v
```

lemma toComplexVector_down_eq_inclCoRealLorentz[\[source\]](#)

```
lemma toComplexVector_down_eq_inclCoRealLorentz (v : Lorentz.CoMod 3) :  
  toComplexVector Color.down v = Lorentz.inclCoRealLorentz v
```

def toComplexPure[\[source\]](#)

The function which turns a real pure tensor into a complex one.

```
noncomputable def toComplexPure {c : Fin n → Color} (p : Pure realLorentzTensor c) :  
  Pure complexLorentzTensor (colorToComplex ∘ c)
```

lemma toComplexPure_component[\[source\]](#)

```
lemma toComplexPure_component {c : Fin n → Color} (p : Pure realLorentzTensor c)  
  (φ : ComponentIdx c) : (toComplexPure p).component (ComponentIdx.complexify φ) =  
  p.component φ
```

lemma actionP_toComplexPure[\[source\]](#)

```
lemma actionP_toComplexPure {n : ℕ} (c : Fin n → Color) (p : Pure realLorentzTensor c)  
  (Λ : SL(2, ℂ)) :  
  Λ • toComplexPure p = toComplexPure (toLorentzGroup Λ • p)
```

lemma toComplex_pure[\[source\]](#)

```
lemma toComplex_pure {n : ℕ} (c : Fin n → Color) (p : Pure realLorentzTensor c) :  
  toComplex p.toTensor = (toComplexPure p).toTensor
```

lemma isReindexing_prodTColorToComplex[\[source\]](#)

```

lemma isReindexing_prodTColorToComplex {n m : ℕ}
  {c : Fin n → realLorentzTensor.Color} {c1 : Fin m → realLorentzTensor.Color} :
  IsReindexing (Fin.append (colorToComplex ∘ c) (colorToComplex ∘ c1))
    (colorToComplex ∘ Fin.append c c1)
    (id : Fin (n + m) → Fin (n + m))

```

2.59 Physlib.Relativity.Tensors.RealTensor.Vector.Basic

[Physlib/Relativity/Tensors/RealTensor/Vector/Basic.lean](#) on GitHub.

lemma eq_of_apply_eq[\[source\]](#)

```

@[ext]
lemma eq_of_apply_eq {d : ℕ} {v w : Vector d} (h : ∀ i, v i = w i) : v = w

```

def ofSpatialComponent[\[source\]](#)

The continuous linear map corresponding to the creation of a Lorentz Vector with only non-zero spatial components.

```

def ofSpatialComponent {d : ℕ} : EuclideanSpace ℝ (Fin d) → L[ℝ] Vector d where

```

2.60 Physlib.Relativity.Tensors.RealTensor.Vector.Representation

[Physlib/Relativity/Tensors/RealTensor/Vector/Representation.lean](#) on GitHub.

def rep[\[source\]](#)

The representation of the Lorentz group on Lorentz vectors.

```

def rep {d : ℕ} : Representation ℝ (LorentzGroup d) (Vector d) where

```

lemma rep_apply_eq_mulVec[\[source\]](#)

```

lemma rep_apply_eq_mulVec (d : ℕ) (Λ : LorentzGroup d) (v : Vector d) :
  rep Λ v = Λ *v v

```

lemma rep_apply_eq_sum[\[source\]](#)

```

lemma rep_apply_eq_sum (d : ℕ) (Λ : LorentzGroup d) (v : Vector d) (k : Fin 1 ⊕ Fin d) :
  rep Λ v k = ∑ j, Λ.1 k j • v j

```

lemma rep_apply_basis[\[source\]](#)

```

lemma rep_apply_basis {d} (μ : Fin 1 ⊕ Fin d) (Λ : LorentzGroup d) :
  rep Λ (basis μ) = ∑ j, Λ.1 j μ • basis j

```

lemma rep_toMatrix[\[source\]](#)

```

lemma rep_toMatrix (d : ℕ) (Λ : LorentzGroup d) :
  LinearMap.toMatrix basis basis (rep Λ) = Λ.1

```

lemma rep_injective[\[source\]](#)

```
lemma rep_injective (d : ℕ) (Λ : LorentzGroup d) : Function.Injective (rep Λ)
```

lemma rep_surjective[\[source\]](#)

```
lemma rep_surjective (d : ℕ) (Λ : LorentzGroup d) : Function.Surjective (rep Λ)
```

lemma rep_bijective[\[source\]](#)

```
lemma rep_bijective (d : ℕ) (Λ : LorentzGroup d) : Function.Bijective (rep Λ)
```

lemma rep_contDiff[\[source\]](#)

```
@[fun_prop]
```

```
lemma rep_contDiff (d : ℕ) {n} (Λ : LorentzGroup d) : ContDiff ℝ n (rep Λ)
```

lemma rep_left_injective[\[source\]](#)

```
lemma rep_left_injective (d : ℕ) : Function.Injective (rep (d := d))
```

2.61 Physlib.Relativity.Tensors.Reindexing

[Physlib/Relativity/Tensors/Reindexing.lean](#) on GitHub.

def IsReindexing[\[source\]](#)

Given two lists of indices $c : \text{Fin } n \rightarrow C$ and $c1 : \text{Fin } m \rightarrow C$ a map $\sigma : \text{Fin } m \rightarrow \text{Fin } n$ satisfies the condition $\text{IsReindexing } c \ c1 \ \sigma$ if it is: - A bijection - Forms a commutative triangle with c and $c1$.

```
def IsReindexing {n m : ℕ} (c : Fin n → C) (c1 : Fin m → C)
  (σ : Fin m → Fin n) : Prop
```

lemma injective[\[source\]](#)

```
lemma injective {n m : ℕ} {c : Fin n → C} {c1 : Fin m → C}
  {σ : Fin m → Fin n} (h : IsReindexing c c1 σ) : Function.Injective σ
```

lemma surjective[\[source\]](#)

```
lemma surjective {n m : ℕ} {c : Fin n → C} {c1 : Fin m → C}
  {σ : Fin m → Fin n} (h : IsReindexing c c1 σ) : Function.Surjective σ
```

lemma auto[\[source\]](#)

```
lemma auto {n m : ℕ} {c : Fin n → C} {c1 : Fin m → C}
  {σ : Fin m → Fin n} (h : IsReindexing c c1 σ := by {simp [IsReindexing]; try decide}) :
  IsReindexing c c1 σ
```

lemma on_id[\[source\]](#)`@[simp]`

```
lemma on_id {n : ℕ} {c c1 : Fin n → C} :
  IsReindexing c c1 (id : Fin n → Fin n) ↔ ∀ i, c i = c1 i
```

lemma on_id_symm[\[source\]](#)

```
lemma on_id_symm {n : ℕ} {c c1 : Fin n → C} (h : IsReindexing c1 c id) :
  IsReindexing c c1 (id : Fin n → Fin n)
```

def inv[\[source\]](#)

For a map σ satisfying $\text{IsReindexing } c \text{ c1 } \sigma$, the inverse of that map.

```
def inv {n m : ℕ} {c : Fin n → C} {c1 : Fin m → C}
  (σ : Fin m → Fin n) (h : IsReindexing c c1 σ) : Fin n → Fin m
```

def toEquiv[\[source\]](#)

For a map $\sigma : \text{Fin } m \rightarrow \text{Fin } n$ satisfying $\text{IsReindexing } c \text{ c1 } \sigma$, that map lifted to an equivalence between $\text{Fin } n$ and $\text{Fin } m$.

```
def toEquiv {n m : ℕ} {c : Fin n → C} {c1 : Fin m → C}
  {σ : Fin m → Fin n} (h : IsReindexing c c1 σ) :
  Fin n ≈ Fin m where
```

lemma apply_inv_apply[\[source\]](#)

```
lemma apply_inv_apply {n m : ℕ} {c : Fin n → C} {c1 : Fin m → C}
  (σ : Fin m → Fin n) (h : IsReindexing c c1 σ) (x : Fin m) :
  h.inv σ (σ x) = x
```

lemma inv_apply_apply[\[source\]](#)

```
lemma inv_apply_apply {n m : ℕ} {c : Fin n → C} {c1 : Fin m → C}
  (σ : Fin m → Fin n) (h : IsReindexing c c1 σ) (x : Fin n) :
  σ (h.inv σ x) = x
```

lemma preserve_color[\[source\]](#)

```
lemma preserve_color {n m : ℕ} {c : Fin n → C} {c1 : Fin m → C}
  {σ : Fin m → Fin n} (h : IsReindexing c c1 σ) :
  ∀ (x : Fin m), c1 x = (c ∘ σ) x
```

lemma inv_perserve_color[\[source\]](#)`@[simp, noint simpVarHead]`

```
lemma inv_perserve_color {n m : ℕ} {c : Fin n → C} {c1 : Fin m → C}
  {σ : Fin m → Fin n} (h : IsReindexing c c1 σ) (x : Fin n) :
  c1 (h.inv σ x) = c x
```


lemma toEquiv_symm_perserve_color[\[source\]](#)`@[simp, nolint simpVarHead]`

```

lemma toEquiv_symm_perserve_color {n m : ℕ} {c : Fin n → C} {c1 : Fin m → C}
  {σ : Fin m → Fin n} (h : IsReindexing c c1 σ) (x : Fin m) :
  c (h.toEquiv.symm x) = c1 x

```

lemma symm[\[source\]](#)

The inverse of a map satisfying IsReindexing c $c1$ σ is a reindexing of $c1$ by c .

```

lemma symm {n m : ℕ} {c : Fin n → C} {c1 : Fin m → C}
  {σ : Fin m → Fin n} (h : IsReindexing c c1 σ) : IsReindexing c1 c (h.inv σ)

```

lemma symm_of_id[\[source\]](#)

```

lemma symm_of_id {n : ℕ} {c c1 : Fin n → C} (h : IsReindexing c c1 id) :
  IsReindexing c1 c id

```

lemma comp[\[source\]](#)

The composition of two maps satisfying IsReindexing also satisfies the IsReindexing.

```

lemma comp {n n1 n2 : ℕ} {c : Fin n → C} {c1 : Fin n1 → C}
  {c2 : Fin n2 → C} {σ : Fin n1 → Fin n} {σ2 : Fin n2 → Fin n1}
  (h : IsReindexing c c1 σ) (h2 : IsReindexing c1 c2 σ2) : IsReindexing c c2 (σ ∘ σ2)

```

lemma append_congr_left[\[source\]](#)

```

lemma append_congr_left {n n' n2 : ℕ} {c : Fin n → C} {c' : Fin n' → C}
  {σ : Fin n' → Fin n} (c2 : Fin n2 → C) (h : IsReindexing c c' σ) :
  IsReindexing (Fin.append c c2) (Fin.append c' c2)
  (Fin.append (Fin.castAdd n2 ∘ σ) (Fin.natAdd n))

```

lemma append_congr_right[\[source\]](#)

```

lemma append_congr_right {n n' n2 : ℕ} {c : Fin n → C} {c' : Fin n' → C}
  {σ : Fin n' → Fin n} (c2 : Fin n2 → C) (h : IsReindexing c c' σ) :
  IsReindexing (Fin.append c2 c) (Fin.append c2 c')
  (Fin.append (Fin.castAdd n) (Fin.natAdd n2 ∘ σ))

```

lemma append_zero_right[\[source\]](#)

```

lemma append_zero_right {n} {c : Fin n → C}
  {c1 : Fin 0 → C} : IsReindexing c (Fin.append c c1) id

```

lemma append_swap[\[source\]](#)

```

lemma append_swap {n n2 : ℕ} {c : Fin n → C} {c2 : Fin n2 → C} :
  IsReindexing (Fin.append c c2) (Fin.append c2 c)
  (Fin.append (Fin.natAdd n) (Fin.castAdd n2))

```

lemma append_assoc_right[\[source\]](#)

```

lemma append_assoc_right {n1 n2 n3 : ℕ} {c : Fin n1 → C} {c2 : Fin n2 → C} {c3 : Fin n3 → C}
  :
  IsReindexing (Fin.append c (Fin.append c2 c3)) (Fin.append (Fin.append c c2) c3)
    (Fin.cast (by grind))

```

lemma append_assoc_left[\[source\]](#)

```

lemma append_assoc_left {n1 n2 n3 : ℕ} {c : Fin n1 → C} {c2 : Fin n2 → C} {c3 : Fin n3 → C} :
  IsReindexing (Fin.append (Fin.append c c2) c3) (Fin.append c (Fin.append c2 c3))
    (Fin.cast (by grind))

```

lemma append_succAbove_natAdd[\[source\]](#)

```

lemma append_succAbove_natAdd {n n1 : ℕ} {c : Fin n → C} {c1 : Fin (n1 + 1) → C}
  (i : Fin (n1 + 1)) :
  IsReindexing (Fin.append c c1 ∘ (Fin.natAdd n i).succAbove)
    (Fin.append c (c1 ∘ i.succAbove)) id

```

lemma append_succAbove_castAdd[\[source\]](#)

```

lemma append_succAbove_castAdd {n n1 : ℕ} {c : Fin (n + 1) → C} {c1 : Fin (n1 + 1) → C}
  (i : Fin (n + 1)) :
  IsReindexing (Fin.append c c1 ∘ (Fin.castAdd (n1 + 1) i).succAbove)
    (Fin.append (c ∘ i.succAbove) c1) (Fin.cast (by grind))

```

lemma append_succSuccAbove_natAdd[\[source\]](#)

*Removing two entries from the right component of $\text{Fin.append } c1 \ c$ commutes with the append: removing the i -th and j -th entries of c and then appending $c1$ matches removing the corresponding entries of $\text{Fin.append } c1 \ c$, via the identity permutation. This is used for the commutation of taking a **product** of tensors with **contraction** of indices.*

```

lemma append_succSuccAbove_natAdd {n n1 : ℕ} {c : Fin (n + 1 + 1) → C}
  {c1 : Fin n1 → C} (i j : Fin (n + 1 + 1)) :
  IsReindexing (Fin.append c1 c ∘ (Fin.natAdd n1 i).succSuccAbove (Fin.natAdd n1 j))
    (Fin.append c1 (c ∘ i.succSuccAbove j)) id

```

lemma succAbove_of_eq_zero[\[source\]](#)

Given a reindexing of c by $c1$ via σ for which the index i is sent to 0 , removing the i -th entry of $c1$ and the first entry of c yields a reindexing of $c \circ \text{Fin.succ}$ by $c1 \circ i.succAbove$ via the map sending j to the predecessor of $\sigma(i.succAbove j)$.

```

lemma succAbove_of_eq_zero {n n1 : ℕ} {c : Fin (n + 1) → C} {c1 : Fin (n1 + 1) → C}
  {σ : Fin (n1 + 1) → Fin (n + 1)} (i : Fin (n1 + 1))
  (h : IsReindexing c c1 σ) (hi : σ i = 0) :
  IsReindexing (c ∘ Fin.succ) (c1 ∘ i.succAbove)
    (fun j => (σ (i.succAbove j)).pred (by simp [← hi, h.injective.eq_iff]))

```

lemma succAbove_of_neq_zero[\[source\]](#)

Given a reindexing of c by $c1$ via σ for which the index i is not sent to 0 , removing the i -th entry of $c1$ and the (σi) -th entry of c yields a reindexing of $c \circ (\sigma i).succAbove$ by $c1 \circ i.succAbove$ via the map $(\sigma i).pred.predAbove \circ \sigma \circ i.succAbove$.

```
lemma succAbove_of_neq_zero {n n1 : ℕ} {c : Fin (n + 1) → C} {c1 : Fin (n1 + 1) → C}
  {σ : Fin (n1 + 1) → Fin (n + 1)} (i : Fin (n1 + 1))
  (h : IsReindexing c c1 σ) (hi : σ i ≠ 0) :
  IsReindexing (c ∘ (σ i).succAbove) (c1 ∘ i.succAbove)
    ((Fin.pred (σ i) hi).predAbove ∘ σ ∘ i.succAbove)
```

lemma succAbove[\[source\]](#)

Given a reindexing of c by $c1$ via σ , removing the i -th entry of $c1$ and the (σi) -th entry of c yields a reindexing of $c \circ (\sigma i).succAbove$ by $c1 \circ i.succAbove$. This unifies `succAbove_of_eq_zero` and `succAbove_of_neq_zero` via a case split on whether $\sigma i = 0$. This is used for the commutation of **permutation** of indices with **evaluation** of indices.

```
lemma succAbove {n n1 : ℕ} {c : Fin (n + 1) → C} {c1 : Fin (n1 + 1) → C}
  {σ : Fin (n1 + 1) → Fin (n + 1)} (i : Fin (n1 + 1))
  (h : IsReindexing c c1 σ) :
  IsReindexing (c ∘ (σ i).succAbove) (c1 ∘ i.succAbove)
    (if hi : σ i = 0 then fun j => (σ (i.succAbove j)).pred (by simp [← hi, h.injective,
eq_iff])
    else (Fin.pred (σ i) hi).predAbove ∘ σ ∘ i.succAbove)
```

lemma succSuccAbove[\[source\]](#)

Given a reindexing of c by $c1$ via σ and two distinct indices $i \neq j$, removing the i -th and j -th entries of $c1$ and the (σi) -th and (σj) -th entries of c yields a reindexing of $c \circ (\sigma i).succSuccAbove (\sigma j)$ by $c1 \circ i.succSuccAbove j$. This is used for the commutation of **permutation** of indices with **contraction** of indices.

```
lemma succSuccAbove {n n1 : ℕ} {c : Fin (n + 1 + 1) → C}
  {c1 : Fin (n1 + 1 + 1) → C}
  (i j : Fin (n1 + 1 + 1)) (hij : i ≠ j)
  {σ : Fin (n1 + 1 + 1) → Fin (n + 1 + 1)} (hσ : IsReindexing c c1 σ) :
  IsReindexing (c ∘ (σ i).succSuccAbove (σ j))
    (c1 ∘ i.succSuccAbove j) (i.funPredPredAbove j hij σ hσ.1)
```

lemma succSuccAbove_comm[\[source\]](#)

Removing two pairs of entries from c in either order gives the same colour list: removing the $i1$ -th and $j1$ -th entries and then the (shifted) $i2$ -th and $j2$ -th entries matches removing the $i2$ -th and $j2$ -th entries first and then the (shifted) $i1$ -th and $j1$ -th entries, via the identity permutation. This is used for the commutation of two **contractions**.

```
lemma succSuccAbove_comm {n : ℕ} {c : Fin (n + 1 + 1 + 1 + 1) → C}
  (i1 j1 : Fin (n + 1 + 1 + 1 + 1)) (i2 j2 : Fin (n + 1 + 1))
  (hij1 : i1 ≠ j1) (hij2 : i2 ≠ j2) :
  let i2'
```

lemma succAbove_succSuccAbove_comm[\[source\]](#)

*Removing one entry and then a pair of entries from c in either order gives the same colour list: removing the k -th entry and then the (shifted) i -th and j -th entries matches removing the i -th and j -th entries first and then the (shifted) k -th entry, via the identity permutation. This is used for the commutation of a **contraction** with an **evaluation**.*

```
lemma succAbove_succSuccAbove_comm {n : ℕ} {c : Fin (n + 1 + 1 + 1) → C}
  (k : Fin (n + 1 + 1 + 1)) (i j : Fin (n + 1 + 1)) (hij : i ≠ j) :
  let i'
```

lemma succAbove_succAbove_comm[\[source\]](#)

*Removing two single entries from c in either order gives the same colour list: removing the k_1 -th entry and then the (shifted) k_2 -th entry matches removing the k_2 -th entry first and then the (shifted) k_1 -th entry, via the identity permutation. This is used for the commutation of two **evaluations** of indices.*

```
lemma succAbove_succAbove_comm {n : ℕ} {c : Fin (n + 1 + 1) → C}
  (k1 : Fin (n + 1 + 1)) (k2 : Fin (n + 1)) :
  let k2'
```

lemma append_succ_last[\[source\]](#)

Splitting a list of colours $c : \text{Fin } (n + 1) \rightarrow C$ into its first n entries and its last entry recovers c : the identity permutation matches $\text{Fin.append } (c \circ (\text{Fin.last } n).\text{succAbove})$ $![c (\text{Fin.last } n)]$ with c .

```
lemma append_succ_last {n : ℕ} (c : Fin (n + 1) → C) :
  IsReindexing (Fin.append (c ∘ (Fin.last n).succAbove) ![c (Fin.last n)]) c id
```

lemma append_of_first[\[source\]](#)

Splitting a list of colours $c : \text{Fin } (n + 1) \rightarrow C$ into its first entry and its remaining n entries recovers c : the canonical reindexing $\text{Fin } (1 + n) \simeq \text{Fin } (n + 1)$ matches $\text{Fin.append } ![c \ 0] (c \circ \text{Fin.succAbove } 0)$ with c .

```
lemma append_of_first {n : ℕ} (c : Fin (n + 1) → C) :
  IsReindexing (Fin.append ![c 0] (c ∘ Fin.succAbove 0)) c (Fin.cast (by grind))
```

lemma fin_cast_isReindexing[\[source\]](#)

Casting the domain along an equality $n_1 = n$ of lengths is a reindexing of c by $c \circ \text{Fin.cast } h$.

```
lemma fin_cast_isReindexing (n n1 : ℕ) {c : Fin n → C} (h : n1 = n) :
  IsReindexing c (c ∘ Fin.cast h) (Fin.cast h)
```

2.62 Physlib.Relativity.Tensors.Tensorial

[Physlib/Relativity/Tensors/Tensorial.lean](#) on GitHub.

lemma prod_eq_sum_eval[\[source\]](#)

Double basis expansion of an element of a tensor product $M \otimes[k] M_2$ of two Tensorial one-index spaces. Given bases b, b_2 of M, M_2 coming from the single-index tensor bases, every $x : M \otimes[k] M_2$ is the double sum over i, j of the iterated evaluation coefficient `toField (evalT 0 j (evalT 0 i (toTensor x)))` times $b\ i \otimes_{\tau} b_2\ j$.

```

lemma prod_eq_sum_eval {c c2 : C} {M2 : Type}
  [Tensorial S ![c] M] [AddCommMonoid M2] [Module k M2]
  [Tensorial S ![c2] M2] {b : Module.Basis (basisIdx c) k M}
  {b2 : Module.Basis (basisIdx c2) k M2}
  (h : b = ((Tensor.basis (S := S) ![c]).map toTensor.symm).reindex ComponentIdx.single)
  (h2 : b2 = ((Tensor.basis (S := S) ![c2]).map toTensor.symm).reindex ComponentIdx.single)
  (x : M  $\otimes[k]$  M2) : x =  $\sum i : (basisIdx\ c), \sum j : (basisIdx\ c2),$ 
    toField (evalT 0 (basisIdxCongr (by rfl) j)
      (evalT 0 (basisIdxCongr (by rfl) i) (toTensor x))) • b i  $\otimes_{\tau}$  b2 j

```

2.63 Physlib.Relativity.Tensors.TensorSpecies.Basic

[Physlib/Relativity/Tensors/TensorSpecies/Basic.lean](#) on GitHub.

lemma basisIdxCongr_heq_arg[\[source\]](#)

basisIdxCongr only depends on its endpoints up to HEq of the arguments: equal target colours and heterogeneously equal labels give equal casts.

```

lemma basisIdxCongr_heq_arg {c1 c2 d : C} (h1 : c1 = d) (h2 : c2 = d)
  {x : basisIdx c1} {y : basisIdx c2} (hxy : HEq x y) :
  basisIdxCongr h1 x = basisIdxCongr h2 y

```

2.64 Physlib.SpaceAndTime.GalileanGroup.Basic

[Physlib/SpaceAndTime/GalileanGroup/Basic.lean](#) on GitHub.

structure GalileanGroup[\[source\]](#)

A Galilean transformation in d spatial dimensions. The fields are, in order, the spatial orthogonal part, boost velocity, spatial translation, and time translation.

```

@[ext]
structure GalileanGroup (d :  $\mathbb{N}$  := 3) where
  rotation : Matrix.orthogonalGroup (Fin d)  $\mathbb{R}$ 
  velocity : EuclideanSpace  $\mathbb{R}$  (Fin d)
  spaceTranslation : EuclideanSpace  $\mathbb{R}$  (Fin d)
  timeTranslation : Time

```

lemma orthogonal_smul_smul[\[source\]](#)

```

lemma orthogonal_smul_smul (R : Matrix.orthogonalGroup (Fin d)  $\mathbb{R}$ ) (c :  $\mathbb{R}$ )
  (v : EuclideanSpace  $\mathbb{R}$  (Fin d)) :
  R • (c • v) = c • (R • v)

```

instance One (GalileanGroup d)[\[source\]](#)

The identity Galilean transformation.

```
instance : One (GalileanGroup d) where
```

lemma one_rotation

[\[source\]](#)

```
@[simp]
lemma one_rotation : (1 : GalileanGroup d).rotation = 1
```

lemma one_velocity

[\[source\]](#)

```
@[simp]
lemma one_velocity : (1 : GalileanGroup d).velocity = 0
```

lemma one_spaceTranslation

[\[source\]](#)

```
@[simp]
lemma one_spaceTranslation : (1 : GalileanGroup d).spaceTranslation = 0
```

lemma one_timeTranslation

[\[source\]](#)

```
@[simp]
lemma one_timeTranslation : (1 : GalileanGroup d).timeTranslation = 0
```

instance Mul (GalileanGroup d)

[\[source\]](#)

*The product whose action is composition: $(g * h) \cdot tx = g \cdot h \cdot tx$.*

```
instance : Mul (GalileanGroup d) where
```

lemma mul_rotation

[\[source\]](#)

```
@[simp]
lemma mul_rotation (g h : GalileanGroup d) :
  (g * h).rotation = g.rotation * h.rotation
```

lemma mul_velocity

[\[source\]](#)

```
@[simp]
lemma mul_velocity (g h : GalileanGroup d) :
  (g * h).velocity = g.rotation • h.velocity + g.velocity
```

lemma mul_spaceTranslation

[\[source\]](#)

```
@[simp]
lemma mul_spaceTranslation (g h : GalileanGroup d) :
  (g * h).spaceTranslation =
    g.spaceTranslation + g.rotation • h.spaceTranslation + h.timeTranslation.val • g.velocity
```

lemma mul_timeTranslation

[\[source\]](#)

```
@[simp]
lemma mul_timeTranslation (g h : GalileanGroup d) :
  (g * h).timeTranslation = g.timeTranslation + h.timeTranslation
```

instance Inv (GalileanGroup d)

[\[source\]](#)

The inverse Galilean transformation.

```
instance : Inv (GalileanGroup d) where
```

lemma inv_rotation

[\[source\]](#)

```
@[simp]
lemma inv_rotation (g : GalileanGroup d) :
  g-1.rotation = g.rotation-1
```

lemma inv_velocity

[\[source\]](#)

The inverse boost velocity formula.

```
@[simp]
lemma inv_velocity (g : GalileanGroup d) :
  g-1.velocity = -(g.rotation-1 • g.velocity)
```

lemma inv_spaceTranslation

[\[source\]](#)

```
@[simp]
lemma inv_spaceTranslation (g : GalileanGroup d) :
  g-1.spaceTranslation =
    -(g.rotation-1 • g.spaceTranslation) +
    g.timeTranslation.val • (g.rotation-1 • g.velocity)
```

lemma inv_timeTranslation

[\[source\]](#)

```
@[simp]
lemma inv_timeTranslation (g : GalileanGroup d) :
  g-1.timeTranslation = -g.timeTranslation
```

instance Group (GalileanGroup d)

[\[source\]](#)

The Galilean transformations form a group under composition.

```
instance : Group (GalileanGroup d) where
```

instance Inhabited (GalileanGroup d)

[\[source\]](#)

```
instance : Inhabited (GalileanGroup d) where
```

def actSpace

[\[source\]](#)

The Galilean action on space at a fixed time.

```
noncomputable def actSpace (g : GalileanGroup d) (t : Time) (x : Space d) : Space d
```

def act

[\[source\]](#)

The Galilean action on $\text{Time} \times \text{Space } d$.

```
noncomputable def act (g : GalileanGroup d) (tx : Time × Space d) : Time × Space d
```

instance MulAction (GalileanGroup d) (Time × Space d)

[\[source\]](#)

```
noncomputable instance : MulAction (GalileanGroup d) (Time × Space d) where
```

lemma smul_fst

[\[source\]](#)

@[simp]

```
lemma smul_fst (g : GalileanGroup d) (tx : Time × Space d) :
  (g • tx).1 = tx.1 + g.timeTranslation
```

lemma smul_snd

[\[source\]](#)

@[simp]

```
lemma smul_snd (g : GalileanGroup d) (tx : Time × Space d) :
  (g • tx).2 = g.actSpace tx.1 tx.2
```

lemma smul_mk

[\[source\]](#)

@[simp]

```
lemma smul_mk (g : GalileanGroup d) (t : Time) (x : Space d) :
  g • ((t, x) : Time × Space d) = (t + g.timeTranslation, g.actSpace t x)
```

lemma actSpace_apply

[\[source\]](#)

@[simp]

```
lemma actSpace_apply (g : GalileanGroup d) (t : Time) (x : Space d) (i : Fin d) :
  g.actSpace t x i =
    (g.rotation • (x -v (0 : Space d))) i + t.val * g.velocity i + g.spaceTranslation i
```

def ofEuclidean

[\[source\]](#)

A Euclidean spatial transformation as a Galilean transformation with zero boost and no time translation.

```
def ofEuclidean (g : EuclideanGroup d) : GalileanGroup d
```

lemma ofEuclidean_rotation

[\[source\]](#)

@[simp]

```
lemma ofEuclidean_rotation (g : EuclideanGroup d) :
  (ofEuclidean g).rotation = g.linear
```


lemma ofEuclidean_velocity[\[source\]](#)

```
@[simp]
lemma ofEuclidean_velocity (g : EuclideanGroup d) :
  (ofEuclidean g).velocity = 0
```

lemma ofEuclidean_spaceTranslation[\[source\]](#)

```
@[simp]
lemma ofEuclidean_spaceTranslation (g : EuclideanGroup d) :
  (ofEuclidean g).spaceTranslation = g.translation
```

lemma ofEuclidean_timeTranslation[\[source\]](#)

```
@[simp]
lemma ofEuclidean_timeTranslation (g : EuclideanGroup d) :
  (ofEuclidean g).timeTranslation = 0
```

def euclidean.incl[\[source\]](#)

Inclusion of the Euclidean group into the Galilean group.

```
def euclidean.incl : EuclideanGroup d →* GalileanGroup d where
```

def ofOrthogonal[\[source\]](#)

A pure orthogonal spatial transformation as a Galilean transformation.

```
def ofOrthogonal (R : Matrix.orthogonalGroup (Fin d) ℝ) : GalileanGroup d
```

lemma ofOrthogonal_rotation[\[source\]](#)

```
@[simp]
lemma ofOrthogonal_rotation (R : Matrix.orthogonalGroup (Fin d) ℝ) :
  (ofOrthogonal R).rotation = R
```

lemma ofOrthogonal_velocity[\[source\]](#)

```
@[simp]
lemma ofOrthogonal_velocity (R : Matrix.orthogonalGroup (Fin d) ℝ) :
  (ofOrthogonal R).velocity = 0
```

lemma ofOrthogonal_spaceTranslation[\[source\]](#)

```
@[simp]
lemma ofOrthogonal_spaceTranslation (R : Matrix.orthogonalGroup (Fin d) ℝ) :
  (ofOrthogonal R).spaceTranslation = 0
```

lemma ofOrthogonal_timeTranslation[\[source\]](#)

```
@[simp]
lemma ofOrthogonal_timeTranslation (R : Matrix.orthogonalGroup (Fin d) ℝ) :
  (ofOrthogonal R).timeTranslation = 0
```

def orthogonal.incl[\[source\]](#)*Inclusion of the orthogonal group into the Galilean group.*

```
def orthogonal.incl : Matrix.orthogonalGroup (Fin d) ℝ →* GalileanGroup d where
```

def rotation.incl[\[source\]](#)*Inclusion of the existing spatial rotation subgroup into the Galilean group.*

```
noncomputable def rotation.incl : EuclideanGroup.RotationGroup d →* GalileanGroup d where
```

def ofSpaceTranslation[\[source\]](#)*A pure spatial translation as a Galilean transformation.*

```
def ofSpaceTranslation (a : EuclideanSpace ℝ (Fin d)) : GalileanGroup d
```

lemma ofSpaceTranslation_rotation[\[source\]](#)

```
@[simp]
```

```
lemma ofSpaceTranslation_rotation (a : EuclideanSpace ℝ (Fin d)) :  
  (ofSpaceTranslation a).rotation = 1
```

lemma ofSpaceTranslation_velocity[\[source\]](#)

```
@[simp]
```

```
lemma ofSpaceTranslation_velocity (a : EuclideanSpace ℝ (Fin d)) :  
  (ofSpaceTranslation a).velocity = 0
```

lemma ofSpaceTranslation_spaceTranslation[\[source\]](#)

```
@[simp]
```

```
lemma ofSpaceTranslation_spaceTranslation (a : EuclideanSpace ℝ (Fin d)) :  
  (ofSpaceTranslation a).spaceTranslation = a
```

lemma ofSpaceTranslation_timeTranslation[\[source\]](#)

```
@[simp]
```

```
lemma ofSpaceTranslation_timeTranslation (a : EuclideanSpace ℝ (Fin d)) :  
  (ofSpaceTranslation a).timeTranslation = 0
```

def spaceTranslation.incl[\[source\]](#)*Inclusion of spatial translations into the Galilean group.*

```
def spaceTranslation.incl :  
  Multiplicative (EuclideanSpace ℝ (Fin d)) →* GalileanGroup d where
```

def spaceTranslationGroup.incl[\[source\]](#)*Inclusion of the existing spatial translation subgroup into the Galilean group.*

```
noncomputable def spaceTranslationGroup.incl :
  EuclideanGroup.TranslationGroup d →* GalileanGroup d
```

def ofTimeTranslation

[\[source\]](#)

A pure time translation as a Galilean transformation.

```
def ofTimeTranslation (b : Time) : GalileanGroup d
```

lemma ofTimeTranslation_rotation

[\[source\]](#)

```
@[simp]
lemma ofTimeTranslation_rotation (b : Time) :
  (ofTimeTranslation (d := d) b).rotation = 1
```

lemma ofTimeTranslation_velocity

[\[source\]](#)

```
@[simp]
lemma ofTimeTranslation_velocity (b : Time) :
  (ofTimeTranslation (d := d) b).velocity = 0
```

lemma ofTimeTranslation_spaceTranslation

[\[source\]](#)

```
@[simp]
lemma ofTimeTranslation_spaceTranslation (b : Time) :
  (ofTimeTranslation (d := d) b).spaceTranslation = 0
```

lemma ofTimeTranslation_timeTranslation

[\[source\]](#)

```
@[simp]
lemma ofTimeTranslation_timeTranslation (b : Time) :
  (ofTimeTranslation (d := d) b).timeTranslation = b
```

def timeTranslation.incl

[\[source\]](#)

Inclusion of time translations into the Galilean group.

```
def timeTranslation.incl : Multiplicative Time →* GalileanGroup d where
```

2.65 Physlib.SpaceAndTime.Space.Basic

[Physlib/SpaceAndTime/Space/Basic.lean](#) on GitHub.

instance instNontrivial

[\[source\]](#)

```
instance instNontrivial {d : ℕ} [NeZero d] : Nontrivial (Space d) where
```

def homEuclideanSpaceSpace

[\[source\]](#)

Space d is homoemorphic to d-dimensional euclidean space. This is used to define the IsManifold instance on Space d.

```
noncomputable def homEuclideanSpaceSpace (d : ℕ) : EuclideanSpace ℝ (Fin d) ≈t Space d where
```

```
instance ChartedSpace (EuclideanSpace ℝ (Fin d)) (Space d) \[source\]
```

```
noncomputable instance (d : ℕ) : ChartedSpace (EuclideanSpace ℝ (Fin d)) (Space d)
```

```
instance IsManifold (ℝ d) τ (Space d) \[source\]
```

```
instance (d : ℕ) : IsManifold (ℝ d) τ (Space d)
```

```
lemma chartAt_eq \[source\]
```

```
lemma chartAt_eq (d : ℕ) (p : Space d) :  
  chartAt (EuclideanSpace ℝ (Fin d)) p =  
  (homEuclideanSpaceSpace d).symm.toOpenPartialHomeomorph
```

2.66 Physlib.SpaceAndTime.Space.Derivatives.Basic

[Physlib/SpaceAndTime/Space/Derivatives/Basic.lean](#) on GitHub.

```
lemma deriv_sub \[source\]
```

Derivatives on space distribute over subtraction.

```
@[to_fun]  
lemma deriv_sub [NormedAddCommGroup M] [NormedSpace ℝ M]  
  (f1 f2 : Space d → M) (hf1 : Differentiable ℝ f1) (hf2 : Differentiable ℝ f2) :  
  ∂[u] (f1 - f2) = ∂[u] f1 - ∂[u] f2
```

2.67 Physlib.SpaceAndTime.Space.Derivatives.Curl

[Physlib/SpaceAndTime/Space/Derivatives/Curl.lean](#) on GitHub.

```
lemma distCurl_coord_apply \[source\]
```

```
lemma distCurl_coord_apply {i : Fin 3}  
  (f : Space →d[ℝ] (EuclideanSpace ℝ (Fin 3))) (η : S(Space, ℝ)) :  
  (∇d × f) η i = - f (SchwartzMap.evalCLM ℝ Space ℝ (basis (i+2)) (fderivCLM ℝ Space ℝ η)) (i+1)  
  + f (SchwartzMap.evalCLM ℝ Space ℝ (basis (i+1)) (fderivCLM ℝ Space ℝ η)) (i+2)
```

2.68 Physlib.SpaceAndTime.Space.Derivatives.Grad

[Physlib/SpaceAndTime/Space/Derivatives/Grad.lean](#) on GitHub.

```
lemma grad_eq_gradient \[source\]
```

```
lemma grad_eq_gradient {d} (f : Space d → ℝ) :  
  ∇ f = basis.repr ∘ gradient f
```

lemma distGrad_const[\[source\]](#)

```
@[simp]
lemma distGrad_const {d} (c : ℝ) :
   $\nabla^d$  (Distribution.const ℝ (Space d) c) = 0
```

2.69 Physlib.SpaceAndTime.Space.Derivatives.Laplacian

[Physlib/SpaceAndTime/Space/Derivatives/Laplacian.lean](#) on GitHub.

lemma laplacian_eq_sum_snd_deriv[\[source\]](#)

```
lemma laplacian_eq_sum_snd_deriv {d} (f : Space d → ℝ) :
   $\Delta f = \text{fun } x \Rightarrow \sum i, \partial[i] (\partial[i] f) x$ 
```

def distLaplacian[\[source\]](#)

The distributional distLaplacian operator.

```
noncomputable def distLaplacian {d} :
  ((Space d) →d [ℝ] ℝ) →l [ℝ] (Space d) →d [ℝ] ℝ
```

lemma distLaplacian_const[\[source\]](#)

```
@[simp]
lemma distLaplacian_const {d : ℕ} (c : ℝ) :
   $\Delta^d$  (Distribution.const ℝ (Space d) c) = 0
```

2.70 Physlib.SpaceAndTime.Space.DistOfFunction

[Physlib/SpaceAndTime/Space/DistOfFunction.lean](#) on GitHub.

lemma distDeriv_distOfFunction[\[source\]](#)

The distributional spatial derivative of the distribution associated to a differentiable function is the distribution associated to the corresponding pointwise spatial derivative.

```
lemma distDeriv_distOfFunction {d : ℕ} (μ : Fin d) (f : Space d → F)
  (hf : IsDistBounded f) (hf_deriv : IsDistBounded (∂[μ] f))
  (hf_diff : Differentiable ℝ f) :
   $\partial^d[\mu] (\text{distOfFunction } f \text{ hf}) = \text{distOfFunction } (\partial[\mu] f) \text{ hf\_deriv}$ 
```

2.71 Physlib.SpaceAndTime.Space.EuclideanGroup.Action

[Physlib/SpaceAndTime/Space/EuclideanGroup/Action.lean](#) on GitHub.

instance MulAction (EuclideanGroup d) (Space d)[\[source\]](#)

The action of the Euclidean group on the affine space of points Space d: $g = \langle t, Q \rangle$ rotates p about the coordinate origin by the orthogonal part Q and then translates by t .

```
noncomputable instance : MulAction (EuclideanGroup d) (Space d) where
```

lemma dist_smul[\[source\]](#)

*The Euclidean group acts on Space d ****by isometries****: every rigid motion preserves distance.*

```
lemma dist_smul (g : EuclideanGroup d) (p q : Space d) :
  dist (g • p) (g • q) = dist p q
```

lemma rotation_smul_vsub_origin[\[source\]](#)

A rotation acts on the displacement from the origin by its orthogonal part, for every p : $(r • p) -_v \text{origin} = Q • (p -_v \text{origin})$.

```
lemma rotation_smul_vsub_origin (r : RotationGroup d) (p : Space d) :
  (r • p) -_v (0 : Space d) = (r : EuclideanGroup d).linear • (p -_v (0 : Space d))
```

lemma rotation_dist_smul[\[source\]](#)

*The rotation group acts on Space d ****by isometries**** (inherited from dist_smul).*

```
lemma rotation_dist_smul (r : RotationGroup d) (p q : Space d) :
  dist (r • p) (r • q) = dist p q
```

lemma chartEuclidean_smul[\[source\]](#)

Under the standard chart, the Euclidean action on Space d is the transport of toAffineIsometryMulEquiv acting on EuclideanSpace: $\text{chart } (g • p) = (\text{toAffineIsometryMulEquiv } g) (\text{chart } p)$.

```
lemma chartEuclidean_smul (g : EuclideanGroup d) (p : Space d) :
  Space.chartEuclidean d (g • p) = toAffineIsometryMulEquiv g (Space.chartEuclidean d p)
```

2.72 Physlib.SpaceAndTime.Space.EuclideanGroup.AffineGroup

[Physlib/SpaceAndTime/Space/EuclideanGroup/AffineGroup.lean](#) on GitHub.

def orthogonalToLinearIsometryEquiv[\[source\]](#)

An orthogonal matrix viewed as a linear isometry equivalence of EuclideanSpace $\mathbb{R}$ (Fin n); the linear ingredient of the first leg toAffineIsometryHom.

```
noncomputable def orthogonalToLinearIsometryEquiv
  (Q : Matrix.orthogonalGroup (Fin n)  $\mathbb{R}$ ) :
  EuclideanSpace  $\mathbb{R}$  (Fin n)  $\approx_{l_1[\mathbb{R}]}$  EuclideanSpace  $\mathbb{R}$  (Fin n)
```

def toAffineIsometryHom[\[source\]](#)

The first leg of the inclusion: $(t, Q) \mapsto (x \mapsto Q x + t)$, bundled as a monoid homomorphism from the Euclidean group into the affine isometry group of EuclideanSpace $\mathbb{R}$ (Fin n).

```
noncomputable def toAffineIsometryHom :
  EuclideanGroup n  $\rightarrow^*$ 
  AffineIsometryEquiv  $\mathbb{R}$  (EuclideanSpace  $\mathbb{R}$  (Fin n)) (EuclideanSpace  $\mathbb{R}$  (Fin n)) where
```

def toAffineEquivHom[\[source\]](#)

The second leg of the inclusion: `AffineIsometryEquiv.toAffineEquiv` bundled as a monoid homomorphism into the affine automorphism group.

```
noncomputable def _root_.AffineIsometryEquiv.toAffineEquivHom :
  AffineIsometryEquiv ℝ (EuclideanSpace ℝ (Fin n)) (EuclideanSpace ℝ (Fin n)) →*
  AffineEquiv ℝ (EuclideanSpace ℝ (Fin n)) (EuclideanSpace ℝ (Fin n)) where
```

def toAffineEquiv[\[source\]](#)

The inclusion of the Euclidean group into Mathlib's affine automorphism group: the composite of the two legs `toAffineIsometryHom` and `AffineIsometryEquiv.toAffineEquivHom`.

```
noncomputable def toAffineEquiv :
  EuclideanGroup n →* AffineEquiv ℝ (EuclideanSpace ℝ (Fin n)) (EuclideanSpace ℝ (Fin n))
```

def linearIsometryEquivToOrthogonal[\[source\]](#)

The `invFun` ingredient of `toAffineIsometryMulEquiv`: a linear isometry equivalence read back as an orthogonal matrix, inverse to the linear bridge `orthogonalToLinearIsometryEquiv` (see the round-trip `@[simp]` lemmas below).

```
noncomputable def linearIsometryEquivToOrthogonal
  (L : EuclideanSpace ℝ (Fin n) ≈Li [ℝ] EuclideanSpace ℝ (Fin n)) :
  Matrix.orthogonalGroup (Fin n) ℝ
```

def toAffineIsometryMulEquiv[\[source\]](#)

`EuclideanGroup n ≈* AffineIsometryEquiv ℝ (EuclideanSpace ℝ (Fin n))` `_`: the Euclidean group is the full group of affine isometries of Euclidean space. This upgrades `toAffineIsometryHom` to a group isomorphism with the same underlying map.

```
noncomputable def toAffineIsometryMulEquiv :
  EuclideanGroup n ≈*
  AffineIsometryEquiv ℝ (EuclideanSpace ℝ (Fin n)) (EuclideanSpace ℝ (Fin n)) where
```

2.73 Physlib.SpaceAndTime.Space.EuclideanGroup.Basic

[Physlib/SpaceAndTime/Space/EuclideanGroup/Basic.lean](#) on GitHub.

structure EuclideanGroup[\[source\]](#)

An n -dimensional Euclidean group is a group of rotations, reflections, and translations.

```
@[ext]
structure EuclideanGroup (n : ℕ) where
  translation : EuclideanSpace ℝ (Fin n)
  linear : Matrix.orthogonalGroup (Fin n) ℝ
```

instance Group (EuclideanGroup n)[\[source\]](#)

*Group structure on $E(n) = \mathbb{R}^n \rtimes O(n)$. The orthogonal component acts on translations by the inherited matrix-vector action on $\text{EuclideanSpace } \mathbb{R} \text{ (Fin } n)$, transported through WithLp from the coordinate action $Q.\text{val} *_{\mathbb{V}} v.\text{ofLp}$.*

```
noncomputable instance : Group (EuclideanGroup n) where
```

lemma det_coe_inv

[\[source\]](#)

```
private lemma det_coe_inv {n : ℕ} (Q : Matrix.orthogonalGroup (Fin n) ℝ) :
  (Q⁻¹).val.det = (Q.val.det)⁻¹
```

lemma coe_inv

[\[source\]](#)

```
private lemma coe_inv {n : ℕ} (Q : Matrix.orthogonalGroup (Fin n) ℝ) :
  (Q⁻¹).val = (Q.val)⁻¹
```

def SpecialEuclideanGroup

[\[source\]](#)

Special Euclidean Group is the subgroup with $\det(Q) = 1$ where $Q \in O(n)$

```
def SpecialEuclideanGroup (n : ℕ) : Subgroup (EuclideanGroup n) where
```

def SpecialEuclideanGroup.incl

[\[source\]](#)

The inclusion of the special Euclidean group into the Euclidean group.

```
noncomputable def SpecialEuclideanGroup.incl (n : ℕ) :
  SpecialEuclideanGroup n →* EuclideanGroup n
```

def TranslationGroup

[\[source\]](#)

Translation is the subgroup with $Q = 1$.

```
def TranslationGroup (n : ℕ) : Subgroup (EuclideanGroup n) where
```

def TranslationGroup.incl

[\[source\]](#)

The inclusion of the translation subgroup into the Euclidean group.

```
noncomputable def TranslationGroup.incl (n : ℕ) :
  TranslationGroup n →* EuclideanGroup n
```

def translationVector.incl

[\[source\]](#)

MonoidHom including a translation vector into the Euclidean Group.

```
def translationVector.incl (n : ℕ) :
  Multiplicative (EuclideanSpace ℝ (Fin n)) →* EuclideanGroup n where
```

lemma incl_range

[\[source\]](#)

An API feature: the translation vector inclusion image is the TranslationGroup carrier.


```
lemma translationVector.incl_range :
  Set.range (@translationVector.incl n) = (TranslationGroup n : Set (EuclideanGroup n))
```

lemma translation_zero

[\[source\]](#)

The translation by the zero vector is the identity of the Euclidean group.

```
lemma translation_zero : translationVector.incl n
  (Multiplicative.ofAdd (0 : EuclideanSpace ℝ (Fin n))) = 1
```

def OriginStabilizer

[\[source\]](#)

*The subgroup of EuclideanGroup n whose elements fix the origin (translation = 0).
This is the copy of $O(n)$ sitting inside $E(n)$.*

```
def OriginStabilizer (n : ℕ) : Subgroup (EuclideanGroup n) where
```

def RotationGroup

[\[source\]](#)

*Rotation Group is the subgroup of $E(n)$ consisting of rotations about the origin:
elements with $\det = 1$ (orientation-preserving) and $\text{translation} = 0$ (origin-fixing).*

```
noncomputable def RotationGroup (n : ℕ) : Subgroup (EuclideanGroup n)
```

def RotationGroup.incl

[\[source\]](#)

The inclusion of the rotation subgroup into the Euclidean group.

```
noncomputable def RotationGroup.incl (n : ℕ) :
  RotationGroup n →* EuclideanGroup n
```

def RotationsAbout

[\[source\]](#)

*The subgroup of rotation about a spatial point $p : \text{EuclideanSpace } \mathbb{R} (\text{Fin } n)$ consists of elements of the form $T(p) * r * T(-p)$ with $T(\cdot) : \text{translationVector.incl } n$ ($\text{Multiplicative.ofAdd } \cdot$) and $r : \text{RotationGroup}$ where r is viewed as a rotation about the origin. Note $T(-p) = T(p)^{-1}$.*

```
noncomputable def RotationsAbout : Subgroup (EuclideanGroup n) where
```

def RotationsAbout.incl

[\[source\]](#)

The inclusion of rotations about p into the Euclidean group.

```
noncomputable def RotationsAbout.incl : RotationsAbout p →* EuclideanGroup n
```

def toOrigin

[\[source\]](#)

Conjugate a rotation about p back to a rotation about the origin.

```
noncomputable def RotationsAbout.toOrigin :
  RotationsAbout p →* RotationGroup n where
```

def fromOrigin[\[source\]](#)

Conjugate a rotation about the origin to a rotation about p .

```
noncomputable def RotationsAbout.fromOrigin :
  RotationGroup n →* RotationsAbout p where
```

lemma fromOrigin_comp_toOrigin[\[source\]](#)

*RotationsAbout.toOrigin p followed by RotationsAbout.fromOrigin p is the identity;
the forward leg of the isomorphism RotationsAboutEquiv.*

```
lemma RotationsAbout.fromOrigin_comp_toOrigin :
  (RotationsAbout.fromOrigin p).comp (RotationsAbout.toOrigin p) =
    MonoidHom.id (RotationsAbout p)
```

lemma toOrigin_comp_fromOrigin[\[source\]](#)

*RotationsAbout.fromOrigin p followed by RotationsAbout.toOrigin p is the identity;
the backward leg of the isomorphism RotationsAboutEquiv.*

```
lemma RotationsAbout.toOrigin_comp_fromOrigin :
  (RotationsAbout.toOrigin p).comp (RotationsAbout.fromOrigin p) =
    MonoidHom.id (RotationGroup n)
```

def RotationsAboutEquiv[\[source\]](#)

*API feature: conjugation by the translation $T(p)$ exhibits the rotations about p as
isomorphic to the rotations about the origin RotationGroup n .*

```
noncomputable def RotationsAboutEquiv : RotationsAbout p ≈* RotationGroup n
```

lemma RotationsAbout_zero[\[source\]](#)

API feature: the degenerate identity that RotationsAbout $\theta = \text{RotationGroup } n$

```
lemma RotationsAbout_zero : RotationsAbout ( $\theta : \text{EuclideanSpace } \mathbb{R} (\text{Fin } n)$ ) = RotationGroup n
```

def specialOrthogonal.incl[\[source\]](#)

*Rotations are members of special orthogonal groups and can be viewed as members
of orthogonal groups.*

```
def specialOrthogonal.incl (n : ℕ) :
  Matrix.specialOrthogonalGroup (Fin n) ℝ →* Matrix.orthogonalGroup (Fin n) ℝ
```

def ofRotation[\[source\]](#)

The Euclidean group element given by a rotation about the origin (zero translation).

```
def ofRotation (Q : Matrix.specialOrthogonalGroup (Fin n) ℝ) :
  EuclideanGroup n
```

def ofRotationTranslation[\[source\]](#)

Specialization to a group element from a rotation and a translation.

```
def ofRotationTranslation (Q : Matrix.specialOrthogonalGroup (Fin n) ℝ)
  (t : EuclideanSpace ℝ (Fin n)) : EuclideanGroup n
```

lemma ofRotationTranslation_translation[\[source\]](#)

The specialization projects back to the translation component.

```
@[simp]
lemma ofRotationTranslation_translation (Q : Matrix.specialOrthogonalGroup (Fin n) ℝ)
  (t : EuclideanSpace ℝ (Fin n)) :
  (ofRotationTranslation Q t).translation = t
```

lemma ofRotationTranslation_linear[\[source\]](#)

The specialization projects back to the linear (rotation) component.

```
@[simp]
lemma ofRotationTranslation_linear (Q : Matrix.specialOrthogonalGroup (Fin n) ℝ)
  (t : EuclideanSpace ℝ (Fin n)) :
  (ofRotationTranslation Q t).linear = specialOrthogonal.incl n Q
```

lemma ofRotationTranslation_decompose[\[source\]](#)

API feature: the inclusion image decomposes as group product.

```
@[simp]
lemma ofRotationTranslation_decompose (Q : Matrix.specialOrthogonalGroup (Fin n) ℝ)
  (t : EuclideanSpace ℝ (Fin n)) :
  (ofRotationTranslation Q t) =
  (translationVector.incl n (Multiplicative.ofAdd t)) * (ofRotation (Q))
```

def toRotation[\[source\]](#)

The isomorphism's forward map: a special orthogonal matrix as a rotation about the origin.

```
noncomputable def specialOrthogonal.toRotation (n : ℕ) :
  Matrix.specialOrthogonalGroup (Fin n) ℝ →* RotationGroup n where
```

def fromRotation[\[source\]](#)

The isomorphism's inverse map: the linear part of a rotation about the origin, as a special orthogonal matrix.

```
noncomputable def specialOrthogonal.fromRotation (n : ℕ) :
  RotationGroup n →* Matrix.specialOrthogonalGroup (Fin n) ℝ where
```

lemma fromRotation_comp_toRotation[\[source\]](#)

specialOrthogonal.toRotation n followed by specialOrthogonal.fromRotation n is the identity; the forward leg of the isomorphism specialOrthogonalEquiv.

```

lemma specialOrthogonal.fromRotation_comp_toRotation :
  (specialOrthogonal.fromRotation n).comp (specialOrthogonal.toRotation n) =
    MonoidHom.id (Matrix.specialOrthogonalGroup (Fin n) ℝ)

```

lemma toRotation_comp_fromRotation

[\[source\]](#)

specialOrthogonal.fromRotation n followed by specialOrthogonal.toRotation n is the identity; the backward leg of the isomorphism specialOrthogonalEquiv.

```

lemma specialOrthogonal.toRotation_comp_fromRotation :
  (specialOrthogonal.toRotation n).comp (specialOrthogonal.fromRotation n) =
    MonoidHom.id (RotationGroup n)

```

def specialOrthogonalEquiv

[\[source\]](#)

API feature: $SO(n) \simeq^ \text{RotationGroup } n$*

```

noncomputable def specialOrthogonalEquiv :
  Matrix.specialOrthogonalGroup (Fin n) ℝ  $\simeq^*$  RotationGroup n

```

2.74 Physlib.SpaceAndTime.Space.Integrals.Basic

[Physlib/SpaceAndTime/Space/Integrals/Basic.lean](#) on GitHub.

lemma integrable_spherical_of_integrable

[\[source\]](#)

```

lemma integrable_spherical_of_integrable {d : ℕ} {F : Type*}
  [NormedAddCommGroup F] [NormedSpace ℝ F] {f : Space d → F}
  (hf : Integrable f volume) :
  Integrable
    (fun x : ↑(Metric.sphere (0 : Space d) 1) × Set.Ioi (0 : ℝ) =>
      f (x.2.1 • x.1.1))
    (volume (α := Space d).toSphere.prod
      (Measure.volumeIoiPow (Module.finrank ℝ (Space d) - 1)))

```

lemma integral_volume_eq_spherical_iterated

[\[source\]](#)

```

lemma integral_volume_eq_spherical_iterated {d : ℕ} [NeZero d] {F : Type*}
  [NormedAddCommGroup F] [NormedSpace ℝ F] (f : Space d → F)
  (hf : Integrable f volume) :
  ∫ x : Space d, f x =
    ∫ n : ↑(Metric.sphere (0 : Space d) 1),
      ∫ r : Set.Ioi (0 : ℝ), f (r.1 • n.1)
      ∂(Measure.volumeIoiPow (Module.finrank ℝ (Space d) - 1))
      ∂(volume (α := Space d).toSphere)

```

lemma integral_volume_eq_spherical_integral

[\[source\]](#)

```

lemma integral_volume_eq_spherical_integral {d : ℕ} [NeZero d] {F : Type*}
  [NormedAddCommGroup F] [NormedSpace ℝ F] (f : Space d → F)
  (hf : Integrable f volume) :
  ∫ x : Space d, f x =
    ∫ n : ↑(Metric.sphere (0 : Space d) 1),
      ∫ r : Set.Ioi (0 : ℝ), (r.1 ^ (d - 1)) • f (r.1 • n.1)

```

```

    ∂(.comap Subtype.val volume)
    ∂(volume (α := Space d).toSphere)

```

2.75 Physlib.SpaceAndTime.Space.Integrals.NormPow

[Physlib/SpaceAndTime/Space/Integrals/NormPow.lean](#) on GitHub.

lemma radial_jacobian_zpow_mul_self

[\[source\]](#)

```

lemma radial_jacobian_zpow_mul_self
  {d p : ℕ} [NeZero d] {q : ℤ} (hp_int : (p : ℤ) = q + (d : ℤ))
  {r : ℝ} (hr : 0 < r) :
  r ^ (d - 1) * (r ^ q * r) = r ^ p

```

lemma radial_jacobian_zpow

[\[source\]](#)

```

private lemma radial_jacobian_zpow
  {d p : ℕ} [NeZero d] {q : ℤ} (hp_int : (p : ℤ) = q + (d : ℤ))
  (hp_pos : 0 < p) {r : ℝ} (hr : 0 < r) :
  r ^ (d - 1) * r ^ q = r ^ (p - 1)

```

lemma radial_norm_power_spherical_integral_eq_space_integral

[\[source\]](#)

```

lemma radial_norm_power_spherical_integral_eq_space_integral
  {d p : ℕ} [NeZero d] {q : ℤ} (hp_int : (p : ℤ) = q + (d : ℤ))
  (hp_pos : 0 < p) (η : ℳ(Space d, ℝ)) :
  ∫ x : Space d, η x * ||x|| ^ q =
  ∫ (n : ↑(Metric.sphere (0 : Space d) 1)),
  ∫ (r : Set.Ioi (0 : ℝ)),
    r.1 ^ (p - 1) * η (r.1 • n.1)
  ∂(.comap Subtype.val volume)
  ∂(volume (α := Space d).toSphere)

```

2.76 Physlib.SpaceAndTime.Space.Module

[Physlib/SpaceAndTime/Space/Module.lean](#) on GitHub.

lemma norm_smul_sphere

[\[source\]](#)

```

lemma norm_smul_sphere {d : ℕ} (n : ↑(Metric.sphere (0 : Space d) 1))
  {r : ℝ} (hr : 0 ≤ r) :
  ||(r • (n : Space d))|| = r

```

2.77 Physlib.SpaceAndTime.Space.Norm.Basic

[Physlib/SpaceAndTime/Space/Norm/Basic.lean](#) on GitHub.

lemma integrable_real_pow_mul_schwartz

[\[source\]](#)

```

private lemma integrable_real_pow_mul_schwartz
  (ψ : ℳ(ℝ, ℝ)) (k : ℕ) :
  Integrable (fun x : ℝ => x ^ k * ψ x) volume

```

lemma radial_power_deriv_integral_by_parts[\[source\]](#)

```

private lemma radial_power_deriv_integral_by_parts
  {d : ℕ} (η : S(Space d, ℝ))
  (n : ↑(Metric.sphere (0 : Space d) 1))
  (p : ℕ) (hp : 0 < p) :
  - ∫ (r : Set.Ioi (0 : ℝ)),
    r.1 ^ p * (←root_.deriv (fun a => η (a • n.1)) r.1)
    ∂(.comap Subtype.val volume)
  =
  (p : ℝ) * ∫ (r : Set.Ioi (0 : ℝ)),
    r.1 ^ (p - 1) * η (r.1 • n.1)
    ∂(.comap Subtype.val volume)

```

lemma distDiv_norm_zpow_smul_repr_self_apply_eq_radial_deriv[\[source\]](#)

```

private lemma distDiv_norm_zpow_smul_repr_self_apply_eq_radial_deriv
  {d p : ℕ} [NeZero d] (q : ℤ) (hq : 0 < q + (d : ℤ))
  (hp_int : (p : ℤ) = q + (d : ℤ))
  (η : S(Space d, ℝ)) :
  (∇d · (distOfFunction (fun x : Space d => ||x|| ^ q • basis.repr x)
    (IsDistBounded.zpow_smul_repr_self q (by omega)))) η =
  - ∫ n, (∫ (r : Set.Ioi (0 : ℝ)),
    r.1 ^ p * (←root_.deriv (fun a => η (a • n.1)) r.1)
    ∂(.comap Subtype.val volume))
  ∂(volume (α := Space d).toSphere)

```

lemma distDiv_norm_zpow_smul_repr_self_eq_smul[\[source\]](#)

```

lemma distDiv_norm_zpow_smul_repr_self_eq_smul
  {d : ℕ} [NeZero d] (q : ℤ) (hq : 0 < q + (d : ℤ)) :
  ∇d · (distOfFunction (fun x : Space d => ||x|| ^ q • basis.repr x)
    (IsDistBounded.zpow_smul_repr_self q (by omega))) =
  (((q + d : ℤ) : ℝ) •
    distOfFunction (fun x : Space d => ||x|| ^ q)
    (IsDistBounded.pow q (by omega)))

```

lemma distLaplacian_distOfFunction_norm_zpow[\[source\]](#)

```

lemma distLaplacian_distOfFunction_norm_zpow {d : ℕ} [NeZero d] (m : ℤ)
  (hdiv : 0 < m - 2 + (d : ℤ)) :
  Δd (distOfFunction (fun x : Space d => ||x|| ^ m)
    (IsDistBounded.pow m (by omega))) =
  (((m : ℝ) * (((m - 2 + d : ℤ) : ℝ))) •
    distOfFunction (fun x : Space d => ||x|| ^ (m - 2))
    (IsDistBounded.pow (m - 2) (by omega)))

```

lemma distLaplacian_fundamentalSolution_norm_zpow[\[source\]](#)

The distributional Laplacian of $\|x\|^{(2-d)}$ is $(2-d) * d * \text{volume}(\text{Metric.ball } 0 \ 1)$ times the Dirac delta at the origin. For $d \geq 3$ this $\|x\|^{(2-d)}$ is the (singular) fundamental solution of the Laplacian, and for $d = 1$ it is $\|x\|$. When $d = 2$ the exponent vanishes, so the identity collapses to the trivial $\Delta^d 1 = 0$; the genuine two-dimensional fundamental solution is the logarithm, proved in `distLaplacian_fundamentalSolution_log_norm`. The statement also holds vacuously for $d = 0$, where the space is trivial.

```

lemma distLaplacian_fundamentalSolution_norm_zpow {d : ℕ} :
  Δd (distOfFunction (fun x : Space d => ||x|| ^ (- ((d : ℤ) - 2)))
    (IsDistBounded.pow _ (by omega))) =
    ((- ((d : ℝ) - 2)) * d *
      (volume (α := Space d)).real (Metric.ball 0 1)) • diracDelta ℝ 0

```

lemma distLaplacian_fundamentalSolution_log_norm

[source]

*In dimension two the fundamental solution of the Laplacian is the logarithm: the distributional Laplacian of $\text{Real.log } ||x||$ is $2 * \text{volume (Metric.ball 0 1)}$ times the Dirac delta at the origin.*

```

lemma distLaplacian_fundamentalSolution_log_norm :
  Δd (distOfFunction (fun x : Space 2 => Real.log ||x||) IsDistBounded.log_norm) =
    (2 * (volume (α := Space 2)).real (Metric.ball 0 1)) • diracDelta ℝ 0

```

2.78 Physlib.SpaceAndTime.Space.Norm.IteratedLaplacian

Physlib/SpaceAndTime/Space/Norm/IteratedLaplacian.lean on GitHub.

def oddNormIteratedLaplacianCoeff

[source]

The scalar factor in the odd-dimensional iterated Laplacian of the norm.

```

noncomputable def oddNormIteratedLaplacianCoeff (m : ℕ) : ℝ

```

lemma oddNormIteratedLaplacianCoeff_factor_ne_zero

[source]

```

private lemma oddNormIteratedLaplacianCoeff_factor_ne_zero {m k : ℕ} (hk : k < m) :
  (((1 : ℤ) - 2 * (k : ℤ) : ℤ) : ℝ) *
    (((1 : ℤ) - 2 * (k : ℤ) - 2 + (2 * m + 1 : ℤ) : ℤ) : ℝ)) ≠ 0

```

lemma oddNormIteratedLaplacianCoeff_ne_zero

[source]

The scalar factor in the odd-dimensional iterated Laplacian of the norm is nonzero.

```

lemma oddNormIteratedLaplacianCoeff_ne_zero (m : ℕ) :
  oddNormIteratedLaplacianCoeff m ≠ 0

```

lemma distLaplacian_norm_zpow_odd_boundary

[source]

```

private lemma distLaplacian_norm_zpow_odd_boundary (m : ℕ) :
  Δd (distOfFunction (fun x : Space (2 * m + 1) =>
    ||x|| ^ ((1 : ℤ) - 2 * (m : ℤ)))
    (IsDistBounded.pow _ (by omega))) =
    (((1 : ℤ) - 2 * (m : ℤ) : ℝ) * (2 * m + 1 : ℝ) *
      (volume (α := Space (2 * m + 1)).real (Metric.ball 0 1)) •
        diracDelta ℝ 0

```

lemma iterated_distLaplacian_norm_zpow_odd_until_boundary

[source]

```

private lemma iterated_distLaplacian_norm_zpow_odd_until_boundary
  (m k : ℕ) (hk : k ≤ m) :

```

```

((distLaplacian (d := 2 * m + 1))^[k])
(distOfFunction (fun x : Space (2 * m + 1) => ||x|| ^ (1 : ℤ))
  (IsDistBounded.pow 1 (by omega))) =
(∏ j ∈ Finset.range k,
  (((1 : ℤ) - 2 * (j : ℤ) : ℤ) : ℝ) *
  (((1 : ℤ) - 2 * (j : ℤ) - 2 + (2 * m + 1 : ℤ) : ℤ) : ℝ))) •
distOfFunction (fun x : Space (2 * m + 1) => ||x|| ^ ((1 : ℤ) - 2 * (k : ℤ)))
  (IsDistBounded.pow _ (by omega))

```

lemma iterated_distLaplacian_norm_zpow_odd_eq_smul_diracDelta

[\[source\]](#)

*In dimension $2 * m + 1$, the $(m + 1)$ -fold distributional Laplacian of the distribution induced by the norm is a nonzero multiple of the Dirac delta at the origin.*

```

lemma iterated_distLaplacian_norm_zpow_odd_eq_smul_diracDelta (m : ℕ) :
  ((distLaplacian (d := 2 * m + 1))^[m + 1])
  (distOfFunction (fun x : Space (2 * m + 1) => ||x|| ^ (1 : ℤ))
    (IsDistBounded.pow 1 (by omega))) =
  oddNormIteratedLaplacianCoeff m • diracDelta ℝ 0

```

2.79 Physlib.SpaceAndTime.Space.Norm.Regularized

[Physlib/SpaceAndTime/Space/Norm/Regularized.lean](#) on GitHub.

def normRegularizedPow

[\[source\]](#)

Power of regularized norm, $(||x||^2 + \varepsilon^2)^{(s/2)}$.

```

def normRegularizedPow (d : ℕ) (ε s : ℝ) : Space d → ℝ

```

lemma normRegularizedPow_eq

[\[source\]](#)

```

lemma normRegularizedPow_eq (d : ℕ) (ε s : ℝ) :
  normRegularizedPow d ε s = fun x ↦ (||x|| ^ 2 + ε ^ 2) ^ (s / 2)

```

lemma norm_sq_add_unit_sq_pos

[\[source\]](#)

For a nonzero regularization parameter, $||x||^2 + \varepsilon^2$ is positive.

```

lemma norm_sq_add_unit_sq_pos {d : ℕ} (ε : ℝ+) (x : Space d) : 0 < ||x|| ^ 2 + ε ^ 2

```

lemma normRegularizedPow_pos

[\[source\]](#)

The regularized norm power is positive for nonzero regularization parameter.

```

lemma normRegularizedPow_pos (d : ℕ) (ε : ℝ+) (s : ℝ) (x : Space d) :
  0 < normRegularizedPow d ε s x

```

lemma normRegularizedPow_hasTemperateGrowth

[\[source\]](#)

The regularized norm power has temperate growth.

```

lemma normRegularizedPow_hasTemperateGrowth (d : ℕ) (ε : ℝ+) (s : ℝ) :
  HasTemperateGrowth (normRegularizedPow d ε s)

```


lemma normRegularizedPow_measurable[\[source\]](#)

```
@[fun_prop]
lemma normRegularizedPow_measurable (d : ℕ) (ε s : ℝ) :
  Measurable (normRegularizedPow d ε s)
```

2.80 Physlib.SpaceAndTime.Space.Origin

[Physlib/SpaceAndTime/Space/Origin.lean](#) on GitHub.

def vectorToSpace[\[source\]](#)

A Euclidean vector, based at the chosen origin, viewed as a point of Space d.

```
noncomputable def vectorToSpace {d : ℕ} (v : EuclideanSpace ℝ (Fin d)) : Space d
```

lemma vectorToSpace_apply[\[source\]](#)

```
@[simp]
lemma vectorToSpace_apply {d : ℕ} (v : EuclideanSpace ℝ (Fin d)) (i : Fin d) :
  vectorToSpace v i = v i
```

lemma vectorToSpace_vsub_zero[\[source\]](#)

```
@[simp]
lemma vectorToSpace_vsub_zero {d : ℕ} (v : EuclideanSpace ℝ (Fin d)) :
  vectorToSpace v -v (0 : Space d) = v
```

def chartEuclidean[\[source\]](#)

The standard chart $\text{Space } d \approx^{a_i} [\mathbb{R}] \text{ EuclideanSpace } \mathbb{R} (\text{Fin } d)$, $p \mapsto p -_v 0$, identifying a point with its coordinate vector relative to the origin (the vector-space zero $(0 : \text{Space } d)$).

```
noncomputable def chartEuclidean (d : ℕ) :
  Space d  $\approx^{a_i} [\mathbb{R}]$  EuclideanSpace ℝ (Fin d)
```

2.81 Physlib.SpaceAndTime.SpaceTime.Derivatives

[Physlib/SpaceAndTime/SpaceTime/Derivatives.lean](#) on GitHub.

def manifoldDeriv[\[source\]](#)

The derivative of a function from SpaceTime d to a manifold along the μ coordinate, as a tangent vector at the value of the function.

```
noncomputable def manifoldDeriv {E H N : Type} [NormedAddCommGroup E] [NormedSpace ℝ E]
  [TopologicalSpace H] (I : ModelWithCorners ℝ E H) [TopologicalSpace N]
  [ChartedSpace H N] {d : ℕ} (μ : Fin 1  $\oplus$  Fin d) (f : SpaceTime d  $\rightarrow$  N) :
  (y : SpaceTime d)  $\rightarrow$  TangentSpace I (f y)
```

lemma manifoldDeriv_eq[\[source\]](#)

```

lemma manifoldDeriv_eq {E H N : Type} [NormedAddCommGroup E] [NormedSpace ℝ E]
  [TopologicalSpace H] (I : ModelWithCorners ℝ E H) [TopologicalSpace N]
  [ChartedSpace H N] {d : ℕ} (μ : Fin 1 ⊕ Fin d) (f : SpaceTime d → N)
  (y : SpaceTime d) :
  manifoldDeriv I μ f y =
  mfderiv ℐ(ℝ, SpaceTime d) I f y
  ((Lorentz.Vector.basis μ : SpaceTime d) : TangentSpace ℐ(ℝ, SpaceTime d) y)

```

lemma deriv_eq_mfderiv[\[source\]](#)

The spacetime derivative is the manifold derivative for functions into normed spaces.

```

lemma deriv_eq_mfderiv {M : Type} [NormedAddCommGroup M] [NormedSpace ℝ M]
  {d : ℕ} (μ : Fin 1 ⊕ Fin d) (f : SpaceTime d → M) (y : SpaceTime d) :
  deriv μ f y =
  mfderiv ℐ(ℝ, SpaceTime d) ℐ(ℝ, M) f y
  ((Lorentz.Vector.basis μ : SpaceTime d) : TangentSpace ℐ(ℝ, SpaceTime d) y)

```

lemma deriv_eq_manifoldDeriv[\[source\]](#)

```

lemma deriv_eq_manifoldDeriv {M : Type} [NormedAddCommGroup M] [NormedSpace ℝ M]
  {d : ℕ} (μ : Fin 1 ⊕ Fin d) (f : SpaceTime d → M) (y : SpaceTime d) :
  deriv μ f y = manifoldDeriv ℐ(ℝ, M) μ f y

```

lemma manifoldDeriv_const[\[source\]](#)

```

@[simp]
lemma manifoldDeriv_const {E H N : Type} [NormedAddCommGroup E] [NormedSpace ℝ E]
  [TopologicalSpace H] (I : ModelWithCorners ℝ E H) [TopologicalSpace N]
  [ChartedSpace H N] {d : ℕ} (μ : Fin 1 ⊕ Fin d) (n : N) (y : SpaceTime d) :
  manifoldDeriv I μ (fun _ : SpaceTime d => n) y = 0

```

lemma contDiff_deriv[\[source\]](#)

If f is C^{n+1} then $\partial_\mu f$ is C^n .

```

@[fun_prop]
lemma contDiff_deriv {M : Type} [NormedAddCommGroup M] [NormedSpace ℝ M] {d : ℕ}
  {n : WithTop ℕ} (μ : Fin 1 ⊕ Fin d) (f : SpaceTime d → M) (hf : ContDiff ℝ (n + 1) f) :
  ContDiff ℝ n (∂_μ f)

```

lemma differentiable_deriv[\[source\]](#)

If f is C^2 then $\partial_\mu f$ is differentiable.

```

@[fun_prop]
lemma differentiable_deriv {M : Type} [NormedAddCommGroup M] [NormedSpace ℝ M] {d : ℕ}
  (μ : Fin 1 ⊕ Fin d) (f : SpaceTime d → M) (hf : ContDiff ℝ 2 f) :
  Differentiable ℝ (∂_μ f)

```

lemma deriv_commute[\[source\]](#)

Derivatives on spacetime commute with one another (Clairaut's theorem).

```

lemma deriv_commute {M : Type} [NormedAddCommGroup M] [NormedSpace ℝ M] {d : ℕ}
  (μ ν : Fin 1 ⊗ Fin d) (f : SpaceTime d → M) (hf : ContDiff ℝ 2 f) :
  ∂_ μ (∂_ ν f) = ∂_ ν (∂_ μ f)

```

2.82 Physlib.SpaceAndTime.Time.Derivatives

[Physlib/SpaceAndTime/Time/Derivatives.lean](#) on GitHub.

def manifoldDeriv

[\[source\]](#)

The time derivative of a function from Time to a manifold, as a tangent vector at the value of the function.

```

noncomputable def manifoldDeriv {E H N : Type} [NormedAddCommGroup E] [NormedSpace ℝ E]
  [TopologicalSpace H] (I : ModelWithCorners ℝ E H) [TopologicalSpace N]
  [ChartedSpace H N] (f : Time → N) : (t : Time) → TangentSpace I (f t)

```

lemma manifoldDeriv_eq

[\[source\]](#)

```

lemma manifoldDeriv_eq {E H N : Type} [NormedAddCommGroup E] [NormedSpace ℝ E]
  [TopologicalSpace H] (I : ModelWithCorners ℝ E H) [TopologicalSpace N]
  [ChartedSpace H N] (f : Time → N) (t : Time) :
  manifoldDeriv I f t =
    mfderiv ℐ(ℝ, Time) I f t ((1 : Time) : TangentSpace ℐ(ℝ, Time) t)

```

lemma deriv_eq_mfderiv

[\[source\]](#)

The time derivative is the manifold derivative for functions into normed spaces.

```

lemma deriv_eq_mfderiv [NormedAddCommGroup M] [NormedSpace ℝ M]
  (f : Time → M) (t : Time) :
  deriv f t =
    mfderiv ℐ(ℝ, Time) ℐ(ℝ, M) f t
    ((1 : Time) : TangentSpace ℐ(ℝ, Time) t)

```

lemma deriv_eq_manifoldDeriv

[\[source\]](#)

```

lemma deriv_eq_manifoldDeriv [NormedAddCommGroup M] [NormedSpace ℝ M]
  (f : Time → M) (t : Time) :
  deriv f t = manifoldDeriv ℐ(ℝ, M) f t

```

lemma manifoldDeriv_const

[\[source\]](#)

```

@[simp]
lemma manifoldDeriv_const {E H N : Type} [NormedAddCommGroup E] [NormedSpace ℝ E]
  [TopologicalSpace H] (I : ModelWithCorners ℝ E H) [TopologicalSpace N]
  [ChartedSpace H N] (n : N) :
  manifoldDeriv I (fun _ : Time => n) t = 0

```

2.83 Physlib.SpaceAndTime.TimeAndSpace.Basic

[Physlib/SpaceAndTime/TimeAndSpace/Basic.lean](#) on GitHub.

abbrev TimeAndSpace[\[source\]](#)*Euclidean spacetime as the product of time and d -dimensional space.***abbrev** TimeAndSpace (d : $\mathbb{N}$:= 3)**def time**[\[source\]](#)*The time-coordinate projection from TimeAndSpace d .***noncomputable def** time {d : $\mathbb{N}$ } : TimeAndSpace d $\rightarrow$ L[$\mathbb{R}$] Time**def space**[\[source\]](#)*The spatial-coordinate projection from TimeAndSpace d .***noncomputable def** space {d : $\mathbb{N}$ } : TimeAndSpace d $\rightarrow$ L[$\mathbb{R}$] Space d**lemma time_apply**[\[source\]](#)

```
@[simp]
lemma time_apply (tx : TimeAndSpace d) :
  time tx = tx.1
```

lemma space_apply[\[source\]](#)

```
@[simp]
lemma space_apply (tx : TimeAndSpace d) :
  space tx = tx.2
```

lemma dist_time_le[\[source\]](#)*The time projection is nonexpanding for the product metric.*

```
lemma dist_time_le (tx ty : TimeAndSpace d) :
  dist (time tx) (time ty)  $\leq$  dist tx ty
```

lemma dist_space_le[\[source\]](#)*The spatial projection is nonexpanding for the product metric.*

```
lemma dist_space_le (tx ty : TimeAndSpace d) :
  dist (space tx) (space ty)  $\leq$  dist tx ty
```

2.84 Physlib.SpaceAndTime.TimeAndSpace.EuclideanGroup.Action

Physlib/SpaceAndTime/TimeAndSpace/EuclideanGroup/Action.lean on GitHub.

instance instMulActionEuclideanGroup[\[source\]](#)*The Euclidean group acts on TimeAndSpace d by fixing time and acting on space.*

```
noncomputable instance instMulActionEuclideanGroup :
  MulAction (EuclideanGroup d) (TimeAndSpace d) where
```

lemma smul_mk[\[source\]](#)*Coordinate formula for the Euclidean-group action on TimeAndSpace d.*

```
@[simp]
lemma smul_mk (g : EuclideanGroup d) (t : Time) (x : Space d) :
  g • ((t, x) : TimeAndSpace d) = (t, g • x)
```

lemma fst_smul[\[source\]](#)*The Euclidean-group action fixes the first coordinate.*

```
@[simp]
lemma fst_smul (g : EuclideanGroup d) (tx : TimeAndSpace d) :
  (g • tx).1 = tx.1
```

lemma snd_smul[\[source\]](#)*The Euclidean-group action applies to the second coordinate.*

```
@[simp]
lemma snd_smul (g : EuclideanGroup d) (tx : TimeAndSpace d) :
  (g • tx).2 = g • tx.2
```

lemma time_smul[\[source\]](#)*The Euclidean-group action fixes the time projection.*

```
lemma time_smul (g : EuclideanGroup d) (tx : TimeAndSpace d) :
  time (g • tx) = time tx
```

lemma space_smul[\[source\]](#)*The Euclidean-group action applies to the space projection.*

```
lemma space_smul (g : EuclideanGroup d) (tx : TimeAndSpace d) :
  space (g • tx) = g • space tx
```

lemma dist_smul[\[source\]](#)*The Euclidean-group action on TimeAndSpace d preserves product distance.*

```
lemma dist_smul (g : EuclideanGroup d) (tx ty : TimeAndSpace d) :
  dist (g • tx) (g • ty) = dist tx ty
```

def actionLinearMap[\[source\]](#)*The linear part of the Euclidean-group action on TimeAndSpace d.*

```
noncomputable def actionLinearMap (g : EuclideanGroup d) :
  TimeAndSpace d → L[ℝ] TimeAndSpace d
```

def actionTranslation[\[source\]](#)

The translation part of the Euclidean-group action on TimeAndSpace d.

```
noncomputable def actionTranslation (g : EuclideanGroup d) : TimeAndSpace d
```

lemma smul_eq_actionLinearMap_add[\[source\]](#)

```
lemma smul_eq_actionLinearMap_add (g : EuclideanGroup d) (tx : TimeAndSpace d) :
  g • tx = actionLinearMap g tx + actionTranslation g
```

lemma smul_hasTemperateGrowth[\[source\]](#)

The Euclidean-group action on TimeAndSpace d has temperate growth.

```
lemma smul_hasTemperateGrowth (g : EuclideanGroup d) :
  Function.HasTemperateGrowth (fun tx : TimeAndSpace d => g • tx)
```

lemma isometry_smul[\[source\]](#)

The Euclidean-group action on TimeAndSpace d is an isometry.

```
lemma isometry_smul (g : EuclideanGroup d) :
  Isometry (fun tx : TimeAndSpace d => g • tx)
```

lemma antilipschitz_smul[\[source\]](#)

The Euclidean-group action on TimeAndSpace d is antilipschitz.

```
lemma antilipschitz_smul (g : EuclideanGroup d) :
  AntilipschitzWith 1 (fun tx : TimeAndSpace d => g • tx)
```

2.85 Physlib.SpaceAndTime.TimeAndSpace.EuclideanGroup.SchwartzAction

[Physlib/SpaceAndTime/TimeAndSpace/EuclideanGroup/SchwartzAction.lean](#) on GitHub.

def schwartzEuclideanAction[\[source\]](#)

The Euclidean-group pullback action on Schwartz maps over TimeAndSpace d, as a continuous representation $\text{ContRepresentation } \mathbb{R} \text{ (EuclideanGroup d) } \mathcal{S}(\text{TimeAndSpace d}, F)$.

ContRepresentation $R \ G \ V$ unfolds to the monoid homomorphism $G \rightarrow^ V \rightarrow^L [R] \ V$ into the **continuous*-linear maps: the builder `SchwartzMap.compCLMOfAntilipschitz` already yields a continuous-linear map for free, and the $\rightarrow^L$ codomain lets the action compose under $\circ^L$. The plain `mathlib Representation` $(G \rightarrow^* V \rightarrow^L [\mathbb{R}] V)$ is recovered via `ContRepresentation.toRepresentation`.**

```
noncomputable def schwartzEuclideanAction {d : ℕ} :
  ContRepresentation ℝ (EuclideanGroup d) S(TimeAndSpace d, F) where
```

lemma schwartzEuclideanAction_apply[\[source\]](#)

Pointwise formula for the monoid-homomorphism form of the Schwartz-map pullback action.

```
@[simp]
lemma schwartzEuclideanAction_apply {d : ℕ} (g : EuclideanGroup d) (η : S(TimeAndSpace d, F))
  (tx : TimeAndSpace d) :
  (schwartzEuclideanAction g η) tx = η (g⁻¹ • tx)
```

instance instMulActionSchwartzMap

[\[source\]](#)

The Euclidean group acts on Schwartz maps over TimeAndSpace d by pullback.

```
noncomputable instance instMulActionSchwartzMap {d : ℕ} :
  MulAction (EuclideanGroup d) S(TimeAndSpace d, F) where
```

lemma smul_schwartzMap_apply

[\[source\]](#)

Pointwise formula for the MulAction instance on Schwartz maps.

```
@[simp]
lemma smul_schwartzMap_apply {d : ℕ} (g : EuclideanGroup d) (η : S(TimeAndSpace d, F))
  (tx : TimeAndSpace d) :
  (g • η) tx = η (g⁻¹ • tx)
```

lemma schwartzEuclideanAction_mul_apply

[\[source\]](#)

*Applying g and then h to a Schwartz map is the pullback action of h * g.*

```
lemma schwartzEuclideanAction_mul_apply {d : ℕ} (g h : EuclideanGroup d)
  (η : S(TimeAndSpace d, F)) :
  schwartzEuclideanAction h (schwartzEuclideanAction g η) =
  schwartzEuclideanAction (h * g) η
```

lemma schwartzEuclideanAction_injective

[\[source\]](#)

Each Euclidean-group pullback action on Schwartz maps is injective.

```
lemma schwartzEuclideanAction_injective {d : ℕ} (g : EuclideanGroup d) :
  Function.Injective (schwartzEuclideanAction (F := F) g)
```

lemma schwartzEuclideanAction_surjective

[\[source\]](#)

Each Euclidean-group pullback action on Schwartz maps is surjective.

```
lemma schwartzEuclideanAction_surjective {d : ℕ} (g : EuclideanGroup d) :
  Function.Surjective (schwartzEuclideanAction (F := F) g)
```

2.86 Physlib.Thermodynamics.Temperature.Basic

[Physlib/Thermodynamics/Temperature/Basic.lean](#) on GitHub.

lemma β_toReal

[\[source\]](#)

```
lemma β_toReal (T : Temperature) : (β T : ℝ) = 1 / (kB * (T : ℝ))
```

lemma ofβ_toReal[\[source\]](#)

```
lemma ofβ_toReal (β : ℝ≥0) : (ofβ β).toReal = 1 / (kB * (β : ℝ))
```

2.87 Physlib.Units.WithDim.Basic

[Physlib/Units/WithDim/Basic.lean](#) on GitHub.

instance Inhabited (WithDim d M)[\[source\]](#)

```
instance (d : Dimension) (M : Type) [Inhabited M] : Inhabited (WithDim d M) where
```

instance Zero (WithDim d M)[\[source\]](#)

```
instance (d : Dimension) (M : Type) [Zero M] : Zero (WithDim d M) where
```

lemma val_zero[\[source\]](#)

```
@[simp]
lemma val_zero {d : Dimension} {M : Type} [Zero M] :
  (0 : WithDim d M).val = 0
```

instance Add (WithDim d M)[\[source\]](#)

```
instance (d : Dimension) (M : Type) [Add M] : Add (WithDim d M) where
```

lemma val_add[\[source\]](#)

```
@[simp]
lemma val_add {d : Dimension} {M : Type} [Add M] (m1 m2 : WithDim d M) :
  (m1 + m2).val = m1.val + m2.val
```

instance Neg (WithDim d M)[\[source\]](#)

```
instance (d : Dimension) (M : Type) [Neg M] : Neg (WithDim d M) where
```

lemma val_neg[\[source\]](#)

```
@[simp]
lemma val_neg {d : Dimension} {M : Type} [Neg M] (m : WithDim d M) :
  (-m).val = -m.val
```

instance Sub (WithDim d M)[\[source\]](#)

```
instance (d : Dimension) (M : Type) [Sub M] : Sub (WithDim d M) where
```

lemma val_sub[\[source\]](#)

```
@[simp]
lemma val_sub {d : Dimension} {M : Type} [Sub M] (m1 m2 : WithDim d M) :
  (m1 - m2).val = m1.val - m2.val
```


instance AddSemigroup (WithDim d M)[\[source\]](#)

```
instance (d : Dimension) (M : Type) [AddSemigroup M] :
  AddSemigroup (WithDim d M) where
```

instance AddCommSemigroup (WithDim d M)[\[source\]](#)

```
instance (d : Dimension) (M : Type) [AddCommSemigroup M] :
  AddCommSemigroup (WithDim d M) where
```

instance AddMonoid (WithDim d M)[\[source\]](#)

```
instance (d : Dimension) (M : Type) [AddMonoid M] :
  AddMonoid (WithDim d M) where
```

instance AddCommMonoid (WithDim d M)[\[source\]](#)

```
instance (d : Dimension) (M : Type) [AddCommMonoid M] :
  AddCommMonoid (WithDim d M) where
```

instance AddGroup (WithDim d M)[\[source\]](#)

```
instance (d : Dimension) (M : Type) [AddGroup M] :
  AddGroup (WithDim d M) where
```

instance AddCommGroup (WithDim d M)[\[source\]](#)

```
instance (d : Dimension) (M : Type) [AddCommGroup M] :
  AddCommGroup (WithDim d M) where
```

instance LE (WithDim d M)[\[source\]](#)

```
instance (d : Dimension) (M : Type) [LE M] : LE (WithDim d M) where
```

lemma le_def[\[source\]](#)

```
@[simp]
lemma le_def {d : Dimension} {M : Type} [LE M] (m1 m2 : WithDim d M) :
  m1 ≤ m2 ↔ m1.val ≤ m2.val
```

instance LT (WithDim d M)[\[source\]](#)

```
instance (d : Dimension) (M : Type) [LT M] : LT (WithDim d M) where
```

lemma lt_def[\[source\]](#)

```
@[simp]
lemma lt_def {d : Dimension} {M : Type} [LT M] (m1 m2 : WithDim d M) :
  m1 < m2 ↔ m1.val < m2.val
```

instance Preorder (WithDim d M)[\[source\]](#)

```
instance (d : Dimension) (M : Type) [Preorder M] :
  Preorder (WithDim d M) where
```

instance PartialOrder (WithDim d M)[\[source\]](#)

```
instance (d : Dimension) (M : Type) [PartialOrder M] :
  PartialOrder (WithDim d M) where
```

2.88 PhyslibAlpha.ClassicalFieldTheory.Local.Action

[PhyslibAlpha/ClassicalFieldTheory/Local/Action.lean](#) on GitHub.

def actionDensity[\[source\]](#)

The action density determined by a local Lagrangian and a field.

```
noncomputable def actionDensity (L : Lagrangian d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m)) : Space d → ℝ
```

def HasFiniteAction[\[source\]](#)

The integrability condition for the action density of a field.

```
def HasFiniteAction (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m)) : Prop
```

def IsAdmissibleForAction[\[source\]](#)

Symmetric admissibility predicate for the action functional: the pair (L, f) is admissible when the field is smooth and the corresponding action density is integrable.

```
def IsAdmissibleForAction (L : Lagrangian d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m)) : Prop
```

def action[\[source\]](#)

The action associated with a local Lagrangian.

```
noncomputable def action (L : Lagrangian d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m)) : ℝ
```

lemma actionDensity_apply[\[source\]](#)

```
@[simp]
lemma actionDensity_apply (L : Lagrangian d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m)) (x : Space d) :
  actionDensity L f x = L (jetAt k f x)
```

lemma action_eq_integral[\[source\]](#)

```
@[simp]
lemma action_eq_integral (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m)) :
  action L f = ∫ x, actionDensity L f x
```

def variedField[\[source\]](#)

The field obtained by varying f in the direction η with parameter s .

```
noncomputable def variedField (f : Space d → EuclideanSpace ℝ (Fin m))
  (η : AdmissibleVariation d (EuclideanSpace ℝ (Fin m)))
  (s : ℝ) :
  Space d → EuclideanSpace ℝ (Fin m)
```

lemma variedField_contDiff[\[source\]](#)

Smoothness of the varied field for smooth f and admissible η .

```
lemma variedField_contDiff (f : Space d → EuclideanSpace ℝ (Fin m))
  (η : AdmissibleVariation d (EuclideanSpace ℝ (Fin m)))
  (s : ℝ)
  (hf : ContDiff ℝ ∞ f) :
  ContDiff ℝ ∞ (variedField f η s)
```

def actionVariation[\[source\]](#)

The action of a field under an admissible variation.

```
noncomputable def actionVariation (L : Lagrangian d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m))
  (η : AdmissibleVariation d (EuclideanSpace ℝ (Fin m))) : ℝ → ℝ
```

def HasFiniteActionVariation[\[source\]](#)

Explicit integrability condition for the family of varied action densities.

```
def HasFiniteActionVariation (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m))
  (η : AdmissibleVariation d (EuclideanSpace ℝ (Fin m))) : Prop
```

def HasCompactlySupportedActionVariationDifference[\[source\]](#)

Locality package for the action density under compactly supported variations: for every variation parameter s , the change in the action density is a continuous compactly supported function. Combined with finiteness of the base action, this implies finiteness of the varied action.

```
def HasCompactlySupportedActionVariationDifference (L : Lagrangian d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m))
  (η : AdmissibleVariation d (EuclideanSpace ℝ (Fin m))) : Prop
```

lemma hasFiniteActionVariation_of_hasFiniteAction_of_compactlySupportedDifference [\[source\]](#)

If the base action is finite and every varied action density differs from the base density by a continuous compactly supported function, then the whole varied family has finite action.

```
lemma hasFiniteActionVariation_of_hasFiniteAction_of_compactlySupportedDifference
  (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m))
  (η : AdmissibleVariation d (EuclideanSpace ℝ (Fin m)))
  (hbase : HasFiniteAction L f)
  (hdiff : HasCompactlySupportedActionVariationDifference L f η) :
```

```
HasFiniteActionVariation L f η
```

lemma variedField_apply[\[source\]](#)

```
@[simp]
lemma variedField_apply (f : Space d → EuclideanSpace ℝ (Fin m))
  (η : AdmissibleVariation d (EuclideanSpace ℝ (Fin m))) (s : ℝ)
  (x : Space d) :
  variedField f η s x = f x + s • η x
```

lemma actionVariation_apply[\[source\]](#)

```
@[simp]
lemma actionVariation_apply (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m))
  (η : AdmissibleVariation d (EuclideanSpace ℝ (Fin m))) (s : ℝ) :
  actionVariation L f η s = action L (variedField f η s)
```

lemma jetAt_variedField_eq_of_notMem_tsupport[\[source\]](#)

```
lemma jetAt_variedField_eq_of_notMem_tsupport
  (k : ℕ) (f : Space d → EuclideanSpace ℝ (Fin m))
  (η : AdmissibleVariation d (EuclideanSpace ℝ (Fin m))) (s : ℝ)
  {x : Space d}
  (hf : ContDiff ℝ ∞ f) (hx : x ∉ tsupport η.toFun) :
  jetAt k (variedField f η s) x = jetAt k f x
```

lemma actionDensity_variedField_eq_of_notMem_tsupport[\[source\]](#)

```
lemma actionDensity_variedField_eq_of_notMem_tsupport
  (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m))
  (η : AdmissibleVariation d (EuclideanSpace ℝ (Fin m)))
  (s : ℝ)
  {x : Space d} (hf : ContDiff ℝ ∞ f) (hx : x ∉ tsupport η.toFun) :
  actionDensity L (variedField f η s) x = actionDensity L f x
```

lemma hasCompactlySupportedActionVariationDifference_of_continuousInCoordinates[\[source\]](#)

```
lemma hasCompactlySupportedActionVariationDifference_of_continuousInCoordinates
  (L : Lagrangian d m k) (hcontL : Lagrangian.ContinuousInCoordinates L)
  (f : Space d → EuclideanSpace ℝ (Fin m)) (hf : ContDiff ℝ ∞ f)
  (η : AdmissibleVariation d (EuclideanSpace ℝ (Fin m))) :
  HasCompactlySupportedActionVariationDifference L f η
```

def IsCritical[\[source\]](#)

A field is critical for a local action if every admissible compactly supported variation has vanishing first derivative at $s = 0$, under the explicit finite-action hypothesis for the varied family.

```
def IsCritical (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m)) : Prop
```

2.89 PhyslibAlpha.ClassicalFieldTheory.Local.EulerLagrange

[PhyslibAlpha/ClassicalFieldTheory/Local/EulerLagrange.lean](#) on GitHub.

def eulerLagrangeTerm

[\[source\]](#)

The summand $(-1)^{|I|} D_I (\partial L / \partial u^a_I)$ in the local Euler-Lagrange formula.

```
noncomputable def eulerLagrangeTerm (L : Lagrangian d m k) (I : DerivativeIndex d k) (a : Fin m)
  (f : Space d → EuclideanSpace ℝ (Fin m)) : Space d → ℝ
```

lemma eulerLagrangeTerm_apply

[\[source\]](#)

```
@[simp]
lemma eulerLagrangeTerm_apply (L : Lagrangian d m k) (I : DerivativeIndex d k) (a : Fin m)
  (f : Space d → EuclideanSpace ℝ (Fin m)) (x : Space d) :
  eulerLagrangeTerm L I a f x =
    (-1 : ℝ) ^ I.1.order * iteratedTotalDerivative k I.1 (L.coordDeriv I a) f x
```

def eulerLagrangeComponent

[\[source\]](#)

The a -th component of the local Euler-Lagrange operator.

```
noncomputable def eulerLagrangeComponent (L : Lagrangian d m k) (a : Fin m)
  (f : Space d → EuclideanSpace ℝ (Fin m)) : Space d → ℝ
```

lemma eulerLagrangeComponent_apply

[\[source\]](#)

```
@[simp]
lemma eulerLagrangeComponent_apply (L : Lagrangian d m k) (a : Fin m)
  (f : Space d → EuclideanSpace ℝ (Fin m)) (x : Space d) :
  eulerLagrangeComponent L a f x = ∑ I : DerivativeIndex d k, eulerLagrangeTerm L I a f x
```

def eulerLagrangeOp

[\[source\]](#)

The local Euler-Lagrange operator attached to a local Lagrangian.

```
noncomputable def eulerLagrangeOp (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m)) :
  Space d → EuclideanSpace ℝ (Fin m)
```

lemma eulerLagrangeOp_apply

[\[source\]](#)

```
lemma eulerLagrangeOp_apply (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m))
  (x : Space d) (a : Fin m) :
  eulerLagrangeOp L f x a = eulerLagrangeComponent L a f x
```

lemma eulerLagrangeOp_eq

[\[source\]](#)

```
@[simp]
lemma eulerLagrangeOp_eq (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m)) :
  eulerLagrangeOp L f = fun x => WithLp.toLp 2 fun a => eulerLagrangeComponent L a f x
```

2.90 PhyslibAlpha.ClassicalFieldTheory.Local.FirstOrder

[PhyslibAlpha/ClassicalFieldTheory/Local/FirstOrder.lean](#) on GitHub.

abbrev FirstOrderJetCoordinates

[\[source\]](#)

Coordinate data for first-order local jet points.

```
abbrev FirstOrderJetCoordinates (d m : ℕ)
```

abbrev FirstOrderJetPoint

[\[source\]](#)

Coordinate-level first-order jet points.

```
abbrev FirstOrderJetPoint (d m : ℕ)
```

abbrev FirstOrderJetFiberData

[\[source\]](#)

Fiber-direction data for first-order jet points.

```
abbrev FirstOrderJetFiberData (d m : ℕ)
```

abbrev FirstOrderLagrangian

[\[source\]](#)

First-order local lagrangians.

```
abbrev FirstOrderLagrangian (d m : ℕ)
```

def firstDerivativeIndex

[\[source\]](#)

The first-derivative coordinate index in the source direction i .

```
def firstDerivativeIndex (d : ℕ) (i : Fin d) : DerivativeIndex d 1
```

lemma firstDerivativeIndex_coe

[\[source\]](#)

```
@[simp]
```

```
lemma firstDerivativeIndex_coe (d : ℕ) (i : Fin d) :  
  ((firstDerivativeIndex d i : DerivativeIndex d 1) : MultiIndex d) =  
    Physlib.MultiIndex.increment 0 i
```

lemma firstDerivativeIndex_order

[\[source\]](#)

```
lemma firstDerivativeIndex_order (d : ℕ) (i : Fin d) :  
  ((firstDerivativeIndex d i : DerivativeIndex d 1) : MultiIndex d).order = 1
```

def firstDerivCoord

[\[source\]](#)

The first derivative coordinate u^a_i of a first-order jet point.

```
def firstDerivCoord (J : FirstOrderJetPoint d m) (i : Fin d) (a : Fin m) : ℝ
```

def firstDerivVector[\[source\]](#)

The bundled target vector of first derivative coordinates in the source direction i .

```
def firstDerivVector (J : FirstOrderJetPoint d m) (i : Fin d) : EuclideanSpace ℝ (Fin m)
```

lemma firstDerivVector_apply[\[source\]](#)

```
@[simp]
```

```
lemma firstDerivVector_apply (J : FirstOrderJetPoint d m) (i : Fin d) (a : Fin m) :  
  J.firstDerivVector i a = J.firstDerivCoord i a
```

lemma firstDerivCoord_ofBaseCoordinates[\[source\]](#)

```
@[simp]
```

```
lemma firstDerivCoord_ofBaseCoordinates (x : Space d) (u : FirstOrderJetCoordinates d m)  
  (i : Fin d) (a : Fin m) :  
  (JetPoint.ofBaseCoordinates x u).firstDerivCoord i a = u (firstDerivativeIndex d i) a
```

abbrev firstOrderJetAt[\[source\]](#)

The first-order coordinate-level jet point of a field at a point.

```
noncomputable abbrev firstOrderJetAt (f : Space d → EuclideanSpace ℝ (Fin m)) (x : Space d) :  
  FirstOrderJetPoint d m
```

lemma firstOrderJetAt_base[\[source\]](#)

```
@[simp]
```

```
lemma firstOrderJetAt_base (f : Space d → EuclideanSpace ℝ (Fin m)) (x : Space d) :  
  (firstOrderJetAt f x).base = x
```

lemma firstOrderJetAt_value[\[source\]](#)

```
lemma firstOrderJetAt_value (f : Space d → EuclideanSpace ℝ (Fin m)) (x : Space d) :  
  (firstOrderJetAt f x).value = f x
```

lemma firstOrderJetAt_firstDerivCoord[\[source\]](#)

```
@[simp]
```

```
lemma firstOrderJetAt_firstDerivCoord (f : Space d → EuclideanSpace ℝ (Fin m))  
  (x : Space d) (i : Fin d) (a : Fin m) :  
  (firstOrderJetAt f x).firstDerivCoord i a =  
    ∂^[Physlib.MultiIndex.increment 0 i] (fun y => (f y) a) x
```

abbrev firstOrderActionDensity[\[source\]](#)

Evaluate a first-order local lagrangian along the first jets of a field.

```
noncomputable abbrev firstOrderActionDensity (L : FirstOrderLagrangian d m)  
  (f : Space d → EuclideanSpace ℝ (Fin m)) : Space d → ℝ
```

abbrev firstOrderAction[\[source\]](#)*The action of a first-order local lagrangian.*

```
noncomputable abbrev firstOrderAction (L : FirstOrderLagrangian d m)
  (f : Space d → EuclideanSpace ℝ (Fin m)) : ℝ
```

abbrev firstOrderEulerLagrangeOp[\[source\]](#)*The Euler-Lagrange operator of a first-order local lagrangian.*

```
noncomputable abbrev firstOrderEulerLagrangeOp (L : FirstOrderLagrangian d m)
  (f : Space d → EuclideanSpace ℝ (Fin m)) : Space d → EuclideanSpace ℝ (Fin m)
```

theorem firstOrder_isCritical_iff_eulerLagrange_zero[\[source\]](#)

```
theorem firstOrder_isCritical_iff_eulerLagrange_zero
  (L : FirstOrderLagrangian d m) (f : Space d → EuclideanSpace ℝ (Fin m))
  (hLf : IsAdmissibleForAction L f)
  (hL : Lagrangian.SmoothInCoordinates L) :
  IsCritical L f ↔ firstOrderEulerLagrangeOp L f = 0
```

2.91 PhyslibAlpha.ClassicalFieldTheory.Local.FirstVariation[PhyslibAlpha/ClassicalFieldTheory/Local/FirstVariation.lean](#) on GitHub.**theorem hasSmoothEulerLagrangeRegularity_of_smoothInCoordinates**[\[source\]](#)

```
theorem hasSmoothEulerLagrangeRegularity_of_smoothInCoordinates
  (L : Lagrangian d m k) (hL : Lagrangian.SmoothInCoordinates L) :
  HasSmoothEulerLagrangeRegularity L
```

theorem isCritical_iff_eulerLagrange_zero_of_contDiff_and_smoothInCoordinates[\[source\]](#)

```
theorem isCritical_iff_eulerLagrange_zero_of_contDiff_and_smoothInCoordinates
  (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m)) (hf : ContDiff ℝ ∞ f)
  (hL : Lagrangian.SmoothInCoordinates L)
  (hbase : HasFiniteAction L f) :
  IsCritical L f ↔ eulerLagrangeOp L f = 0
```

theorem isCritical_iff_eulerLagrange_zero_of_admissibleForAction_and_smoothInCoordinates[\[source\]](#)

```
theorem isCritical_iff_eulerLagrange_zero_of_admissibleForAction_and_smoothInCoordinates
  (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m))
  (hLf : IsAdmissibleForAction L f)
  (hL : Lagrangian.SmoothInCoordinates L) :
  IsCritical L f ↔ eulerLagrangeOp L f = 0
```

2.92 PhyslibAlpha.ClassicalFieldTheory.Local.FirstVariation.Basic[PhyslibAlpha/ClassicalFieldTheory/Local/FirstVariation/Basic.lean](#) on GitHub.

def firstVariationDensityTerm[\[source\]](#)

A single term in the linearized first-variation density before integration by parts.

```
noncomputable def firstVariationDensityTerm (L : Lagrangian d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m))
  (η : AdmissibleVariation d (EuclideanSpace ℝ (Fin m))) (I : DerivativeIndex d k)
  (a : Fin m) :
  Space d → ℝ
```

def firstVariationDensity[\[source\]](#)

The pointwise linearized first-variation density before integration by parts.

```
noncomputable def firstVariationDensity (L : Lagrangian d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m))
  (η : AdmissibleVariation d (EuclideanSpace ℝ (Fin m))) : Space d → ℝ
```

def firstVariationValue[\[source\]](#)

The value predicted by the first-variation formula for an admissible variation.

```
noncomputable def firstVariationValue (L : Lagrangian d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m))
  (η : AdmissibleVariation d (EuclideanSpace ℝ (Fin m))) : ℝ
```

lemma firstVariationDensityTerm_apply[\[source\]](#)

```
@[simp]
lemma firstVariationDensityTerm_apply (L : Lagrangian d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m))
  (η : AdmissibleVariation d (EuclideanSpace ℝ (Fin m))) (I : DerivativeIndex d k)
  (a : Fin m) (x : Space d) :
  firstVariationDensityTerm L f η I a x =
    L.coordDeriv I a (jetAt k f x) * ∂^[I.1] (fun y => (η y) a) x
```

lemma firstVariationDensity_apply[\[source\]](#)

```
@[simp]
lemma firstVariationDensity_apply (L : Lagrangian d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m))
  (η : AdmissibleVariation d (EuclideanSpace ℝ (Fin m))) (x : Space d) :
  firstVariationDensity L f η x =
    ∑ I : DerivativeIndex d k, ∑ a : Fin m, firstVariationDensityTerm L f η I a x
```

lemma firstVariationValue_eq_integral[\[source\]](#)

```
@[simp]
lemma firstVariationValue_eq_integral (L : Lagrangian d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m))
  (η : AdmissibleVariation d (EuclideanSpace ℝ (Fin m))) :
  firstVariationValue L f η = ∫ x, ⟨⟨eulerLagrangeOp L f x, η x⟩⟩_ℝ
```

lemma firstVariationValue_eq_zero_of_eulerLagrange_zero[\[source\]](#)

```

lemma firstVariationValue_eq_zero_of_eulerLagrange_zero
  (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m))
  (η : AdmissibleVariation d (EuclideanSpace ℝ (Fin m))) (hEuler : eulerLagrangeOp L f = 0) :
  firstVariationValue L f η = 0

```

2.93 PhyslibAlpha.ClassicalFieldTheory.Local.FirstVariation.Criterion

[PhyslibAlpha/ClassicalFieldTheory/Local/FirstVariation/Criterion.lean](#) on GitHub.

def AllVariationsHaveFiniteAction[\[source\]](#)

Explicit global finiteness hypothesis for all admissible variations.

```

def AllVariationsHaveFiniteAction (L : Lagrangian d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m)) : Prop

```

def HasFirstVariationFormula[\[source\]](#)

The first-variation formula for the local action, packaged as a reusable hypothesis.

```

def HasFirstVariationFormula (L : Lagrangian d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m)) : Prop

```

lemma isCritical_of_eulerLagrange_zero[\[source\]](#)

```

lemma isCritical_of_eulerLagrange_zero (L : Lagrangian d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m))
  (hfirst : HasFirstVariationFormula L f) (hEuler : eulerLagrangeOp L f = 0) :
  IsCritical L f

```

lemma eulerLagrange_zero_of_isCritical[\[source\]](#)

```

lemma eulerLagrange_zero_of_isCritical (L : Lagrangian d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m))
  (hfirst : HasFirstVariationFormula L f)
  (hfin : AllVariationsHaveFiniteAction L f)
  (hcont : Continuous (eulerLagrangeOp L f))
  (hcrit : IsCritical L f) :
  eulerLagrangeOp L f = 0

```

theorem isCritical_iff_eulerLagrange_zero[\[source\]](#)

```

theorem isCritical_iff_eulerLagrange_zero (L : Lagrangian d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m))
  (hfirst : HasFirstVariationFormula L f)
  (hfin : AllVariationsHaveFiniteAction L f)
  (hcont : Continuous (eulerLagrangeOp L f)) :
  IsCritical L f ↔ eulerLagrangeOp L f = 0

```

lemma hasFirstVariationFormula_of_underIntegral_linearized_and_parts[\[source\]](#)

```
private lemma hasFirstVariationFormula_of_underIntegral_linearized_and_parts
  (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m))
  (hint : HasActionVariationDerivativeUnderIntegral L f)
  (hpoint : HasPointwiseLinearizedDensityFormula L f)
  (hibp : HasIntegratedByPartsFormula L f) :
  HasFirstVariationFormula L f
```

lemma hasFirstVariationFormula_of_contDiff_underIntegral_and_parts

[\[source\]](#)

```
private lemma hasFirstVariationFormula_of_contDiff_underIntegral_and_parts
  (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m)) (hf : ContDiff ℝ ∞ f)
  (hint : HasActionVariationDerivativeUnderIntegral L f)
  (hibp : HasIntegratedByPartsFormula L f) :
  HasFirstVariationFormula L f
```

lemma hasFirstVariationFormula_of_underIntegral_linearized_and_termwise

[\[source\]](#)

```
private lemma hasFirstVariationFormula_of_underIntegral_linearized_and_termwise
  (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m))
  (hint : HasActionVariationDerivativeUnderIntegral L f)
  (hpoint : HasPointwiseLinearizedDensityFormula L f)
  (hterm : HasTermwiseIntegratedByPartsFormula L f) :
  HasFirstVariationFormula L f
```

lemma hasFirstVariationFormula_of_contDiff_underIntegral_and_termwise

[\[source\]](#)

```
private lemma hasFirstVariationFormula_of_contDiff_underIntegral_and_termwise
  (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m)) (hf : ContDiff ℝ ∞ f)
  (hint : HasActionVariationDerivativeUnderIntegral L f)
  (hterm : HasTermwiseIntegratedByPartsFormula L f) :
  HasFirstVariationFormula L f
```

theorem isCritical_iff_eulerLagrange_zero_of_underIntegral_linearized_and_parts [\[source\]](#)

The local Euler-Lagrange criterion obtained from the packaged analytic ingredients of the first-variation formula. This is the current formalized form of Theorem 5.2.

```
private theorem isCritical_iff_eulerLagrange_zero_of_underIntegral_linearized_and_parts
  (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m))
  (hint : HasActionVariationDerivativeUnderIntegral L f)
  (hpoint : HasPointwiseLinearizedDensityFormula L f)
  (hibp : HasIntegratedByPartsFormula L f)
  (hfin : AllVariationsHaveFiniteAction L f)
  (hcont : Continuous (eulerLagrangeOp L f)) :
  IsCritical L f ↔ eulerLagrangeOp L f = 0
```

theorem isCritical_iff_eulerLagrange_zero_of_underIntegral_linearized_and_termwise

[\[source\]](#)

Variant of the local Euler-Lagrange criterion where the integration-by-parts input is reduced to a termwise hypothesis.

```
private theorem isCritical_iff_eulerLagrange_zero_of_underIntegral_linearized_and_termwise
  (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m))
  (hint : HasActionVariationDerivativeUnderIntegral L f)
```

```

(hpoint : HasPointwiseLinearizedDensityFormula L f)
(hterm : HasTermwiseIntegratedByPartsFormula L f)
(hfin : AllVariationsHaveFiniteAction L f)
(hcont : Continuous (eulerLagrangeOp L f)) :
IsCritical L f  $\leftrightarrow$  eulerLagrangeOp L f = 0

```

theorem isCritical_iff_eulerLagrange_zero_of_contDiff_underIntegral_and_termwise [\[source\]](#)

```

private theorem isCritical_iff_eulerLagrange_zero_of_contDiff_underIntegral_and_termwise
  (L : Lagrangian d m k) (f : Space d  $\rightarrow$  EuclideanSpace  $\mathbb{R}$  (Fin m)) (hf : ContDiff  $\mathbb{R} \infty$  f)
  (hint : HasActionVariationDerivativeUnderIntegral L f)
  (hterm : HasTermwiseIntegratedByPartsFormula L f)
  (hfin : AllVariationsHaveFiniteAction L f)
  (hcont : Continuous (eulerLagrangeOp L f)) :
IsCritical L f  $\leftrightarrow$  eulerLagrangeOp L f = 0

```

theorem isCritical_iff_eulerLagrange_zero_of_contDiff_continuous_coordDeriv_and_termwise [\[source\]](#)

```

private theorem
  isCritical_iff_eulerLagrange_zero_of_contDiff_continuous_coordDeriv_and_termwise
  (L : Lagrangian d m k) (f : Space d  $\rightarrow$  EuclideanSpace  $\mathbb{R}$  (Fin m)) (hf : ContDiff  $\mathbb{R} \infty$  f)
  (hcontVar :  $\forall \eta$  : AdmissibleVariation d (EuclideanSpace  $\mathbb{R}$  (Fin m)),
     $\forall I$  : DerivativeIndex d k,  $\forall a$  : Fin m,
    Continuous ( $\text{fun } p : \mathbb{R} \times \text{Space } d \Rightarrow$ 
      L.coordDeriv I a (jetAt k (variedField f  $\eta$  p.1) p.2)))
  (hterm : HasTermwiseIntegratedByPartsFormula L f)
  (hfin : AllVariationsHaveFiniteAction L f)
  (hcont : Continuous (eulerLagrangeOp L f)) :
IsCritical L f  $\leftrightarrow$  eulerLagrangeOp L f = 0

```

theorem isCritical_iff_eulerLagrange_zero_of_contDiff_and_regular [\[source\]](#)

```

private theorem isCritical_iff_eulerLagrange_zero_of_contDiff_and_regular
  (L : Lagrangian d m k) (f : Space d  $\rightarrow$  EuclideanSpace  $\mathbb{R}$  (Fin m)) (hf : ContDiff  $\mathbb{R} \infty$  f)
  (hcontVar :  $\forall \eta$  : AdmissibleVariation d (EuclideanSpace  $\mathbb{R}$  (Fin m)),
    Lagrangian.ContinuousCoordDerivAlongFamily L ( $\text{fun } s : \mathbb{R} \Rightarrow \text{variedField } f \eta s$ ))
  (hcoeff : Lagrangian.ContDiffCoordDerivAlongField L f)
  (hfin : AllVariationsHaveFiniteAction L f)
  (hcont : Continuous (eulerLagrangeOp L f)) :
IsCritical L f  $\leftrightarrow$  eulerLagrangeOp L f = 0

```

theorem isCritical_iff_eulerLagrange_zero_of_contDiff_and_regularityAt [\[source\]](#)

```

private theorem isCritical_iff_eulerLagrange_zero_of_contDiff_and_regularityAt
  (L : Lagrangian d m k) (f : Space d  $\rightarrow$  EuclideanSpace  $\mathbb{R}$  (Fin m)) (hf : ContDiff  $\mathbb{R} \infty$  f)
  (hreg : HasEulerLagrangeRegularityAt L f)
  (hfin : AllVariationsHaveFiniteAction L f)
  (hcont : Continuous (eulerLagrangeOp L f)) :
IsCritical L f  $\leftrightarrow$  eulerLagrangeOp L f = 0

```

theorem isCritical_iff_eulerLagrange_zero_of_contDiff_and_smoothRegularity [\[source\]](#)

```

private theorem isCritical_iff_eulerLagrange_zero_of_contDiff_and_smoothRegularity

```

```

(L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m)) (hf : ContDiff ℝ ∞ f)
(hreg : HasSmoothEulerLagrangeRegularity L)
(hfin : AllVariationsHaveFiniteAction L f)
(hcont : Continuous (eulerLagrangeOp L f)) :
IsCritical L f ↔ eulerLagrangeOp L f = 0

```

theorem isCritical_iff_eulerLagrange_zero_of_contDiff_and_coordinateRegularity [\[source\]](#)

```

private theorem isCritical_iff_eulerLagrange_zero_of_contDiff_and_coordinateRegularity
(L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m)) (hf : ContDiff ℝ ∞ f)
(hcoord : Lagrangian.ContDiffCoordDerivInCoordinates L)
(hfin : AllVariationsHaveFiniteAction L f)
(hcont : Continuous (eulerLagrangeOp L f)) :
IsCritical L f ↔ eulerLagrangeOp L f = 0

```

theorem isCritical_iff_eulerLagrange_zero_of_contDiff_and_regularityAt' [\[source\]](#)

```

private theorem isCritical_iff_eulerLagrange_zero_of_contDiff_and_regularityAt'
(L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m)) (hf : ContDiff ℝ ∞ f)
(hreg : HasEulerLagrangeRegularityAt L f)
(hfin : AllVariationsHaveFiniteAction L f) :
IsCritical L f ↔ eulerLagrangeOp L f = 0

```

theorem isCritical_iff_eulerLagrange_zero_of_contDiff_and_smoothRegularity' [\[source\]](#)

```

private theorem isCritical_iff_eulerLagrange_zero_of_contDiff_and_smoothRegularity'
(L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m)) (hf : ContDiff ℝ ∞ f)
(hreg : HasSmoothEulerLagrangeRegularity L)
(hfin : AllVariationsHaveFiniteAction L f) :
IsCritical L f ↔ eulerLagrangeOp L f = 0

```

theorem isCritical_iff_eulerLagrange_zero_of_contDiff_and_coordinateRegularity' [\[source\]](#)

```

private theorem isCritical_iff_eulerLagrange_zero_of_contDiff_and_coordinateRegularity'
(L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m)) (hf : ContDiff ℝ ∞ f)
(hcoord : Lagrangian.ContDiffCoordDerivInCoordinates L)
(hfin : AllVariationsHaveFiniteAction L f) :
IsCritical L f ↔ eulerLagrangeOp L f = 0

```

theorem «theorem@4a640fdf» [\[source\]](#)

```

private theorem
isCritical_iff_eulerLagrange_zero_of_contDiff_and_coordinateRegularity_of_hasFiniteAction
(L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m)) (hf : ContDiff ℝ ∞ f)
(hcoord : Lagrangian.ContDiffCoordDerivInCoordinates L)
(hbase : HasFiniteAction L f)
(hlocal :
  ∀ η : AdmissibleVariation d (EuclideanSpace ℝ (Fin m)),
    HasCompactlySupportedActionVariationDifference L f η) :
IsCritical L f ↔ eulerLagrangeOp L f = 0

```

theorem isCritical_iff_eulerLagrange_zero_of_hasFiniteAction_and_continuousInCoordinates [\[source\]](#)

```
theorem isCritical_iff_eulerLagrange_zero_of_hasFiniteAction_and_continuousInCoordinates
  (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m)) (hf : ContDiff ℝ ∞ f)
  (hcoord : Lagrangian.ContDiffCoordDerivInCoordinates L)
  (hcontL : Lagrangian.ContinuousInCoordinates L)
  (hbase : HasFiniteAction L f) :
  IsCritical L f ↔ eulerLagrangeOp L f = 0
```

2.94 PhyslibAlpha.ClassicalFieldTheory.Local.FirstVariation.Density

[PhyslibAlpha/ClassicalFieldTheory/Local/FirstVariation/Density.lean](#) on GitHub.

def HasActionVariationDerivativeUnderIntegral

[\[source\]](#)

Differentiation of the varied action under the integral sign, packaged as a hypothesis.

```
def HasActionVariationDerivativeUnderIntegral (L : Lagrangian d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m)) :
  Prop
```

def HasPointwiseLinearizedDensityFormula

[\[source\]](#)

Pointwise linearization of the action density before integration by parts.

```
def HasPointwiseLinearizedDensityFormula (L : Lagrangian d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m)) : Prop
```

lemma actionVariation_hasDerivAt_of_dominated_loc_of_deriv_le

[\[source\]](#)

A direct bridge from Mathlib's dominated differentiation-under-the-integral theorem to the local action variation. This isolates the measure-theoretic input needed to prove HasActionVariationDerivativeUnderIntegral.

```
lemma actionVariation_hasDerivAt_of_dominated_loc_of_deriv_le
  (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m))
  (η : AdmissibleVariation d (EuclideanSpace ℝ (Fin m)))
  {F' : ℝ → Space d → ℝ} {bound : Space d → ℝ} {ε : ℝ} (hε : 0 < ε)
  (hmeas :
    ∀f s in nhds (0 : ℝ),
      AESTronglyMeasurable (actionDensity L (variedField f η s)) volume)
  (hfinite0 : Integrable (actionDensity L (variedField f η 0)))
  (hF'_meas : AESTronglyMeasurable (F' 0) volume)
  (hbound :
    ∀m x ∂volume, ∀ s ∈ Metric.ball (0 : ℝ) ε, ||F' s x|| ≤ bound x)
  (hbound_int : Integrable bound volume)
  (hderiv :
    ∀m x ∂volume, ∀ s ∈ Metric.ball (0 : ℝ) ε,
      HasDerivAt (fun r : ℝ => actionDensity L (variedField f η r) x) (F' s x) s) :
  HasDerivAt (actionVariation L f η) (∫ x, F' 0 x) 0
```

def HasDominatedVariationDerivativeNear

[\[source\]](#)

A dominated bound for the pointwise first variation along the varied-field family near $s = 0$. This packages the analytic domination needed to differentiate the action under the integral sign.

```
def HasDominatedVariationDerivativeNear (L : Lagrangian d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m))
  (η : AdmissibleVariation d (EuclideanSpace ℝ (Fin m))) : Prop
```

lemma continuous_coordDeriv_at_zero

[\[source\]](#)

```
private lemma continuous_coordDeriv_at_zero
  (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m))
  (η : AdmissibleVariation d (EuclideanSpace ℝ (Fin m)))
  (hcont : ∀ I : DerivativeIndex d k, ∀ a : Fin m,
    Continuous (fun p : ℝ × Space d =>
      L.coordDeriv I a (jetAt k (variedField f η p.1) p.2)))
  (I : DerivativeIndex d k) (a : Fin m) :
  Continuous (fun x : Space d => L.coordDeriv I a (jetAt k f x))
```

lemma firstVariationDensityTerm_norm_le_of_coeff_bound

[\[source\]](#)

```
private lemma firstVariationDensityTerm_norm_le_of_coeff_bound
  (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m))
  (η : AdmissibleVariation d (EuclideanSpace ℝ (Fin m)))
  (ψ : DerivativeIndex d k → Fin m → Space d → ℝ)
  (hψ : ∀ I : DerivativeIndex d k, ∀ a : Fin m, ∀ x : Space d,
    ψ I a x = ∂^[I.1] (fun y => (η y) a) x)
  (C : DerivativeIndex d k → Fin m → ℝ)
  (K : DerivativeIndex d k → Fin m → Set (Space d))
  (hKzero : ∀ I : DerivativeIndex d k, ∀ a : Fin m, ∀ x, x ∉ K I a → ψ I a x = 0)
  (hC : ∀ I : DerivativeIndex d k, ∀ a : Fin m,
    ∀ p ∈ Metric.closedBall (0 : ℝ) 1 ×s K I a,
    ||L.coordDeriv I a (jetAt k (variedField f η p.1) p.2)|| ≤ C I a)
  {x : Space d} {s : ℝ} (hs : s ∈ Metric.ball (0 : ℝ) 1)
  (I : DerivativeIndex d k) (a : Fin m) :
  ||firstVariationDensityTerm L (variedField f η s) η I a x|| ≤ C I a * ||ψ I a x||
```

lemma norm_firstVariationDensity_le_bound

[\[source\]](#)

```
private lemma norm_firstVariationDensity_le_bound
  (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m))
  (η : AdmissibleVariation d (EuclideanSpace ℝ (Fin m)))
  (ψ : DerivativeIndex d k → Fin m → Space d → ℝ)
  (hψ : ∀ I : DerivativeIndex d k, ∀ a : Fin m, ∀ x : Space d,
    ψ I a x = ∂^[I.1] (fun y => (η y) a) x)
  (C : DerivativeIndex d k → Fin m → ℝ)
  (K : DerivativeIndex d k → Fin m → Set (Space d))
  (hKzero : ∀ I : DerivativeIndex d k, ∀ a : Fin m, ∀ x, x ∉ K I a → ψ I a x = 0)
  (hC : ∀ I : DerivativeIndex d k, ∀ a : Fin m,
    ∀ p ∈ Metric.closedBall (0 : ℝ) 1 ×s K I a,
    ||L.coordDeriv I a (jetAt k (variedField f η p.1) p.2)|| ≤ C I a)
  {x : Space d} {s : ℝ} (hs : s ∈ Metric.ball (0 : ℝ) 1) :
  ||firstVariationDensity L (variedField f η s) η x||
  ≤ ∑ I : DerivativeIndex d k, ∑ a : Fin m, C I a * ||ψ I a x||
```

lemma hasDominatedVariationDerivativeNear_of_continuous_coordDeriv

[\[source\]](#)

A concrete dominated bound for the varied first-variation density, obtained from

continuity of the jet-coordinate coefficient family in (s, x) . The compact support of the iterated derivatives of the admissible variation supplies the integrable majorant.

```
lemma hasDominatedVariationDerivativeNear_of_continuous_coordDeriv
  (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m))
  (η : AdmissibleVariation d (EuclideanSpace ℝ (Fin m)))
  (hcont : ∀ I : DerivativeIndex d k, ∀ a : Fin m,
    Continuous (fun p : ℝ × Space d =>
      L.coordDeriv I a (jetAt k (variedField f η p.1) p.2))) :
  HasDominatedVariationDerivativeNear L f η
```

lemma hasDerivAt_actionDensity_variedField_zero_of_contDiff

[\[source\]](#)

```
lemma hasDerivAt_actionDensity_variedField_zero_of_contDiff
  (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m)) (hf : ContDiff ℝ ∞ f) :
  ∀ η : AdmissibleVariation d (EuclideanSpace ℝ (Fin m)), ∀ x : Space d,
  HasDerivAt (fun s : ℝ => actionDensity L (variedField f η s) x)
    (firstVariationDensity L f η x) 0
```

lemma hasDerivAt_actionDensity_variedField_of_contDiff

[\[source\]](#)

```
lemma hasDerivAt_actionDensity_variedField_of_contDiff
  (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m)) (hf : ContDiff ℝ ∞ f)
  (η : AdmissibleVariation d (EuclideanSpace ℝ (Fin m))) (x : Space d) (s : ℝ) :
  HasDerivAt (fun r : ℝ => actionDensity L (variedField f η r) x)
    (firstVariationDensity L (variedField f η s) η x) s
```

lemma hasPointwiseLinearizedDensityFormula_of_contDiff

[\[source\]](#)

```
lemma hasPointwiseLinearizedDensityFormula_of_contDiff
  (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m)) (hf : ContDiff ℝ ∞ f) :
  HasPointwiseLinearizedDensityFormula L f
```

lemma hasActionVariationDerivativeUnderIntegral_of_contDiff_of_dominatedVariation [\[source\]](#)

```
lemma hasActionVariationDerivativeUnderIntegral_of_contDiff_of_dominatedVariation
  (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m)) (hf : ContDiff ℝ ∞ f)
  (hdom : ∀ η : AdmissibleVariation d (EuclideanSpace ℝ (Fin m)),
    HasFiniteActionVariation L f η →
    HasDominatedVariationDerivativeNear L f η) :
  HasActionVariationDerivativeUnderIntegral L f
```

lemma hasActionVariationDerivativeUnderIntegral_of_contDiff_of_continuous_coordDeriv [\[source\]](#)

```
lemma hasActionVariationDerivativeUnderIntegral_of_contDiff_of_continuous_coordDeriv
  (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m)) (hf : ContDiff ℝ ∞ f)
  (hcontVar : ∀ η : AdmissibleVariation d (EuclideanSpace ℝ (Fin m)),
    ∀ I : DerivativeIndex d k, ∀ a : Fin m,
    Continuous (fun p : ℝ × Space d =>
      L.coordDeriv I a (jetAt k (variedField f η p.1) p.2))) :
  HasActionVariationDerivativeUnderIntegral L f
```

lemma hasActionVariationDerivativeUnderIntegral_of_contDiff_of_regular

[\[source\]](#)


```

lemma hasActionVariationDerivativeUnderIntegral_of_contDiff_of_regular
  (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m)) (hf : ContDiff ℝ ∞ f)
  (hcontVar : ∀ η : AdmissibleVariation d (EuclideanSpace ℝ (Fin m)),
    Lagrangian.ContinuousCoordDerivAlongFamily L (fun s : ℝ => variedField f η s)) :
    HasActionVariationDerivativeUnderIntegral L f

```

2.95 PhyslibAlpha.ClassicalFieldTheory.Local.FirstVariation.IntegrationByParts

[PhyslibAlpha/ClassicalFieldTheory/Local/FirstVariation/IntegrationByParts.lean](#) on GitHub.

def HasIntegratedByPartsFormula

[\[source\]](#)

The integration-by-parts step sending the linearized density to the Euler-Lagrange pairing.

```

def HasIntegratedByPartsFormula (L : Lagrangian d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m)) : Prop

```

def HasTermwiseIntegratedByPartsFormula

[\[source\]](#)

Termwise integration-by-parts data for the linearized first-variation density.

```

def HasTermwiseIntegratedByPartsFormula (L : Lagrangian d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m)) : Prop

```

lemma integral_mul_space_deriv_eq_neg_deriv_mul

[\[source\]](#)

```

private lemma integral_mul_space_deriv_eq_neg_deriv_mul (i : Fin d) {g h : Space d → ℝ}
  (hg : ContDiff ℝ ∞ g) (hh : IsTestFunction h) :
  ∫ x, g x * ∂[i] h x = ∫ x, (-∂[i] g x) * h x

```

lemma integral_mul_iteratedDerivList_eq_sign

[\[source\]](#)

```

private lemma integral_mul_iteratedDerivList_eq_sign (L : List (Fin d)) {g h : Space d → ℝ}
  (hg : ContDiff ℝ ∞ g) (hh : IsTestFunction h) :
  ∫ x, g x * L.foldr (fun i f => ∂[i] f) h x =
  ∫ x, (((-1 : ℝ) ^ L.length) * L.foldr (fun i f => ∂[i] f) g x) * h x

```

lemma integral_mul_iteratedDeriv_eq_sign

[\[source\]](#)

Repeated integration by parts for iterated coordinate derivatives against a test function.

```

lemma integral_mul_iteratedDeriv_eq_sign {g h : Space d → ℝ} (I : MultiIndex d)
  (hg : ContDiff ℝ ∞ g) (hh : IsTestFunction h) :
  ∫ x, g x * ∂^[I] h x =
  ∫ x, (((-1 : ℝ) ^ I.order) * ∂^[I] g x) * h x

```

lemma hasTermwiseIntegratedByPartsFormula_of_contDiff_coordDeriv

[\[source\]](#)

Concrete termwise integration by parts for the first-variation density, assuming the jet coordinate coefficient functions are smooth along the field.

```

lemma hasTermwiseIntegratedByPartsFormula_of_contDiff_coordDeriv
  (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m))
  (hcoeff : ∀ I : DerivativeIndex d k, ∀ a : Fin m,
    ContDiff ℝ ∞ (fun x => L.coordDeriv I a (jetAt k f x))) :
  HasTermwiseIntegratedByPartsFormula L f

```

lemma hasTermwiseIntegratedByPartsFormula_of_regular

[\[source\]](#)

```

lemma hasTermwiseIntegratedByPartsFormula_of_regular
  (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m))
  (hcoeff : Lagrangian.ContDiffCoordDerivAlongField L f) :
  HasTermwiseIntegratedByPartsFormula L f

```

lemma integral_firstVariationDensity_eq_termwise_sum

[\[source\]](#)

```

private lemma integral_firstVariationDensity_eq_termwise_sum
  (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m))
  (η : AdmissibleVariation d (EuclideanSpace ℝ (Fin m)))
  (hterm : HasTermwiseIntegratedByPartsFormula L f) :
  ∫ x, firstVariationDensity L f η x =
    ∑ I : DerivativeIndex d k, ∑ a : Fin m,
      ∫ x, firstVariationDensityTerm L f η I a x

```

lemma integral_eulerLagrange_termwise_sum_eq_pairing

[\[source\]](#)

```

private lemma integral_eulerLagrange_termwise_sum_eq_pairing
  (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m))
  (η : AdmissibleVariation d (EuclideanSpace ℝ (Fin m)))
  (hterm : HasTermwiseIntegratedByPartsFormula L f) :
  ∑ I : DerivativeIndex d k, ∑ a : Fin m, ∫ x, eulerLagrangeTerm L I a f x * (η x) a
    = firstVariationValue L f η

```

lemma hasIntegratedByPartsFormula_of_termwise

[\[source\]](#)

```

lemma hasIntegratedByPartsFormula_of_termwise (L : Lagrangian d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m))
  (hterm : HasTermwiseIntegratedByPartsFormula L f) :
  HasIntegratedByPartsFormula L f

```

2.96 PhyslibAlpha.ClassicalFieldTheory.Local.FirstVariation.Regularity

[PhyslibAlpha/ClassicalFieldTheory/Local/FirstVariation/Regularity.lean](#) on GitHub.

def HasEulerLagrangeRegularityAt

[\[source\]](#)

The combined regularity package currently needed to derive the local Euler-Lagrange criterion for a field f : smooth coefficient functions along f , and continuity of the corresponding coefficient families along every admissible varied-field family based at f .

```

def HasEulerLagrangeRegularityAt (L : Lagrangian d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m)) : Prop

```

def HasSmoothEulerLagrangeRegularity[\[source\]](#)

A local Lagrangian has smooth Euler-Lagrange regularity if every smooth field satisfies the regularity package currently needed by the local first-variation proof.

```
def HasSmoothEulerLagrangeRegularity (L : Lagrangian d m k) : Prop
```

lemma continuous_eulerLagrangeTerm_of_regular[\[source\]](#)

```
lemma continuous_eulerLagrangeTerm_of_regular
  (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m))
  (hcoeff : Lagrangian.ContDiffCoordDerivAlongField L f)
  (I : DerivativeIndex d k) (a : Fin m) :
  Continuous (eulerLagrangeTerm L I a f)
```

lemma continuous_eulerLagrangeComponent_of_regular[\[source\]](#)

```
lemma continuous_eulerLagrangeComponent_of_regular
  (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m))
  (hcoeff : Lagrangian.ContDiffCoordDerivAlongField L f)
  (a : Fin m) :
  Continuous (eulerLagrangeComponent L a f)
```

lemma continuous_eulerLagrangeOp_of_regular[\[source\]](#)

```
lemma continuous_eulerLagrangeOp_of_regular
  (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m))
  (hcoeff : Lagrangian.ContDiffCoordDerivAlongField L f) :
  Continuous (eulerLagrangeOp L f)
```

theorem hasSmoothEulerLagrangeRegularity_of_contDiffCoordDerivInCoordinates[\[source\]](#)

```
theorem hasSmoothEulerLagrangeRegularity_of_contDiffCoordDerivInCoordinates
  (L : Lagrangian d m k) (hcoord : Lagrangian.ContDiffCoordDerivInCoordinates L) :
  HasSmoothEulerLagrangeRegularity L
```

2.97 PhyslibAlpha.ClassicalFieldTheory.Local.FirstVariation.Support

[PhyslibAlpha/ClassicalFieldTheory/Local/FirstVariation/Support.lean](#) on GitHub.

lemma variedField_zero[\[source\]](#)

```
@[simp]
lemma variedField_zero (f : Space d → EuclideanSpace ℝ (Fin m))
  (η : AdmissibleVariation d (EuclideanSpace ℝ (Fin m))) :
  variedField f η 0 = f
```

lemma variedField_variedField[\[source\]](#)

```
@[simp]
lemma variedField_variedField (f : Space d → EuclideanSpace ℝ (Fin m))
  (η : AdmissibleVariation d (EuclideanSpace ℝ (Fin m)))
  (s t : ℝ) :
  variedField (variedField f η s) η t = variedField f η (s + t)
```

lemma contDiff_space_deriv[\[source\]](#)

```
lemma contDiff_space_deriv {g : Space d → ℝ} (hg : ContDiff ℝ ∞ g)
  (i : Fin d) : ContDiff ℝ ∞ (∂[i] g)
```

lemma isTestFunction_space_deriv[\[source\]](#)

```
lemma isTestFunction_space_deriv {g : Space d → ℝ} (hg : IsTestFunction g) (i : Fin d) :
  IsTestFunction (∂[i] g)
```

lemma iteratedDerivList_contDiff[\[source\]](#)

```
lemma iteratedDerivList_contDiff (L : List (Fin d)) {g : Space d → ℝ}
  (hg : ContDiff ℝ ∞ g) :
  ContDiff ℝ ∞ (L.foldr (fun i h => ∂[i] h) g)
```

lemma iteratedDerivList_isTestFunction[\[source\]](#)

```
lemma iteratedDerivList_isTestFunction (L : List (Fin d)) {g : Space d → ℝ}
  (hg : IsTestFunction g) :
  IsTestFunction (L.foldr (fun i h => ∂[i] h) g)
```

lemma iteratedDerivList_commute_deriv[\[source\]](#)

```
lemma iteratedDerivList_commute_deriv (L : List (Fin d)) (i : Fin d)
  {g : Space d → ℝ} (hg : ContDiff ℝ ∞ g) :
  L.foldr (fun j h => ∂[j] h) (∂[i] g) = ∂[i] (L.foldr (fun j h => ∂[j] h) g)
```

lemma iteratedDeriv_coord_isTestFunction[\[source\]](#)

```
lemma iteratedDeriv_coord_isTestFunction (η : AdmissibleVariation d (EuclideanSpace ℝ (Fin m)))
  (I : DerivativeIndex d k) (a : Fin m) :
  IsTestFunction (fun x => ∂^[I.1] (fun y => (η y) a) x)
```

lemma firstVariationDensityTerm_integrable_of_continuous_coordDeriv[\[source\]](#)

```
lemma firstVariationDensityTerm_integrable_of_continuous_coordDeriv
  (L : Lagrangian d m k) (f : Space d → EuclideanSpace ℝ (Fin m))
  (η : AdmissibleVariation d (EuclideanSpace ℝ (Fin m)))
  (I : DerivativeIndex d k) (a : Fin m)
  (hcont : Continuous (fun x : Space d =>
    L.coordDeriv I a (jetAt k f x))) :
  Integrable (firstVariationDensityTerm L f η I a)
```

lemma continuous_jetBaseCoordinates_variedField[\[source\]](#)

```
lemma continuous_jetBaseCoordinates_variedField
  (k : ℕ) (f : Space d → EuclideanSpace ℝ (Fin m))
  (η : AdmissibleVariation d (EuclideanSpace ℝ (Fin m)))
  (hf : ContDiff ℝ ∞ f) :
  Continuous
    (fun p : ℝ × Space d => (p.2, jetCoordinatesAt k (variedField f η p.1) p.2))
```

2.98 PhyslibAlpha.ClassicalFieldTheory.Local.JetPoint

[PhyslibAlpha/ClassicalFieldTheory/Local/JetPoint.lean](#) on GitHub.

abbrev JetCoordinates

[\[source\]](#)

Coordinate data u^a_I for k -th order jet points.

For each derivative index I , these are the scalar components of the corresponding EuclideanSpace $\mathbb{R}$ (Fin m) fiber coordinate. We keep this componentwise form because the local Euler-Lagrange formulas are indexed by both I and a .

```
abbrev JetCoordinates (d m k : ℕ)
```

structure JetPoint

[\[source\]](#)

The coordinate-level points of the locally trivialized k -jet bundle for fields Space $d \rightarrow$ EuclideanSpace $\mathbb{R}$ (Fin m).

```
structure JetPoint (d m k : ℕ) where
  base : Space d
  fiber : JetCoordinates d m k
```

def coord

[\[source\]](#)

The coordinate u^a_I of a jet point.

```
def coord (J : JetPoint d m k) (I : DerivativeIndex d k) (a : Fin m) : ℝ
```

def value

[\[source\]](#)

The zero-th order value encoded by a jet point.

```
def value (J : JetPoint d m k) : EuclideanSpace ℝ (Fin m)
```

lemma value_apply

[\[source\]](#)

```
@[simp]
lemma value_apply (J : JetPoint d m k) (a : Fin m) :
  J.value a = J.coord 0 a
```

lemma ext

[\[source\]](#)

```
@[ext]
lemma ext (J K : JetPoint d m k) (hbase : J.base = K.base)
  (hcoord : ∀ I a, J.coord I a = K.coord I a) :
  J = K
```

def ofBaseCoordinates

[\[source\]](#)

Build a jet point from a base point and its jet coordinates.

```
def ofBaseCoordinates (x : Space d) (u : JetCoordinates d m k) : JetPoint d m k where
```

def toBaseCoordinates[\[source\]](#)*The base point and jet coordinates of a jet point.*

```
def toBaseCoordinates (J : JetPoint d m k) : Space d × JetCoordinates d m k
```

lemma ofBaseCoordinates_base[\[source\]](#)

```
@[simp]
```

```
lemma ofBaseCoordinates_base (x : Space d) (u : JetCoordinates d m k) :  
  (ofBaseCoordinates x u).base = x
```

lemma ofBaseCoordinates_fiber[\[source\]](#)

```
@[simp]
```

```
lemma ofBaseCoordinates_fiber (x : Space d) (u : JetCoordinates d m k) :  
  (ofBaseCoordinates x u).fiber = u
```

lemma ofBaseCoordinates_coord[\[source\]](#)

```
@[simp]
```

```
lemma ofBaseCoordinates_coord (x : Space d) (u : JetCoordinates d m k)  
  (I : DerivativeIndex d k) (a : Fin m) :  
  (ofBaseCoordinates x u).coord I a = u I a
```

lemma ofBaseCoordinates_base_fiber[\[source\]](#)

```
@[simp]
```

```
lemma ofBaseCoordinates_base_fiber (J : JetPoint d m k) :  
  ofBaseCoordinates J.base J.fiber = J
```

lemma toBaseCoordinates_ofBaseCoordinates[\[source\]](#)

```
@[simp]
```

```
lemma toBaseCoordinates_ofBaseCoordinates (x : Space d) (u : JetCoordinates d m k) :  
  (ofBaseCoordinates x u).toBaseCoordinates = (x, u)
```

def jetCoordinatesAt[\[source\]](#)*The jet-coordinate function determined by a field g.*

```
noncomputable def jetCoordinatesAt (k : ℕ) (g : Space d → EuclideanSpace ℝ (Fin m))  
  (x : Space d) :  
  JetCoordinates d m k
```

def jetAt[\[source\]](#)*The coordinate-level k-jet point of a field f at the point x.*

```
noncomputable def jetAt (k : ℕ) (f : Space d → EuclideanSpace ℝ (Fin m)) (x : Space d) :  
  JetPoint d m k where
```

lemma jetAt_base[\[source\]](#)

```
@[simp]
lemma jetAt_base (k : ℕ) (f : Space d → EuclideanSpace ℝ (Fin m)) (x : Space d) :
  (jetAt k f x).base = x
```

lemma jetAt_value[\[source\]](#)

```
@[simp]
lemma jetAt_value (k : ℕ) (f : Space d → EuclideanSpace ℝ (Fin m)) (x : Space d) :
  (jetAt k f x).value = f x
```

lemma jetAt_coord[\[source\]](#)

```
@[simp]
lemma jetAt_coord (k : ℕ) (f : Space d → EuclideanSpace ℝ (Fin m)) (x : Space d)
  (I : DerivativeIndex d k) (a : Fin m) :
  (jetAt k f x).coord I a = ∂^[I.1] (fun y => (f y) a) x
```

lemma jetAt_coord_zero[\[source\]](#)

```
lemma jetAt_coord_zero (k : ℕ) (f : Space d → EuclideanSpace ℝ (Fin m)) (x : Space d)
  (a : Fin m) :
  (jetAt k f x).coord 0 a = (f x) a
```

lemma jetCoordinatesAt_eq[\[source\]](#)

```
@[simp]
lemma jetCoordinatesAt_eq (k : ℕ) (g : Space d → EuclideanSpace ℝ (Fin m)) (x : Space d)
  (I : DerivativeIndex d k) (a : Fin m) :
  jetCoordinatesAt k g x I a = ∂^[I.1] (fun y => (g y) a) x
```

lemma jetCoordinatesAt_zero[\[source\]](#)

```
lemma jetCoordinatesAt_zero (k : ℕ) (g : Space d → EuclideanSpace ℝ (Fin m))
  (x : Space d) (a : Fin m) :
  jetCoordinatesAt k g x 0 a = (g x) a
```

lemma jetAt_eq_ofBaseCoordinates[\[source\]](#)

```
lemma jetAt_eq_ofBaseCoordinates (k : ℕ) (f : Space d → EuclideanSpace ℝ (Fin m))
  (x : Space d) :
  jetAt k f x = JetPoint.ofBaseCoordinates x (jetCoordinatesAt k f x)
```

2.99 PhyslibAlpha.ClassicalFieldTheory.Local.JetPointFiber

[PhyslibAlpha/ClassicalFieldTheory/Local/JetPointFiber.lean](#) on GitHub.

structure JetFiberData[\[source\]](#)

Fiber-coordinate data for jet points of order k . This records only the coordinates u^a_I , not the base point in Space d .

```
structure JetFiberData (d m k : ℕ) where
  coord : JetCoordinates d m k
```

```
instance CoeFun (JetFiberData d m k) (fun _ => DerivativeIndex d k → Fin m → ℝ) \[source\]
```

```
instance : CoeFun (JetFiberData d m k) (fun _ => DerivativeIndex d k → Fin m → ℝ) where
```

```
lemma ext \[source\]
```

```
@[ext]
```

```
lemma ext (V W : JetFiberData d m k) (hcoord : ∀ I a, V.coord I a = W.coord I a) : V = W
```

```
def value \[source\]
```

The zero-th order component of a jet-fiber direction, corresponding to the field value.

```
def value (V : JetFiberData d m k) : EuclideanSpace ℝ (Fin m)
```

```
lemma value_apply \[source\]
```

```
@[simp]
```

```
lemma value_apply (V : JetFiberData d m k) (a : Fin m) :
  V.value a = V.coord 0 a
```

```
instance Zero (JetFiberData d m k) \[source\]
```

```
instance : Zero (JetFiberData d m k) where
```

```
instance Add (JetFiberData d m k) \[source\]
```

```
instance : Add (JetFiberData d m k) where
```

```
instance SMul ℝ (JetFiberData d m k) \[source\]
```

```
instance : SMul ℝ (JetFiberData d m k) where
```

```
lemma zero_coord \[source\]
```

```
@[simp]
```

```
lemma zero_coord (I : DerivativeIndex d k) (a : Fin m) :
  (0 : JetFiberData d m k).coord I a = 0
```

```
lemma add_coord \[source\]
```

```
@[simp]
```

```
lemma add_coord (V W : JetFiberData d m k) (I : DerivativeIndex d k) (a : Fin m) :
  (V + W).coord I a = V.coord I a + W.coord I a
```


lemma smul_coord[\[source\]](#)

```
@[simp]
lemma smul_coord (c : ℝ) (V : JetFiberData d m k) (I : DerivativeIndex d k) (a : Fin m) :
  (c • V).coord I a = c * V.coord I a
```

def addFiber[\[source\]](#)

Translate a jet point by a fiber-direction increment.

```
def addFiber (J : JetPoint d m k) (V : JetFiberData d m k) : JetPoint d m k where
```

def lineMap[\[source\]](#)

The affine line in jet space through J in the fiber direction V.

```
def lineMap (J : JetPoint d m k) (V : JetFiberData d m k) (s : ℝ) : JetPoint d m k
```

lemma addFiber_base[\[source\]](#)

```
@[simp]
lemma addFiber_base (J : JetPoint d m k) (V : JetFiberData d m k) :
  (J.addFiber V).base = J.base
```

lemma addFiber_value[\[source\]](#)

```
@[simp]
lemma addFiber_value (J : JetPoint d m k) (V : JetFiberData d m k) :
  (J.addFiber V).value = J.value + V.value
```

lemma addFiber_coord[\[source\]](#)

```
@[simp]
lemma addFiber_coord (J : JetPoint d m k) (V : JetFiberData d m k)
  (I : DerivativeIndex d k) (a : Fin m) :
  (J.addFiber V).coord I a = J.coord I a + V.coord I a
```

lemma lineMap_base[\[source\]](#)

```
@[simp]
lemma lineMap_base (J : JetPoint d m k) (V : JetFiberData d m k) (s : ℝ) :
  (J.lineMap V s).base = J.base
```

lemma lineMap_coord[\[source\]](#)

```
@[simp]
lemma lineMap_coord (J : JetPoint d m k) (V : JetFiberData d m k) (s : ℝ)
  (I : DerivativeIndex d k) (a : Fin m) :
  (J.lineMap V s).coord I a = J.coord I a + s * V.coord I a
```

def jetDirectionAt[\[source\]](#)

The jet-fiber direction determined by a field g at a point x .

```
noncomputable def jetDirectionAt (k : ℕ) (g : Space d → EuclideanSpace ℝ (Fin m))
  (x : Space d) :
  JetFiberData d m k where
```

lemma jetDirectionAt_coord[\[source\]](#)

```
@[simp]
lemma jetDirectionAt_coord (k : ℕ) (g : Space d → EuclideanSpace ℝ (Fin m)) (x : Space d)
  (I : DerivativeIndex d k) (a : Fin m) :
  (jetDirectionAt k g x).coord I a = ∂^[I.1] (fun y => (g y) a) x
```

lemma jetDirectionAt_coord_zero[\[source\]](#)

```
lemma jetDirectionAt_coord_zero (k : ℕ) (g : Space d → EuclideanSpace ℝ (Fin m))
  (x : Space d) (a : Fin m) :
  (jetDirectionAt k g x).coord 0 a = (g x) a
```

lemma jetCoordinatesAt_add_smul[\[source\]](#)

```
lemma jetCoordinatesAt_add_smul (k : ℕ) (f g : Space d → EuclideanSpace ℝ (Fin m))
  (x : Space d) (s : ℝ)
  (hf : ContDiff ℝ ∞ f) (hg : ContDiff ℝ ∞ g) :
  jetCoordinatesAt k (fun y => f y + s • g y) x =
  jetCoordinatesAt k f x + s • jetCoordinatesAt k g x
```

lemma jetAt_add_smul[\[source\]](#)

```
lemma jetAt_add_smul (k : ℕ) (f g : Space d → EuclideanSpace ℝ (Fin m))
  (x : Space d) (s : ℝ)
  (hf : ContDiff ℝ ∞ f) (hg : ContDiff ℝ ∞ g) :
  jetAt k (fun y => f y + s • g y) x = (jetAt k f x).lineMap (jetDirectionAt k g x) s
```

2.100 PhyslibAlpha.ClassicalFieldTheory.Local.JetPointRegularity

[PhyslibAlpha/ClassicalFieldTheory/Local/JetPointRegularity.lean](#) on GitHub.

lemma jetCoordinatesAt_contDiff[\[source\]](#)

Smoothness of the jet-coordinate map of a smooth field.

```
lemma jetCoordinatesAt_contDiff (k : ℕ) (g : Space d → EuclideanSpace ℝ (Fin m))
  (hg : ContDiff ℝ ∞ g) :
  ContDiff ℝ ∞ (jetCoordinatesAt k g)
```

lemma jetBaseCoordinates_contDiff[\[source\]](#)

Smoothness of the base-plus-coordinate jet map of a smooth field.

```
lemma jetBaseCoordinates_contDiff (k : ℕ) (g : Space d → EuclideanSpace ℝ (Fin m))
  (hg : ContDiff ℝ ∞ g) :
  ContDiff ℝ ∞ (fun x : Space d => (x, jetCoordinatesAt k g x))
```

lemma notMem_tsupport_coord[\[source\]](#)

```
private lemma notMem_tsupport_coord {g : Space d → EuclideanSpace ℝ (Fin m)} {x : Space d}
  (hx : x ∉ tsupport g) (a : Fin m) :
  x ∉ tsupport (fun y => (g y) a)
```

lemma jetCoordinatesAt_eq_zero_of_notMem_tsupport[\[source\]](#)

Outside the topological support of a field, all of its jet coordinates vanish.

```
lemma jetCoordinatesAt_eq_zero_of_notMem_tsupport (k : ℕ)
  (g : Space d → EuclideanSpace ℝ (Fin m)) {x : Space d}
  (hx : x ∉ tsupport g) :
  jetCoordinatesAt k g x = 0
```

lemma jetDirectionAt_eq_zero_of_notMem_tsupport[\[source\]](#)

Outside the topological support of a field, the corresponding jet-fiber direction vanishes.

```
lemma jetDirectionAt_eq_zero_of_notMem_tsupport (k : ℕ)
  (g : Space d → EuclideanSpace ℝ (Fin m)) {x : Space d}
  (hx : x ∉ tsupport g) :
  jetDirectionAt k g x = 0
```

2.101 PhyslibAlpha.ClassicalFieldTheory.Local.Lagrangian

[PhyslibAlpha/ClassicalFieldTheory/Local/Lagrangian.lean](#) on GitHub.

structure Lagrangian[\[source\]](#)

A local k -th order Lagrangian for fields $\text{Space } d \rightarrow \text{EuclideanSpace } \mathbb{R} \text{ (Fin } m \text{)}$.

```
structure Lagrangian (d m k : ℕ) where
  toFun : JetPoint d m k → ℝ
  coordDeriv : DerivativeIndex d k → Fin m → JetPoint d m k → ℝ
  hasDerivAlongFiber :
    ∀ J : JetPoint d m k, ∀ V : JetFiberData d m k,
    HasDerivAt (fun s : ℝ => toFun (J.lineMap V s))
      (∑ I : DerivativeIndex d k, ∑ a : Fin m, coordDeriv I a J * V.coord I a) 0
```

instance CoeFun (Lagrangian d m k) (fun _ => JetPoint d m k → ℝ)[\[source\]](#)

```
instance : CoeFun (Lagrangian d m k) (fun _ => JetPoint d m k → ℝ) where
```

def fiberDerivative[\[source\]](#)

The derivative of L along the fiber-jet direction V based at J .

```
noncomputable def fiberDerivative (L : Lagrangian d m k) (J : JetPoint d m k)
  (V : JetFiberData d m k) : ℝ
```

lemma hasDerivAt_lineMap[\[source\]](#)

```
lemma hasDerivAt_lineMap (L : Lagrangian d m k) (J : JetPoint d m k) (V : JetFiberData d m k) :
  HasDerivAt (fun s : ℝ => L (J.lineMap V s)) (L.fiberDerivative J V) 0
```

def ContinuousInCoordinates[\[source\]](#)

Continuity of a local lagrangian in the explicit local jet coordinates (x, u^a_I) .

```
def ContinuousInCoordinates (L : Lagrangian d m k) : Prop
```

def ContDiffCoordDerivInCoordinates[\[source\]](#)

Intrinsic smoothness of the jet-coordinate derivatives of a local Lagrangian in the explicit local jet coordinates (x, u^a_I) .

```
def ContDiffCoordDerivInCoordinates (L : Lagrangian d m k) : Prop
```

def SmoothInCoordinates[\[source\]](#)

Public regularity package for a smooth local lagrangian in explicit local jet coordinates: continuity of the lagrangian itself together with smoothness of all jet-coordinate derivatives.

```
def SmoothInCoordinates (L : Lagrangian d m k) : Prop
```

def ContDiffCoordDerivAlongField[\[source\]](#)

Smoothness of all coefficient functions $x \mapsto \partial L / \partial u^a_I (j^k f(x))$ along a field f .

```
def ContDiffCoordDerivAlongField (L : Lagrangian d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m)) : Prop
```

def ContinuousCoordDerivAlongFamily[\[source\]](#)

Continuity of the coefficient family $(s, x) \mapsto \partial L / \partial u^a_I (j^k(F s)(x))$ along a one-parameter family of fields $F : \mathbb{R} \rightarrow \text{Space } d \rightarrow \text{EuclideanSpace } \mathbb{R} \text{ (Fin } m)$.

```
def ContinuousCoordDerivAlongFamily (L : Lagrangian d m k)
  (F : ℝ → Space d → EuclideanSpace ℝ (Fin m)) : Prop
```

lemma continuousAlongField_of_inCoordinates[\[source\]](#)

```
lemma continuousAlongField_of_inCoordinates (L : Lagrangian d m k)
  (hcont : ContinuousInCoordinates L) (f : Space d → EuclideanSpace ℝ (Fin m))
  (hf : ContDiff ℝ ∞ f) :
  Continuous (fun x : Space d => L (jetAt k f x))
```

lemma contDiffCoordDerivAlongField_of_inCoordinates[\[source\]](#)

```
lemma contDiffCoordDerivAlongField_of_inCoordinates (L : Lagrangian d m k)
  (hcoord : ContDiffCoordDerivInCoordinates L) (f : Space d → EuclideanSpace ℝ (Fin m))
  (hf : ContDiff ℝ ∞ f) :
  ContDiffCoordDerivAlongField L f
```

lemma continuousCoordDerivAlongFamily_of_inCoordinates[\[source\]](#)

```

lemma continuousCoordDerivAlongFamily_of_inCoordinates (L : Lagrangian d m k)
  (hcoord : ContDiffCoordDerivInCoordinates L)
  (F :  $\mathbb{R} \rightarrow \text{Space } d \rightarrow \text{EuclideanSpace } \mathbb{R} \text{ (Fin } m)$ )
  (hjet : Continuous (fun p :  $\mathbb{R} \times \text{Space } d \Rightarrow (p.2, \text{jetCoordinatesAt } k \text{ (F p.1) p.2))$ ) :
  ContinuousCoordDerivAlongFamily L F

```

def alongField[\[source\]](#)

Evaluate a local Lagrangian along the k -jets of a field.

```

noncomputable def alongField (L : Lagrangian d m k)
  (f :  $\text{Space } d \rightarrow \text{EuclideanSpace } \mathbb{R} \text{ (Fin } m)$ ) :  $\text{Space } d \rightarrow \mathbb{R}$ 

```

lemma alongField_apply[\[source\]](#)

```

@[simp]
lemma alongField_apply (L : Lagrangian d m k)
  (f :  $\text{Space } d \rightarrow \text{EuclideanSpace } \mathbb{R} \text{ (Fin } m)$ ) (x :  $\text{Space } d$ ) :
  L.alongField f x = L (jetAt k f x)

```

2.102 PhyslibAlpha.ClassicalFieldTheory.Local.TotalDerivative

[PhyslibAlpha/ClassicalFieldTheory/Local/TotalDerivative.lean](#) on GitHub.

def evalOnJet[\[source\]](#)

Evaluate a local jet-dependent function along the k -jet of a field.

```

noncomputable def evalOnJet (k :  $\mathbb{N}$ ) (F :  $\text{JetPoint } d \text{ m } k \rightarrow \mathbb{R}$ )
  (f :  $\text{Space } d \rightarrow \text{EuclideanSpace } \mathbb{R} \text{ (Fin } m)$ ) :
   $\text{Space } d \rightarrow \mathbb{R}$ 

```

def totalDerivative[\[source\]](#)

The total derivative of a jet-dependent function in the coordinate direction i .

```

noncomputable def totalDerivative (k :  $\mathbb{N}$ ) (i : Fin d) (F :  $\text{JetPoint } d \text{ m } k \rightarrow \mathbb{R}$ )
  (f :  $\text{Space } d \rightarrow \text{EuclideanSpace } \mathbb{R} \text{ (Fin } m)$ ) :  $\text{Space } d \rightarrow \mathbb{R}$ 

```

def iteratedTotalDerivative[\[source\]](#)

The iterated total derivative indexed by a multi-index.

```

noncomputable def iteratedTotalDerivative (k :  $\mathbb{N}$ ) (I : MultiIndex d) (F :  $\text{JetPoint } d \text{ m } k \rightarrow \mathbb{R}$ )
  (f :  $\text{Space } d \rightarrow \text{EuclideanSpace } \mathbb{R} \text{ (Fin } m)$ ) :  $\text{Space } d \rightarrow \mathbb{R}$ 

```

lemma evalOnJet_apply[\[source\]](#)

```

@[simp]
lemma evalOnJet_apply (k :  $\mathbb{N}$ ) (F :  $\text{JetPoint } d \text{ m } k \rightarrow \mathbb{R}$ )
  (f :  $\text{Space } d \rightarrow \text{EuclideanSpace } \mathbb{R} \text{ (Fin } m)$ ) (x :  $\text{Space } d$ ) :
  evalOnJet k F f x = F (jetAt k f x)

```

lemma totalDerivative_eq[\[source\]](#)

```
@[simp]
lemma totalDerivative_eq (k : ℕ) (i : Fin d) (F : JetPoint d m k → ℝ)
  (f : Space d → EuclideanSpace ℝ (Fin m)) :
  totalDerivative k i F f = ∂[i] (fun x => F (jetAt k f x))
```

lemma iteratedTotalDerivative_eq[\[source\]](#)

```
@[simp]
lemma iteratedTotalDerivative_eq (k : ℕ) (I : MultiIndex d) (F : JetPoint d m k → ℝ)
  (f : Space d → EuclideanSpace ℝ (Fin m)) :
  iteratedTotalDerivative k I F f = ∂^[I] (fun x => F (jetAt k f x))
```

lemma iteratedTotalDerivative_zero[\[source\]](#)

```
lemma iteratedTotalDerivative_zero (k : ℕ) (F : JetPoint d m k → ℝ)
  (f : Space d → EuclideanSpace ℝ (Fin m)) :
  iteratedTotalDerivative k (0 : MultiIndex d) F f = evalOnJet k F f
```

lemma iteratedTotalDerivative_single[\[source\]](#)

```
lemma iteratedTotalDerivative_single (k : ℕ) (i : Fin d) (F : JetPoint d m k → ℝ)
  (f : Space d → EuclideanSpace ℝ (Fin m)) :
  iteratedTotalDerivative k (Physlib.MultiIndex.increment 0 i) F f =
    totalDerivative k i F f
```

2.103 PhyslibAlpha.ClassicalFieldTheory.Local.TotalDivergence

[PhyslibAlpha/ClassicalFieldTheory/Local/TotalDivergence.lean](#) on GitHub.

def HasTotalDivergenceDensity[\[source\]](#)

A lagrangian density has total-divergence form if, along every field, it is the sum of the total derivatives of a jet-dependent current. A current of order k gives a density of order $k + 1$.

```
def HasTotalDivergenceDensity (L : Lagrangian d m (k + 1))
  (current : Fin d → JetPoint d m k → ℝ) : Prop
```

def IsEulerLagrangeTrivial[\[source\]](#)

A local lagrangian is Euler-Lagrange-trivial if its Euler-Lagrange operator vanishes along every field.

```
def IsEulerLagrangeTrivial (L : Lagrangian d m k) : Prop
```

structure TotalDivergence[\[source\]](#)

A packaged local total divergence.

The current has order k , while the associated lagrangian has order $k + 1$, matching the coordinate expression $L = \sum_i \dot{B}_i \cdot D_i B_i$. The Euler-Lagrange-triviality field records the variationally trivial direction needed by later equivalence results.

```

structure TotalDivergence (d m k : ℕ) where
  lagrangian : Lagrangian d m (k + 1)
  current : Fin d → JetPoint d m k → ℝ
  hasTotalDivergenceDensity : HasTotalDivergenceDensity lagrangian current
  isEulerLagrangeTrivial : IsEulerLagrangeTrivial lagrangian

```

```

instance CoeFun (TotalDivergence d m k) (fun _ => Lagrangian d m (k + 1)) [source]

```

```

instance : CoeFun (TotalDivergence d m k) (fun _ => Lagrangian d m (k + 1)) where

```

```

lemma actionDensity_eq [source]

```

```

@[simp]
lemma actionDensity_eq (T : TotalDivergence d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m)) :
  actionDensity T.lagrangian f =
    fun x => ∑ i : Fin d, totalDerivative k i (T.current i) f x

```

```

lemma eulerLagrange0p_eq_zero [source]

```

```

lemma eulerLagrange0p_eq_zero (T : TotalDivergence d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m)) :
  eulerLagrange0p T.lagrangian f = 0

```

```

lemma isCritical [source]

```

```

lemma isCritical (T : TotalDivergence d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m))
  (hTf : IsAdmissibleForAction T.lagrangian f)
  (hT : Lagrangian.SmoothInCoordinates T.lagrangian) :
  IsCritical T.lagrangian f

```

2.104 PhyslibAlpha.ClassicalFieldTheory.Local.TotalDivergenceEquivalence

[PhyslibAlpha/ClassicalFieldTheory/Local/TotalDivergenceEquivalence.lean](#) on GitHub.

```

def HasTotalDivergenceDifference [source]

```

A lagrangian target differs from source by the density of a packaged total divergence.

The total divergence has current order k , hence its associated lagrangian and both compared lagrangians have order $k + 1$.

```

def HasTotalDivergenceDifference (source target : Lagrangian d m (k + 1))
  (T : TotalDivergence d m k) : Prop

```

```

def IsEulerLagrangeEquivalent [source]

```

Two local lagrangians are Euler-Lagrange equivalent when they determine the same local Euler-Lagrange operator along every field.

```

def IsEulerLagrangeEquivalent (source target : Lagrangian d m k) : Prop

```

lemma refl[\[source\]](#)

```
lemma refl (L : Lagrangian d m k) : IsEulerLagrangeEquivalent L L
```

lemma symm[\[source\]](#)

```
lemma symm {L1 L2 : Lagrangian d m k} (h : IsEulerLagrangeEquivalent L1 L2) :
  IsEulerLagrangeEquivalent L2 L1
```

lemma trans[\[source\]](#)

```
lemma trans {L1 L2 L3 : Lagrangian d m k}
  (h12 : IsEulerLagrangeEquivalent L1 L2) (h23 : IsEulerLagrangeEquivalent L2 L3) :
  IsEulerLagrangeEquivalent L1 L3
```

lemma IsEulerLagrangeEquivalent.eulerLagrangeOp_eq[\[source\]](#)

```
lemma eulerLagrangeOp_eq {source target : Lagrangian d m k}
  (h : IsEulerLagrangeEquivalent source target)
  (f : Space d → EuclideanSpace ℝ (Fin m)) :
  eulerLagrangeOp target f = eulerLagrangeOp source f
```

lemma IsEulerLagrangeEquivalent.eulerLagrangeOp_eq_zero_iff[\[source\]](#)

```
lemma eulerLagrangeOp_eq_zero_iff {source target : Lagrangian d m k}
  (h : IsEulerLagrangeEquivalent source target)
  (f : Space d → EuclideanSpace ℝ (Fin m)) :
  eulerLagrangeOp target f = 0 ↔ eulerLagrangeOp source f = 0
```

lemma IsEulerLagrangeEquivalent.isCritical_iff[\[source\]](#)

```
lemma isCritical_iff {source target : Lagrangian d m k}
  (h : IsEulerLagrangeEquivalent source target)
  (f : Space d → EuclideanSpace ℝ (Fin m))
  (hsourcef : IsAdmissibleForAction source f)
  (htargetf : IsAdmissibleForAction target f)
  (hsource : Lagrangian.SmoothInCoordinates source)
  (htarget : Lagrangian.SmoothInCoordinates target) :
  IsCritical target f ↔ IsCritical source f
```

lemma IsEulerLagrangeEquivalent.isCritical_target_of_source[\[source\]](#)

```
lemma isCritical_target_of_source {source target : Lagrangian d m k}
  (h : IsEulerLagrangeEquivalent source target)
  (f : Space d → EuclideanSpace ℝ (Fin m))
  (hsourcef : IsAdmissibleForAction source f)
  (htargetf : IsAdmissibleForAction target f)
  (hsource : Lagrangian.SmoothInCoordinates source)
  (htarget : Lagrangian.SmoothInCoordinates target)
  (hcrit : IsCritical source f) :
  IsCritical target f
```


lemma IsEulerLagrangeEquivalent.isCritical_source_of_target[\[source\]](#)

```

lemma isCritical_source_of_target {source target : Lagrangian d m k}
  (h : IsEulerLagrangeEquivalent source target)
  (f : Space d → EuclideanSpace ℝ (Fin m))
  (hsourcef : IsAdmissibleForAction source f)
  (htargetf : IsAdmissibleForAction target f)
  (hsource : Lagrangian.SmoothInCoordinates source)
  (htarget : Lagrangian.SmoothInCoordinates target)
  (hcrit : IsCritical target f) :
  IsCritical source f

```

structure TotalDivergenceEquivalence[\[source\]](#)

A packaged equivalence between two local lagrangians whose densities differ by a total divergence.

The field `sameEulerLagrangeOp` records the variational consequence in the current Alpha API. It will become a theorem once the stack has enough symbolic calculus for coordinate derivatives of total divergences.

```

structure TotalDivergenceEquivalence (d m k : ℕ) where
  source : Lagrangian d m (k + 1)
  target : Lagrangian d m (k + 1)
  divergence : TotalDivergence d m k
  hasTotalDivergenceDifference :
    HasTotalDivergenceDifference source target divergence
  sameEulerLagrangeOp : IsEulerLagrangeEquivalent source target

```

lemma actionDensity_eq[\[source\]](#)

```

@[simp]
lemma actionDensity_eq (E : TotalDivergenceEquivalence d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m)) :
  actionDensity E.target f =
    fun x => actionDensity E.source f x + actionDensity E.divergence.lagrangian f x

```

lemma TotalDivergenceEquivalence.eulerLagrangeOp_eq[\[source\]](#)

```

lemma eulerLagrangeOp_eq (E : TotalDivergenceEquivalence d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m)) :
  eulerLagrangeOp E.target f = eulerLagrangeOp E.source f

```

lemma TotalDivergenceEquivalence.eulerLagrangeOp_eq_zero_iff[\[source\]](#)

```

lemma eulerLagrangeOp_eq_zero_iff (E : TotalDivergenceEquivalence d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m)) :
  eulerLagrangeOp E.target f = 0 ↔ eulerLagrangeOp E.source f = 0

```

lemma TotalDivergenceEquivalence.isCritical_iff[\[source\]](#)

```

lemma isCritical_iff (E : TotalDivergenceEquivalence d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m))
  (hsourcef : IsAdmissibleForAction E.source f)
  (htargetf : IsAdmissibleForAction E.target f)

```

```
(hsource : Lagrangian.SmoothInCoordinates E.source)
(htarget : Lagrangian.SmoothInCoordinates E.target) :
IsCritical E.target f ↔ IsCritical E.source f
```

lemma TotalDivergenceEquivalence.isCritical_target_of_source

[\[source\]](#)

```
lemma isCritical_target_of_source (E : TotalDivergenceEquivalence d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m))
  (hsourcef : IsAdmissibleForAction E.source f)
  (htargetf : IsAdmissibleForAction E.target f)
  (hsource : Lagrangian.SmoothInCoordinates E.source)
  (htarget : Lagrangian.SmoothInCoordinates E.target)
  (hcrit : IsCritical E.source f) :
  IsCritical E.target f
```

lemma TotalDivergenceEquivalence.isCritical_source_of_target

[\[source\]](#)

```
lemma isCritical_source_of_target (E : TotalDivergenceEquivalence d m k)
  (f : Space d → EuclideanSpace ℝ (Fin m))
  (hsourcef : IsAdmissibleForAction E.source f)
  (htargetf : IsAdmissibleForAction E.target f)
  (hsource : Lagrangian.SmoothInCoordinates E.source)
  (htarget : Lagrangian.SmoothInCoordinates E.target)
  (hcrit : IsCritical E.target f) :
  IsCritical E.source f
```

2.105 PhyslibAlpha.ClassicalFieldTheory.Local.Variation

[PhyslibAlpha/ClassicalFieldTheory/Local/Variation.lean](#) on GitHub.

lemma coord_euclidean

[\[source\]](#)

A Euclidean component of an admissible variation is again a test function.

```
@[fun_prop]
lemma coord_euclidean (η : AdmissibleVariation d (EuclideanSpace ℝ (Fin m))) (a : Fin m) :
  IsTestFunction (fun x => (η.toFun x) a)
```

2.106 PhyslibAlpha.ClassicalMechanics.CoupledSpringPotential

[PhyslibAlpha/ClassicalMechanics/CoupledSpringPotential.lean](#) on GitHub.

def couplingPotential

[\[source\]](#)

The potential energy of a pair of coupled springs.

```
noncomputable def couplingPotential (x : EuclideanSpace ℝ (Fin 2)) : ℝ
```

lemma couplingPotential_analyticAt

[\[source\]](#)

```
lemma couplingPotential_analyticAt (z : EuclideanSpace ℝ (Fin 2)) :
  AnalyticAt ℝ couplingPotential z
```

lemma couplingPotential_hasFDerivAt[\[source\]](#)

```

lemma couplingPotential_hasFDerivAt (x : EuclideanSpace ℝ (Fin 2)) :
  HasFDerivAt couplingPotential
    ((2 * x 0 + x 1) • (EuclideanSpace.proj (ℓk := ℝ) 0)
     + (x 0 + 2 * x 1) • (EuclideanSpace.proj (ℓk := ℝ) 1)) x

```

lemma couplingPotential_gradient_zero[\[source\]](#)

```

lemma couplingPotential_gradient_zero :
  gradient couplingPotential (!z[0, 0] : EuclideanSpace ℝ (Fin 2)) = 0

```

lemma couplingPotential_iteratedFDeriv_two[\[source\]](#)

```

lemma couplingPotential_iteratedFDeriv_two (z : EuclideanSpace ℝ (Fin 2))
  (a b : EuclideanSpace ℝ (Fin 2)) :
  iteratedFDeriv ℝ 2 couplingPotential z ![a, b]
    = 2 * a 0 * b 0 + a 0 * b 1 + a 1 * b 0 + 2 * a 1 * b 1

```

lemma couplingPotential_posDef[\[source\]](#)

```

lemma couplingPotential_posDef :
  (iteratedFDerivQuadraticMap couplingPotential
    (!z[0, 0] : EuclideanSpace ℝ (Fin 2))).PosDef

```

lemma coupled_spring_potential[\[source\]](#)

```

lemma coupled_spring_potential :
  let U

```

2.107 PhyslibAlpha.Mathematics.PartialDerivativeTest

[PhyslibAlpha/Mathematics/PartialDerivativeTest.lean](#) on GitHub.

lemma update₀[\[source\]](#)

Update a vector of length 2 in coordinate 0.

```

@[simp]
lemma Function.update₀ {α : Type*} {a b c : α} : Function.update ![a,b] 0 c = ![c,b]

```

lemma update₁[\[source\]](#)

Update a vector of length 2 in coordinate 1.

```

@[simp]
lemma Function.update₁ {α : Type*} {a b c : α} : Function.update ![a,b] 1 c = ![a,c]

```

def hessianBilinearCompanion[\[source\]](#)

The Hessian companion as a bilinear map.

```

noncomputable def hessianBilinearCompanion {V : Type*} [NormedAddCommGroup V]
  [NormedSpace ℝ V] (f : V → ℝ) (x₀ : V) : V →ₗ[ℝ] V →ₗ[ℝ] ℝ

```

def iteratedFDerivQuadraticMap[\[source\]](#)

The second iterated Frechét derivative as a quadratic map.

```
noncomputable def iteratedFDerivQuadraticMap {V : Type*} [NormedAddCommGroup V]
  [NormedSpace ℝ V] (f : V → ℝ) (x₀ : V) : QuadraticMap ℝ V ℝ
```

def continuousBilinearMapOfContinuousMultilinearMap[\[source\]](#)

A continuous multilinear map is bilinear.

```
noncomputable def continuousBilinearMapOfContinuousMultilinearMap
  {ℓ : Type*} [NontriviallyNormedField ℓ] [CompleteSpace ℓ]
  {V : Type*}
  [NormedAddCommGroup V] [NormedSpace ℓ V] [FiniteDimensional ℓ V]
  (g : ContinuousMultilinearMap ℓ (fun _ : Fin 2 => V) ℓ) : V → L[ℓ] V → L[ℓ] ℓ
```

def toMultilinearMap[\[source\]](#)

```
def QuadraticMap.toMultilinearMap {V : Type*} [AddCommGroup V] [Module ℝ V]
  (Q : QuadraticMap ℝ V ℝ) : MultilinearMap ℝ (fun _ : Fin 2 => V) ℝ
```

def toMultilinearMapHalfPolarBilin[\[source\]](#)

```
noncomputable def QuadraticMap.toMultilinearMapHalfPolarBilin
  {V : Type*} [AddCommGroup V] [Module ℝ V]
  (Q : QuadraticMap ℝ V ℝ) :
  MultilinearMap ℝ (fun _ : Fin 2 => V) ℝ
```

theorem toMultilinearMap_continuous[\[source\]](#)

The multilinear map associated to a quadratic map is continuous.

```
theorem QuadraticMap.toMultilinearMap_continuous {V : Type*}
  [NormedAddCommGroup V] [NormedSpace ℝ V]
  [FiniteDimensional ℝ V] (Q : QuadraticMap ℝ V ℝ) : Continuous Q.toMultilinearMap
```

theorem toMultilinearMapHalfPolarBilin_continuous[\[source\]](#)

```
theorem QuadraticMap.toMultilinearMapHalfPolarBilin_continuous {V : Type*}
  [NormedAddCommGroup V] [NormedSpace ℝ V]
  [FiniteDimensional ℝ V] (Q : QuadraticMap ℝ V ℝ) :
  Continuous Q.toMultilinearMapHalfPolarBilin
```

def toContinuousMultilinearMap[\[source\]](#)

Construct a continuous multilinear map from a quadratic map on a finite dimensional space.

```
def QuadraticMap.toContinuousMultilinearMap {V : Type*} [NormedAddCommGroup V] [NormedSpace ℝ V]
  [FiniteDimensional ℝ V] (Q : QuadraticMap ℝ V ℝ) :
  ContinuousMultilinearMap ℝ (fun _ : Fin 2 → V) ℝ
```

def toContinuousMultilinearMapHalfPolarBilin

[\[source\]](#)

```
noncomputable def QuadraticMap.toContinuousMultilinearMapHalfPolarBilin {V : Type*}
  [NormedAddCommGroup V] [NormedSpace ℝ V] [FiniteDimensional ℝ V] (Q : QuadraticMap ℝ V ℝ) :
  ContinuousMultilinearMap ℝ (fun _ : Fin 2 → V) ℝ
```

theorem toContinuousMultilinearMap_apply

[\[source\]](#)

The constructed continuous multilinear map agrees with the polar bilinear form.

```
theorem QuadraticMap.toContinuousMultilinearMap_apply {V : Type*} [NormedAddCommGroup V]
  [NormedSpace ℝ V] [FiniteDimensional ℝ V] (Q : QuadraticMap ℝ V ℝ) (x y : V) :
  Q.toContinuousMultilinearMap ![x, y] = Q.polarBilin x y
```

theorem toContinuousMultilinearMap_applyHalf

[\[source\]](#)

```
theorem QuadraticMap.toContinuousMultilinearMap_applyHalf {V : Type*} [NormedAddCommGroup V]
  [NormedSpace ℝ V] [FiniteDimensional ℝ V] (Q : QuadraticMap ℝ V ℝ) (x y : V) :
  Q.toContinuousMultilinearMapHalfPolarBilin ![x, y] = (1/2) * Q.polarBilin x y
```

lemma coercive_of_posdefHalf

[\[source\]](#)

```
lemma coercive_of_posdefHalf {V : Type*} [NormedAddCommGroup V] [NormedSpace ℝ V]
  [FiniteDimensional ℝ V] {F : QuadraticMap ℝ V ℝ}
  (hf' : F.PosDef) :
  IsCoercive (continuousBilinearMapOfContinuousMultilinearMap
    F.toContinuousMultilinearMapHalfPolarBilin)
```

lemma polarBilin_iteratedFDerivQuadraticMap

[\[source\]](#)

The polar bilinear form of the second-derivative quadratic map is the symmetrized Hessian.

```
lemma polarBilin_iteratedFDerivQuadraticMap {V : Type*} [NormedAddCommGroup V] [NormedSpace ℝ V]
  (f : V → ℝ) (x₀ : V) (x y : V) :
  (iteratedFDerivQuadraticMap f x₀).polarBilin x y
  = iteratedFDeriv ℝ 2 f x₀ ![x, y] + iteratedFDeriv ℝ 2 f x₀ ![y, x]
```

lemma iteratedFDeriv_two_swap

[\[source\]](#)

Symmetry of the second iterated Fréchet derivative (Schwarz's theorem).

```

lemma iteratedFDeriv_two_swap {V : Type*} [NormedAddCommGroup V] [NormedSpace ℝ V]
  {f : V → ℝ} {x₀ : V} (hf : ContDiffAt ℝ 1 f x₀) (x y : V) :
  iteratedFDeriv ℝ 2 f x₀ ![y, x] = iteratedFDeriv ℝ 2 f x₀ ![x, y]

```

lemma coercive_of_posdef_of_contdiff

[\[source\]](#)

Positive definiteness implies coercivity. The proof uses the general fact `coercive_of_posdefHalf` but requires a differentiability assumption on f .

```

lemma coercive_of_posdef_of_contdiff {V : Type*}
  [NormedAddCommGroup V] [NormedSpace ℝ V]
  [FiniteDimensional ℝ V] {f : V → ℝ} {x₀ : V}
  (hf : ContDiffAt ℝ 1 f x₀)
  (hf' : (iteratedFDerivQuadraticMap f x₀).PosDef) :
  IsCoercive (continuousBilinearMapOfContinuousMultilinearMap
    (iteratedFDeriv ℝ 2 f x₀))

```

lemma coercive_of_posdef

[\[source\]](#)

Positive definiteness implies coercivity.

```

lemma coercive_of_posdef {V : Type*} [NormedAddCommGroup V] [NormedSpace ℝ V]
  [FiniteDimensional ℝ V] {f : V → ℝ} {x₀ : V}
  (hf' : (iteratedFDerivQuadraticMap f x₀).PosDef) :
  IsCoercive (continuousBilinearMapOfContinuousMultilinearMap
    (iteratedFDeriv ℝ 2 f x₀))

```

theorem le_of_little0

[\[source\]](#)

```

theorem le_of_little0 {V : Type*}
  [NormedAddCommGroup V] [InnerProductSpace ℝ V]
  {f : V → ℝ} {x₀ x : V} {C : ℝ}
  (hx : C * (||x - x₀|| ^ 2) ≤ iteratedFDeriv ℝ 2 f x₀ fun _ => x - x₀)
  (hx₀ : fderiv ℝ f x₀ x = fderiv ℝ f x₀ x₀)
  (h₁ : ||f x - ∑ i ∈ range 3, 1 / i! * iteratedFDeriv ℝ i f x₀ fun _ => x - x₀||
    ≤ C / 2 * ||x - x₀|| ^ 2) :
  f x₀ ≤ f x

```

theorem isLocalMin_of_posDef_of_little0

[\[source\]](#)

Second partial derivative test, "little oh" form.

```

theorem isLocalMin_of_posDef_of_little0 {V : Type*} [NormedAddCommGroup V]
  [InnerProductSpace ℝ V] [FiniteDimensional ℝ V] {f : V → ℝ} {x₀ : V}
  (h : (fun x => f x - ∑ i ∈ range 3, 1 / (i)! * iteratedFDeriv ℝ i f x₀ fun _ => x - x₀)
    =o[nhds x₀] fun x => ||x - x₀|| ^ 2) (h₀ : gradient f x₀ = 0) (hcd : ContDiffAt ℝ 1 f x₀)
  (hf : (iteratedFDerivQuadraticMap f x₀).PosDef) : IsLocalMin f x₀

```

lemma little0_of_powerseries

[\[source\]](#)

Having a power series implies quadratic approximation.

```

lemma little0_of_powerseries {V : Type*} [NormedAddCommGroup V] [NormedSpace ℝ V]
  {f : V → ℝ} {x₀ : V} {p : FormalMultilinearSeries ℝ V ℝ}
  {r : NNReal} (h : HasFPowerSeriesOnBall f p x₀ r) :
  (fun x => f x - p.partialSum 3 (x - x₀)) =o[nhds x₀] fun x => ||x - x₀|| ^ 2

```

lemma of_subsingleton

[\[source\]](#)

```

@[nontriviality]
lemma isLocalMin.of_subsingleton {V : Type*} [TopologicalSpace V]
  [Subsingleton V] {f : V → ℝ} {x₀ : V} : IsLocalMin f x₀

```

theorem second_derivative_test

[\[source\]](#)

The second partial derivative test.

```

theorem second_derivative_test {V : Type*} [NormedAddCommGroup V] [InnerProductSpace ℝ V]
  [FiniteDimensional ℝ V] {f : V → ℝ} {x₀ : V} {p : FormalMultilinearSeries ℝ V ℝ}
  (h₀ : gradient f x₀ = 0) {r : NNReal}
  (h₁ : HasFPowerSeriesOnBall f p x₀ r)
  (hf : (iteratedFDerivQuadraticMap f x₀).PosDef) : IsLocalMin f x₀

```

theorem second_derivative_test_analyticAt

[\[source\]](#)

A more convenient form of the second partial derivative test: it suffices that f is analytic at x_0 (so that an explicit power series need not be produced by hand).

```

theorem second_derivative_test_analyticAt {V : Type*} [NormedAddCommGroup V]
  [InnerProductSpace ℝ V] [FiniteDimensional ℝ V] {f : V → ℝ} {x₀ : V}
  (h₀ : gradient f x₀ = 0) (han : AnalyticAt ℝ f x₀)
  (hf : (iteratedFDerivQuadraticMap f x₀).PosDef) : IsLocalMin f x₀

```

2.108 PhyslibAlpha.QuantumMechanics.QuantumHarmonicOscillator

[PhyslibAlpha/QuantumMechanics/QuantumHarmonicOscillator.lean](#) on GitHub.

def a

[\[source\]](#)

Annihilation operator.

```

def a (x : ℕ → ℂ) : ℕ → ℂ

```

def aLin

[\[source\]](#)

```

def aLin : (ℕ → ℂ) →l[ℂ] (ℕ → ℂ)

```

def a_dag

[\[source\]](#)

Creation operator.

```

def a_dag (x : ℕ → ℂ) : ℕ → ℂ

```

def a_dagLin[\[source\]](#)

```
def a_dagLin : (ℕ → ℂ) →l[ℂ] (ℕ → ℂ)
```

def ε[\[source\]](#)

```
def ε (n : ℕ) (c : ℂ) : ℕ → ℂ
```

lemma a_dag_eq[\[source\]](#)

Verify that a_dag really is the transpose of a.

```
lemma a_dag_eq (i j : ℕ) : a_dag (ε i 1) j = star (a (ε j 1) i)
```

lemma verify_a[\[source\]](#)

Verify that $a |n+1\rangle = \sqrt{n+1} |n\rangle$

```
lemma verify_a (n : ℕ) :
  a (ε (n + 1) 1) =
    ε n (√(n + 1))
```

lemma verify_a_dag[\[source\]](#)

Verify that $a^\dagger |n\rangle = \sqrt{n+1} |n+1\rangle$.

```
lemma verify_a_dag (n : ℕ) :
  a_dag (ε n 1) = ε (n + 1) (√(n + 1))
```

lemma verify_a_dag_a[\[source\]](#)

```
lemma verify_a_dag_a (n : ℕ) (x : ℕ → ℂ) :
  a_dag (a x) n = n * x n
```

lemma verify_a_a_dag[\[source\]](#)

```
lemma verify_a_a_dag (n : ℕ) (x : ℕ → ℂ) :
  a (a_dag x) n = (n + 1) * x n
```

lemma commutation_relation[\[source\]](#)

```
lemma commutation_relation :
  a ∘ a_dag - a_dag ∘ a = id
```

def coherentState[\[source\]](#)

```
def coherentState (α : ℂ) : ℕ → ℂ
```

def probabilityOf[\[source\]](#)

```
def probabilityOf (n : ℕ) (α : ℂ) : NNReal
```


lemma probabilityOf_eq_poisson_C[\[source\]](#)*Coherent state has a Poisson distribution.*

```
lemma probabilityOf_eq_poisson_C (n : ℕ) (α : ℂ) :
  let Λ
```

lemma coherentState_only_eigenvector[\[source\]](#)*The only eigenvectors of a are the coherent states.*

```
lemma coherentState_only_eigenvector (α : ℂ) (v : ℕ → ℂ) :
  a v = α • v ↔
  v = (Complex.exp (1/||α|| ^ 2 / 2) * v 0) • coherentState α
```

lemma eigenvector_coherentState[\[source\]](#)

```
lemma eigenvector_coherentState (α : ℂ) :
  a (coherentState α) = α • coherentState α
```

lemma distinct_eigenvectors_a[\[source\]](#)*None of the coherentState eigenvectors are proportional.*

```
lemma distinct_eigenvectors_a (α β c : ℂ)
  (hc : coherentState α = c • coherentState β) : α = β
```

lemma a_dagalmost_eigenvector[\[source\]](#)*Formal eigenvectors for a_dag (not in $\ell^2(\mathbb{C})$) (fails at $n=0$) .*

```
lemma a_dagalmost_eigenvector {α : ℂ} (hα : α ≠ 0) {n : ℕ} (hn : n ≠ 0) :
  a_dag (fun n => √(n.factorial) / α^n) n =
  α • (fun n => √(n.factorial) / α^n) n
```

lemma a_dag_nullspace[\[source\]](#)

```
lemma a_dag_nullspace
  {v : ℕ → ℂ} (hv : a_dag v = 0) : v = 0
```

lemma no_a_dag_eigenvector[\[source\]](#)

```
lemma no_a_dag_eigenvector (α : ℂ) (v : ℕ → ℂ) :
  a_dag v = α • v ↔ v = 0
```

lemma commutationRelation[\[source\]](#)

```
lemma commutationRelation : {aLin, a_dagLin} = 1
```

def coherentState_ℓ2[\[source\]](#)*The coherent state belongs to $\ell^2(\mathbb{C})$.*

```
def coherentState_ℓ2 (α : ℂ) : lp (fun _ : ℕ => ℂ) 2
```

2.109 PhyslibAlpha.QuantumMechanics.StinespringDilation

[PhyslibAlpha/QuantumMechanics/StinespringDilation.lean](#) on GitHub.

def krausApply

[\[source\]](#)

Completely positive map given by a (not necessarily minimal) Kraus family.

```
def krausApply {R : Type*} [Mul R] [Star R] [AddCommMonoid R]
  {q r : Type*} [Fintype q] [Fintype r]
  (K : r → Matrix q q R) (ρ : Matrix q q R) : Matrix q q R
```

lemma posSemidef

[\[source\]](#)

Kraus operator preserves PSD property.

```
lemma krausApply.posSemidef {R : Type*} [Ring R] [PartialOrder R] [StarRing R]
  [AddLeftMono R]
  {q r : Type*} [Fintype q] [Fintype r]
  (K : r → Matrix q q R)
  {ρ : Matrix q q R} (hp : ρ.PosSemidef) :
  (krausApply K ρ).PosSemidef
```

def QuantumChannel

[\[source\]](#)

Quantum channel.

```
def QuantumChannel {R : Type*} [Mul R] [One R] [Star R] [AddCommMonoid R]
  {q r : Type*} [Fintype q] [Fintype r] [DecidableEq q]
  (K : r → Matrix q q R)
```

def QuantumOperation

[\[source\]](#)

Quantum operation.

```
def QuantumOperation {R : Type*} [RCLike R]
  {q r : Type*} [Fintype q] [Fintype r] [DecidableEq q]
  (K : r → Matrix q q R)
```

def densityMatrix

[\[source\]](#)

Density matrix.

```
def densityMatrix {R : Type*} [Ring R] [PartialOrder R] [StarRing R] (d : Type*) [Fintype d]
```

def convex_comb

[\[source\]](#)

Density matrices are closed under real convex combinations.

```
def densityMatrix.convex_comb {R : Type*} [RCLike R]
  {d : ℕ} (ρ₀ ρ₁ : densityMatrix (Fin d) (R := R)) {t : R}
  (hp₀ : 0 ≤ t) (hp₁ : 0 ≤ 1 - t) : densityMatrix (Fin d) (R := R)
```

def tr₂[\[source\]](#)*Also known as `partialTraceRight`.*

```
def tr2 {R : Type*} [Ring R] {m n m' : Type*} [Fintype n]
  (ρ : Matrix (m × n) (m' × n) R) : Matrix m m' R
```

def stinespringOp[\[source\]](#)*stinespringOp is often written as `V`.*

```
def stinespringOp {R : Type*} [Ring R]
  {m r : Type*} [Fintype r] [DecidableEq r]
  (K : r → Matrix m m R) : Matrix (m × r) m R
```

def stinespringDilation[\[source\]](#)*The Stinespring dilation.*

```
def stinespringDilation {R : Type*} [Ring R] [StarRing R]
  {m r : Type*} [Fintype r] [DecidableEq r] [Fintype m]
  (K : r → Matrix m m R)
  (ρ : Matrix m m R)
```

def stinespringForm[\[source\]](#)*The partial trace of the Stinespring dilation.*

```
def stinespringForm {R : Type*} [Ring R] [StarRing R]
  {m r : Type*} [Fintype r] [DecidableEq r] [Fintype m]
  (K : r → Matrix m m R)
```

lemma stinespringOp_adjoint_mul_self[\[source\]](#)*A useful identity for Stinespring dilations.*

```
lemma stinespringOp_adjoint_mul_self {R : Type*} [Ring R] [StarRing R]
  {m r : Type*} [Fintype r] [DecidableEq r] [Fintype m]
  (K : r → Matrix m m R) :
  ∑ i, star K i * K i = (stinespringOp K)H * stinespringOp K
```

lemma stinespringForm_CPTNI[\[source\]](#)*A useful identity for completely positive, trace non-increasing maps.*

```
lemma stinespringForm_CPTNI {R : Type*} [RCLike R]
  {m r : Type*} [Fintype r] [DecidableEq r] [Fintype m] [DecidableEq m]
  (K : r → Matrix m m R)
  (hK : ∑ i, (K i)H * K i ≤ 1) :
  (stinespringOp K)H * (stinespringOp K) ≤ 1
```

lemma stinespringForm_CPTP_isometry[\[source\]](#)*The Stinespring operator of a completely positive, trace-preserving maps is an isometry. (Note that `stinespringOp K` is not a square matrix in general.)*

```

lemma stinespringForm_CPTP_isometry {R : Type*} [Ring R] [StarRing R]
  {m r : Type*} [Fintype r] [DecidableEq r] [Fintype m] [DecidableEq m]
  {K : r → Matrix m m R}
  (hK :  $\sum i, (K i)^H * K i = 1$ ) :
  (stinespringOp K)H * (stinespringOp K) = 1

```

lemma stinespringOrtho

[\[source\]](#)

Proving the columns of $V = \text{stinespringOp } K$ are independent is a step on the way to constructing the unitary dilation.

```

lemma stinespringOrtho {R : Type*} [RCLike R]
  {m r : Type*} [Fintype r] [DecidableEq r] [Fintype m] [DecidableEq m]
  {K : r → Matrix m m R}
  (hK :  $\sum i, (K i)^H * K i = 1$ ) :
  Orthonormal ( $\mathbb{K} := R$ )
  fun j : m => WithLp.toLp 2 fun i : m × r => stinespringOp K i j

```

lemma basisCard

[\[source\]](#)

m will of course be finite and bounded by n here, but no need to assume or prove that.

```

lemma basisCard {R : Type*} [RCLike R] {n m : Type*} [Fintype n] {s : Matrix n m R}
  (ho : Orthonormal R fun j ↦ WithLp.toLp 2 fun i ↦ s i j) :
  Fintype.card n =
  ho.toSubtypeRange.exists_orthonormalBasis_extension.choose.card

```

lemma stinespringCard

[\[source\]](#)

Calculating the cardinality of the orthonormal basis obtained by extending the Stinespring orthonormal set, the columns of $\text{stinespringOp } K$.

```

lemma stinespringCard {R : Type*} [RCLike R]
  {m r : Type*} [Fintype r] [DecidableEq r] [Fintype m] [DecidableEq m]
  {K : r → Matrix m m R}
  (hK :  $\sum i, (K i)^H * K i = 1$ ) :
  Fintype.card (m × r) = (stinespringOrtho
  hK).toSubtypeRange.exists_orthonormalBasis_extension.choose.card

```

theorem complCard

[\[source\]](#)

We need the 1 matrix, which we don't seem to have for an arbitrary [Fintype m]. Since we are comparing $\text{Fin } r$ and $\text{Fin } (r-1)$ we also cannot too easily use an arbitrary [Fintype r] [Zero r].

```

theorem complCard {R : Type*} [RCLike R] {m r : ℕ}
  {K : Fin r → Matrix (Fin m) (Fin m) R}
  (hK :  $\sum i, (K i)^H * K i = 1$ ) (z : Fin r) :
  let 0

```

def predAbove_of_ne

[\[source\]](#)

See discussion at <https://leanprover.zulipchat.com/#narrow/channel/217875-Is-there-code-for-X.3F/topic/succAbove.20and.20predAbove.20lemmas/with/584270574>

```
def Fin.predAbove_of_ne {n : ℕ} {k i : Fin n}
  (h : i ≠ k) : Fin (n - 1)
```

lemma predAbove_of_ne_injective

[\[source\]](#)

A "missing lemma" for Fin types.

```
lemma Fin.predAbove_of_ne_injective (n : ℕ) (k x y : Fin n)
  (hx : x ≠ k) (hy : y ≠ k)
  (heq : Fin.predAbove_of_ne hx = Fin.predAbove_of_ne hy) : x = y
```

def onbPart

[\[source\]](#)

The way this is written, Fin r and Fin (r-1) both occur so it is tricky to go to a general Fintype.

```
def onbPart {R : Type*} [RCLike R]
  {m r : ℕ} {K : Fin r → Matrix (Fin m) (Fin m) R}
  (hK : ∑ i, (K i)H * K i = 1) (x : Fin m × Fin r) {z : Fin r} (hx : ¬x.2 = z) :
```

lemma onbPart_inner

[\[source\]](#)

```
lemma onbPart_inner {R : Type*} [RCLike R] {m r : ℕ} {K : Fin r → Matrix (Fin m) (Fin m) R}
  (hK : ∑ i, (K i)H * K i = 1) {z : Fin r}
  {y : Fin m × Fin r} (hy : ¬y.2 = z)
  {x : Fin m × Fin r} (hx : ¬x.2 = z)
  (h : y ≠ x) :
  inner R (WithLp.toLp 2 <| onbPart hK y hy)
    (WithLp.toLp 2 <| onbPart hK x hx) = 0
```

lemma onbPart_norm

[\[source\]](#)

The vectors in the Stinespring orthonormal basis have norm 1.

```
lemma onbPart_norm {R : Type*} [RCLike R] {m r : ℕ} {K : Fin r → Matrix (Fin m) (Fin m) R}
  (hK : ∑ i, (K i)H * K i = 1) (x : Fin m × Fin r)
  {z : Fin r} (hx : ¬x.2 = z) :
  ||WithLp.toLp 2 <| onbPart hK x hx|| = 1
```

def Ud

[\[source\]](#)

Also known as unitaryDilation. Respects x,y order.

```
def Ud {R : Type*} [RCLike R] {m r : ℕ}
  {K : Fin r → Matrix (Fin m) (Fin m) R}
  (hK : ∑ i, (K i)H * K i = 1) (z : Fin r) :
  Matrix (Fin m × Fin r) (Fin m × Fin r) R
```

def general_dilation

[\[source\]](#)

This generalization of Stinespring dilation has the right "shape" but otherwise nothing specific to it.

```
def general_dilation {R : Type*}
  {m r : Type*} [DecidableEq r]
  (z : r)
  (S : Matrix (m × r) m R)
  (M : Matrix (m × r) (m × r) R) :
  Matrix (m × r) (m × r) R
```

def dilation

[\[source\]](#)

A general, not necessarily unitary, dilation.

```
def dilation {R : Type*} [Ring R]
  {m r : Type*} [Fintype r] [DecidableEq r]
  (K : r → Matrix m m R) (z : r) (M : Matrix (m × r) (m × r) R) :
  Matrix (m × r) (m × r) R
```

theorem Ud_orthonormal₁

[\[source\]](#)

One version of orthonormality of stinespringOp.

```
theorem Ud_orthonormal1 {R : Type*} [RCLike R] {m r : ℕ} {K : Fin r → Matrix (Fin m) (Fin m) R}
  (hK : ∑ i, (K i)H * K i = 1) (z : Fin r) :
  Orthonormal R fun y ↦ if hy : y.2 = z then WithLp.toLp 2 fun i ↦ stinespringOp K i y.1
  else WithLp.toLp 2 fun i ↦ onbPart hK y hy i
```

theorem Ud_orthonormal₂

[\[source\]](#)

The Stinespring dilation columns form an orthonormal basis.

```
theorem Ud_orthonormal2 {R : Type*} [RCLike R]
  {m r : ℕ} {K : Fin r → Matrix (Fin m) (Fin m) R}
  (hK : ∑ i, (K i)H * K i = 1) (z : Fin r) :
  Orthonormal R fun y ↦
    WithLp.toLp 2 fun i ↦ if hy : y.2 = z then stinespringOp K i y.1 else onbPart hK y hy i
```

lemma smul_self_one_of_norm_one

[\[source\]](#)

If β has length 1 then the dot product of β with itself is 1.

```
lemma smul_self_one_of_norm_one {R : Type*} [RCLike R]
  {t : Type*} [Fintype t] {β : t → R} (hj : ||WithLp.toLp 2 β|| = 1) :
  ∑ x, (starRingEnd R) (β x) * β x = 1
```

theorem unitary_of_orthonormal

[\[source\]](#)

A matrix whose columns are orthonormal is unitary.

```
theorem unitary_of_orthonormal {R : Type*} [RCLike R]
  {m r : Type*} [Fintype r] [DecidableEq r] [Fintype m] [DecidableEq m]
  (α : Matrix (m × r) (m × r) R)
  (hα : Orthonormal R fun i ↦ WithLp.toLp 2 (α i)) : α * star α = 1
```

lemma Ud_unitaryT[\[source\]](#)

The transpose of the unitary dilation is unitary.

```
lemma Ud_unitaryT {R : Type*} [RCLike R]
  {m r : ℕ} {K : Fin r → Matrix (Fin m) (Fin m) R}
  (hK :  $\sum i, (K i)^H * K i = 1$ ) (z : Fin r) :
  (Ud hK z)T ∈ unitary _
```

lemma Ud_unitary[\[source\]](#)

The unitary dilation Ud is in fact unitary.

```
lemma Ud_unitary {R : Type*} [RCLike R]
  {m r : ℕ} {K : Fin r → Matrix (Fin m) (Fin m) R}
  (hK :  $\sum i, (K i)^H * K i = 1$ ) (z : Fin r) :
  (Ud hK z) ∈ unitary _
```

lemma tr2_e⊗Xe[\[source\]](#)

Taking the partial trace of a tensor product with a matrix of trace 1 is the identity map.

```
lemma tr2_e⊗Xe {R : Type*} [RCLike R]
  {m w : Type*} [Fintype w] [Zero w]
  (e : Matrix w w R) (htr : e.trace = 1)
  (ρ : Matrix m m R) :
  tr2 (ρ ⊗ e) = ρ
```

def stinespringUnitaryForm[\[source\]](#)

The Stinespring unitary form.

```
def stinespringUnitaryForm {R : Type*} [RCLike R] {m r : ℕ}
  {K : Fin r → Matrix (Fin m) (Fin m) R}
  (hK :  $\sum i, (K i)^H * K i = 1$ ) (z : Fin r)
  (ρ : Matrix (Fin m) (Fin m) R) :
  (Matrix (Fin m) (Fin m) R)
```

def stinespringUnitaryForm_e[\[source\]](#)

The Stinespring unitary form, general version.

```
def stinespringUnitaryForm_e {R : Type*} [RCLike R] {m r : ℕ}
  {K : Fin r → Matrix (Fin m) (Fin m) R}
  (hK :  $\sum i, (K i)^H * K i = 1$ ) (z : Fin r) (e : Matrix (Fin r) (Fin r) R)
  (ρ : Matrix (Fin m) (Fin m) R) :
  (Matrix (Fin m) (Fin m) R)
```

theorem tracefree_version[\[source\]](#)

Trace-free version of the Stinespring Dilation Theorem.

```
theorem tracefree_version {R : Type*} [RCLike R]
  {m r : Type*} [Fintype r] [DecidableEq r] [Zero r] [Fintype m] [DecidableEq m]
  {K : r → Matrix m m R}
```

```
(p : Matrix m m R) :
let K'
```

theorem heisenberg_schrödinger[\[source\]](#)

A Heisberg picture / Schrödinger picture view of the Stinespring dilation.

```
theorem heisenberg_schrödinger {R : Type*} [RCLike R]
  {m r : Type*} [Fintype r] [DecidableEq r] [Zero r] [Fintype m] [DecidableEq m]
  {K : r → Matrix m m R}
  (p : Matrix m m R) :
let K'
```

def generalForm[\[source\]](#)

A further generalization of stinespringGeneralForm.

```
def generalForm {R : Type*} [RCLike R]
  {m r : Type*} [Fintype r] [DecidableEq r] [Fintype m]
  (z : r)
  (S : Matrix (m × r) m R)
  (M : Matrix (m × r) (m × r) R)
```

def stinespringGeneralForm[\[source\]](#)

General form of the Stinespring dilation.

```
def stinespringGeneralForm {R : Type*} [RCLike R]
  {m r : Type*} [Fintype r] [DecidableEq r] [Fintype m]
  (K : r → Matrix m m R) (z : r)
  (M : Matrix (m × r) (m × r) R)
```

def stinespringGeneralForm_e[\[source\]](#)

Even more general form of the Stinespring dilation.

```
def stinespringGeneralForm_e {R : Type*} [RCLike R]
  {m r : Type*} [Fintype r] [DecidableEq r] [Fintype m]
  (K : r → Matrix m m R) (z : r) (e : Matrix r r R)
  (M : Matrix (m × r) (m × r) R)
```

theorem unitaryForm_of_general[\[source\]](#)

When we plug in $M = Ud \ hK$ into the general stinespringGeneralForm, then we do get stinespringUnitaryForm hK.

```
theorem unitaryForm_of_general {R : Type*} [RCLike R] {m r : ℕ}
  {K : Fin r → Matrix (Fin m) (Fin m) R}
  (hK :  $\sum i, (K i)^H * K i = 1$ ) (z : Fin r) :
  stinespringGeneralForm K z (Ud hK z) =
  stinespringUnitaryForm hK z
```

theorem unitaryForm_of_general_e[\[source\]](#)

The Stinespring unitary form as a general form applied to the unitary dilation.


```

theorem unitaryForm_of_general_e {R : Type*} [RCLike R] {m r : ℕ}
  {K : Fin r → Matrix (Fin m) (Fin m) R}
  (hK :  $\sum i, (K i)^H * K i = 1$ ) (z : Fin r) (e : Matrix (Fin r) (Fin r) R) :
  stinespringGeneralForm_e K z e (Ud hK z) =
  stinespringUnitaryForm_e hK z e

```

lemma stinespringGeneralForm_works

[\[source\]](#)

Note we don't need any special properties of M , and we don't need K to be CPTP.

Uses Fin types because of the use of $\text{Fin.sum_univ_succAbove}$ in the proof.

```

lemma stinespringGeneralForm_works {R : Type*} [RCLike R] {m r : ℕ}
  (K : Fin r → Matrix (Fin m) (Fin m) R) (z : Fin r)
  (M : Matrix (Fin m × Fin r) (Fin m × Fin r) R) :
  stinespringGeneralForm K z M = krausApply K

```

lemma stinespringUnitaryForm_works

[\[source\]](#)

Notice that unitarity is a side property, it is not why the Stinespring form works.

Here z is the coordinate used for the ancilla.

```

lemma stinespringUnitaryForm_works {R : Type*} [RCLike R] {m r : ℕ}
  {K : Fin r → Matrix (Fin m) (Fin m) R}
  (hK :  $\sum i, (K i)^H * K i = 1$ ) (z : Fin r) :
  stinespringUnitaryForm hK z = krausApply K

```

def krausCompletion

[\[source\]](#)

The "orthogonal" CPTP completion of a CPTNI map. vtilde is an alternative name for krausCompletion .

```

def krausCompletion {R : Type*} [RCLike R] {m r : ℕ}
  (K : Fin r → Matrix (Fin m) (Fin m) R) :
  Matrix (Fin m × Fin (r+1)) (Fin m) R

```

theorem stinespringOp_apply

[\[source\]](#)

Entrywise formula for the Stinespring isometry: its $((x_1, x_2), y)$ entry is $K x_2 x_1 y$.

```

theorem stinespringOp_apply {R : Type*} [Ring R] {m r : Type*} [Fintype r] [DecidableEq r]
  [Fintype m] [DecidableEq m] (K : r → Matrix m m R) (x : m × r) (y : m) :
  stinespringOp K x y = K x.2 x.1 y

```

theorem stinespringOp_gram

[\[source\]](#)

*The Gram matrix of the Stinespring isometry is $\sum i, (K i)^H * K i$.*

```

theorem stinespringOp_gram {R : Type*} [RCLike R] {m r : ℕ}
  (K : Fin r → Matrix (Fin m) (Fin m) R) :
  (stinespringOp K)^H * stinespringOp K =  $\sum i, \text{star } (K i) * K i$ 

```

lemma krausCompletion_isometry_of_TNI[\[source\]](#)

Mar 14, 2026 by Bjørn for 4.27 June 13, 2026 by Aristotle for 4.31 including stinespringOp_gram and stinespringOp_apply.

```
lemma krausCompletion_isometry_of_TNI {R : Type*} [RCLike R] {m r : ℕ}
  {K : Fin r → Matrix (Fin m) (Fin m) R}
  (hK : ∑ i, star K i * K i ≤ 1) :
  (krausCompletion K)H * krausCompletion K = 1
```

def unital[\[source\]](#)

A unital operator.

```
def unital {R : Type*} [RCLike R] {m r : ℕ}
  (K : Fin r → Matrix (Fin m) (Fin m) R)
```

def subunital[\[source\]](#)

A subunital operator.

```
def subunital {R : Type*} [RCLike R] {m r : ℕ}
  (K : Fin r → Matrix (Fin m) (Fin m) R)
```

lemma partialTrace_tensor[\[source\]](#)

The identity $Tr_B (A \otimes B) = Tr(B) \cdot A$

```
lemma partialTrace_tensor {R : Type*} [RCLike R] {m n : ℕ}
  (A : Matrix (Fin m) (Fin m) R) (B : Matrix (Fin n) (Fin n) R) :
  tr₂ (A ⊗ₖ B) = (trace B) • A
```

lemma krausApply_of_tensor[\[source\]](#)

A unitary dilation view of the application of a Kraus operator.

```
lemma krausApply_of_tensor {R : Type*} [RCLike R] {m r : ℕ}
  {K : Fin r → Matrix (Fin m) (Fin m) R}
  (hK : ∑ i, (K i)H * K i = 1) (z : Fin r)
  (ρ α : Matrix (Fin m) (Fin m) R) (β : Matrix (Fin r) (Fin r) R)
  (hβ : β.trace = 1) (h : (Ud hK z) * (ρ ⊗ₖ (single z z 1)) * (Ud hK z)H = α ⊗ₖ β) :
  krausApply K ρ = α
```

lemma trace_tr₂[\[source\]](#)

Trace of partial trace equals trace.

```
lemma trace_tr₂ {R : Type*} [RCLike R] {m n : ℕ}
  (ρ : Matrix (Fin m × Fin n) (Fin m × Fin n) R) :
  trace ρ = trace (tr₂ ρ)
```

def krausCompletionChannelMap[\[source\]](#)

The Kraus completion as a map from operations to channels.

```
def krausCompletionChannelMap {R : Type*} [RCLike R] {q r : ℕ}
  {K : Fin r → Matrix (Fin q) (Fin q) R} (hK : QuantumOperation K) :
  {K : Fin (r+1) → Matrix (Fin q) (Fin q) R | QuantumChannel K}
```

lemma CPTP_of_CPTNI

[\[source\]](#)

The "not orthogonal" CPTP completion of a CPTNI map.

```
lemma CPTP_of_CPTNI {R : Type*} [RCLike R]
  {q r : ℕ}
  {K : Fin r → Matrix (Fin q) (Fin q) R}
  (hq : QuantumOperation K) :
  ∃ K' : Fin (r+1) → Matrix (Fin q) (Fin q) R,
  QuantumChannel K' ∧
  ∀ i, ∀ H : i ≠ Fin.last r, K' i = K (i.1, Fin.val_lt_last H)
```

def partialTraceLeft

[\[source\]](#)

Partial trace on the left of a tensor product.

```
def partialTraceLeft {R : Type*} [RCLike R]
  {m n : Type*} [Fintype m]
  (ρ : Matrix (m × n)
    (m × n) R) : Matrix (n) (n) R
```

theorem stinespringForm_eq

[\[source\]](#)

A version of the Stinespring Dilation Theorem.

```
theorem stinespringForm_eq {R : Type*} [RCLike R] {m r : ℕ}
  (K : Fin r → Matrix (Fin m) (Fin m) R)
  (ρ : Matrix (Fin m) (Fin m) R) :
  tr₂ (stinespringDilation K ρ) = krausApply K ρ
```

2.110 PhyslibAlpha.SpaceAndTime.Space.Surfaces.HalfPlane

[PhyslibAlpha/SpaceAndTime/Space/Surfaces/HalfPlane.lean](#) on GitHub.

def halfPlaneDomain

[\[source\]](#)

The domain of the half-plane inside Space 2.

```
def halfPlaneDomain : Set (Space 2)
```

def halfPlane

[\[source\]](#)

The coordinate plane embedding used for the half-plane surface in Space 3.

```
def halfPlane : Space 2 → Space 3
```

lemma halfPlane_eq

[\[source\]](#)

```
lemma halfPlane_eq : halfPlane = (slice (2 : Fin 3)).symm ∘ (fun x : Space 2 => (0, x))
```

lemma halfPlane_injective[\[source\]](#)

```
lemma halfPlane_injective : Function.Injective halfPlane
```

lemma halfPlane_continuous[\[source\]](#)

```
@[fun_prop]
lemma halfPlane_continuous : Continuous halfPlane
```

lemma halfPlane_measurableEmbedding[\[source\]](#)

```
lemma halfPlane_measurableEmbedding : MeasurableEmbedding halfPlane
```

lemma norm_halfPlane[\[source\]](#)

```
@[simp]
lemma norm_halfPlane (x : Space 2) : ||halfPlane x|| = ||x||
```

def halfPlaneMeasure[\[source\]](#)

The measure on Space 3 corresponding to integration over a half-plane.

```
def halfPlaneMeasure : Measure (Space 3)
```

instance halfPlaneMeasure_hasTemperateGrowth[\[source\]](#)

```
instance halfPlaneMeasure_hasTemperateGrowth :
  halfPlaneMeasure.HasTemperateGrowth
```

instance halfPlaneMeasure_sFinite[\[source\]](#)

```
instance halfPlaneMeasure_sFinite : SFinite halfPlaneMeasure
```

def halfPlaneDist[\[source\]](#)

The distribution on Space 3 corresponding to integration over a half-plane.

```
def halfPlaneDist : (Space 3) →d[ℝ] ℝ
```

lemma halfPlaneDist_apply_eq_integral_halfPlaneMeasure[\[source\]](#)

```
lemma halfPlaneDist_apply_eq_integral_halfPlaneMeasure (f : S(Space 3, ℝ)) :
  halfPlaneDist f = ∫ x, f x ∂halfPlaneMeasure
```

lemma halfPlaneDist_apply_eq_integral_volume[\[source\]](#)

```
lemma halfPlaneDist_apply_eq_integral_volume (f : S(Space 3, ℝ)) :
  halfPlaneDist f = ∫ x, f (halfPlane x) ∂(volume.restrict halfPlaneDomain)
```

def halfPlaneSubmodule[\[source\]](#)*The coordinate plane containing the half-plane in Space 3.***def** halfPlaneSubmodule : Submodule $\mathbb{R}$ (Space 3) **where****lemma halfPlane_mem_halfPlaneSubmodule**[\[source\]](#)**lemma** halfPlane_mem_halfPlaneSubmodule (x : Space 2) : halfPlane x $\in$ halfPlaneSubmodule**lemma halfPlane_image_domain_subset_halfPlaneSubmodule**[\[source\]](#)**lemma** halfPlane_image_domain_subset_halfPlaneSubmodule :
halfPlane '' halfPlaneDomain $\subseteq$ (halfPlaneSubmodule : Set (Space 3))**lemma halfPlaneSubmodule_ne_top**[\[source\]](#)**lemma** halfPlaneSubmodule_ne_top : halfPlaneSubmodule $\neq$ $\top$ **lemma volume_halfPlane_image_domain**[\[source\]](#)**lemma** volume_halfPlane_image_domain :
volume (halfPlane '' halfPlaneDomain : Set (Space 3)) = 0**2.111 PhyslibAlpha.SpaceAndTime.Space.Surfaces.Line**

PhyslibAlpha/SpaceAndTime/Space/Surfaces/Line.lean on GitHub.

def line[\[source\]](#)*The coordinate line embedded in Space d.***def** line (d : $\mathbb{N}$) [NeZero d] : $\mathbb{R} \rightarrow$ Space d**lemma line_eq_smul_basis**[\[source\]](#)**lemma** line_eq_smul_basis (d : $\mathbb{N}$) [NeZero d] :
line d = **fun** r => r • basis (0 : Fin d)**lemma line_injective**[\[source\]](#)**lemma** line_injective (d : $\mathbb{N}$) [NeZero d] : Function.Injective (line d)**lemma line_continuous**[\[source\]](#)

@[fun_prop]

lemma line_continuous (d : $\mathbb{N}$) [NeZero d] : Continuous (line d)**lemma line_measurableEmbedding**[\[source\]](#)**lemma** line_measurableEmbedding (d : $\mathbb{N}$) [NeZero d] : MeasurableEmbedding (line d)

lemma norm_line[\[source\]](#)`@[simp]``lemma norm_line (d : ℕ) [NeZero d] (r : ℝ) : ||line d r|| = ||r||`**def lineMeasure**[\[source\]](#)*The measure on Space d corresponding to integration along a coordinate line.*`def lineMeasure (d : ℕ) [NeZero d] : Measure (Space d)`**instance lineMeasure\_hasTemperateGrowth**[\[source\]](#)`instance lineMeasure_hasTemperateGrowth (d : ℕ) [NeZero d] :
 (lineMeasure d).HasTemperateGrowth`**def lineDist**[\[source\]](#)*The distribution on Space d corresponding to integration along a coordinate line.
One can roughly think of this distribution as taking a test function f to its integral
against a mass, charge or current density concentrated on a line.*`def lineDist (d : ℕ) [NeZero d] : (Space d) →d[ℝ] ℝ`**lemma lineDist_apply_eq_integral_lineMeasure**[\[source\]](#)`lemma lineDist_apply_eq_integral_lineMeasure (d : ℕ) [NeZero d] (f : ℒ(Space d, ℝ)) :
 lineDist d f = ∫ x, f x ∂lineMeasure d`**lemma lineDist\_apply\_eq\_integral\_volume**[\[source\]](#)`lemma lineDist_apply_eq_integral_volume (d : ℕ) [NeZero d] (f : ℒ(Space d, ℝ)) :
 lineDist d f = ∫ r : ℝ, f (line d r)`**def lineSubmodule**[\[source\]](#)*The linear subspace spanned by the coordinate line in Space d.*`def lineSubmodule (d : ℕ) [NeZero d] : Submodule ℝ (Space d)`**lemma line\_mem\_lineSubmodule**[\[source\]](#)`lemma line_mem_lineSubmodule (d : ℕ) [NeZero d] (r : ℝ) : line d r ∈ lineSubmodule d`**lemma range\_line\_subset\_lineSubmodule**[\[source\]](#)`lemma range_line_subset_lineSubmodule (d : ℕ) [NeZero d] :
 Set.range (line d) ⊆ (lineSubmodule d : Set (Space d))`**lemma lineSubmodule\_ne\_top**[\[source\]](#)`lemma lineSubmodule_ne_top (d : ℕ) [NeZero d] (hd : 2 ≤ d) : lineSubmodule d ≠ ⊤`

lemma volume_line_range[\[source\]](#)

```
lemma volume_line_range (d : ℕ) [NeZero d] (hd : 2 ≤ d) :
  volume (Set.range (line d) : Set (Space d)) = 0
```

2.112 PhyslibAlpha.SpaceAndTime.Space.Surfaces.Ring

[PhyslibAlpha/SpaceAndTime/Space/Surfaces/Ring.lean](#) on GitHub.

def ring[\[source\]](#)

The embedding of the unit ring (a circle S^1 in Space 2) into Space 3.

```
def ring : Metric.sphere (0 : Space 2) 1 → Space 3
```

lemma ring_eq[\[source\]](#)

```
lemma ring_eq : ring = (slice 2).symm ∘ (fun x => (0, sphericalShell 2 x))
```

lemma ring_injective[\[source\]](#)

```
lemma ring_injective : Function.Injective ring
```

lemma ring_continuous[\[source\]](#)

```
@[fun_prop]
lemma ring_continuous : Continuous ring
```

lemma ring_measurableEmbedding[\[source\]](#)

```
lemma ring_measurableEmbedding : MeasurableEmbedding ring
```

def ringMeasure[\[source\]](#)

The measure on Space 3 corresponding to integration around a ring.

```
def ringMeasure : Measure (Space 3)
```

instance ringMeasure_hasTemperateGrowth[\[source\]](#)

```
instance ringMeasure_hasTemperateGrowth :
  ringMeasure.HasTemperateGrowth
```

instance ringMeasure_prod_volume_hasTemperateGrowth[\[source\]](#)

```
instance ringMeasure_prod_volume_hasTemperateGrowth :
  (ringMeasure.prod (volume (α := Space))).HasTemperateGrowth
```

instance ringMeasure_sFinite[\[source\]](#)

```
instance ringMeasure_sFinite: SFinite ringMeasure
```

instance ringMeasure_finite[\[source\]](#)

```
instance ringMeasure_finite: IsFiniteMeasure ringMeasure
```

lemma integrable_ringMeasure_of_continuous[\[source\]](#)

```
lemma integrable_ringMeasure_of_continuous (f : Space → ℝ) (hf : Continuous (f ∘ ring)) :
  Integrable f ringMeasure
```

lemma integrable_ringMeasure_of_continuous_euclid[\[source\]](#)

```
lemma integrable_ringMeasure_of_continuous_euclid (f : Space → EuclideanSpace ℝ (Fin n))
  (hf : Continuous (f ∘ ring)) :
  Integrable f ringMeasure
```

lemma ringMeasure_prod_volume_map[\[source\]](#)

```
lemma ringMeasure_prod_volume_map :
  (ringMeasure.prod (volume (α := Space))).map (fun x : Space × Space => (x.1, x.2 + x.1))
  = (ringMeasure.prod (volume (α := Space)))
```

lemma ringMeasure_univ[\[source\]](#)

```
@[simp]
lemma ringMeasure_univ : ringMeasure Set.univ = ENNReal.ofReal ((2 : ℝ) * π)
```

def ringDist[\[source\]](#)

The distribution on Space 3 corresponding to integration around a ring.

```
def ringDist : (Space 3) → d[ℝ] ℝ
```

lemma ringDist_apply_eq_integral_ringMeasure[\[source\]](#)

```
lemma ringDist_apply_eq_integral_ringMeasure (f : S(Space 3, ℝ)) :
  ringDist f = ∫ x, f x ∂ringMeasure
```

lemma ringDist_eq_integral_delta[\[source\]](#)

```
lemma ringDist_eq_integral_delta (f : S(Space 3, ℝ)) :
  ringDist f = ∫ z, diracDelta ℝ z f ∂ringMeasure
```

2.113 PhyslibAlpha.SpaceAndTime.Space.Surfaces.SolidCylinder

[PhyslibAlpha/SpaceAndTime/Space/Surfaces/SolidCylinder.lean](#) on GitHub.

def solidCylinder[\[source\]](#)

The map embedding a cross-sectional disk extruded along the axis into Space 3. The disk is cut out by the solid-cylinder measure, which is supported on the closed unit disk.


```
def solidCylinder : Space 2 × ℝ → Space 3
```

lemma solidCylinder_eq [\[source\]](#)

```
lemma solidCylinder_eq :
  solidCylinder = (slice 2).symm ∘ (fun x : Space 2 × ℝ => (x.2, x.1))
```

lemma solidCylinder_injective [\[source\]](#)

```
lemma solidCylinder_injective : Function.Injective solidCylinder
```

lemma solidCylinder_continuous [\[source\]](#)

```
@[fun_prop]
lemma solidCylinder_continuous : Continuous solidCylinder
```

lemma solidCylinder_measurableEmbedding [\[source\]](#)

```
lemma solidCylinder_measurableEmbedding : MeasurableEmbedding solidCylinder
```

lemma norm_solidCylinder [\[source\]](#)

```
@[simp]
lemma norm_solidCylinder (x : Space 2 × ℝ) :
  ||solidCylinder x|| = √(||x.2|| ^ 2 + ||x.1|| ^ 2)
```

def solidCylinderMeasure [\[source\]](#)

The measure on Space 3 corresponding to integration over a solid cylinder, i.e. the pushforward of the product of the solid-sphere (closed unit disk) measure on Space 2 with the line measure along the axis.

```
def solidCylinderMeasure : Measure (Space 3)
```

instance solidCylinderMeasure_hasTemperateGrowth [\[source\]](#)

```
instance solidCylinderMeasure_hasTemperateGrowth :
  solidCylinderMeasure.HasTemperateGrowth
```

instance solidCylinderMeasure_sFinite [\[source\]](#)

```
instance solidCylinderMeasure_sFinite : SFinite solidCylinderMeasure
```

def solidCylinderDist [\[source\]](#)

The distribution on Space 3 corresponding to integration over a solid cylinder. One can roughly think of this distribution as taking a test function f to its integral against a mass, charge or current density spread over a solid cylinder.

```
def solidCylinderDist : (Space 3) → d[ℝ] ℝ
```

lemma solidCylinderDist_apply_eq_integral_solidCylinderMeasure[\[source\]](#)

```
lemma solidCylinderDist_apply_eq_integral_solidCylinderMeasure (f :  $\mathcal{S}(\text{Space } 3, \mathbb{R})$ ) :
  solidCylinderDist f =  $\int x, f x \partial \text{solidCylinderMeasure}$ 
```

lemma solidCylinderDist_apply_eq_integral_disk_volume[\[source\]](#)

```
lemma solidCylinderDist_apply_eq_integral_disk_volume (f :  $\mathcal{S}(\text{Space } 3, \mathbb{R})$ ) :
  solidCylinderDist f =
     $\int x, f (\text{solidCylinder } x) \partial ((\text{solidSphereMeasure } 2).prod (\text{volume } (\alpha := \mathbb{R})))$ 
```

lemma solidCylinderMeasure_univ_pos[\[source\]](#)

```
lemma solidCylinderMeasure_univ_pos : 0 < solidCylinderMeasure Set.univ
```

2.114 PhyslibAlpha.SpaceAndTime.Space.Surfaces.SolidSphere

[PhyslibAlpha/SpaceAndTime/Space/Surfaces/SolidSphere.lean](#) on GitHub.

def solidSphere[\[source\]](#)

The inclusion into Space d of its closed unit ball B^d (the subtype coercion $x \mapsto x.1$).

```
def solidSphere (d :  $\mathbb{N}$ ) : Metric.closedBall (0 : Space d) 1  $\rightarrow$  Space d
```

lemma solidSphere_injective[\[source\]](#)

```
lemma solidSphere_injective (d :  $\mathbb{N}$ ) : Function.Injective (solidSphere d)
```

lemma solidSphere_continuous[\[source\]](#)

```
lemma solidSphere_continuous (d :  $\mathbb{N}$ ) : Continuous (solidSphere d)
```

lemma solidSphere_measurableEmbedding[\[source\]](#)

```
lemma solidSphere_measurableEmbedding (d :  $\mathbb{N}$ ) : MeasurableEmbedding (solidSphere d)
```

lemma norm_solidSphere_le[\[source\]](#)

```
@[simp]
lemma norm_solidSphere_le (d :  $\mathbb{N}$ ) (x : Metric.closedBall (0 : Space d) 1) :
  ||solidSphere d x||  $\leq$  1
```

def solidSphereMeasure[\[source\]](#)

The measure on Space d corresponding to integration over a solid sphere, i.e. the ambient volume measure restricted to the closed unit ball.

```
def solidSphereMeasure (d :  $\mathbb{N}$ ) : Measure (Space d)
```

instance solidSphereMeasure_isFiniteMeasure [\[source\]](#)

```
instance solidSphereMeasure_isFiniteMeasure (d : ℕ) :
  IsFiniteMeasure (solidSphereMeasure d)
```

instance solidSphereMeasure_hasTemperateGrowth [\[source\]](#)

```
instance solidSphereMeasure_hasTemperateGrowth (d : ℕ) :
  (solidSphereMeasure d).HasTemperateGrowth
```

def solidSphereDist [\[source\]](#)

The distribution on Space d corresponding to integration over a solid sphere. One can roughly think of this distribution as taking a test function f to its integral against a mass, charge or current density spread over a solid ball.

```
def solidSphereDist (d : ℕ) : (Space d) → d[ℝ] ℝ
```

lemma solidSphereDist_apply_eq_integral_solidSphereMeasure [\[source\]](#)

```
lemma solidSphereDist_apply_eq_integral_solidSphereMeasure (d : ℕ) (f : S(Space d, ℝ)) :
  solidSphereDist d f = ∫ x, f x ∂solidSphereMeasure d
```

lemma solidSphereDist_apply_eq_integral_closedBall [\[source\]](#)

```
lemma solidSphereDist_apply_eq_integral_closedBall (d : ℕ) (f : S(Space d, ℝ)) :
  solidSphereDist d f = ∫ x in Metric.closedBall (0 : Space d) 1, f x
```

lemma solidSphere_volume_pos [\[source\]](#)

```
lemma solidSphere_volume_pos (d : ℕ) :
  0 < volume (Metric.closedBall (0 : Space d) 1)
```

lemma solidSphereMeasure_univ_pos [\[source\]](#)

```
lemma solidSphereMeasure_univ_pos (d : ℕ) : 0 < solidSphereMeasure d Set.univ
```

2.115 PhyslibAlpha.SpaceAndTime.Space.Surfaces.SphericalCylinder

[PhyslibAlpha/SpaceAndTime/Space/Surfaces/SphericalCylinder.lean](#) on GitHub.

def sphericalCylinder [\[source\]](#)

The map embedding the unit circular shell extruded along the axis into Space 3.

```
def sphericalCylinder : Metric.sphere (0 : Space 2) 1 × ℝ → Space 3
```

lemma sphericalCylinder_eq [\[source\]](#)

```
lemma sphericalCylinder_eq :
  sphericalCylinder = (slice 2).symm ∘ (fun x => (x.2, sphericalShell 2 x.1))
```

lemma sphericalCylinder_injective[\[source\]](#)

```
lemma sphericalCylinder_injective : Function.Injective sphericalCylinder
```

lemma sphericalCylinder_continuous[\[source\]](#)

```
@[fun_prop]
```

```
lemma sphericalCylinder_continuous : Continuous sphericalCylinder
```

lemma sphericalCylinder_measurableEmbedding[\[source\]](#)

```
lemma sphericalCylinder_measurableEmbedding : MeasurableEmbedding sphericalCylinder
```

lemma norm_sphericalCylinder[\[source\]](#)

```
@[simp]
```

```
lemma norm_sphericalCylinder (x : Metric.sphere (0 : Space 2) 1 × ℝ) :  
  ||sphericalCylinder x|| = √(||x.2|| ^ 2 + 1)
```

def sphericalCylinderMeasure[\[source\]](#)

The measure on Space 3 corresponding to integration over a spherical cylinder.

```
def sphericalCylinderMeasure : Measure (Space 3)
```

instance sphericalCylinderMeasure_hasTemperateGrowth[\[source\]](#)

```
instance sphericalCylinderMeasure_hasTemperateGrowth :  
  sphericalCylinderMeasure.HasTemperateGrowth
```

instance sphericalCylinderMeasure_sFinite[\[source\]](#)

```
instance sphericalCylinderMeasure_sFinite : SFinite sphericalCylinderMeasure
```

def sphericalCylinderDist[\[source\]](#)

The distribution on Space 3 corresponding to integration over a spherical cylinder.

```
def sphericalCylinderDist : (Space 3) → d[ℝ] ℝ
```

lemma sphericalCylinderDist_apply_eq_integral_sphericalCylinderMeasure[\[source\]](#)

```
lemma sphericalCylinderDist_apply_eq_integral_sphericalCylinderMeasure (f : S(Space 3, ℝ)) :  
  sphericalCylinderDist f = ∫ x, f x ∂sphericalCylinderMeasure
```

lemma sphericalCylinderDist_apply_eq_integral_sphere_volume[\[source\]](#)

```
lemma sphericalCylinderDist_apply_eq_integral_sphere_volume (f : S(Space 3, ℝ)) :  
  sphericalCylinderDist f =  
  ∫ x, f (sphericalCylinder x)  
  ∂((MeasureTheory.Measure.toSphere volume).prod (volume (α := ℝ)))
```

2.116 PhyslibAlpha.SpaceAndTime.Space.Surfaces.SphericalShell

[PhyslibAlpha/SpaceAndTime/Space/Surfaces/SphericalShell.lean](#) on GitHub.

def sphericalShell [\[source\]](#)

The inclusion into Space d of its unit sphere $S^{\{d-1\}}$ (the subtype coercion $x \mapsto x.1$).

```
def sphericalShell (d : ℕ) : Metric.sphere (0 : Space d) 1 → Space d
```

lemma sphericalShell_injective [\[source\]](#)

```
lemma sphericalShell_injective (d : ℕ) : Function.Injective (sphericalShell d)
```

lemma sphericalShell_continuous [\[source\]](#)

```
lemma sphericalShell_continuous (d : ℕ) : Continuous (sphericalShell d)
```

lemma sphericalShell_measurableEmbedding [\[source\]](#)

```
lemma sphericalShell_measurableEmbedding (d : ℕ) : MeasurableEmbedding (sphericalShell d)
```

lemma norm_sphericalShell [\[source\]](#)

```
@[simp]
lemma norm_sphericalShell (d : ℕ) (x : Metric.sphere (0 : Space d) 1) :
  ||sphericalShell d x|| = 1
```

def sphericalShellMeasure [\[source\]](#)

The measure on Space d corresponding to integration around a spherical shell.

```
def sphericalShellMeasure (d : ℕ) : Measure (Space d)
```

instance sphericalShellMeasure_hasTemperateGrowth [\[source\]](#)

```
instance sphericalShellMeasure_hasTemperateGrowth (d : ℕ) :
  (sphericalShellMeasure d).HasTemperateGrowth
```

def sphericalShellDist [\[source\]](#)

The distribution associated with a spherical shell. One can roughly think of this distribution as the distribution which takes test functions $f(r)$ to $\int d^3r f(r) \rho(r)$ where $\rho(r)$ is the mass, charge or current etc. distribution.

```
def sphericalShellDist (d : ℕ) : (Space d) → d[ℝ] ℝ
```

lemma sphericalShellDist_apply_eq_integral_sphericalShellMeasure [\[source\]](#)

```
lemma sphericalShellDist_apply_eq_integral_sphericalShellMeasure (d : ℕ) (f : S(Space d, ℝ)) :
  sphericalShellDist d f = ∫ x, f x ∂sphericalShellMeasure d
```

lemma sphericalShellDist_apply_eq_integral_sphere_volume[\[source\]](#)

```

lemma sphericalShellDist_apply_eq_integral_sphere_volume (d : ℕ) (f :  $\mathcal{S}(\text{Space } d, \mathbb{R})$ ) :
  sphericalShellDist d f =
   $\int x, f (\text{sphericalShell } d \ x) \ \partial(\text{MeasureTheory.Measure.toSphere volume})$ 

```

2.117 QuantumInfo.States.Mixed.MState

[QuantumInfo/States/Mixed/MState.lean](#) on GitHub.

theorem traceNorm_eq_one[\[source\]](#)

```

@[simp]
theorem traceNorm_eq_one (p : MState d) : p.m.traceNorm = 1

```

2.118 scripts.MetaPrograms.module_doc_lint

[scripts/MetaPrograms/module_doc_lint.lean](#) on GitHub.

def linterExemptions[\[source\]](#)

*The array of modules exempt from all linters, read from scripts/LinterExemption.txt.
This is used to lint QuantumInfo file-by-file.*

```

def linterExemptions : IO (Array FilePath)

```

def importedFilePaths[\[source\]](#)

The file paths of the modules imported into the module mods (e.g. Physlib).

```

def importedFilePaths (mods : Name) : IO (Array FilePath)

```

2.119 scripts.MetaPrograms.runPhyslibLinters

[scripts/MetaPrograms/runPhyslibLinters.lean](#) on GitHub.

def linterExemptions[\[source\]](#)

*The file paths of modules exempt from all linters, read from scripts/LinterExemption.txt.
This is used to lint QuantumInfo file-by-file.*

```

def linterExemptions : IO (Array String)

```

def moduleToFilePathString[\[source\]](#)

The file path (as a string e.g. QuantumInfo/Foo/Bar.lean) of the module mod.

```

def moduleToFilePathString (mod : Name) : String

```

2.120 scripts.MetaPrograms.style_lint

[scripts/MetaPrograms/style_lint.lean](#) on GitHub.

def importedFilePaths[\[source\]](#)

The file paths of the modules imported into the module mods (e.g. Physlib), found by reading the module's .olean file.

```
def importedFilePaths (mods : Name) : IO (Array FilePath)
```

def linterExemptions[\[source\]](#)

The file paths of modules which should be skipped by the linters, read from scripts/LinterExemption.txt. This is used to lint QuantumInfo file-by-file.

```
def linterExemptions : IO (Array String)
```